

LECTURES ON

Static and Dynamic Game Theory

Foundations for Multi-Agent Systems and Control



TAMER BAŞAR
QUANYAN ZHU

Lectures on Static and Dynamic Game Theory: Foundations for Multi-Agent Systems and Control

Tamer Başar and Quanyan Zhu

Sample: Part I — Fundamentals of Game Theory

September 25, 2026

Contents

List of Figures	xiii
List of Tables	xv
List of Algorithms	xvii
Preface	xix
I Fundamentals of Game Theory	1
1 Introduction to Game Theory	3
1.1 Introduction to Game Theory	3
1.2 Games as Models of Strategic Situations	4
1.3 The Past and the Present	5
1.4 A First Dive into Game Theory	7
1.5 Why Study Game Theory?	12
1.6 Structure of These Lecture Notes	13
1.7 Game Theory in Action: Modern Applications and Their Theoretical Foundations	17
1.8 Further Notes and Reading	25
1.9 Exercises	26
2 Matrix Games	29
2.1 Two-Person Zero-Sum Games	29
2.2 Saddle-point Equilibrium	32
2.3 Mixed Strategies	34
2.4 Computing Saddle-Point Equilibria	37
2.5 Fictitious Play	44
2.6 Further Notes and Reading	49
2.7 Exercises	52
3 Nash Equilibrium: Concepts and Definition	55
3.1 Finite N -Person Nonzero-Sum Static Games	55
3.2 Nash Equilibrium	58
3.3 Examples Illustrating Nash Equilibria	58
3.4 Strategic Equivalence	62
3.5 Properties of Nash Equilibria	63
3.6 Further Notes and Reading	77

3.7	Exercises	77
4	Nash Equilibrium: Properties and Computation	83
4.1	Computational Methods for Mixed-Strategy Nash Equilibrium	83
4.2	Special Classes of Games	92
4.3	From Dominance to Rationalizability to Nash Equilibrium: Motivation and Interpretation	99
4.4	Further Notes and Reading	102
4.5	Exercises	103
5	Extensive-Form Games	107
5.1	Introduction to Extensive-Form Games	107
5.2	From Extensive-Form to Normal-Form Games	117
5.3	Mixed and Behavioral Strategies in Extensive-Form Games	121
5.4	Kuhn's Theorem and Perfect Recall	132
5.5	Further Notes and Reading	136
5.6	Exercises	137
6	Stackelberg (Leader-Follower) Games	143
6.1	The Extensive-Form View	144
6.2	From the Tree to the Normal Form	145
6.3	Stackelberg Equilibria in Pure Strategies	148
6.4	Stackelberg Equilibria in Mixed Strategies	152
6.5	The Bilevel View and the Computation of Optimal Commitments	157
6.6	Stackelberg Equilibria in Behavioral Strategies	158
6.7	An Application: Stackelberg Security Games	161
6.8	A Case Study: Stackelberg Network Design and Defense	164
6.9	One Leader, Many Followers: Designing the Followers' Equilibrium	168
6.10	Further Notes and Reading	171
6.11	Exercises	173
7	Equilibrium Refinement	177
7.1	Motivation for Refinements of Nash Equilibrium	177
7.2	Trembling-Hand Perfect Equilibrium	179
7.3	ε -Perfect Equilibrium	185
7.4	Proper Equilibrium	188
7.5	Other Refinements	192
7.6	Further Notes and Reading	197
7.7	Exercises	198
8	Correlated Equilibrium	203
8.1	Motivation: Coordination in the Game of Chicken	203
8.2	Correlated Equilibrium	205
8.3	Extensive-Form Interpretation of Correlated Equilibrium	212
8.4	Bayesian Interpretation of Correlated Equilibrium	215
8.5	Properties of Correlated Equilibria	216
8.6	No-Regret Learning	219
8.7	From No-Regret Learning to Equilibrium	221
8.8	Regret Matching in the Game of Chicken	223
8.9	Further Notes and Reading	226
8.10	Exercises	226

II	Continuous-Kernel Nash and Stackelberg Games	231
9	Infinite and Continuous-Kernel Games	233
9.1	From Finite Games to Continuous-Kernel Games	233
9.2	Existence of Pure-Strategy Nash Equilibria	235
9.3	Mixed-Strategy Nash Equilibria	243
9.4	Stability and Computation of Nash Equilibria	245
9.5	Constrained Games and Generalized Nash Equilibria	255
9.6	Further Notes and Reading	260
9.7	Exercises	262
10	Continuous-Kernel Stackelberg Games	265
10.1	Hierarchical Decision Making with Continuous Action Sets	265
10.2	ϵ -Stackelberg Strategies and Existence of Equilibria	267
10.3	The Stackelberg Duopoly	272
10.4	Mixed-Strategy Stackelberg Solutions	275
10.5	Bilevel Programming and Mathematical Programs with Equilibrium Con- straints	278
10.6	Inverse Stackelberg Games: Committing to a Policy	280
10.7	Case Study: Demand Response Management in the Smart Grid	283
10.8	Further Notes and Reading	286
10.9	Exercises	288
11	Quadratic and Network Games	293
11.1	Quadratic Games: Formulation	293
11.2	Nash Equilibria: Existence, Uniqueness, and Computation	294
11.3	Zero-Sum Quadratic Games	298
11.4	Team Problems	299
11.5	The Stackelberg Solution	300
11.6	Stochastic Quadratic Games	301
11.7	Does More Information Help?	307
11.8	Quadratic Games over Networks	312
11.9	Further Notes and Reading	316
11.10	Exercises	318
III	Games with Incomplete Information	321
12	Bayesian Games	323
12.1	Harsanyi's Bayesian Games	324
12.2	Incomplete Information as a Foundation for Mixed Strategies	333
12.3	Aumann Model of Incomplete Information	335
12.4	Further Notes and Reading	345
12.5	Exercises	346
13	Auction Games	351
13.1	Auctions	351
13.2	First-Price Auction	354
13.3	Second-Price Auction	357
13.4	Third-Price Auction	358
13.5	Revenue Equivalence	360

13.6	Non-Bayesian Frameworks	366
13.7	Further Notes and Reading	368
13.8	Exercises	369
14	Revenue-Optimal Auction Design	371
14.1	Single-Item Auction Design	371
14.2	Optimal Mechanism Design	380
14.3	Further Investigation	385
14.4	Further Notes and Reading	392
14.5	Exercises	392
15	Mechanism Design Theory	397
15.1	Mechanism Design for Complete-Information Games	398
15.2	Mechanism Design for Bayesian Games	404
15.3	Contract Design as a Bayesian Mechanism	409
15.4	Information Design	414
15.5	Further Notes and Reading	419
15.6	Exercises	420
IV	Dynamic and Repeated Games	423
16	Extensive-Form Dynamic Games	425
16.1	Subgame Perfect Equilibrium	425
16.2	Sequential Equilibrium	439
16.3	Further Notes and Reading	447
16.4	Exercises	448
17	Principal-Agent Games	453
17.1	Principal-Agent Model: Signaling	454
17.2	Principal-Agent Model: Screening	469
17.3	Principal-Agent Model: Moral Hazard	477
17.4	Further Notes and Reading	484
17.5	Exercises	485
18	Repeated Games	489
18.1	Basic Models of Repeated Games	489
18.2	Equilibrium Concepts under Information Structures	496
18.3	One-Stage Deviation Principle	500
18.4	Folk Theorem under Perfect Monitoring	504
18.5	Dynamic Programming Principle: Finite Horizon	509
18.6	Dynamic Programming Principle (Infinite Horizon Case)	513
18.7	Further Notes and Reading	516
18.8	Exercises	516
19	Markov Games	519
19.1	Markov Games as Extensive-Form Games	519
19.2	Infinite-Horizon Discounted Markov Games with Complete Information	525
19.3	Computing Stationary Markov Perfect Equilibria	535
19.4	One-Sided Partially Observable Markov Games	537
19.5	Further Notes and Reading	545

19.6	Exercises	546
20	Dynamic Infinite Games	551
20.1	General Formulation	551
20.2	Informational Non-Uniqueness of Nash Equilibria	558
20.3	Open-Loop and Closed-Loop Feedback Nash Equilibria	563
20.4	Removal of Informational Non-Uniqueness by Infinitesimal Random Per- turbations	572
20.5	Further Notes and Reading	576
20.6	Exercises	577
V	Differential and Stochastic Games	581
21	Differential Games	583
21.1	Formulation of Differential Games	583
21.2	Open-Loop Nash Equilibrium	590
21.3	Closed-Loop Feedback Nash Equilibrium	593
21.4	Zero-Sum Differential Games and the Isaacs Equation	598
21.5	Differential Games with Structural and Modal Uncertainties	603
21.6	An Illustrative Application: Multiuser Flow Control	610
21.7	Further Notes and Reading	614
21.8	Exercises	615
22	Dynamic Stackelberg Games	619
22.1	Open-Loop Stackelberg Solutions	620
22.2	Feedback Stackelberg Solutions	622
22.3	Global Stackelberg Solutions and the Power of Dynamic Information	624
22.4	Deterministic Incentives	630
22.5	Stochastic Dynamic Stackelberg Games	637
22.6	Stochastic Incentives	639
22.7	Stackelberg Differential Games	642
22.8	From Incentives to Mechanism Design	645
22.9	Further Notes and Reading	646
22.10	Exercises	647
23	Stochastic Differential Games	653
23.1	Stochastic Differential Games: The Risk-Neutral Baseline	653
23.2	Problem Formulation: Risk-Sensitive Stochastic Differential Games	665
23.3	Risk-Sensitive Stochastic Control	667
23.4	The Risk-Sensitive Game: Nash Equilibrium and Its Expansions	670
23.5	Minimax Estimation as a Zero-Sum Dynamic Game	674
23.6	Further Notes and Reading	683
23.7	Exercises	684
24	Mean-Field Games	687
24.1	Motivation: Why Mean-Field Games?	687
24.2	The Finite-Population Game	690
24.3	The Mean-Field Limit	691
24.4	Risk-Neutral Mean-Field Games	692
24.5	Mean-Field Games versus Mean-Field Control	705

24.6	Risk-Sensitive Mean-Field Games	709
24.7	Further Notes and Reading	717
24.8	Exercises	719
VI	Cooperative and Matching Games	723
25	Nash Bargaining	725
25.1	The Bargaining Problem	725
25.2	The Nash Bargaining Problem	732
25.3	Extensions and Generalizations of the Nash Bargaining Solution	739
25.4	Further Notes and Reading	741
25.5	Exercises	741
26	Coalition Games	745
26.1	Coalition Games	745
26.2	The Nucleolus	754
26.3	Shapley Value	756
26.4	Further Notes and Reading	764
26.5	Exercises	765
27	Matching Games	769
27.1	Gale-Shapley Matching	770
27.2	Many-to-One Matching	779
27.3	Top Trading Cycles (TTC)	782
27.4	Matching Markets and Platform Design	786
27.5	Further Notes and Reading	790
27.6	Exercises	791
VII	Advanced Topics	795
28	Routing Games	797
28.1	Routing Games	797
28.2	Wardrop Equilibrium	799
28.3	Social Optimum	802
28.4	Price of Anarchy	803
28.5	Braess's Paradox	806
28.6	Mean-Field View of Wardrop Equilibrium	810
28.7	Further Notes and Reading	817
28.8	Exercises	818
29	Evolutionary Games	821
29.1	Hawk-Dove Game as a Motivating Example	821
29.2	Population Game Model and Evolutionarily Stable Strategy	825
29.3	Replicator Dynamics	827
29.4	Connection Between Nash Equilibrium and ESS	832
29.5	Further Notes and Reading	836
29.6	Exercises	837
30	Learning in Games	841

30.1	Introduction to General Forms of Learning in Games	841
30.2	Regularized Fictitious Play	844
30.3	Evolutionary Games and Dynamics	852
30.4	Learning in Markov Games	856
30.5	Further Notes and Reading	865
30.6	Exercises	866
A	Elements of Optimal Control Theory	871
A.1	Dynamic Programming for Discrete-Time Systems	871
A.2	Dynamic Programming for Continuous Time: The HJB Equation	873
A.3	The Minimum Principle	876
A.4	Representations of Strategies and Time Consistency	879
A.5	Further Notes and Reading	881
A.6	Exercises	882
B	Mathematical Background	885
B.1	Vector Spaces, Norms, and Completeness	885
B.2	The Banach Fixed Point Theorem	886
B.3	Convex Sets and Convex Functions	887
B.4	Tangent Cones, Normal Cones, and Optimality Conditions	888
B.5	Variational Inequalities	889
B.6	Saddle Points and Minimax	891
B.7	Itô Stochastic Differential Equations	892
B.8	Further Notes and Reading	896
B.9	Exercises	896
	Bibliography	899
	Index	923

List of Figures

1.1	A brief timeline of game theory, from early precursors through its formal foundations to its recognition across the sciences and its integration into modern artificial intelligence	8
1.2	Normal Form (Prisoner's Dilemma)	9
1.3	Extensive Form with Information Set	9
1.4	Roadmap of the seven parts	17
1.5	AlphaGo Zero as a self-play loop	19
1.6	Pac-Man as a multi-agent stochastic (Markov) game	20
1.7	The Colonel Blotto game	21
1.8	The pursuer is restricted to motion along the circular boundary of radius R , shown as an arc between its initial position and the potential exit point E . . .	22
1.9	The Shapley value as feature attribution	23
1.10	Evolutionary game theory in cancer therapy	25
2.1	Graphical method for P1	38
2.2	Graphical method for P2	39
2.3	Graphical solution of the 2×3 game	40
2.4	Fictitious play in the 3×3 zero-sum game	46
2.5	Fictitious play in the 3×3 zero-sum game	47
2.6	Running average of realized stage losses $\bar{\ell}_t$ converging to the game value $v^* = 1$.	47
2.7	Diagnostic	48
3.1	The correspondence $F(x) = [0, 1 - x]$ illustrated by shading the feasible region	64
3.2	The jump correspondence $F(x) = \{0\}$ for $x < 0$, $F(0) = \{0, 1\}$, $F(x) = \{1\}$ for $x > 0$	65
3.3	Graph of the correspondence F	66
3.4	One-dimensional intuition	70
3.5	Two-dimensional intuition for Brouwer's Fixed Point Theorem	71
3.6	Payoff table for the first few columns	73
3.7	P2's payoff increases with column j according to $(j - 1)/j$ and approaches 1 as $j \rightarrow \infty$, but never reaches it	74
3.8	P1's payoff as a function of a_1 when $a_2 = 0.6$	76
3.9	The correspondence $F : [0, 1] \rightrightarrows [0, 1]$	80
3.10	Annulus $1 \leq x^2 + y^2 \leq 2$ with the mapping f that projects every point to the inner circle	81
4.1	Best-response correspondences for the Battle of the Sexes	85

5.1	Extensive-form representation in which P1 moves first and P2 does not observe Player 1's action.	108
5.2	Strategically equivalent extensive-form representation in which P2 moves first and P1 does not observe Player 2's action.	109
5.3	An extensive-form game with a payoff-relevant and information-relevant move by Nature	110
5.4	A complete-information extensive-form game with a chance move	110
5.5	Modified Matching Pennies	117
5.6	Zero-sum extensive-form game	125
5.7	Dynamic game tree with shaded, color-coded information sets	130
5.8	Absent-minded driver game	133
5.9	Mapping a mixed strategy to an equivalent behavioral strategy under imperfect recall, with decision vertices (solid) and terminal vertices (hollow).	135
5.10	Extensive-form game with a clearly visible dotted information set enclosing the two Mugger nodes under <i>Not notice</i> and <i>Take</i>	138
5.11	Game 1: P1's two right-branch nodes are in the same information set.	140
5.12	Game 2: All three of P1's decision nodes aligned horizontally within a single information set (imperfect recall).	141
6.1	Extensive form of the entry-deterrence game	145
6.2	The leader's pessimistic (worst-case) average cost in Example 6.18, as a function of her mixing probability y_1	155
6.3	The three polar architectures of Example 6.31 ($n = 7, k = 2$)	168
7.1	A simple two-stage extensive-form game.	184
7.2	Extensive-form representation of the game with labeled decision nodes and terminal histories.	199
8.1	Geometry of the <i>correlated equilibrium</i> (CE) set in the Game of Chicken, shown via a projection of the four-dimensional probability simplex	208
8.2	Geometry of the correlated equilibrium (CE) and coarse correlated equilibrium (CCE) sets in the Game of Chicken, shown in the same projection of the probability simplex as Figure 8.1	212
8.3	Extensive-form representation of the Game of Chicken with a correlation device	213
8.4	Bayesian interpretation of a correlated equilibrium	217
8.5	Internal-regret matching dynamics in the Game of Chicken for $\varepsilon = 0.02$ over $T = 20,000$ rounds	225
9.1	Reaction curves of the symmetric Cournot duopoly ($a - c = 1, b = 1$)	239
9.2	The minimizer's worst case in the game $J = (u_1 - u_2)^2$ on $[0, 2] \times [0, 2]$	244
9.3	Cobweb diagram of sequential best-response dynamics in the symmetric Cournot duopoly ($a - c = 1, b = 1$)	249
9.4	Reaction curves of a two-user power-control game with spreading gain $G = 4$ and $a_1 = 1$, in received-power coordinates	253
9.5	Parallel best-response updates in the symmetric power-control game with $G = 4$, $a_i = 1$, and common initial power $y(0) = 0.1$ (by symmetry, all users' iterates coincide, and hence a single trajectory is shown for each case)	255
9.6	Geometry of the shared-budget game of Example 9.41	259
10.1	The leader's worst-case cost in the game of Remark 10.10	269
10.2	The Stackelberg duopoly with $a - c = 1, b = 1$	274

10.3	The demand response management game of [72]	283
11.1	The star network of Example 11.20 with $a_i = 1$ and $\phi = 1/2$	315
11.2	Equilibrium efforts in the star network of Example 11.20 as functions of the complementarity strength ϕ (with $a_i = 1$)	316
12.1	Timeline of informational stages in a Bayesian game	329
12.2	Information graph of states	339
13.1	Equilibrium bidding in first-, second-, and third-price auctions. Bidding functions $\mu(\theta)$ are shown for three value distributions with $N = 6$ bidders.	361
14.1	Virtual valuation functions $\varphi(\theta)$ for eight distributions	379
14.2	Case study of Myerson ironing. Panel (a) shows the non-monotone virtual value $\varphi(\theta)$	392
15.1	Structure of an indirect mechanism	406
15.2	Structure of a direct mechanism	406
15.3	Direct vs. indirect mechanisms. In an indirect mechanism, types θ map to bids b via strategies β , and the allocation and payments are given by (π, q)	407
16.1	Extensive form of the Entry-Deterrence Game Γ_{ED}	426
16.2	Four-turn alternating game	431
16.3	Signaling game	432
16.4	Two-person extensive-form game with terminal nodes $o_1, \dots, o_6$ and payoffs yielding a mixed strategy for Player A.	442
16.5	Player I chooses T or B	444
17.1	Information flow in a basic signaling game.	453
17.2	Extensive form of the two-type signaling game	460
17.3	The Rothschild-Stiglitz separating equilibrium in the state-contingent consumption plane for the parameters of Example 17.16	476
17.4	The incentive-insurance tradeoff in Example 17.32	484
17.5	The Beer-Quiche signaling game.	486
18.1	The grim-trigger incentive constraint in the repeated Prisoner's Dilemma of Example 18.14	504
18.2	Normalized discounted payoff $v(\delta)$ of the deterministic 100-stage cycle of Example 18.15, as a function of the discount factor	507
18.3	Feasible payoff set F (blue) of the repeated Prisoner's Dilemma and its individually rational subset F^{IR} (orange), the intersection of F with the minmax lower bounds $v_i \geq 1$ (gray dashed lines)	508
19.1	Timing structure of a single round in the Markov game	520
19.2	Extensive-form representation of a dynamic game with imperfect information	521
19.3	Shapley value iteration for the two-state zero-sum Markov game with $\delta = \frac{3}{4}$, started from $V^0(1) = 0$	532
19.4	Asymmetric LQ Markov game	534
19.5	Belief-dependent equilibrium policies. (a) Value function $V(b)$	545
19.6	Algorithmic convergence of value iteration. (a) Sup-norm change in the value function $\ V^k - V^{k-1}\ _\infty$	546

20.1	Price stickiness as memory in the duopoly of Example 20.1	553
20.2	The discrete-time dynamic game of Definition 20.2 as a closed loop	554
20.3	Open-loop versus feedback equilibria in Example 20.23: trajectories and cost decomposition	571
20.4	Noise selects the disturbance-robust equilibrium	573
21.1	Pursuit and evasion in the plane: geometry, a representative chase, and the effect of the information pattern	585
21.2	Open-loop versus feedback equilibrium gains in Example 21.18 with $T = 1$. . .	597
21.3	The three regimes of the scalar game Riccati equation of Example 21.20, plotted against the time to go	603
21.4	The multiuser flow-control model	611
21.5	Free riding in the symmetric flow-control game (normalized to $c = 1$)	613
22.1	Geometry of the global Stackelberg solution of Example 22.11, drawn for $x_1 = 1$ in the (u^2, u^1) -plane	626
22.2	Geometry of the incentive in Example 22.16	633
22.3	Geometry of Example 22.19	635
22.4	Information structure of the two-stage stochastic games of Examples 22.21 and 22.22	639
22.5	The scalar linear-quadratic Stackelberg differential game of Example 22.26 . . .	645
23.1	The open-loop equilibrium of Example 23.4	658
23.2	The two halves of the equilibrium value $W(t, x) = \frac{1}{2}Z(t)x^2 + \ell(t)$ in Example 23.10 ($r = 0.5, t_f = 5$)	664
23.3	Closed-loop equilibrium trajectories $dx = -NZ(t)x dt + \sqrt{\epsilon}e^{-at}db$ in Example 23.10 ($N = 2, r = 0.5, t_f = 5, \epsilon = 0.09, x_0 = 1$), driven by the <i>same</i> Brownian sample path under two noise decay rates	664
23.4	The risk-sensitive criterion of Example 23.11	666
23.5	Solutions of the LEQG Riccati equation in Example 23.12 with $q_f = 5, t_f = 1$, integrated backward from $Z(t_f) = 5$	669
23.6	The worst-case error ratio $\rho(k) = \sqrt{(1-k)^2 + k^2/r}$ over linear estimator gains for the scalar channel of Example 23.17, for three noise weights r	678
23.7	Certification of the filtering level in Example 23.19	682
24.1	The mean-field game loop	688
24.2	One simulated path of the empirical mean M_t^N of the N -player game of Example 24.15 under the mean-field equilibrium feedback, for $N \in \{5, 50, 500\}$, against the mean-field prediction $\bar{m}_t \equiv 1$ (dashed)	700
24.3	Convergence of the empirical measure to the mean field in Wasserstein distance, at the mid-horizon time $t = T/2$, as a function of the population size (log-log scale; each point averages several hundred independent populations) . .	701
24.4	The population distribution itself converging to the mean field	701
24.5	Equilibrium effort field of the rank-based competition (Example 24.16; $R(x, r) = r, \sigma = 1, c = \frac{1}{2}, T = 1$), computed by Picard iteration on the fixed-point map Φ 704	
24.6	The equilibrium terminal distribution of the rank-based competition (bars) against the law the population would have if nobody exerted effort (red curve) .	704
24.7	From N players to the mean field	705
24.8	The mean channel of Example 24.19 for $\kappa = \frac{3}{4}, T = 1, \bar{m}_0 = 1$	709

24.9	Equilibrium tracking gains in the risk-sensitive mean-field game of Example 24.26, horizon $T = 1$	714
24.10	Equilibrium population variance (24.23) in the risk-sensitive mean-field game of Example 24.26 ($T = 3, \epsilon = 0.4, v_0 = 0.25$)	715
24.11	Evolution of the equilibrium mean field $m(t, \cdot)$ in the risk-sensitive mean-field game of Example 24.26 ($T = 3, \epsilon = 0.4, v_0 = 0.25$)	716
24.12	The equilibrium value function (24.22) of Example 24.26 ($T = 3, \epsilon = 0.4$), on identical axes	716
24.13	The equilibrium feedback $\mu_\delta^*(t, x) = -Z^\delta(t)(x - \bar{m}_0)$ of Example 24.26 at four times ($T = 3, \epsilon = 0.4$), on identical axes	717
25.1	Split-the-pie (ultimatum) game with a shaded half-pie	727
25.2	Two-stage split-the-pie bargaining game	728
25.3	Mapping of allocation space X to a convex utility space U via $u_1(\cdot), u_2(\cdot)$	733
25.4	Larger convex bargaining set U (shaded) with a concave Pareto frontier ∂U (label raised)	740
26.1	Network formulation	761
27.1	Round 1 of the deferred acceptance process, consistent with Example 27.1	773
27.2	Round 2: After m_2 's rejection by w_1 , he proposes to w_3 , who accepts. The tentative matching is now (m_1, w_1) , (m_2, w_3) , and (m_3, w_2)	773
27.3	The resulting directed graph G^0 in Round 1 of the TTC algorithm for the four-agent housing market	784
28.1	A simple network with two s - t routes	799
28.2	Braess's network	806
28.3	Five-link directed network serving two origin-destination pairs.	818
29.1	Fitness of Hawks and Doves as a function of the population composition p in the numerical Hawk-Dove game ($V = 8, C = 10$)	823
29.2	Bifurcation diagram of the Hawk-Dove game as the resource-to-cost ratio $r = V/C$ varies	824
29.3	Replicator dynamics for the Hawk-Dove game	829
29.4	Numerical solutions of the replicator ODE $\dot{p} = p(1-p)(1-5p)$ for different initial conditions	830
29.5	Replicator orbits of the zero-sum Rock-Paper-Scissors game on the simplex $\Delta(S)$, computed numerically from the initial conditions $(0.5, 0.25, 0.25)$, $(0.7, 0.15, 0.15)$, and $(0.88, 0.06, 0.06)$	831
30.1	Learning dynamics in Shapley's game	851
30.2	Hawk-Dove dynamics. Replicator ODE (dashed) and finite-population stochastic imitation (solid, mean $\pm$ s.d., $N = 800$) under time rescaling $t = (\alpha/N)k$	854
30.3	Zero-sum Markov game learning dynamics under Softmax exploration	859
A.1	Time-optimal synthesis for the double integrator (Example A.11)	878
B.1	Tangent and normal cones	889
B.2	Brownian motion, and why squared increments behave like dt	893
B.3	Itô's correction term is not optional	895

List of Tables

2.1	Early rounds of fictitious play in the 3×3 zero-sum game with initial counts $\text{counts}_p^0 = (5, 0, 5)$ and $\text{counts}_q^0 = (10, 0, 0)$	45
3.1	P1's payoff matrix in the 3×3 discretized game.	75
4.1	P1's payoff table in the three-level joint-effort game.	97
5.1	Matching Pennies	108
5.2	Normal form for the modified Matching Pennies game with perfect information for Player 2.	119
10.1	Cournot-Nash versus Stackelberg outcomes in the symmetric linear duopoly.	273
14.1	CDF $F(\theta)$, density $f(\theta)$, and virtual valuation $\varphi(\theta)$ for commonly used distributions in auction theory	380
17.1	Contract menu and expected utilities in Example 17.16	477
17.2	Zero-profit insurance arrangements in Example 17.32	483
24.1	Mean-field games versus mean-field control	707

List of Algorithms

Algorithm 8.1	Internal-Regret Matching for the Game of Chicken (Player i)	224
Algorithm 30.1	Regularized Fictitious Play	845

Preface

Part I

Fundamentals of Game Theory

Chapter 1

Introduction to Game Theory

1.1 ■ Introduction to Game Theory

Game theory is the mathematical study of strategic interactions among rational decision-makers. It provides a systematic framework for analyzing situations in which the outcome for each participant depends not only on her own choices but also on the actions, beliefs, and potential reactions of others. In contrast to classical decision theory, which models agents as isolated optimizers operating in a fixed and unresponsive environment, game theory emphasizes *interdependence*. Each player must form expectations about the behavior of others, often reasoning about how others are reasoning about themselves, creating a recursive structure of strategic beliefs. This recursive reasoning makes game theory indispensable for understanding environments where interaction is competitive, cooperative, or adversarial in nature.

In game-theoretic terminology, the decision-makers are referred to as *players* (or *agents*). Each player has an *objective function* that encodes her preferences over possible outcomes. When the objective function is to be maximized, it is referred to as a *utility* or *benefit* function; when it is to be minimized, it is called a *cost* or *loss* function. In these Lecture Notes, finite and static games are typically formulated in terms of utility functions u_i that players maximize, whereas continuous-kernel and differential games are formulated in terms of cost functions (denoted J_i or L^i) that players minimize; each lecture states its convention explicitly. In non-trivial strategic settings, at least one player's objective depends on the actions, also called *decision variables*, chosen by other players. This dependence creates a coupling between decisions: no player can determine her optimal course of action without taking into account the likely decisions of others, even in the absence of formal collaboration.

When players can enter into binding agreements and coordinate their strategies for mutual benefit, the situation falls under the domain of *cooperative game theory*. This branch examines how coalitions are formed, how joint strategies are agreed upon, and how the resulting surplus is distributed among participants [91, 142, 40]. In contrast, *non-cooperative game theory* studies situations in which binding agreements are either impossible or unenforceable, and each player acts independently to maximize her own payoff. The central solution concept in this setting is the *Nash equilibrium*, defined as a profile of strategies from which no player can improve her outcome by unilaterally deviating. John Nash's seminal work established that such equilibria exist in all finite games [59, 60], that is, games in which each player has a finite number of actions, providing a foundational result that underpins much of modern strategic analysis.

Non-cooperative games can be classified in several ways [12]. One such distinction is between *zero-sum* and *nonzero-sum* games. In zero-sum games, there are two players, and the

gains of one player are exactly offset by the losses of the other, possibly after affine transformations of the payoffs; these games capture purely competitive situations such as certain military engagements or sports matches. Nonzero-sum games, by contrast, allow for both conflict and cooperation, as improvements in one player's outcome need not come entirely at the expense of others. Another distinction is between *finite* and *infinite* games. A game is finite if each player has a finite set of possible actions, a structure often represented in matrix form; infinite games arise when the action sets are continuous or unbounded, and in some cases payoffs depend continuously on these actions, as in so-called continuous-kernel games.

Games may also be classified as *deterministic* or *stochastic*. In deterministic games, the outcome is entirely determined by the players' chosen strategies or actions, with no external randomness. Stochastic games introduce uncertainty through random events, such as nature's moves or probabilistic state transitions, making them especially relevant for modeling real-world environments with incomplete control over outcomes. The distinction between *complete* and *incomplete* information concerns the players' knowledge about the structure of the game. In complete information games, all relevant details, including payoff functions and available strategies, are common knowledge. In incomplete-information games, players may possess private information or hidden types, requiring the use of beliefs and Bayesian reasoning to model strategic choices.

Another important distinction is whether the game is *static* or *dynamic*. In static games, players make their decisions simultaneously or without any knowledge of others' moves, whereas in dynamic games, actions unfold over time and may be contingent on earlier moves, allowing for adaptation, signaling, and the building of reputations. Relatedly, games can be *single-act* or *multi-act*. In single-act games, each player makes one move only; in multi-act games, players make a sequence of decisions, often responding to evolving circumstances. Dynamic multi-act games can be modeled in either discrete or continuous time. The continuous-time case, known as a *differential game*, is particularly relevant in control theory and engineering, where the evolution of system states is governed by differential equations.

By articulating these distinctions and modeling assumptions, game theory provides a versatile and precise language for describing strategic interactions in a vast range of contexts, from economic markets and political negotiations to biological competition and multi-agent artificial intelligence systems. Its analytical framework allows researchers not only to predict behavior, but also to design mechanisms that guide strategic interaction toward desirable outcomes.

1.2 ■ Games as Models of Strategic Situations

At its core, a game is an abstract model of a strategic situation in which multiple decision-makers choose actions that jointly determine the system outcome and, consequently, their individual payoffs. These models serve as idealized laboratories for analyzing behavior, predicting outcomes, and designing mechanisms in settings where interests may align, conflict, or both.

A static game is typically specified by three elements: (i) a set of *players*, who may represent individuals, firms, robots, software agents, or larger collectives; (ii) a set of *actions* available to each player; and (iii) a *payoff function* mapping every action profile to a numerical value representing each player's utility. Players could randomize over their action sets according to some probability distributions. Action sets can be finite or infinite, depending on the problem at hand.

The defining feature of game-theoretic models is *mutual interdependence*: a player's payoff depends not only on her own strategy but also on the strategies chosen by others. Anticipating and responding to others' behavior, *strategic reasoning*, is therefore fundamental. This applies equally to competitive settings, where players' objectives conflict, and to cooperative contexts, where coordinated action can increase collective welfare.

Games vary in structure and information patterns. In static games, formally introduced above,

players act simultaneously or without knowledge of others' actions; a more general class of games is dynamic games, which are defined over a time horizon and where information structure (pattern) is another important element, governing the information each player acquires over past actions of other players, and strategies are causal mappings from information sets to action sets. Complete-information games fully disclose payoffs and game structure to all players, while incomplete-information games require probabilistic beliefs and Bayesian updating to handle hidden information.

The scope of applications of game theory is vast. In economics, games model market competition, pricing, and negotiation. In political science, they describe voting systems, coalition building, and international relations. In engineering and computer science, they underlie resource allocation, protocol design, robust designs, and coordination in distributed systems. In artificial intelligence, game-theoretic models provide the foundations for multi-agent learning, strategic planning, and adversarial robustness.

By distilling complex strategic environments into formal mathematical objects, game theory enables both analysis and design. It not only predicts behavior under rationality assumptions, but also informs the creation of mechanisms that align individual incentives with socially desirable outcomes, making it a central pillar in the study and engineering of intelligent, interactive systems.

1.3 ■ The Past and the Present

The modern scientific study of game theory began with John von Neumann and Oskar Morgenstern's landmark 1944 work, *Theory of Games and Economic Behavior* [138], which fused mathematical rigor with economic modeling to create a unified framework for strategic analysis. This work followed von Neumann's earlier 1928 *Minimax Theorem* [137], which proved that every finite two-player zero-sum game admits a saddle point in mixed strategies. By extending the scope of analysis beyond zero-sum interactions to include non-zero-sum games, cooperative agreements, and broader economic applications, the 1944 book marked the formal emergence of game theory as a distinct discipline.

Although 1944 is often regarded as the starting point, earlier contributions had already laid important groundwork. Émile Borel's pioneering studies on strategic games [16], André-Marie Ampère's 1782 essay on the mathematics of games [3], and Georges-Louis Leclerc de Buffon's 1777 writings on probabilistic reasoning [29] all anticipated key ideas that would later be formalized within game theory.

The influence of game theory is reflected in repeated recognition by the Nobel Prize in Economic Sciences. In 1994, John C. Harsanyi, John F. Nash Jr., and Reinhard Selten were honored for their pioneering analysis of equilibria in the theory of non-cooperative games [126]. Nash introduced the now-fundamental concept of Nash equilibrium and proved its existence in mixed strategies in all finite games; Harsanyi developed the framework for analyzing games of incomplete information, commonly known as Bayesian games; and Selten refined equilibrium concepts for dynamic environments through the notion of subgame perfection.

In 2005, Robert J. Aumann and Thomas C. Schelling were recognized by the Nobel Committee for advancing our understanding of conflict and cooperation through game-theoretic analysis [128]. Aumann's work on repeated games demonstrated how cooperation can be sustained even without binding agreements, while Schelling's insights into focal points, commitment strategies, and deterrence had wide-ranging influence in economics, political science, and international relations.

In 2007, Leonid Hurwicz, Eric S. Maskin, and Roger B. Myerson received the prize for establishing the foundations of mechanism design theory [129]. Mechanism design studies how to construct rules and institutions that induce self-interested agents, operating under asymmetric

information, to achieve socially desirable outcomes, and it provides a unifying framework for markets, voting systems, auctions, and organizational design.

This trajectory continued in 2012, when Alvin E. Roth and Lloyd S. Shapley were awarded the Nobel Prize for the theory of stable allocations and the practice of market design [130]. Building on cooperative game theory and matching theory, their work demonstrated how equilibrium and stability concepts can be translated into real-world institutional designs, including kidney exchange programs, school choice mechanisms, and labor markets.

Beyond prizes explicitly centered on game theory, several Nobel recognitions in economics highlight closely related developments in which strategic interaction, incentives, and equilibrium reasoning play a central role. In 2004, Finn E. Kydland and Edward C. Prescott were awarded the prize for their analysis of time consistency in economic policy [127], a concept closely connected to dynamic games between policymakers and private agents. In 2016, Oliver Hart and Bengt Holmström were recognized for their contributions to contract theory [131], which formalizes incentive alignment and strategic behavior under incomplete information. In 2017, Richard Thaler received the prize for behavioral economics [132], introducing systematic deviations from classical rationality that have since been incorporated into behavioral and bounded-rational game-theoretic models. In 2020, Paul R. Milgrom and Robert B. Wilson were honored for advances in auction theory [133], a domain in which equilibrium analysis, information structure, and strategic bidding are central.

Beyond economics, game theory has also been recognized at the highest level in other scientific fields. In 1999, the Crafoord Prize in Biological Sciences was awarded to John Maynard Smith, together with Ernst Mayr and G. Williams, for their role in introducing evolutionary game theory to biology. Maynard Smith's concept of the *evolutionarily stable strategy* (ESS) provided a rigorous mathematical basis for understanding the dynamics of competition and cooperation in biological populations, influencing research from animal behavior to epidemiology [89].

1.3.1 ■ From Foundations to Recent Advances

Following the foundational contributions of Nash, Selten, and Aumann, game theory expanded well beyond its original economic context, impacting political science, biology, engineering, and computer science. This broadening of scope was accompanied by several important methodological and conceptual developments. One such development is *algorithmic game theory*, which merges the principles of game theory with computational complexity theory to study how equilibria can be computed efficiently in large, structured games [86, 67]. This field addresses practical challenges such as equilibrium computation in massive networks, the design of online markets, and the allocation of resources in large-scale distributed systems. Its methods underpin much of modern market design for digital platforms, online advertising auctions, and cloud resource management.

Another significant advance is *behavioral game theory*, which draws from psychology and experimental economics to develop more realistic models of decision-making [22, 34]. Whereas classical game theory assumes perfect rationality, behavioral approaches account for bounded rationality, social preferences, fairness concerns, and reciprocity, providing richer explanations for observed deviations from equilibrium predictions. This line of work has provided experimental insight into trust, cooperation, and strategic deception, which influences policy design and human-machine interaction protocols.

Evolutionary game theory, originally formulated in biology [119], has found wide application in the social sciences and technology domains. Models explain how strategies evolve over time through processes similar to natural selection, allowing the study of cultural evolution, the diffusion of innovations, and the emergence of conventions in large populations. In network science, evolutionary models explain how cooperation, competition, and coordination patterns

form and persist in complex social and technological systems, including online communities and peer-to-peer networks [87].

A foundational strand of this development is *differential game theory*, introduced by Rufus Isaacs at the RAND Corporation in the 1950s and synthesized in his 1965 monograph *Differential Games* [57]. Motivated by military pursuit–evasion problems, Isaacs cast dynamic conflicts as games played out in continuous time and derived the equation—now known as the Hamilton–Jacobi–Isaacs equation—that characterizes the value of a two-player zero-sum differential game. Together with the optimal-control theory of Pontryagin and the dynamic programming of Bellman, his work laid the mathematical foundations of dynamic and differential games, later systematized by Başar and Olsder [12] and extended to stochastic and mean-field settings. These developments are taken up in Lectures 20–24, most directly in Lecture 21 (Differential Games) and Lecture 23 (Stochastic Differential Games).

In engineering and systems science, *dynamic and differential game theory* has been applied to the analysis and control of systems with multiple interacting agents [12, 32, 145]. These methods have informed the design of algorithms for network resource allocation, wireless and communication systems, cyber-physical system security, network security, and cooperative or competitive control in autonomous robotics [46, 45, 2, 94, 26, 150, 148, 62]. Applications range from optimal power grid management to the coordination of fleets of autonomous vehicles and drones. Network security and cyber-physical resilience have been especially active areas for game-theoretic modeling, including survey-level treatments of network security games and dynamic-game approaches to secure control [74, 147, 56].

We have also seen important applications of the framework of dynamic game theory (particularly zero-sum games) to the design of robust policies for systems with unknown, deterministic uncertainty, where the uncertainty is modeled as being controlled by an adversary. Originally developed for control systems in the design of robust control policies under full as well as partial state information [11], this framework has subsequently been extended to economic decision making when the decision maker is faced with misspecified models or does not fully trust the model at hand [48].

The integration of game theory into *artificial intelligence* represents one of its most transformative recent developments. In multi-agent reinforcement learning, agents adapt their strategies in response to the evolving strategies of others, creating environments that closely mirror repeated or stochastic games [71, 111]. Generative adversarial networks (GANs) cast the training process as a two-player zero-sum game between a generator and a discriminator, demonstrating how adversarial setups can drive the creation of realistic synthetic data [43]. *Stackelberg security games* have been deployed to optimize resource allocation in defense and security contexts, from airport screening to wildlife protection [124]. Finally, *mechanism design*, when implemented computationally, guides the creation of decentralized protocols and marketplaces in which self-interested agents can interact efficiently and fairly without centralized control [82]. Through these applications, game theory has moved from a largely descriptive science of strategic behavior to an engineering discipline capable of designing robust, fair, and efficient interactions in complex, high-dimensional, and often adversarial environments.

Figure 1.1 places these developments on a single timeline, from early precursors to the discipline’s integration into modern artificial intelligence.

1.4 ■ A First Dive into Game Theory

Game theory provides a structured language for describing situations in which multiple decision-makers interact strategically. To analyze such environments, it is essential to understand several foundational distinctions that determine how a game is represented, what players know, and how they may interact. Before introducing formal mathematical definitions in later lectures, this

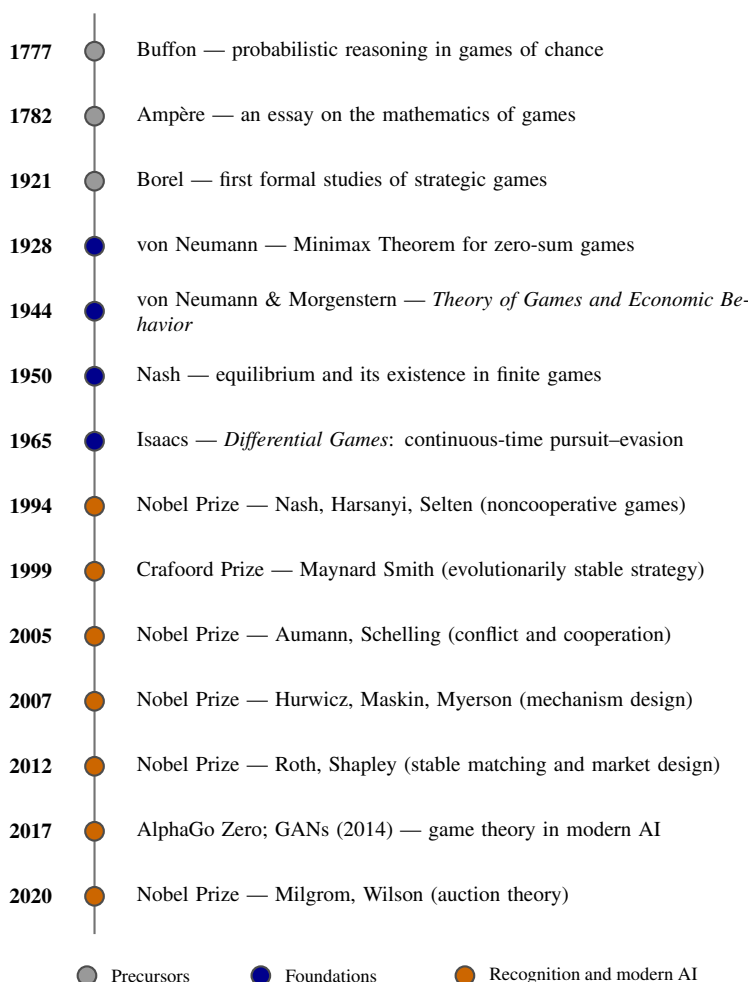


Figure 1.1: A brief timeline of game theory, from early precursors through its formal foundations to its recognition across the sciences and its integration into modern artificial intelligence. Years for the Nobel Memorial Prize in Economic Sciences (and the 1999 Crafoord Prize in biology) mark the award years.

section offers a high-level introduction to these conceptual building blocks. These include the difference between normal-form and extensive-form representations, the horizon of the interaction, the role of complete versus incomplete information, and the distinction between cooperative and noncooperative behavior. Each of these dimensions will be developed in depth in subsequent lectures, and together they form the conceptual backbone of the theory.

1.4.1 ■ Normal-Form vs. Extensive-Form Games

The normal form and the extensive form are two standard representations of strategic interactions, each emphasizing different features of a game, and they serve as complementary tools in game-theoretic analysis. The *normal form* (or strategic form) describes a game using a table or matrix. Each row of the table corresponds to a strategy available to one player, and each column corresponds to a strategy of another player. The entries of the matrix show the payoffs received

by all players if those strategies are chosen. For instance, in the classical *Prisoner's Dilemma*, the two prisoners each have two strategies: “Cooperate” or “Defect.” In Figure 1.2, a 2×2 table lists the payoffs for every possible pair of choices. This representation displays clearly how outcomes depend on simultaneous decisions. It is particularly useful for static games with complete information, meaning that all players choose at the same time and each knows the rules of the game, the available strategies, and the payoffs for everyone.

P1	P2	
	Cooperate (C)	Defect (D)
Cooperate (C)	$(-1, -1)$	$(-3, 0)$
Defect (D)	$(0, -3)$	$(-2, -2)$

Figure 1.2: Normal Form (Prisoner's Dilemma): a 2×2 payoff matrix showing simultaneous choices.

The *extensive form* presents a game as a tree. Each node represents a point at which a player must choose an action, and the branches leaving the node correspond to the available actions. Following these branches leads to terminal nodes, where the resulting payoffs are specified. This representation makes the timing of decisions explicit: who moves first, what each player observes, and how later choices may depend on earlier ones.

To represent situations in which a player does not know exactly which decision node she is at, the extensive form includes the concept of an *information set*. An information set groups together decision nodes that a player cannot distinguish between when making a move.

In the Prisoner's Dilemma depicted in Figure 1.3, Player 1 moves first by choosing either *C* or *D*. Player 2 then chooses between *C* and *D*, but does *not* observe Player 1's earlier action. Consequently, Player 2's two decision nodes, including one reached after Player 1 chooses *C*, and one reached after Player 1 chooses *D*, are combined into a single information set, represented by the dashed box in the figure. This means Player 2 must choose an action without knowing Player 1's choice, thereby capturing the simultaneous-move structure of the normal-form Prisoner's Dilemma within an extensive-form representation.

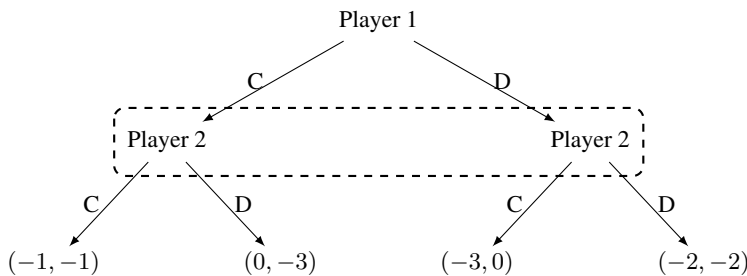


Figure 1.3: Extensive Form with Information Set: Player 1 moves first; Player 2 does not observe Player 1's choice (dashed box).

These two representations emphasize different aspects of strategic interaction. They form the foundation for the equilibrium concepts developed in later lectures in Part I. While these representations are straightforward when the set of available actions is finite, many strategic environments involve infinitely many or even a continuum of choices. In such cases, neither a

payoff matrix nor a finite game tree can be written down explicitly. Numerous settings illustrate this naturally. In an auction, bidders submit real-valued bids rather than choosing from a handful of discrete options. In a bargaining problem, two parties deciding how to divide a resource choose any split between 0 and 1, producing a continuum of possible demands. In engineering applications, a robot that chooses a motion direction or a control input is not restricted to a finite set of maneuvers, but moves in a continuous action space such as an angle, a thrust level, or a velocity vector. Firms in a competitive market may choose production quantities or prices across a continuum rather than discrete levels, and investors allocate portfolios by choosing real-valued positions in multiple assets. Infinite or continuous action spaces arise routinely in engineering, control, and robotics, where agents must choose real-valued control inputs, trajectories, or resource allocations.

When action spaces become continuous, the mathematical structure of the game changes: best responses may require calculus, optimization over convex sets, or differential equations; equilibria may not be guaranteed without assumptions such as continuity, compactness, or quasiconcavity; and strategies may themselves become functions rather than discrete choices. These richer models allow game theory to capture fine-grained decisions that arise in realistic strategic situations, but they also require more sophisticated analytical tools.

The formal study of games with infinite or continuous action spaces, including the tools needed to analyze them (e.g., convex analysis, fixed-point arguments, dynamic programming, and differential game techniques) will be developed later in Parts II and V of these Lecture Notes; the necessary mathematical background on convexity and fixed-point theorems is collected in Appendix B (Mathematical Background), and the elements of optimal control theory underlying differential games are reviewed in Appendix A (Elements of Optimal Control Theory). There we will see how the foundational ideas introduced in the finite setting extend to richer environments, and how the structure of continuous strategy spaces can produce new equilibrium behaviors, enable sophisticated feedback strategies, and reveal deeper connections between game theory, optimization, and control.

1.4.2 ■ Complete-Information vs. Incomplete-Information Games

A second fundamental distinction concerns what players know about the strategic environment. A classic illustration comes from Akerlof's "market for lemons" problem [1]. Consider a used-car market in which sellers know the quality of their own cars (whether each vehicle is a high-quality "peach" or a defective "lemon"), but buyers cannot directly observe this quality. If buyers could perfectly assess the quality of every car, the interaction would be a game of *complete information* in which all players know the full structure of the game, including the payoff-relevant attributes and preferences of every participant. Prices would then reflect quality, and high- and low-quality cars would trade at appropriate values.

However, in reality, buyers only see the exterior appearance of a car and the asking price of the seller, not whether the car is a lemon or a peach. Sellers hold private information that buyers do not possess, and buyers must therefore form expectations about the likely quality of the car offered to them. This creates a game of *incomplete information*, where hidden characteristics matter and strategic behavior must account for informational asymmetries. In the modern game-theoretic formulation introduced by Harsanyi [50], such hidden information is modeled using *types* that encode the seller's private knowledge, and buyers hold beliefs over these types. Strategic interactions unfold not only through actions, such as setting prices or deciding whether to purchase or not, but also through the beliefs players form and update.

This informational asymmetry leads to phenomena that do not arise under complete information. High-quality car sellers can behave differently from lemon sellers, and buyers can interpret prices or warranties as potential *signals* of underlying quality. Market institutions such as certi-

fication, inspections, or guarantees serve as *screening* mechanisms designed to separate different types of sellers. Over time, repeated interactions can allow honest sellers to build a *reputation*, influencing buyers' beliefs, and improving market efficiency. The market for lemons thus demonstrates how incomplete information can fundamentally change strategic behavior, market outcomes, and the design of effective mechanisms.

The formal analysis of incomplete-information games, Bayesian Nash equilibrium, signaling models, and mechanism design will be developed extensively in Part III of these Lecture Notes.

1.4.3 ■ Static vs. Dynamic Games

A third important distinction concerns the horizon of strategic interactions. In a *static game*, all players choose their actions simultaneously, or at least without observing the choices of others. Even if decisions occur at different physical times, what matters is that no player gains information about another's choice before acting. Consider two technology firms deciding whether to launch a new product line in the coming year. Each firm knows that its profits will depend on what the other one does, but neither firm can observe the rival's decision before committing. The strategic problem is therefore static: each firm must reason about the likely behavior of the other and choose its own action accordingly. Normal-form representations are particularly well suited for studying such settings because the entire interaction can be summarized by a matrix of strategies and payoffs.

In many strategic environments, however, decisions unfold sequentially and players observe each other's earlier actions. These settings give rise to *dynamic games*. Returning to the product-launch example, imagine now that one firm moves first by announcing its launch plans months ahead of time, while the second firm responds after observing this announcement. The timing of moves transforms the strategic structure: the follower can tailor its decision to the leader's action, and the leader may choose aggressively or conservatively in anticipation of how the follower will react. Dynamic games require players to think not only about immediate outcomes but also about how their actions influence future behavior. Extensive-form representations capture this sequential logic through decision trees, information sets, and outcome paths.

An illustrative example is the classical Prisoner's Dilemma. In the one-shot static game, as illustrated in Figure 1.2, mutual defection is the only rational outcome, and cooperation cannot be sustained no matter how desirable it would be for both players. Yet when the same players interact repeatedly in a dynamic setting, the strategic landscape changes dramatically. Today's action affects tomorrow's payoffs, and players can reward cooperation or punish deviation across time. Strategies such as "grim trigger" or "tit-for-tat" can make cooperation credible, even though it is impossible to sustain in a single play. In repeated games, trust, reciprocity, and reputation can emerge endogenously from forward-looking incentives. The formal analysis of these phenomena and the conditions under which cooperation can be supported appear later in the Lecture Notes in Lecture 18 on repeated games and folk theorems, with extensions to stochastic and general dynamic settings in Lectures 19 and 20.

Static and dynamic formulations often lead to different predictions and behaviors, even when the underlying payoffs are similar. Dynamic models emphasize credibility, strategic foresight, and contingent planning, while static models emphasize simultaneous incentives and best responses. Understanding when a situation should be modeled as static or dynamic is therefore fundamental to accurate strategic analysis.

The formal development of dynamic games begins in Part IV of these Lecture Notes, where we introduce extensive-form games, subgame perfection, and backward induction in Lecture 16, followed by principal-agent models in Lecture 17. The deeper study of repeated interaction, cooperation, and reputation-building in dynamic environments follows in Lectures 18 to 20. These lectures show how the addition of temporal structure can radically alter strategic possibilities and

sustain outcomes that static models rule out entirely.

1.4.4 ■ Noncooperative vs. Cooperative Games

A fourth important distinction is between *noncooperative* and *cooperative* games. Noncooperative game theory focuses on environments where players choose their strategies independently and cannot form binding agreements. A useful narrative example is two firms competing in a price-setting market. Each firm acts in its own interest, taking the competitor's behavior as given, and strategic interaction is captured through concepts such as best response and Nash equilibrium. The analysis reveals how competitive pressures, incentives to undercut rivals, and the inability to credibly commit to joint behavior shape the resulting market outcomes.

Cooperative game theory, in contrast, studies situations where players *can* coordinate, form binding agreements, or jointly commit to plans of action. Consider a group of municipalities deciding whether to build a shared water-treatment facility. No single municipality may find it cost-effective to build the facility alone, but together they can achieve substantial economies of scale. The central question becomes how to allocate the total cost or surplus among the participants so that every coalition is satisfied and no subgroup has an incentive to break away. Concepts such as the *core*, which identifies stable allocations, the *Shapley value*, which distributes surplus according to marginal contributions, and cooperative bargaining solutions, which model negotiated outcomes, provide tools for describing fairness and stability in cooperative arrangements.

Bargaining problems illustrate especially clearly how cooperative and noncooperative perspectives offer complementary insights into the same strategic situation. Imagine two firms negotiating how to divide the surplus from a joint venture. From a cooperative point of view, or “top-down,” one can directly seek an allocation that satisfies the natural axioms of fairness and efficiency. The Nash bargaining solution, for example, prescribes a unique outcome determined entirely by these normative principles, without reference to the details of the negotiation process. From a noncooperative, or “bottom-up,” perspective, the bargaining procedure itself becomes the object of analysis. Rubinstein's alternating-offers model shows how the eventual agreement can emerge endogenously from strategic timing, patience, and credible threats. Intriguingly, as the time between offers shrinks, the noncooperative equilibrium outcome converges to the Nash bargaining solution, indicating that cooperative principles can arise naturally from decentralized strategic behavior.

Cooperative frameworks arise naturally in problems of resource sharing, team formation, and joint infrastructure development, both in economic and engineering applications. The formal study of cooperative game theory, bargaining, and coalition formation will appear later in these Lecture Notes in Part VI (Lectures 25 to 27), where we examine axiomatic bargaining solutions, strategic bargaining models, the core, and the Shapley value in detail. These lectures show how cooperative and noncooperative perspectives are linked mathematically and how they illuminate the structure of negotiation, fairness, and joint decision-making in multi-agent systems.

1.5 ■ Why Study Game Theory?

The study of game theory offers a foundational lens through which we can understand, model, and influence the behavior of rational agents engaged in strategic interactions. In environments where multiple decision-makers, human or artificial, act with potentially conflicting interests, outcomes are shaped not only by individual actions but also by mutual expectations and interdependencies. Game theory provides a formal language and a rich set of mathematical tools to dissect such scenarios, enabling us to characterize equilibrium behaviors, explore optimal strategies, and anticipate the effects of information, incentives, and uncertainty.

Strategic reasoning pervades diverse domains. In economics, game theory explains market

dynamics, price formation, and oligopolistic competition. In political science, it clarifies coalition formation, voting behavior, and deterrence strategies. In evolutionary biology, it models the emergence of cooperation among self-interested organisms. In cybersecurity, it underpins defense mechanisms against adaptive adversaries in cyber-physical systems. Across these fields, game theory illuminates the underlying structure of conflict and cooperation.

From a practical perspective, game theory is not only descriptive but also prescriptive. It is employed to design mechanisms, rules, and algorithms that promote socially desirable outcomes in settings where agents act independently and often selfishly. Mechanism design, the “inverse” of game theory, uses these insights to engineer environments in which agents are incentivized to act in ways that align with collective objectives. Classic applications include the design of voting systems that reflect true preferences, auctions that encourage truthful bidding, market protocols that prevent manipulation, and contracts that balance risk and reward among stakeholders.

In the modern era, game theory has become indispensable in the development of artificial intelligence and multi-agent systems. As intelligent agents proliferate in domains such as robotics, autonomous driving, online platforms, and distributed sensor networks, the ability to model strategic interactions is vital. Game-theoretic frameworks support the design of negotiation protocols, consensus algorithms, and adversarial learning architectures. In adversarial machine learning and AI safety, game theory provides a rigorous foundation for anticipating and minimizing harmful behaviors in environments where AI agents may have conflicting objectives or operate under asymmetric information.

Beyond applications, the study of game theory deepens our understanding of decision-making itself. It highlights the roles of belief formation, learning, commitment, communication, and bounded rationality in shaping behavior. Recent advances in large language models (LLMs) and agentic AI open new opportunities for exploring multi-agent interactions and offer a bridge between human cognition and artificial decision-making. These developments extend the relevance of game theory to the design and governance of complex socio-technical systems, where the goal is not only to understand but also to shape behaviors, guide cooperation, and achieve resilient and equitable outcomes.

1.6 ■ Structure of These Lecture Notes

These Lecture Notes develop the theory of games along several complementary dimensions, with the dual aim of providing rigorous foundations and demonstrating their relevance in engineering, economics, and the sciences. The presentation progresses systematically, beginning with the most fundamental concepts and building toward advanced topics, algorithmic methods, and real-world applications.

Part I: Fundamentals of Game Theory

Part I develops the foundational principles of strategic interaction by moving systematically from elementary static environments to increasingly refined equilibrium concepts. The discussion begins in Lecture 2 with matrix games, the simplest representations of strategic conflict. Here we introduce saddle points, mixed strategies, and von Neumann’s minimax theorem, which together establish the mathematical structure of zero-sum strategic behavior. These ideas set the stage for the general analysis of nonzero-sum situations developed in Lectures 3 and 4, where Nash equilibrium emerges as the central solution concept. Lecture 3 builds intuition through illustrative examples that reveal how best-response reasoning leads to equilibrium, while Lecture 4 formalizes the concept through fixed-point theorems, for which Appendix B (Mathematical Background) provides the relevant background, and explores its computational aspects, including best-response dynamics, bilinear programming formulations, and special classes such as

potential and supermodular games.

The framework expands in Lecture 5 to extensive-form representations, which introduce sequential structure and information sets while remaining within the static, finite-horizon setting. Extensive-form games expose issues of credibility and belief formation. Lecture 6 takes up the first of these issues directly, in the hierarchical (Stackelberg) games in which one player commits to a strategy before the other responds; it develops the solution concept first on the game tree and then as a bilevel optimization problem, and it examines existence, computation, and the incentive effects of commitment, contrasting hierarchical interaction with simultaneous-move Nash equilibria. The credibility questions raised there motivate the need for equilibrium refinements more broadly. Lecture 7 develops these refinements (trembling-hand perfection, proper equilibrium, quasi-perfect equilibrium, and sequential equilibrium), each adding layers of robustness and internal consistency to the basic Nash prediction. The part concludes with Lecture 8, which generalizes the notion of independent strategies by introducing correlated equilibrium. This concept, grounded in ideas of Bayesian updating and regret-based learning, highlights the role of shared signals and information structures in enabling coordination beyond what Nash equilibrium predicts.

Part II: Continuous-Kernel Nash and Stackelberg Games

Part II moves from finite action spaces to games with infinite action sets, introducing mathematical tools well suited to control and engineering problems. Lecture 9 develops *continuous-kernel games*, where payoffs are defined by integral operators rather than matrices. This functional-analytic framework allows the study of games involving continuous resources, trajectories, and distributed decision variables, and it highlights the geometric and variational principles underlying equilibrium behavior.

Lecture 10 carries the hierarchical theory of Lecture 6 into this setting, developing *continuous-kernel Stackelberg games*, in which both action spaces and interaction kernels are continuous. These models connect naturally to optimal control, signaling, pricing, and other engineering settings where one agent shapes the environment faced by others.

Lecture 11 concludes the part with *quadratic and network games*. Quadratic structures yield tractable equilibrium characterizations and capture a broad class of distributed optimization and resource-allocation problems, while network games model strategic behavior over graphs. Taken together, Lectures 9 to 11 provide a unified framework for continuous-action games, bridging game theory with functional analysis, control, and network modeling and setting the stage for the study of incomplete information and dynamic interaction in later parts.

Part III: Games with Incomplete Information

Part III introduces uncertainty into strategic interaction, moving from settings of complete information to environments where players hold private types, unknown payoffs, or beliefs about hidden states. Lecture 12 establishes the framework through Harsanyi's Bayesian model, which embeds types and beliefs into an expanded game and identifies Bayesian Nash equilibrium as the central solution concept. Examples such as Cournot competition with uncertain costs and security games with attackers of unknown capability illustrate how private information shapes strategic behavior. The lecture also clarifies the link between the Harsanyi transformation and Aumann's common-knowledge framework, showing how belief hierarchies are encoded within Bayesian games.

Lecture 13 develops auction theory, where private valuations generate a rich array of strategic outcomes. First-price, second-price, and third-price auctions are analyzed under Bayesian Nash equilibrium, and the revenue equivalence theorem provides a unifying principle across formats.

Lecture 14 presents Myerson's optimal auction theory, demonstrating how incentive compatibility and monotonicity yield revenue-maximizing mechanisms through virtual valuations and ironing. These results highlight the interplay among information, incentives, and optimal allocation.

Lecture 15 broadens the scope to mechanism design under complete and incomplete information. Beyond auctions, the treatment includes Vickrey-Clarke-Groves mechanisms, contract design, and information-design problems in which a principal shapes agents' beliefs to influence behavior. Collectively, Lectures 12 to 15 show how uncertainty and private information reshape equilibrium structure, how mechanisms must be engineered to elicit truthful behavior, and how optimal designs emerge when information is dispersed across strategic agents.

Part IV: Dynamic and Repeated Games

Part IV extends strategic analysis from one-shot interactions to environments where decisions evolve over time. Lecture 16 introduces extensive-form dynamic games, emphasizing how histories, information sets, and timing structure shape strategic behavior. Subgame perfect equilibrium arises through backward induction as the core solution concept, with sequential equilibrium providing a belief-consistent refinement that captures credibility in dynamic settings. Lecture 17 develops principal-agent models, where asymmetric information and intertemporal incentives drive long-term relationships. Signaling, screening, and moral hazard are treated within a unified framework, illustrated through canonical examples such as insurance and contracting. These models show how contracts and mechanisms must be designed when actions or types are hidden over time.

Lecture 18 turns to repeated games, revealing how cooperation, punishment, and reputation emerge through repeated interaction. Classical folk theorems in both finite- and infinite-horizon settings characterize the range of payoffs that can be sustained in equilibrium, highlighting dynamic incentives absent in one-shot games. Lecture 19 introduces Markov games, which combine stochastic state transitions with strategic choice. Markov perfect equilibrium serves as the natural counterpart to subgame perfection when the system evolves over time, and extensions to partially observable environments incorporate inference about hidden states. Lecture 20 concludes the part with general dynamic infinite games, providing the mathematical foundations for continuous-time and open-horizon formulations.

Part V: Differential and Stochastic Games

Part V moves from discrete-time dynamics to continuous-time strategic interactions, beginning in Lecture 21 with differential games. These games model situations in which the system state evolves according to differential equations whose trajectories depend on players' control inputs. The lecture develops the Hamilton-Jacobi-Isaacs framework, value functions, and saddle-point conditions, illustrating applications in pursuit-evasion, robotics, and optimal control; Appendix A (Elements of Optimal Control Theory) collects the requisite background on dynamic programming, the minimum principle, and Riccati equations. The analysis of hierarchical interaction continues in Lecture 22 with dynamic Stackelberg games, where a leader-follower structure is embedded within a continuous-time system, revealing how commitment and information shape control strategies over time.

Uncertainty enters the picture in Lecture 23 through stochastic differential games, in which the state dynamics evolve under stochastic disturbances modeled by diffusion processes. These settings require tools from stochastic calculus and dynamic programming, and they encompass applications ranging from security models with random shocks to financial engineering. Lecture 24 concludes the part by introducing mean-field games, which analyze strategic behavior

in large-scale populations of interacting agents. By leveraging the asymptotics of large systems, mean-field games provide tractable formulations for decentralized control, social dynamics, and security applications in networked environments.

Part VI: Cooperative and Matching Games

Part VI shifts from noncooperative interaction to environments in which players can form binding agreements and coordinate their strategies. Lecture 25 begins with Nash bargaining, presenting axiomatic foundations for bilateral negotiation and characterizing how fairness and efficiency jointly determine outcomes. Lecture 26 develops the theory of coalition games, where groups of players can form coalitions to accomplish tasks collectively. The Shapley value, core, and nucleolus are introduced as central concepts capturing fairness, stability, and balancedness, with linear programming techniques used to analyze feasible payoff allocations. Examples drawn from network cooperation, resource sharing, and cost allocation highlight how coalition theory applies to modern engineering and economic systems.

Lecture 27 turns to matching theory, which studies markets where binding agreements must respect two-sided preferences. The classical marriage model and the Gale-Shapley deferred acceptance algorithm provide the foundation for analyzing stability, incentives, and optimality in matching markets. Extensions to school choice, labor markets, and assignment problems demonstrate how matching theory shapes institutional design in environments where monetary transfers are limited or undesirable.

Part VII: Advanced Topics

Part VII introduces several advanced themes that lie at the intersection of game theory, networks, and dynamical systems. Lecture 28 explores routing games, in which selfish agents traverse networked systems and create congestion externalities. The price of anarchy quantifies the efficiency loss due to noncooperative behavior, and Braess's paradox illustrates how adding resources to a network can counterintuitively worsen performance. These phenomena connect game theory to traffic engineering, communication networks, and decentralized resource allocation.

Lectures 29 and 30 conclude the Lecture Notes with evolutionary games and learning in games. Replicator dynamics provide a continuous-time evolutionary framework in which population shares adjust according to payoff performance, linking Nash equilibria to evolutionarily stable strategies. Lecture 30 develops learning dynamics, such as fictitious play, reinforcement learning, and regret-minimizing algorithms, that describe how agents adapt behavior over time without perfect rationality or complete information. These models reveal how equilibrium behavior can emerge from decentralized adaptation, providing a bridge between game theory, machine learning, and multi-agent systems.

Overall Perspective

The seven parts of the Lecture Notes trace a unified progression through the landscape of game theory, from static matrix games to differential and stochastic games, from incomplete-information auctions to evolutionary and learning dynamics. Throughout, the exposition integrates mathematical models with algorithmic methods and illustrative applications, emphasizing both analytical rigor and practical relevance. The overarching aim is twofold. First, the Lecture Notes equip readers with the technical machinery needed to analyze equilibrium behavior across a wide range of strategic settings, drawing on tools from economics, mathematics, systems science, control theory, and computer science. Second, they highlight the broader implications of strategic reasoning in systems composed of interacting, adaptive, and intelligent agents, including AI

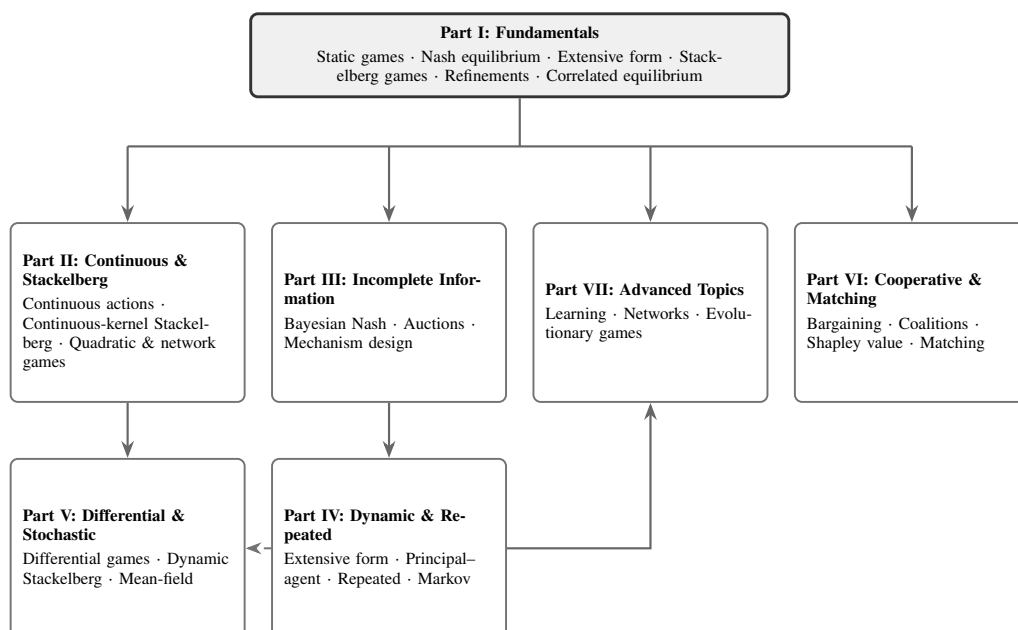


Figure 1.4: Roadmap of the seven parts. Part I (Fundamentals) is the core, from which four primary branches emanate: continuous and Stackelberg games (Part II), incomplete-information games (Part III), cooperative and matching games (Part VI), and advanced topics (Part VII). Part II extends to the differential and stochastic games of Part V, and Part III to the dynamic and repeated games of Part IV. The dynamic-game methods of Part IV in turn feed the advanced topics of Part VII and, via the dashed link, also the differential and stochastic games of Part V. Together the parts provide a progression from foundations to applications in static and dynamic multi-agent decision making.

systems, cybersecurity and communication networks, socio-technical infrastructures, and other complex environments. By weaving together theory, computation, and application, the Lecture Notes present game theory as a foundational framework for understanding and designing modern multi-agent systems.

1.7 ■ Game Theory in Action: Modern Applications and Their Theoretical Foundations

Game theory serves not only as a theoretical discipline but also as a practical modeling framework for analyzing strategic interactions across a wide range of real-world scenarios. While applications in economics are the most familiar, the influence of game theory extends far beyond this domain. In what follows, we introduce several applications whose impact spans artificial intelligence, machine learning, security, robotics, and the biomedical sciences.

1.7.1 ■ AlphaGo Zero: Self-Play and Reinforcement Learning as Game Dynamics

One of the most significant milestones in the recent history of artificial intelligence is the development of AlphaGo Zero, a system that achieved superhuman proficiency in the board game Go en-

tirely through self-play, without any reliance on human data or expert demonstrations [115, 114]. In contrast to its predecessor AlphaGo, which was trained initially via supervised learning on human games and subsequently refined via reinforcement learning [113], AlphaGo Zero discarded human priors altogether. Instead, it learned from scratch by playing games against itself in a closed-loop environment.

This learning process is game-theoretically profound. At its core, AlphaGo Zero models learning as a repeated, adversarial interaction with an evolving copy of itself (Figure 1.5). At each iteration, the agent plays games against itself, using the current network to guide a Monte Carlo tree search, and refines its policy to perform well against the strongest play it has produced so far, including its own earlier versions. Game-theoretically, this is a two-player zero-sum interaction in which both players are instantiated by the learning agent at different stages of its evolution, and each update pushes the policy toward a best response to the strategies it has encountered.

This mechanism closely resembles a *fictitious play* dynamic [21], where each agent forms a belief about the opponent's play by observing past behavior and chooses its best response accordingly. Over time, the empirical frequency of play converges to a Nash equilibrium under certain conditions. In the case of AlphaGo Zero, the repeated play environment induces convergence towards a minimax policy, one that minimizes the maximum possible loss, assuming the worst-case opponent strategy.

Formally, let π^t denote the policy of the agent at training iteration t , and let π^{-t} represent a distribution over past policies. Define the expected payoff when policy π plays against policy π' as:

$$V(\pi, \pi') = \mathbb{E}_{s \sim \mathcal{D}} \left[\sum_{t=0}^T r_t \mid a_t \sim \pi, a'_t \sim \pi' \right],$$

where r_t is the reward at time t , $\mathbb{E}$ denotes expectation, and $\mathcal{D}$ is a distribution over initial game states. The objective is to find a policy π^* that solves the following minimax (more precisely, maximin) problem:

$$\pi^* = \arg \max_{\pi} \min_{\pi'} V(\pi, \pi').$$

This optimization problem reflects a saddle-point formulation, where the goal is to find a strategy that performs optimally against the worst-case counter-strategy.

Reinforcement learning methods such as policy gradient or Monte Carlo tree search (MCTS) guide this optimization. In AlphaGo Zero, a neural network is used to approximate both the policy and the value function, and MCTS is used to generate self-play data that improves the network iteratively. Through many iterations of self-play, the policy network converges toward a fixed point in the space of strategies, one that is empirically unbeatable by any of its prior incarnations.

The success of AlphaGo Zero thus illustrates how self-play in adversarial games naturally embodies a form of strategic reasoning, grounded in game-theoretic principles. The training dynamics emulate the convergence properties of best-response sequences in zero-sum games, while the architecture and algorithms leverage powerful function approximators to scale this reasoning to extremely large state and action spaces, such as those encountered in Go. The best-response and fictitious-play dynamics sketched here, and their convergence to minimax and Nash solutions, are developed in Lecture 30 (Learning in Games), building on the zero-sum foundations of Lecture 2.

1.7.2 ■ Pac-Man as a Multi-Agent Stochastic Game

The classic arcade game **Pac-Man** serves as a compelling case study for modeling interactive, multi-agent decision-making using game-theoretic tools. In this game, the central character Pac-

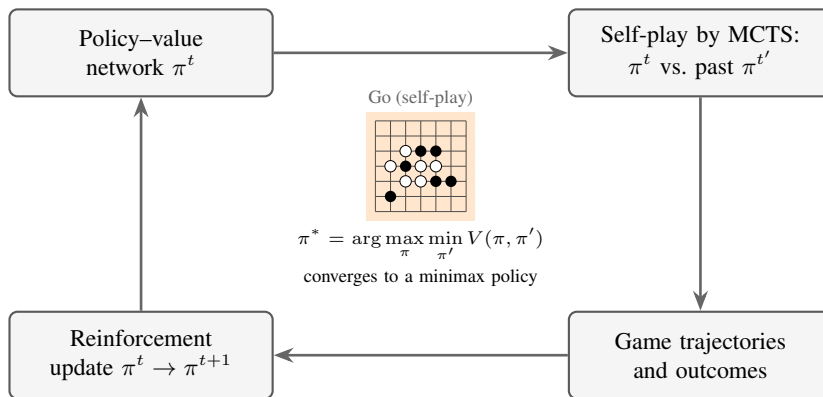


Figure 1.5: AlphaGo Zero as a self-play loop. The current policy–value network repeatedly plays Go against its own past versions (center board); the resulting games train an improved network. This best-response-to-past-play dynamic mirrors fictitious play and drives the policy toward the maximin (minimax) solution of the underlying zero-sum game.

Man must navigate a maze while avoiding capture by four ghost agents. Although the original design does not encode strategic reasoning in a formal sense, the structural dynamics of the game naturally align with the framework of *stochastic games* (also known as *Markov games*), which generalize Markov decision processes to multiple agents with potentially conflicting goals.

In this setting, the full configuration of the game at any given time defines the *state space*. A single state captures the locations and movement directions of Pac-Man and each ghost, the positions of uneaten pellets, the status of power-ups, and auxiliary game parameters such as score and remaining lives. Since the game unfolds over discrete time steps, each agent (Pac-Man and each ghost) makes a move by selecting an action from a fixed set of options (e.g., up, down, left, right). These actions are applied simultaneously or in quick succession, and the resulting state transition can be either deterministic or stochastic depending on the specific mechanics of the game engine.

Each agent is associated with a distinct reward function. Pac-Man earns points by consuming pellets and power-ups, and receives bonus rewards for eating ghosts when empowered by a power-up. He incurs a penalty when captured. The ghosts, by contrast, are rewarded for successfully intercepting Pac-Man and may be penalized for being eaten during his powered-up state. This setup yields an adversarial structure: Pac-Man attempts to maximize cumulative reward through pellet collection and survival, while ghosts aim to minimize that reward by restricting his mobility and ultimately ending the game.

From a game-theoretic standpoint, Pac-Man can be framed as a multi-agent stochastic game with competing objectives (Figure 1.6). The agents operate in a shared environment and adapt their strategies in response to evolving positions and anticipated future moves. In the formal model, the objective of each agent is to optimize an expected return over a long horizon, taking into account both immediate payoffs and the strategic reactions of other agents. The resulting dynamics resemble repeated general-sum games, in which equilibrium concepts such as Nash equilibrium or correlated equilibrium characterize long-term optimal behavior.

Solving this game involves learning or computing policies that map observed states to optimal actions. In artificial intelligence, this has been approached using algorithms from multi-agent reinforcement learning (MARL), including independent Q-learning, centralized training with decentralized execution, or policy gradient methods adapted for multi-agent systems. These methods allow agents to incrementally adapt their policies based on observed outcomes and

inferred behaviors of other agents, potentially converging to a stable equilibrium.

Modeling Pac-Man in this way not only provides a theoretical grounding for understanding the strategic structure of the game but also demonstrates how classical game theory can inform modern AI architectures. It illustrates how the combination of state transitions, reward asymmetries, and adaptive behavior gives rise to rich emergent dynamics that are amenable to rigorous analysis using game-theoretic principles. Such stochastic (Markov) games are the subject of Lecture 19, and the multi-agent reinforcement-learning methods noted above are developed in Lecture 30 (Learning in Games).

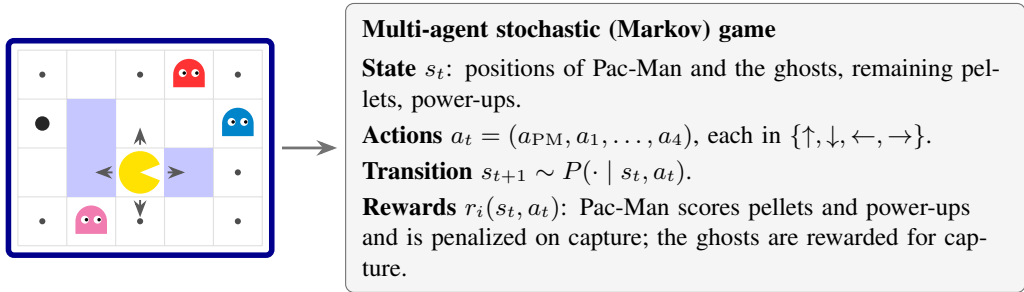


Figure 1.6: Pac-Man as a multi-agent stochastic (Markov) game. The maze configuration is the state; Pac-Man and the four ghosts simultaneously choose movement actions; the state evolves under a transition kernel; and each agent has its own reward, yielding an adversarial, general-sum stochastic game.

1.7.3 ■ Colonel Blotto as a Strategic Resource Allocation Game

The **Colonel Blotto game** is a seminal model in non-cooperative game theory that captures the essential features of strategic resource allocation under competitive conditions [17, 100]. In its classical form, two players, often represented as military commanders or colonels, must simultaneously distribute a finite amount of resources (such as troops, funds, or effort) across a set of battlefields. Each battlefield is won by the player who allocates more resources to it, and the overall objective is to win the majority of these contests. Crucially, the players act without knowledge of their opponent's allocation, creating a setting of simultaneous play with imperfect information (Figure 1.7).

From a game-theoretic perspective, the Blotto game is a *zero-sum game*, meaning that the gain of one player is precisely the loss of the other [44]. This fundamental antagonism implies that equilibrium strategies must protect against worst-case scenarios: each player must optimize her allocation in a way that minimizes the maximum advantage the opponent could exploit. Given the combinatorial complexity of distributing resources across multiple fronts, equilibrium strategies are often not pure (deterministic) but rather *mixed strategies*, where players randomize their allocations according to specific probability distributions. The use of mixed strategies serves to preserve unpredictability and prevent the opponent from easily countering one's plan.

Analytical work on the Colonel Blotto game has yielded characterizations of equilibrium strategies in both symmetric (equal resources and valuation across battlefields) and asymmetric (differing capacities or battlefield values) settings. In symmetric cases, the optimal strategy often involves spreading resources in such a way that no individual battlefield is consistently over- or under-defended. Theoretical tools such as linear programming, duality theory, and measure-theoretic probability have been used to describe these equilibria, often involving continuous distributions over feasible allocations.

Beyond its military metaphor, the Colonel Blotto game has wide-ranging applications in political science, economics, and cybersecurity. In political campaigning, for instance, candidates allocate budgets across different constituencies to maximize electoral wins. In marketing, firms distribute advertising efforts across markets or customer segments. In cybersecurity, defenders allocate monitoring or protective resources across attack surfaces or system vulnerabilities. In each of these domains, the Blotto framework models the tension between strategic planning and the unpredictability of a rational adversary and offers principled guidance for optimal allocation under uncertainty. The zero-sum structure and mixed-strategy equilibria that underlie Blotto games are developed in Lecture 2, within the broader theory of mixed-strategy Nash equilibria of Lecture 3.

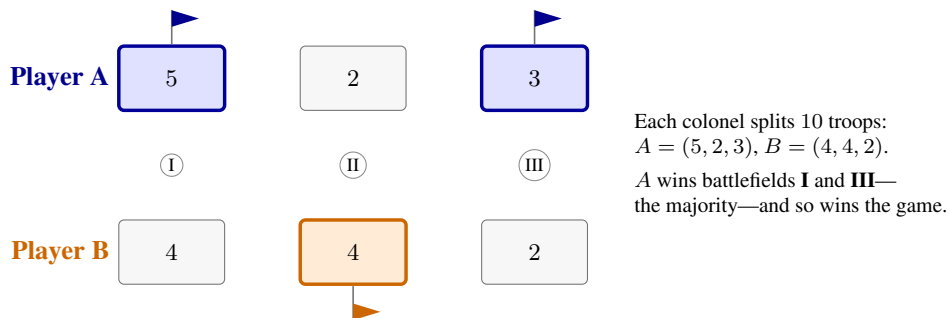


Figure 1.7: The Colonel Blotto game. Two players simultaneously divide a fixed budget of resources across several battlefields; each battlefield is won by whoever commits more, and the player who wins the majority of battlefields wins. Because allocations are chosen without knowledge of the opponent's, equilibrium play is typically in mixed (randomized) strategies.

1.7.4 ■ Lady in the Lake as a Differential Game of Pursuit and Evasion

The **Lady in the Lake** problem is a canonical example of a *differential game*, a class of games in which multiple players interact through the control of continuous-time dynamical systems [57]. As illustrated in Figure 1.8, the game models a two-player adversarial interaction between an evader, located within a circular lake, and a pursuer, constrained to move along the lake's perimeter. The evader, representing the “lady,” seeks to escape the lake by reaching the boundary without interception, while the pursuer's objective is to intercept the evader precisely at the moment she reaches the shore.

This dynamic interaction is governed by continuous-time control inputs. The evader controls the direction of swimming within the disk, typically with a bounded speed constraint. Meanwhile, the pursuer is constrained to move tangentially along the circular boundary and adjusts their angular velocity to position themselves optimally along the perimeter. Both agents make strategic decisions about their motion paths, with the understanding that the final outcome, a successful escape or interception, depends on their relative speeds, initial positions, and the timing of control changes.

Game theory offers a powerful lens through which to analyze such problems. The Lady in the Lake game can be formulated as a two-player zero-sum differential game, and its solution often involves tools from optimal control theory, such as Pontryagin's Maximum Principle [95]; see Appendix A (Elements of Optimal Control Theory) for the relevant background. These tools help characterize optimal control trajectories for both the pursuer and the evader, allowing one to compute reachable sets, interception paths, and escape cones.

One of the key insights game theory brings to this scenario is the identification of *critical speed ratios*. For example, it can be shown that if the maximum speed of the evader is less than a certain critical threshold relative to the tangential speed of the pursuer, the pursuer has a guaranteed winning strategy: there exists a deterministic path that ensures interception regardless of the evader's evasive maneuvering [57]. On the other hand, if the evader's speed exceeds this threshold, then she can escape with probability one, even under optimal pursuit. These results demonstrate how geometric and dynamic constraints translate into strategic advantages or limitations.

Moreover, the Lady in the Lake model reveals the strategic value of *information and prediction* in dynamic games. A pursuer with real-time knowledge of the evader's position and velocity can adjust more effectively and pre-empt evasive maneuvers. In contrast, an evader who feints or delays commitment to a trajectory can induce suboptimal responses from the pursuer. These information asymmetries are central themes in broader applications of differential games, from missile defense to autonomous vehicle coordination [12]. The theory of two-player zero-sum differential games, including the Hamilton–Jacobi–Isaacs equation introduced by Isaacs that characterizes their value, is developed in Lecture 21 (Differential Games), with the stochastic case treated in Lecture 23.

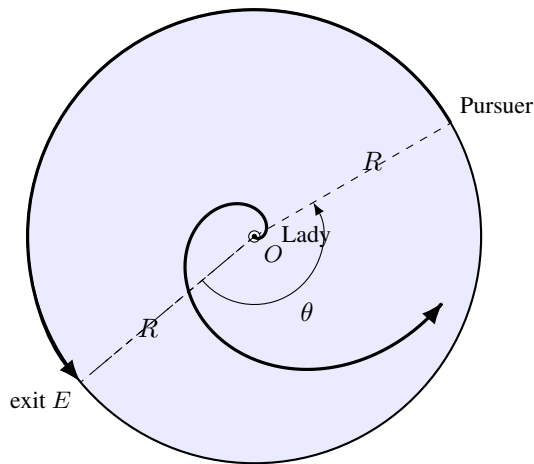


Figure 1.8: The pursuer is restricted to motion along the circular boundary of radius R , shown as an arc between its initial position and the potential exit point E . The evader (lady), initially located near the center O , moves along an interior trajectory (e.g., a spiral-like curve) toward the boundary. The central angle θ represents the separation between the pursuer and the exit point, which determines the pursuer's ability to intercept.

1.7.5 ■ Shapley Value as an Interpretive Framework for Learning Systems

The Shapley value provides a principled way of explaining the predictions of nonlinear models common in the field of machine learning. By interpreting a model trained on a set of features as a value function on a coalition of players, Shapley values offer a natural mechanism to quantify the contribution of each feature to a prediction or to the uncertainty associated with that prediction. This perspective frames the prediction task as a cooperative game, where features are the players and the model output is the total payoff to be distributed.

One of the strengths of the Shapley framework is that it offers a unified foundation for several widely used interpretability methods. Techniques such as Locally Interpretable Model-Agnostic Explanations (LIME) [99], DeepLIFT [112], and Layer-Wise Relevance Propagation (LRP) can be seen as specific instances or approximations of the Shapley value principle. This unification underscores the conceptual robustness of Shapley-based explanations: rather than relying on ad hoc heuristics, the method derives from axioms of fairness in cooperative game theory.

In practice, applying Shapley values to nonlinear models, including ensemble methods, deep neural networks, and kernel-based predictors, provides a transparent decomposition of the output into additive contributions from each feature (Figure 1.9). These contributions can be aggregated over instances to yield global importance measures or examined on a case-by-case basis for local interpretability. Furthermore, by extending the formulation to measure contributions to predictive uncertainty, Shapley values can inform feature selection, active learning strategies, and risk-aware decision-making.

This interpretive lens is especially valuable in high-stakes domains such as finance, healthcare, and security, where understanding *why* a model makes a prediction is as critical as the prediction itself. In such settings, the Shapley value framework bridges the gap between model accuracy and trust, providing explanations grounded in rigorous cooperative game-theoretic reasoning. The Shapley value and other cooperative solution concepts are developed in Lecture 26 (Coalition Games).

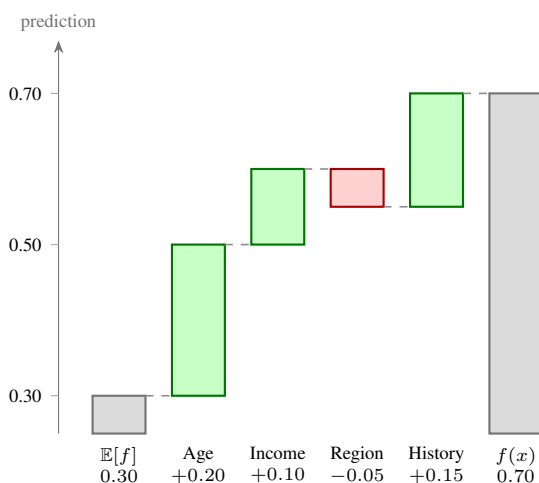


Figure 1.9: The Shapley value as feature attribution. Interpreting a trained model as a cooperative game—features are the players and the prediction is the value to be shared—each feature’s Shapley value φ_i is its fair additive contribution, decomposing the output $f(x)$ from the baseline $\mathbb{E}[f]$ into per-feature effects (positive in green, negative in red).

1.7.6 ■ Evolutionary Games and Their Impact on Biomedical Sciences

Evolutionary game theory extends classical game theory to settings in which the strategic interactions among players are shaped by evolutionary processes rather than deliberate rational choice [120, 119, 55, 143]. Instead of assuming that players are perfectly rational agents who compute best responses, evolutionary models describe populations of agents whose strategies evolve over time through mechanisms such as reproduction, imitation, and selection.

In the evolutionary framework, the payoff of a strategy is interpreted as a measure of its

fitness, which indicates the expected reproductive success of an individual using that strategy. Strategies that yield higher-than-average payoffs tend to increase in frequency within the population, while those with lower payoffs diminish. This process is often modeled using *replicator dynamics* [125, 55], a set of differential equations that describe how the proportion of the population adopting each strategy changes over time.

Applications to biology and the biomedical sciences are extensive. Models such as the *Hawk-Dove game* explain the coexistence of aggressive and peaceful behaviors in animal populations, showing how frequency-dependent selection can maintain behavioral diversity [119]. Games like the *Prisoner's Dilemma* and *Public Goods games*, when analyzed in evolutionary terms, reveal conditions under which cooperative traits can persist despite short-term incentives to defect [88]. Evolutionary game models have also been used to predict the allocation of parental care in species where males and females face different reproductive trade-offs [119]. In the biomedical domain, modeling the interactions between immune systems and pathogens as evolutionary games helps explain the persistence of certain virulence levels and immune strategies [55, 121]. A prominent clinical example is *cancer*. A tumor is a heterogeneous population of competing cell types, and *adaptive therapy* [41] exploits the evolutionary interplay between drug-sensitive and drug-resistant cells. Rather than administering the maximum tolerated dose—which eradicates the sensitive cells and releases the resistant ones from competition, hastening a resistant relapse—the dose is modulated to retain a controlled population of sensitive cells that competitively suppress the resistant subpopulation (Figure 1.10). Treating the tumor as an evolving adversary rather than a fixed target is a direct application of evolutionary game theory in oncology [122]. Crucially, the approach has moved from theory to the clinic: in a pilot trial for metastatic castration-resistant prostate cancer, patients receiving adaptive abiraterone dosing achieved a substantially longer median time to progression while using markedly less cumulative drug than under standard continuous therapy [146]. This success story illustrates how equilibrium and evolutionary reasoning can reshape real-world treatment design. This perspective extends to the study of *self genes*, genes whose primary function is to ensure the survival and replication of the host organism. The immune system's recognition of “self” versus “non-self” can be viewed as the outcome of an evolutionary game, where self genes promote cooperative stability within the body, while pathogens engage in strategies to evade or exploit this recognition.

The oncological application also brings a second, less obvious game-theoretic structure to the surface. Once the physician is treated as a player rather than as a protocol fixed in advance, the two sides of the interaction are asymmetric in a specific way: the tumor's cells inherit their traits and can only adapt to the treatment already administered, whereas the physician can anticipate that adaptation and act ahead of it. This is precisely the leader-follower asymmetry of the Stackelberg games of Lectures 6 and 10, played here against an evolving population rather than against a rational opponent, and models of this kind are called *Stackelberg evolutionary games*. Their ranking of solution concepts matters to the patient. Dosing at the maximum tolerated level is never better, and is usually much worse, than the Nash outcome in which the physician merely reacts to the resistance level it currently observes; and that Nash outcome is in turn never better, and usually much worse, than the Stackelberg outcome in which the physician anticipates the evolutionary response and steers it. The survey [144] develops this ranking together with the *eco-evolutionary* models on which it rests, in which population sizes and heritable resistance traits evolve as coupled dynamics, and observes that the same leader-follower structure governs pest management, fisheries, and the control of infectious disease. The episode illustrates a theme running through these lectures: what a solution concept is worth depends on what each side knows and on who moves first, and here that difference is measured in the patient's time to progression and quality of life.

A central concept in evolutionary game theory is the *evolutionarily stable strategy* (ESS) [120]. An ESS is a strategy which, if adopted by almost all members of a population, cannot be

invaded by any rare mutant strategy. This concept generalizes the Nash equilibrium to account for stability under evolutionary dynamics, ensuring that the population resists shifts away from the equilibrium due to small perturbations.

The impact of evolutionary game theory extends beyond biology into economics, social sciences, and computer science. In biology and biomedical sciences, it has transformed the study of adaptation by providing a quantitative framework for understanding how strategic interactions at the individual level can shape population-level outcomes over generations. It has shed light on the evolution of signaling systems, the emergence of cooperation, and the maintenance of biodiversity [87, 55]. By linking game-theoretic reasoning with Darwinian selection, evolutionary games bridge the gap between individual behavior and population-level phenomena, making them a cornerstone of modern theoretical biology and biomedical research. Replicator dynamics, the evolutionarily stable strategy, and their relation to Nash equilibrium are developed in Lecture 29 (Evolutionary Games), with connections to learning dynamics in Lecture 30; the leader-follower structure just described is the subject of Lectures 6, 10, and 22.

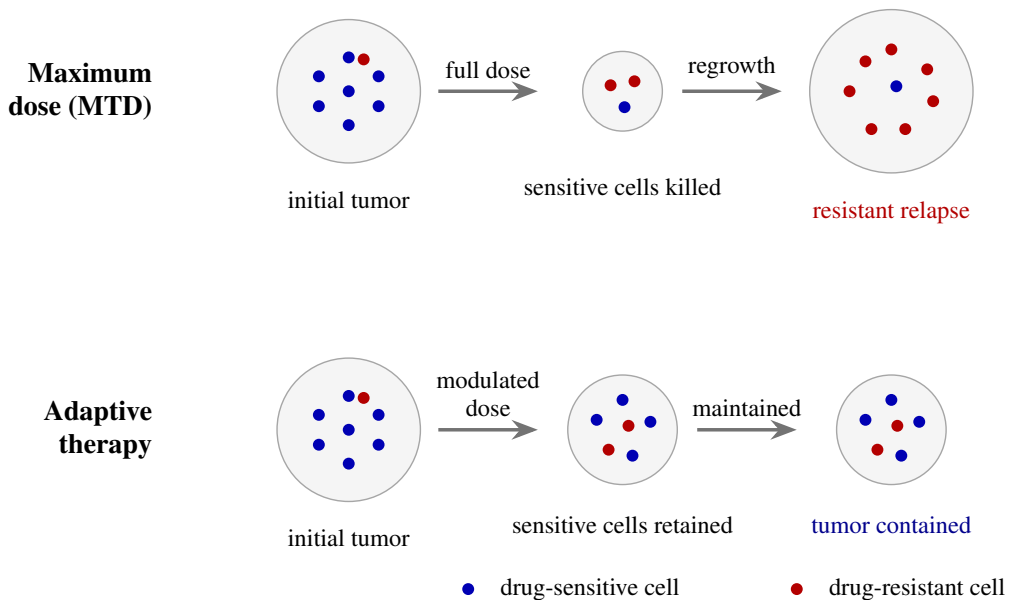


Figure 1.10: Evolutionary game theory in cancer therapy. A tumor is a heterogeneous population of drug-sensitive (blue) and drug-resistant (red) cells. *Top*: maximum-tolerated-dose therapy eliminates the sensitive cells but releases the resistant cells from competition, and hence the tumor returns as a resistant relapse. *Bottom*: *adaptive therapy* modulates the dose to preserve sensitive cells that competitively suppress the resistant subpopulation, containing the tumor for far longer. Treating the tumor as an evolving adversary rather than a fixed target is a direct application of evolutionary game theory in oncology.

1.8 ■ Further Notes and Reading

The mathematical foundations of game theory were laid by von Neumann and Morgenstern in their treatise [138], which developed the theory of zero-sum games together with the expected-utility framework, and by Nash, whose papers [59, 60] introduced the equilibrium concept that now bears his name and established its existence in finite games. The extension of equilibrium

analysis to dynamic settings originates with Selten [106], whose notion of subgame perfection is developed in Lectures 5 and 7, while the treatment of incomplete information is due to Harsanyi [50], whose Bayesian framework underlies Lectures 12 to 15. Schelling's classic study [105] remains an influential and largely nontechnical account of commitment, credibility, and coordination in strategic interactions.

Several textbooks complement these lectures. Comprehensive graduate treatments of noncooperative game theory include Fudenberg and Tirole [40] and Osborne and Rubinstein [90], while Owen [91] covers the noncooperative and cooperative branches of the theory in a unified manner. The book by Başar and Olsder [12], closest in spirit to the dynamic portions of these Notes, develops static and dynamic noncooperative game theory in both discrete and continuous time, and Isaacs [58] is the pioneering reference on differential games, taken up in Lectures 21 to 23. The two-volume handbook edited by Başar and Zaccour [7] surveys the modern state of dynamic game theory, spanning its theoretical foundations and its applications. For the computational and algorithmic perspective that recurs throughout these lectures, the collection edited by Nisan, Roughgarden, Tardos, and Vazirani [86] provides a broad entry point. Evolutionary game theory, introduced informally earlier in this lecture and developed in Lecture 29, is treated in depth by Maynard Smith [119] and Weibull [143].

Each of the following lectures closes with a section of this kind, pointing to the primary sources and to specialized surveys for the topics developed there.

1.9 ■ Exercises

Exercise 1.1 (Two-Period Prisoner's Dilemma). The Prisoner's Dilemma in Figure 1.2 and Figure 1.3 depicts a one-shot interaction in which each player chooses between cooperation (C) and defection (D), and in which defection strictly dominates cooperation. Real-world strategic relationships, however, rarely end after a single encounter: repeated interaction may enable reputation building, conditional strategies, or attempts to induce cooperation over time. To examine this possibility, suppose that the same two players face the Prisoner's Dilemma of Figure 1.2 *twice* in sequence, that both players observe the complete outcome of the first stage before the second stage begins, and that each player's total payoff is the sum of the two stage payoffs.

- (i) Explain how the timing of moves in this two-period game differs from that of the one-shot game. State precisely what each player knows at each decision point, and describe how second-period choices may depend on the first-period outcome.
- (ii) Construct the extensive-form game tree of the two-period game. Numerical payoffs need not be attached to the terminal nodes; label every action (C or D), display the sequence of moves, and indicate how the second-period decision nodes follow from each of the four possible first-period outcomes.
- (iii) Identify the information sets of Player 2 under the assumption that both players observe all previous actions, and mark them on the tree from part (ii). Explain why the information structure is simple in this case, and describe how it would change if the stage moves were simultaneous or if histories were only partially observed.
- (iv) Implement, in Python or MATLAB, a backward-induction solver for this two-period game, using the stage payoffs of Figure 1.2: for each of the four first-period outcomes, determine the second-period play by eliminating dominated actions, and then solve the resulting first-period game. Report the equilibrium path of play and the total payoff vector. The expected output is defection by both players in both periods, with total payoffs $(-4, -4)$.

Exercise 1.2 (Modeling a Coordination Game in Normal Form). Many social and economic environments require players not to outmaneuver one another but to *align their actions*. The classical “stag hunt” captures this idea. Two hunters independently choose whether to hunt a *stag*, which provides a high payoff only if both hunters cooperate, or to hunt a *hare*, which yields a smaller but guaranteed payoff regardless of the partner’s choice. This scenario models situations such as technology adoption, collaborative research, and team production, where success hinges on mutual commitment. For the computational parts below, use the numerical instance in which the payoff pairs are (4, 4) if both hunt stag, (3, 3) if both hunt hare, and (0, 3) (respectively (3, 0)) if the first (respectively second) hunter hunts stag while the other hunts hare.

- (i) Explain the strategic tension in this game: state what makes hunting stag attractive, what makes it risky, and why a hunter may choose the safer hare option even though coordinated stag hunting is collectively superior.
- (ii) Formulate the normal-form representation of the stag-hunt game. Name the two players, list their available actions (STAG, HARE), and express the payoffs symbolically (for example, “High,” “Low,” “Safe”) to emphasize structure over specific values.
- (iii) Explain why this game represents *coordination* rather than *conflict*: the players’ preferences are aligned, since both prefer the coordinated high-payoff outcome and both wish to avoid miscoordination, and hence the difficulty arises not from opposed interests but from forming consistent expectations about the other’s choice.
- (iv) Implement, in Python or MATLAB, a routine that enumerates the four pure action profiles of the numerical instance above and checks, for each profile, whether either player can strictly improve their payoff by a unilateral deviation. Report which profiles survive this test. The expected output is that exactly two pure Nash equilibria exist, namely (STAG, STAG) and (HARE, HARE).
- (v) Plot, for $q \in [0, 1]$ denoting the probability that the partner hunts stag, the expected payoff of choosing STAG and the expected payoff of choosing HARE in the numerical instance above, on the same axes. Compute the crossing point of the two lines, and explain its interpretation as the symmetric mixed-strategy Nash equilibrium in which each hunter chooses STAG with probability $q^* = 3/4$.

Chapter 2

Matrix Games

Following Lecture 1, this second lecture aims to present the formal structure and solution concepts of matrix games. It focuses on two-person zero-sum interactions, emphasizing matrix representations, security strategies, saddle-point equilibria, and mixed-strategy solutions. It develops both theoretical foundations, including the minimax theorem and saddle-point characterization, and practical methods, including graphical techniques, linear programming formulations, and learning dynamics such as fictitious play.

2.1 ■ Two-Person Zero-Sum Games

Two-person zero-sum games constitute one of the most fundamental classes of strategic interactions. To describe such a game formally, we begin with its basic components. A *finite two-player normal-form game* is defined by the triplet $(\mathcal{N}, A, u)$, where $\mathcal{N} = \{1, 2\}$ is the player set, each player $i \in \mathcal{N}$ has a finite action set A_i , and the joint action space is $A = A_1 \times A_2$. An element $a = (a_1, a_2) \in A$ is a *pure action profile*. Each player i evaluates outcomes through a payoff function

$$u_i : A \rightarrow \mathbb{R},$$

which specifies the utility obtained when the action profile a is played.

A game is called *zero-sum* if the players' payoffs are exactly opposed at every outcome. In this case, there exists a single payoff function $u : A \rightarrow \mathbb{R}$ such that

$$u_1(a) = u(a), \quad u_2(a) = -u(a), \quad \text{for all } a \in A.$$

Thus, the payoff of Player 2 (P2) is fully determined by that of Player 1 (P1), and the entire strategic environment can be analyzed through the single function u .

In the convention adopted here, *P1 is the minimizer* and seeks to make the payoff u as small as possible, while *P2 is the maximizer* and seeks to make u as large as possible. The zero-sum structure therefore expresses perfect adversarial alignment between the two players.

To distinguish between payoffs defined on pure actions and those induced by mixed strategies, we introduce the expected payoff function $U : \Delta(A_1) \times \Delta(A_2) \rightarrow \mathbb{R}$, defined by

$$U(\sigma_1, \sigma_2) := \mathbb{E}_{a_1 \sim \sigma_1, a_2 \sim \sigma_2} [u(a_1, a_2)] = \sum_{a_1 \in A_1} \sum_{a_2 \in A_2} \sigma_1(a_1) \sigma_2(a_2) u(a_1, a_2),$$

where $\Delta(A_i)$ denotes the set of all probability distributions over the finite action set A_i , $i = 1, 2$.

This structural antagonism has powerful implications for equilibrium analysis. A mixed-strategy profile (σ_1^*, σ_2^*) is a *saddle-point equilibrium*, if and only if, it satisfies the minimax/maximin identities

$$\min_{\sigma_1 \in \Delta(A_1)} \max_{\sigma_2 \in \Delta(A_2)} U(\sigma_1, \sigma_2) = U(\sigma_1^*, \sigma_2^*) = \max_{\sigma_2 \in \Delta(A_2)} \min_{\sigma_1 \in \Delta(A_1)} U(\sigma_1, \sigma_2).$$

The left-hand side reflects P1's ability to secure the smallest possible expected payoff against optimal play by P2, while the right-hand side reflects P2's ability to secure the largest possible expected payoff against optimal play by P1. Von Neumann's minimax theorem guarantees that such saddle points always exist in finite zero-sum games. If σ_1^* and σ_2^* are one-point distributions, then we say that the zero-sum game admits a *pure-strategy saddle point*.

Equilibrium computation in this setting is intimately connected with convex optimization and saddle-point theory. These principles also underpin many modern adversarial learning methods used in machine learning, cybersecurity, and control systems.

2.1.1 ■ Matrix Games

A *matrix game* is a standard and tractable representation of a two-person zero-sum game in which the strategic interaction between players is captured in a tabular format. In this formulation, P1 (the row player) chooses among a finite set of actions indexed by the rows of a matrix, while P2 (the column player) simultaneously selects from a finite set of actions indexed by the columns. The outcome of the game is determined by the entry in the matrix corresponding to the selected row and column.

Formally, the game is described by a real-valued matrix $A = [a_{ij}] \in \mathbb{R}^{m \times n}$, where m is the number of pure strategies available to P1 and n is the number of pure strategies available to P2. If P1 chooses row $i \in \{1, \dots, m\}$ and P2 chooses column $j \in \{1, \dots, n\}$, the resulting payoff is a_{ij} . This value represents the cost/loss to P1 and, by the zero-sum property, the corresponding gain to P2, whose payoff is $-a_{ij}$.

The matrix structure provides a compact and visually intuitive framework for analyzing the game. Each cell of the matrix encodes a possible outcome, allowing players to reason about the best responses, dominant strategies, and worst-case guarantees. Moreover, matrix games serve as the foundation for key solution concepts in zero-sum settings, including security strategies, saddle points, and the application of the minimax theorem. The matrix form also facilitates the use of algorithmic and computational tools, such as linear programming and graphical methods, to compute optimal strategies and game values.

2.1.2 ■ Security Strategies

In zero-sum games, each player seeks to guarantee an acceptable outcome, regardless of how the opponent acts. This leads to the concept of *security strategies*, which are strategies that protect a player against the worst-case behavior of the opponent. For P1, who aims to minimize losses, the objective is to secure the smallest possible maximum payoff the opponent might impose. In contrast, P2, aiming to maximize gains, seeks to secure the largest possible minimum payoff the opponent might impose. These strategies are rooted in a conservative outlook: each player plans against the most unfavorable response of the opponent.

P1 must commit to an action before P2 makes a choice. In this sequential setting, P1 is at a disadvantage, since P2 can observe P1's action and respond optimally. Anticipating this, P1 evaluates each available strategy and selects the one that minimizes the worst-case loss that P2 can impose.

This leads to the *minimax strategy*. P1 selects a row i^* such that the maximum entry in that row is as small as possible. Formally, the resulting value is

$$\bar{V} = \min_{i \in \{1, \dots, m\}} \max_{j \in \{1, \dots, n\}} a_{ij} = \max_{j \in \{1, \dots, n\}} a_{i^*j},$$

and the quantity $\bar{V}$ is called the *upper value* of the game. The action i^* that attains this value is P1's *security strategy*, guaranteeing that P1's payoff does not exceed $\bar{V}$ regardless of P2's response.

Now consider the reverse order of play, in which P2 must commit to an action before P1 makes a choice. In this case, the advantage goes to P1, who can observe P2's action and respond optimally. Anticipating this, P2 selects a strategy that maximizes the minimum payoff she can guarantee, regardless of P1's response.

This leads to the *maximin strategy*. P2 selects a column j^* that maximizes the smallest entry in that column, yielding

$$\underline{V} = \max_{j \in \{1, \dots, n\}} \min_{i \in \{1, \dots, m\}} a_{ij} = \min_{i \in \{1, \dots, m\}} a_{ij^*},$$

and the quantity $\underline{V}$ is called the *lower value* of the game. The action j^* that attains this value is P2's *security strategy*, guaranteeing that P2's payoff is at least $\underline{V}$ regardless of P1's response.

The upper and lower values satisfy the fundamental inequality $\underline{V} \leq \bar{V}$, which reflects the fact that the maximum payoff the maximizer can guarantee by committing first, represented by the lower value, can never exceed the minimum payoff the minimizer can force when committing first, represented by the upper value. If $\underline{V} \neq \bar{V}$, then the game does not admit a saddle point in pure strategies. In such cases, the players must resort to *mixed strategies*, that is, randomized strategies, to secure equilibrium outcomes.

However, when the upper and lower values coincide, that is, $\underline{V} = \bar{V} =: V$, the game admits a *pure-strategy saddle point*. At this saddle point, both players' security strategies are mutual best responses, and neither player has an incentive to deviate (see Definition 2.3 in the next section). The value V represents the equilibrium outcome of the game and is called the *value of the game*. In this setting, the notions of security strategies and best responses coincide, and the game reaches a stable outcome in pure strategies.

Example 2.1 A 2×2 Zero-Sum Game without a Pure-Strategy Saddle Point.

Consider the payoff matrix

$$A = \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix},$$

where P1 (the row player) seeks to minimize the payoff and P2 (the column player) seeks to maximize it. Neither row strictly dominates the other, and neither column strictly dominates the other.

We first compute the upper value by examining P1's minimax problem. For each row, the worst-case outcome is

$$\max_j a_{1j} = \max\{2, 0\} = 2, \quad \max_j a_{2j} = \max\{1, 3\} = 3.$$

P1 selects the row that minimizes this quantity, yielding $\bar{V} = \min\{2, 3\} = 2$.

Next, we compute the lower value by examining P2's maximin problem. For each column, the guaranteed outcome is

$$\min_i a_{i1} = \min\{2, 1\} = 1, \quad \min_i a_{i2} = \min\{0, 3\} = 0.$$

P2 selects the column that maximizes this quantity, yielding $\underline{V} = \max\{1, 0\} = 1$.

Since $\underline{V} < \overline{V}$, the game does not admit a saddle point in pure strategies. No action profile simultaneously minimizes P1's worst-case loss and maximizes P2's guaranteed payoff. This example illustrates that, in the absence of a pure-strategy saddle point, the security levels of the two players do not coincide. As a consequence, pure strategies are insufficient to characterize equilibrium behavior, motivating the introduction of mixed strategies and the minimax theorem in the next section. $\square$

Example 2.2 Matching Pennies.

Consider the payoff matrix of the *matching pennies* game:

$$A = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix},$$

where P1 (the row player) wins +1 if the coins match and loses −1 if they differ. P2's payoff is the negative of P1's payoff. For each row, take the maximum entry:

$$\max_j a_{1j} = \max\{1, -1\} = 1, \quad \max_j a_{2j} = \max\{-1, 1\} = 1.$$

Then $\overline{V} = \min\{1, 1\} = 1$. For each column, take the minimum entry:

$$\min_i a_{i1} = \min\{1, -1\} = -1, \quad \min_i a_{i2} = \min\{-1, 1\} = -1.$$

Then, $\underline{V} = \max\{-1, -1\} = -1$. We find $\underline{V} = -1 < 1 = \overline{V}$, and thus the game has no pure-strategy saddle point. Intuitively, each player can always undermine the other's choice: if P1 attempts to secure +1 by choosing row 1, P2 can respond with column 2 and inflict −1, and an analogous response defeats every other pure choice. This cycling prevents a stable pure-strategy outcome. $\square$

2.2 ■ Saddle-point Equilibrium

A central solution concept in two-person zero-sum games is the *saddle-point equilibrium*. This concept characterizes a strategy profile in which each player is playing her best response to the other, and neither player can improve their outcome by unilaterally deviating. The existence of a saddle point implies strategic stability: once reached, no player has an incentive to change her strategy.

Definition 2.3 (Pure-Strategy Saddle-Point Equilibrium and Value of the Game). Let $A = (a_{ij}) \in \mathbb{R}^{m \times n}$ be the payoff matrix of a two-player zero-sum game, where a_{ij} represents the

loss or cost to P1 (the row player) when P1 chooses row i and P2 (the column player) chooses column j .

An entry $a_{i^*j^*}$ in A is called a saddle-point equilibrium in pure strategies if it satisfies

$$a_{i^*j} \leq a_{i^*j^*} \leq a_{ij^*}, \quad \text{for all } i \in \{1, \dots, m\}, j \in \{1, \dots, n\}.$$

At such a saddle point, the action profile (i^*, j^*) constitutes a saddle-point equilibrium in pure strategies: P1 can guarantee at least the payoff $a_{i^*j^*}$ regardless of P2's choice, while P2 can restrict P1 to at most the same payoff regardless of P1's choice. The common value $V = a_{i^*j^*}$ is called the value of the game. If a saddle point exists, both players can play their equilibrium pure strategies with certainty, and the outcome of the game is uniquely determined by V .

This definition states that the entry $a_{i^*j^*}$ is simultaneously the maximum in its row and the minimum in its column. In other words, P1 cannot reduce her loss by switching to another row, and P2 cannot improve her gain by switching to another column. Thus, the strategy profile (i^*, j^*) constitutes a pair of mutual best responses. The entry $a_{i^*j^*}$ is then called the *value of the game* under pure strategies, and the corresponding actions i^* and j^* are actions in saddle-point equilibrium, which can also be viewed as *security strategies* of the players. The existence of such a saddle point is closely tied to the concepts of the game's upper and lower values. The following theorem formalizes this relationship.

Theorem 2.4 (Security Levels and Pure-Strategy Saddle-Point Equilibrium in Matrix Games).

Let $A = (a_{ij}) \in \mathbb{R}^{m \times n}$ be the payoff matrix of a finite two-player zero-sum matrix game in which **P1** is the minimizer (row player, choosing rows i) and **P2** is the maximizer (column player, choosing columns j). Define the security levels

$$\underline{V}(A) := \max_j \min_i a_{ij} \quad \text{and} \quad \overline{V}(A) := \min_i \max_j a_{ij}.$$

Then:

- (i) (Uniqueness of levels) The values $\underline{V}(A)$ and $\overline{V}(A)$ are uniquely defined.¹
- (ii) (Existence of security strategies) There exist an index $i^* \in \arg \min_i \max_j a_{ij}$ (a security strategy for P1) and an index $j^* \in \arg \max_j \min_i a_{ij}$ (a security strategy for P2).
- (iii) (Minimax inequality) The maximizer's security level never exceeds the minimizer's, i.e.,

$$\underline{V}(A) \leq \overline{V}(A).$$

- (iv) (saddle-point equivalence) The following are equivalent:

- (a) $\underline{V}(A) = \overline{V}(A)$;
- (b) there exists a (pure) saddle-point equilibrium (i^*, j^*) with

$$a_{i^*j^*} = \min_i a_{ij^*} = \max_j a_{i^*j} = \min_i \max_j a_{ij} = \max_j \min_i a_{ij}.$$

In this case, $a_{i^*j^*}$ is the value of the game, and i^* and j^* are security strategies for P1 and P2, respectively.

¹Uniqueness here refers to the values; the sets of strategies attaining them need not be unique.

Proof. Parts (i) and (ii) hold because the finite maxima and minima are attained. For (iii), observe that for every pair (k, ℓ) ,

$$\min_i a_{i\ell} \leq a_{k\ell} \leq \max_j a_{kj}.$$

Taking $\max_\ell$ on the left inequality and $\min_k$ on the right yields $\max_\ell \min_i a_{i\ell} \leq \min_k \max_j a_{kj}$, i.e., $\underline{V}(A) \leq \overline{V}(A)$.

For (iv), if a saddle point (i^*, j^*) exists, then $a_{i^*j} \leq a_{i^*j^*} \leq a_{ij^*}$ for all i, j , hence $a_{i^*j^*} = \min_i a_{ij^*} = \max_j a_{i^*j}$ and consequently $\underline{V}(A) = \overline{V}(A) = a_{i^*j^*}$. Conversely, if $\underline{V}(A) = \overline{V}(A) =: V$, choose $i^* \in \arg \min_i \max_j a_{ij}$ and $j^* \in \arg \max_j \min_i a_{ij}$. Then

$$a_{i^*j^*} \leq \max_j a_{i^*j} = \overline{V} = \underline{V} = \min_i a_{ij^*} \leq a_{i^*j^*},$$

forcing equalities throughout and leading to the saddle-point inequalities, and thus (i^*, j^*) is a saddle-point with value V . $\square$

Corollary 2.5 (Ordered Interchangeability). *In a matrix game A , suppose that both (i_1, j_1) and (i_2, j_2) are saddle-point strategy pairs. Then the mixed pairs (i_1, j_2) and (i_2, j_1) are also saddle-point strategy pairs.*

Proof. The result follows from Theorem 2.4(iv). Since (i_1, j_1) and (i_2, j_2) are saddle points, both i_1 and i_2 are security strategies for P1, while j_1 and j_2 are security strategies for P2. Therefore, any ordered combination (i_k, j_ℓ) with $k, \ell \in \{1, 2\}$ constitutes a saddle point. $\square$

This interchangeability shows that when multiple saddle points exist, a player need not anticipate the opponent's specific choice among them. All such strategies are mutually compatible and guarantee the same game value, ensuring robustness of equilibrium play.

Not all matrix games have pure-strategy saddle points, as we have seen earlier. When $\overline{V} \neq \underline{V}$, the game lacks such a pure-strategy solution. In this case, players must resort to *mixed strategies*, probability distributions over their actions, to reach equilibrium. The celebrated *Minimax Theorem* guarantees that every finite zero-sum game has a saddle-point equilibrium in mixed strategies, ensuring strategic predictability even in the absence of pure equilibria.

2.3 ■ Mixed Strategies

In the absence of a pure-strategy saddle point, players may adopt *mixed strategies* to achieve strategic stability; i.e., once both players adopt their equilibrium mixed strategies, neither has an incentive to deviate unilaterally. A mixed strategy assigns probabilities to pure strategies (equivalently, actions), allowing players to randomly select their decisions. The idea is to make one's behavior unpredictable while still securing the best worst-case outcome.

Definition 2.6 (Mixed Strategy). *A mixed strategy for a player is a probability distribution over her finite set of pure strategies. For P1 (the row player with m actions), a mixed strategy is a vector $p = (p_1, \dots, p_m)^T \in \Delta_m$, where*

$$\Delta_m = \left\{ p \in \mathbb{R}^m : p_i \geq 0 \text{ for all } i, \quad \sum_{i=1}^m p_i = 1 \right\}.$$

Similarly, for P2 (the column player with n actions), a mixed strategy is $q = (q_1, \dots, q_n)^T \in \Delta_n$,

where

$$\Delta_n = \left\{ q \in \mathbb{R}^n : q_j \geq 0 \text{ for all } j, \quad \sum_{j=1}^n q_j = 1 \right\}.$$

Identifying the action sets of the two players with the row and column indices, $A_1 = \{1, \dots, m\}$ and $A_2 = \{1, \dots, n\}$, the simplices above are precisely $\Delta_m = \Delta(A_1)$ and $\Delta_n = \Delta(A_2)$, introduced earlier, so that $p \in \Delta(A_1)$ and $q \in \Delta(A_2)$. Throughout this lecture we retain the dimension-explicit notation Δ_m and Δ_n , which is convenient for matrix-vector computations.

Given mixed strategies $p \in \Delta_m$ and $q \in \Delta_n$, the expected loss for P1 (or the expected payoff for P2) is

$$J(p, q) = p^T A q.$$

This is the bilinear form induced by the matrix A , reflecting the expected outcome when both players randomize according to p and q .

Definition 2.7 (Mixed Security Strategies and Security Levels). Let $A \in \mathbb{R}^{m \times n}$ be the payoff matrix of a finite two-player zero-sum game. Denote by Δ_m and Δ_n the simplices of mixed strategies for P1 (row player) and P2 (column player), respectively.

The lower value of the game (in mixed strategies) is defined as

$$\underline{V}_{\text{ms}}(A) := \max_{q \in \Delta_n} \min_{p \in \Delta_m} p^T A q,$$

and the upper value of the game (in mixed strategies) is defined as

$$\overline{V}_{\text{ms}}(A) := \min_{p \in \Delta_m} \max_{q \in \Delta_n} p^T A q.$$

In general, one always has $\underline{V}_{\text{ms}}(A) \leq \overline{V}_{\text{ms}}(A)$.

A strategy $p^* \in \Delta_m$ is called a mixed security strategy for P1 if

$$\max_{q \in \Delta_n} (p^*)^T A q = \overline{V}_{\text{ms}}(A),$$

and similarly, $q^* \in \Delta_n$ is a mixed security strategy for P2 if

$$\min_{p \in \Delta_m} p^T A q^* = \underline{V}_{\text{ms}}(A).$$

Definition 2.8 (Mixed-Strategy Saddle Point Equilibrium). A pair $(p^*, q^*) \in \Delta_m \times \Delta_n$ is called a mixed-strategy saddle point (or mixed-strategy equilibrium) if

$$p^T A q^* \leq (p^*)^T A q^* \leq (p^*)^T A q \quad \text{for all } p \in \Delta_m, q \in \Delta_n.$$

Equivalently, (p^*, q^*) is a pair of mutual best responses, and the common value

$$(p^*)^T A q^* = \underline{V}_{\text{ms}}(A) = \overline{V}_{\text{ms}}(A)$$

is called the value of the game (in mixed strategies).

Theorem 2.9 (Minimax Theorem [138]). Let $A \in \mathbb{R}^{m \times n}$ be the payoff matrix of a finite two-player zero-sum game, and let

$$\overline{V}_{\text{ms}}(A) := \min_{p \in \Delta_m} \max_{q \in \Delta_n} p^T A q, \quad \underline{V}_{\text{ms}}(A) := \max_{q \in \Delta_n} \min_{p \in \Delta_m} p^T A q.$$

Then,

$$\overline{V}_{\text{ms}}(A) = \underline{V}_{\text{ms}}(A).$$

Moreover, there exists a pair $(p^*, q^*) \in \Delta_m \times \Delta_n$ attaining this common value and satisfying the saddle-point inequalities

$$p^\top A q^* \leq (p^*)^\top A q^* \leq (p^*)^\top A q, \quad \forall p \in \Delta_m, q \in \Delta_n.$$

Proof. Since Δ_n is the convex hull of the unit vectors $\{e_j\}$ and $p^\top A(\cdot)$ is linear, we have

$$\max_{q \in \Delta_n} p^\top A q = \max_j (A^\top p)_j.$$

Dually, since $\Delta_m = \text{conv}\{e_i\}$ and $(\cdot)^\top A q$ is linear,

$$\min_{p \in \Delta_m} p^\top A q = \min_i (A q)_i.$$

By Gordan's theorem of the alternative, for any A exactly one of the systems

$$\exists p \in \Delta_m : A^\top p \leq 0 \quad \text{or} \quad \exists q \in \Delta_n : A q \geq 0$$

is feasible. If $A^\top p^\circ \leq 0$ for some p° , then

$$\overline{V}_{\text{ms}}(A) = \min_p \max_q p^\top A q \leq \max_q p^{\circ\top} A q = \max_j (A^\top p^\circ)_j \leq 0.$$

If instead $A q^\circ \geq 0$ for some q° , then

$$\underline{V}_{\text{ms}}(A) = \max_q \min_p p^\top A q \geq \min_p p^\top A q^\circ = \min_i (A q^\circ)_i \geq 0.$$

Hence, for every A , at least one of

$$\overline{V}_{\text{ms}}(A) \leq 0 \quad \text{or} \quad \underline{V}_{\text{ms}}(A) \geq 0 \tag{2.1}$$

must hold.

Now fix $c \in \mathbb{R}$ and set $B = A - c \mathbf{1}\mathbf{1}^\top$. For any $(p, q) \in \Delta_m \times \Delta_n$ we have

$$p^\top B q = p^\top A q - c,$$

and hence

$$\overline{V}_{\text{ms}}(B) = \overline{V}_{\text{ms}}(A) - c, \quad \underline{V}_{\text{ms}}(B) = \underline{V}_{\text{ms}}(A) - c.$$

Applying (2.1) to B gives: for every c ,

$$\overline{V}_{\text{ms}}(A) \leq c \quad \text{or} \quad \underline{V}_{\text{ms}}(A) \geq c. \tag{2.2}$$

If $\overline{V}_{\text{ms}}(A) > \underline{V}_{\text{ms}}(A)$, choose

$$c = \frac{1}{2}(\overline{V}_{\text{ms}}(A) + \underline{V}_{\text{ms}}(A)).$$

Then, $\underline{V}_{\text{ms}}(A) < c < \overline{V}_{\text{ms}}(A)$, and thus neither disjunct in (2.2) holds, which leads to a contradiction. Therefore

$$\overline{V}_{\text{ms}}(A) = \underline{V}_{\text{ms}}(A) =: V.$$

Define $\phi(p) = \max_{q \in \Delta_n} p^\top Aq$ and $\psi(q) = \min_{p \in \Delta_m} p^\top Aq$. By the formulas above, both are continuous on the compact simplices. Hence there exist

$$p^* \in \arg \min_p \phi(p), \quad q^* \in \arg \max_q \psi(q).$$

Using $\phi(p^*) = \psi(q^*) = V$, we obtain

$$p^\top Aq^* \leq \psi(q^*) = V = \phi(p^*) \leq (p^*)^\top Aq \quad \forall p, q,$$

which is precisely the saddle-point condition. In particular, $(p^*)^\top Aq^* = V$. $\square$

The Minimax Theorem ensures that even when no pure-strategy saddle point exists (i.e., when $\min_i \max_j a_{ij} > \max_j \min_i a_{ij}$), equilibrium still exists in mixed strategies. This result lies at the foundation of modern game theory, and it establishes that all finite zero-sum games admit stable, optimal solutions in the form of mixed strategies; general minimax and saddle-point results beyond the matrix case, including Sion's minimax theorem, are reviewed in Appendix B (Mathematical Background).

2.4 ■ Computing Saddle-Point Equilibria

2.4.1 ■ Graphical Methods

Graphical methods provide an intuitive and visual approach to computing saddle-point equilibria in two-person zero-sum games when one player has only two pure strategies. These methods are most effective for small games of the form $2 \times n$ or $m \times 2$. The central idea is that the expected payoff, as a function of the opponent's mixing probability, is piecewise linear. By plotting these lines and identifying their upper or lower envelopes, one can visually locate the equilibrium strategy and game value.

Consider a $2 \times n$ matrix game with payoff matrix $A = (a_{ij})$, where P1 has two pure strategies and P2 has n . If P1 mixes with

$$p = (y, 1 - y)^\top, \quad y \in [0, 1],$$

then P1's expected payoff against the choice of column j by P2 is

$$J_j(y) = y a_{1j} + (1 - y) a_{2j} = (a_{1j} - a_{2j})y + a_{2j}.$$

Each $J_j(y)$ is a straight line in y .

P1 seeks to minimize the maximum payoff across P2's pure strategies, $J(y) = \max_j J_j(y)$. The minimizer y^* of this upper envelope yields the optimal mixed strategy $p^* = (y^*, 1 - y^*)^\top$.

By symmetry, in an $m \times 2$ game, P2 mixes as $q = (z, 1 - z)^\top$, the loss to P1's pure row i is

$$J_i(z) = z a_{i1} + (1 - z) a_{i2},$$

and the lower envelope $\min_i J_i(z)$ determines P2's best response.

Example 2.10 A 2×2 Game.

Consider

$$A = \begin{bmatrix} 3 & 0 \\ -1 & 1 \end{bmatrix}.$$

Compute the lower and upper values:

$$\underline{V} = \max_j \min_i a_{ij} = \max(-1, 0) = 0, \quad \bar{V} = \min_i \max_j a_{ij} = \min(3, 1) = 1.$$

Since $\underline{V} \neq \bar{V}$, no pure-strategy saddle point exists.

If P2 chooses column 1, P1's loss is

$$J_1(y) = 3y + (-1)(1 - y) = 4y - 1.$$

If P2 chooses column 2, then

$$J_2(y) = 0 \cdot y + 1(1 - y) = 1 - y.$$

The equilibrium occurs where the two lines intersect:

$$4y - 1 = 1 - y \Rightarrow y^* = \frac{2}{5}.$$

Thus P1 mixes as $p^* = (\frac{2}{5}, \frac{3}{5})$. At this point, $J(y^*) = 0.6$.

Let $q = (z, 1 - z)$. P2's payoffs of choosing z against P1's row choices $i = 1, 2$ are

$$J'_1(z) = 3z, \quad J'_2(z) = -z + (1 - z) = 1 - 2z.$$

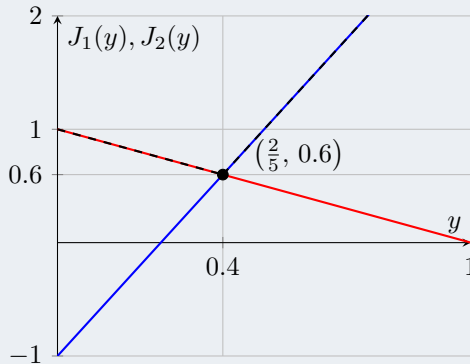
Taking the lower envelope, the intersection occurs at

$$3z = 1 - 2z \Rightarrow z^* = \frac{1}{5}.$$

So P2 mixes as $q^* = (\frac{1}{5}, \frac{4}{5})$. The value is $V = 0.6$, consistent with P1's calculation.

The unique mixed-strategy saddle point is thus

$$p^* = \left(\frac{2}{5}, \frac{3}{5}\right), \quad q^* = \left(\frac{1}{5}, \frac{4}{5}\right), \quad V = 0.6.$$



$$\text{---} J_1(y) = 4y - 1 \quad \text{---} J_2(y) = 1 - y$$

Figure 2.1: Graphical method for P1. The equilibrium occurs at the intersection of the two payoff lines, minimizing the upper envelope.

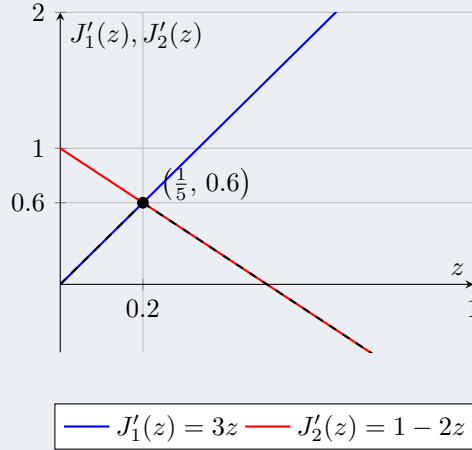


Figure 2.2: Graphical method for P2. The equilibrium occurs at the intersection of the two payoff lines, maximizing the lower envelope.

Figures 2.1 and 2.2 illustrate the dual perspectives of the players. P1 evaluates the losses from each of P2's pure-strategy choices and seeks to minimize the *upper envelope* of these linear functions. This corresponds to hedging against the worst possible column choice. Conversely, P2 evaluates the payoffs from each of P1's pure-strategy choices and seeks to maximize the *lower envelope*, ensuring protection against the worst possible row choice.

At equilibrium, both players' strategies are chosen so that no unilateral deviation can improve their outcome. In particular, each player's mixed strategy makes the opponent indifferent among those pure actions that are assigned positive probability in equilibrium. Specifically, P1's mixing probability $y^* = \frac{2}{5}$ equalizes P2's payoffs from column 1 and column 2, since $J_1(y^*) = J_2(y^*) = 0.6$. As a result, P2 is indifferent between these two columns, both of which belong to the support of P2's equilibrium mixed strategy. Likewise, P2's mixing probability $z^* = \frac{1}{5}$ equalizes P1's losses from row 1 and row 2, $J'_1(z^*) = J'_2(z^*) = 0.6$, making P1 indifferent between these rows, which constitute the support of P1's equilibrium mixed strategy.

More generally, in a mixed-strategy saddle-point equilibrium, indifference need only hold across actions that receive positive probability. Actions assigned zero probability may yield strictly worse payoffs and are therefore excluded from the support. This support-based indifference condition is the essential mechanism ensuring equilibrium stability in mixed strategies.

□

Example 2.11 A 2×3 Game.

Consider

$$A = \begin{bmatrix} 15 & 0 & 11 \\ 5 & 15 & 6 \end{bmatrix}.$$

If P1 mixes with $p = (y, 1 - y)^\top$, the loss against each column is

$$\begin{aligned} J_1(y) &= (15 - 5)y + 5 = 10y + 5, \\ J_2(y) &= (0 - 15)y + 15 = 15 - 15y, \\ J_3(y) &= (11 - 6)y + 6 = 5y + 6. \end{aligned}$$

The pairwise intersections occur at

$$\begin{aligned} J_1 &= J_2 : y = \frac{2}{5}, J = 9, \\ J_1 &= J_3 : y = \frac{1}{5}, J = 7, \\ J_2 &= J_3 : y = \frac{9}{20}, J = \frac{33}{4} = 8.25. \end{aligned}$$

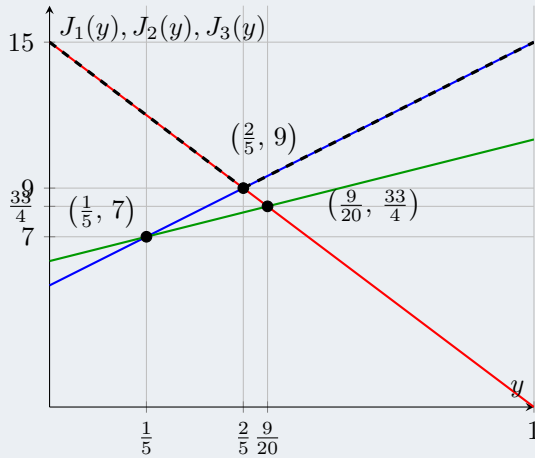
The upper envelope $J(y) = \max\{J_1(y), J_2(y), J_3(y)\}$ is dominated by J_2 in $[0, 0.4]$ and by J_1 on $[0.4, 1]$ (note that J_3 never exceeds the maximum in $[0, 1]$). Hence, the minimizer of the upper envelope is the intersection of the active lines J_1 and J_2 at

$$y^* = \frac{2}{5}, \quad V = J(y^*) = 9.$$

Therefore, P1's optimal strategy is $p^* = (\frac{2}{5}, \frac{3}{5})$. At the optimum, the active columns are C_1 and C_2 ; P2 mixes over them to equalize P1's rows:

$$\begin{aligned} \text{Let } q &= (\alpha, \beta, 0), \quad \alpha + \beta = 1, \\ 15\alpha &= 5\alpha + 15\beta \quad \Rightarrow \quad \alpha = \frac{3}{5}, \beta = \frac{2}{5}. \end{aligned}$$

Thus, $q^* = (\frac{3}{5}, \frac{2}{5}, 0)$ and the game value is $V = 9$.



$$\text{--- } J_1(y) = 10y + 5 \quad \text{--- } J_2(y) = 15 - 15y \quad \text{--- } J_3(y) = 5y + 6$$

Figure 2.3: Graphical solution of the 2×3 game: three lines intersect at distinct points; the upper envelope switches from J_2 to J_1 at $y^* = \frac{2}{5}$, yielding value $V = 9$.

□

Graphical methods thus provide a powerful and visual tool to solve small matrix games. However, their utility is limited to low-dimensional settings, and for higher-dimensional games, one must rely on linear programming or algorithmic methods such as the simplex or iterative methods (e.g., fictitious play or gradient descent-based approaches, as discussed next).

2.4.2 ■ Linear Programming Formulation

Graphical methods are intuitive and effective for small matrix games, but they become infeasible as the game size grows. A more scalable and general technique for solving mixed-strategy equilibria is based on *linear programming* (LP). This approach leverages the fact that the min-max optimization problem in two-person zero-sum games is equivalent to solving a pair of dual LPs; see Appendix B (Mathematical Background) for the relevant background on Lagrangian and linear programming duality.

Let $A = [a_{ij}] \in \mathbb{R}^{m \times n}$ be the payoff matrix of a two-person zero-sum game, where P1 chooses rows and P2 chooses columns. Assume, without loss of generality, that $a_{ij} > 0$ for all i, j . This assumption simplifies the LP formulation and can always be ensured by adding a constant to all entries of the matrix (see Lemma 2.14 below).

The value of the game under mixed strategies is given by the min-max and max-min equality:

$$V_{\text{ms}}(A) = \min_{p \in \Delta_m} \max_{q \in \Delta_n} p^T A q = \max_{q \in \Delta_n} \min_{p \in \Delta_m} p^T A q.$$

Let us first consider the problem from P1's perspective.

P1's LP: Minimizing the Maximum Loss

Let $V_1(p) = \max_{q \in \Delta_n} p^T A q$ be the worst-case expected loss for P1 when using strategy $p \in \Delta_m$. Then, the min-max optimization becomes:

$$\min_{p \in \Delta_m} \max_{q \in \Delta_n} p^T A q = \min_{p \in \Delta_m} V_1(p). \quad (2.3)$$

Note that for any $q \in \Delta_n$, we have $V_1(p) \geq p^T A q$, and thus:

$$V_1(p) \cdot \mathbf{1}_n \geq A^T p.$$

Introduce a variable substitution: let $\tilde{p} = \frac{p}{V_1(p)}$. The preceding inequality then becomes:

$$A^T \tilde{p} \leq \mathbf{1}_n.$$

Since $p \in \Delta_m$, we have $\sum_i p_i = 1 \Rightarrow \sum_i \tilde{p}_i = \frac{1}{V_1(p)}$. Therefore, solving the following LP gives $\tilde{p}$, from which we recover $V_{\text{ms}}(A) = \frac{1}{\sum_i \tilde{p}_i}$:

$$\begin{aligned} & \text{maximize} && \tilde{p}^T \mathbf{1}_m \\ & \text{subject to} && A^T \tilde{p} \leq \mathbf{1}_n, \\ & && \tilde{p} \geq 0. \end{aligned}$$

P2's LP: Maximizing the Minimum Gain

Symmetrically, P2 wishes to maximize the minimum expected gain:

$$V_2(q) = \min_{p \in \Delta_m} p^T Aq.$$

For fixed q , we have

$$V_2(q) \leq p^T Aq, \quad \forall p \in \Delta_m.$$

This implies $Aq \geq V_2(q) \cdot \mathbf{1}_m$, or equivalently:

$$A\tilde{q} \geq \mathbf{1}_m, \quad \text{where } \tilde{q} = \frac{q}{V_2(q)}.$$

The dual LP becomes

$$\begin{aligned} & \text{minimize} && \tilde{q}^T \mathbf{1}_n \\ & \text{subject to} && A\tilde{q} \geq \mathbf{1}_m, \\ & && \tilde{q} \geq 0. \end{aligned}$$

Theorem 2.12 (Linear Programming Formulation of Zero-Sum Matrix Games). *Let $A \in \mathbb{R}^{m \times n}$ be the payoff matrix in a finite two-player zero-sum game, with mixed-strategy simplices $\Delta_m \subset \mathbb{R}^m$ and $\Delta_n \subset \mathbb{R}^n$. Since all entries of A are strictly positive by assumption, the (minimax) game value $V_{\text{ms}}(A)$ is automatically strictly positive. Then, the equilibrium can be obtained from the primal-dual LP pair*

$$(P) \text{ P1: } \max_{\tilde{p} \in \mathbb{R}_{\geq 0}^m} \mathbf{1}_m^T \tilde{p} \quad \text{s.t.} \quad A^T \tilde{p} \leq \mathbf{1}_n, \quad (2.4)$$

$$(D) \text{ P2: } \min_{\tilde{q} \in \mathbb{R}_{\geq 0}^n} \mathbf{1}_n^T \tilde{q} \quad \text{s.t.} \quad A\tilde{q} \geq \mathbf{1}_m. \quad (2.5)$$

Both programs are feasible and bounded; their optimal values coincide by LP duality:

$$\text{OPT}(P) = \text{OPT}(D) = \frac{1}{V_{\text{ms}}(A)}.$$

Moreover, from any optimal solutions $\tilde{p}^*$ and $\tilde{q}^*$, the unique mixed-strategy saddle point (up to ties) and the game value are recovered as

$$p^* = \frac{\tilde{p}^*}{\mathbf{1}_m^T \tilde{p}^*}, \quad q^* = \frac{\tilde{q}^*}{\mathbf{1}_n^T \tilde{q}^*}, \quad V_{\text{ms}}(A) = \frac{1}{\mathbf{1}_m^T \tilde{p}^*} = \frac{1}{\mathbf{1}_n^T \tilde{q}^*}.$$

Proof. Define $V_1(p) = \max_{q \in \Delta_n} p^T Aq$ for $p \in \Delta_m$. For any $q \in \Delta_n$ we have $V_1(p) \geq p^T Aq$, and hence $V_1(p)\mathbf{1}_n \geq A^T p$. If $V_1(p) > 0$, scale by $\tilde{p} := p/V_1(p)$ to obtain $A^T \tilde{p} \leq \mathbf{1}_n$ with $\tilde{p} \geq 0$. Since $\sum_i p_i = 1$, we have $\mathbf{1}_m^T \tilde{p} = 1/V_1(p)$. Minimizing $V_1(p)$ over $p \in \Delta_m$ is thus equivalent to maximizing $\mathbf{1}_m^T \tilde{p}$ subject to $A^T \tilde{p} \leq \mathbf{1}_n$, $\tilde{p} \geq 0$, i.e., program (2.4). At the maximizer $\tilde{p}^*$, the objective value equals $1/V_{\text{ms}}(A)$, and thus $V_{\text{ms}}(A) = 1/(\mathbf{1}_m^T \tilde{p}^*)$ and $p^* = \tilde{p}^*/(\mathbf{1}_m^T \tilde{p}^*)$.

Dually, define $V_2(q) = \min_{p \in \Delta_m} p^T Aq$. For all $p \in \Delta_m$ we have $p^T Aq \geq V_2(q)$, which implies $Aq \geq V_2(q)\mathbf{1}_m$. With $\tilde{q} := q/V_2(q)$ (assuming $V_2(q) > 0$) we get $A\tilde{q} \geq \mathbf{1}_m$, $\tilde{q} \geq 0$, and $\mathbf{1}_n^T \tilde{q} = 1/V_2(q)$. Maximizing $V_2(q)$ over $q \in \Delta_n$ is therefore equivalent to minimizing $\mathbf{1}_n^T \tilde{q}$ subject to $A\tilde{q} \geq \mathbf{1}_m$, $\tilde{q} \geq 0$, i.e., program (2.5). Standard LP duality yields $\text{OPT}(P) = \text{OPT}(D)$, and hence both equal $1/V_{\text{ms}}(A)$. Normalizing $\tilde{q}^*$ recovers q^* and $V_{\text{ms}}(A)$ as stated.

Feasibility follows since, after a positive shift if necessary, the constraints admit strictly feasible points. Boundedness is ensured because scaling up $\tilde{p}$ or $\tilde{q}$ would violate $A^\top \tilde{p} \leq \mathbf{1}_n$ or $A\tilde{q} \geq \mathbf{1}_m$ respectively. Complementary slackness aligns supports of p^* and q^* with tight inequalities in (2.4) and (2.5), yielding the saddle-point conditions. $\square$

Example 2.13 Linear Programming Solution for the 2×2 Game in Example 2.10.

We revisit the 2×2 zero-sum matrix game introduced earlier in Example 2.10. This game admits no pure-strategy saddle point but has a mixed-strategy saddle-point equilibrium. We now compute the equilibrium using the linear programming formulation. The primal LP for P1 (the minimizer) is

$$\text{maximize } \tilde{p}_1 + \tilde{p}_2, \quad \text{subject to } \begin{cases} 3\tilde{p}_1 - \tilde{p}_2 \leq 1, \\ \tilde{p}_2 \leq 1, \\ \tilde{p}_1, \tilde{p}_2 \geq 0. \end{cases}$$

Solving this LP yields

$$\tilde{p}^* = \begin{pmatrix} \frac{2}{3} \\ \frac{1}{3} \end{pmatrix}, \quad V_{\text{ms}}(A) = \frac{1}{\tilde{p}_1^* + \tilde{p}_2^*} = 0.6.$$

Normalizing, we recover P1's equilibrium mixed strategy

$$p^* = \frac{\tilde{p}^*}{\tilde{p}_1^* + \tilde{p}_2^*} = \begin{pmatrix} \frac{2}{5} \\ \frac{3}{5} \end{pmatrix}.$$

Applying the dual LP for P2 yields the equilibrium mixed strategy

$$q^* = \begin{pmatrix} \frac{1}{5} \\ \frac{4}{5} \end{pmatrix}.$$

The value of the game is

$$V = p^{*\top} A q^* = 0.6,$$

which coincides with the value obtained earlier via the graphical method. $\square$

Sometimes a matrix A contains negative entries. We can shift all values by a constant $c > 0$ to ensure strict positivity. The following lemma formalizes the relationship between the original and shifted games.

Lemma 2.14. *Let $A, B \in \mathbb{R}^{m \times n}$ such that $A = B + c \cdot \mathbf{1}_m \mathbf{1}_n^\top$ for some constant c . Then:*

- (i) *Any mixed-strategy equilibrium of game A is also a mixed-strategy equilibrium of game B , and vice versa.*
- (ii) *The values of the games satisfy: $V_{\text{ms}}(A) = V_{\text{ms}}(B) + c$.*

Proof. Since $A = B + c \cdot \mathbf{1}_m \mathbf{1}_n^T$, we have:

$$p^T A q = p^T B q + c.$$

Taking the min-max over all $p \in \Delta_m, q \in \Delta_n$, we obtain:

$$V_{\text{ms}}(A) = \min_p \max_q p^T B q + c = V_{\text{ms}}(B) + c.$$

The strategies that optimize each remain unchanged, as the offset does not affect the maximization/minimization behavior. $\square$

2.5 ■ Fictitious Play

While graphical and linear programming methods offer efficient *offline* computation of equilibria, they are often impractical or unavailable in dynamic, repeated-play scenarios. In many realistic environments, players may not know the full game structure or the opponent's strategy in advance. Instead, they must adaptively learn their strategies through interaction, observing only realized actions or payoffs. This motivates the study of *learning in games*, where strategies evolve gradually as players build up experience over time rather than being solved in one-shot, static form; a systematic treatment of learning dynamics, including the computation of minimax solutions through learning, is given in Lecture 30.

One of the foundational models for learning in games is *fictitious play* (FP), introduced by Brown (1951) [21]. The central idea is conceptually simple yet powerful: each player repeatedly participates in the game, records the history of her opponent's actions, and interprets the empirical distribution of those actions as if it were a mixed strategy. Players then best-respond to this empirical belief. This creates a feedback loop where beliefs are updated from past plays and new best responses generate the actions that shape future beliefs.

Before proceeding, we introduce the indicator function notation. For any logical statement or event E , define

$$\mathbf{1}\{E\} = \begin{cases} 1, & \text{if } E \text{ is true,} \\ 0, & \text{otherwise.} \end{cases}$$

This notation allows us to express empirical frequencies compactly.

Definition 2.15 (Fictitious Play). Consider a two-player zero-sum game with payoff matrix $A \in \mathbb{R}^{m \times n}$. Let $a^t \in \{1, \dots, m\}$ and $b^t \in \{1, \dots, n\}$ denote, respectively, the actions of P1 and P2 at time t . Then, fictitious play proceeds as follows:

- At time t , each player computes the empirical frequency of the opponent's past actions:

$$\hat{q}_j^t = \frac{1}{t} \sum_{\tau=1}^t \mathbf{1}\{b^\tau = j\}, \quad \hat{p}_i^t = \frac{1}{t} \sum_{\tau=1}^t \mathbf{1}\{a^\tau = i\}.$$

- P1 plays a best response to $\hat{q}^t$: choose

$$a^{t+1} \in \arg \min_i \sum_{j=1}^n a_{ij} \hat{q}_j^t.$$

- P2 plays a best response to $\hat{p}^t$: choose

$$b^{t+1} \in \arg \max_j \sum_{i=1}^m \hat{p}_i^t a_{ij}.$$

Intuitively, fictitious play models a form of *myopic rationality*: at every stage, each player assumes that the opponent is committed to the empirical mixed strategy observed so far and chooses the best response against it. In practice, this can be interpreted as an iterative adjustment process in which strategies are refined by experience.

Example 2.16 3×3 Zero-sum Game.

Let P1 choose rows and P2 choose columns, with loss matrix to P1:

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 1 & 2 & 0 \\ 0 & 1 & 2 \end{bmatrix}.$$

At each time t , we record the cumulative pure-strategy counts

$$\text{counts}_p^t(i) = \sum_{\tau=1}^t \mathbf{1}\{a^\tau = i\}, \quad \text{counts}_q^t(j) = \sum_{\tau=1}^t \mathbf{1}\{b^\tau = j\},$$

and the corresponding empirical frequencies

$$\hat{p}^t = \frac{\text{counts}_p^t}{t}, \quad \hat{q}^t = \frac{\text{counts}_q^t}{t}.$$

P1 chooses $a^{t+1} \in \arg \min_i (A\hat{q}^t)_i$, where $A\hat{q}^t$ gives the *expected losses by row*, and P2 chooses $b^{t+1} \in \arg \max_j (-\hat{p}^t)^\top A_j$, where $(\hat{p}^t)^\top A$ gives *P1's expected losses by column* and $-\hat{p}^t)^\top A$ gives *P2's expected payoffs by column*. Ties are resolved by randomization among the tied choices.

The following table illustrates the early stages of fictitious play, starting from the *initial counts* at time $t = 0$:

$$\text{counts}_p^0 = (5, 0, 5), \quad \text{counts}_q^0 = (10, 0, 0).$$

These initial counts are common knowledge to both players and determine the empirical distributions that each player uses to compute her best responses.

t	a^t	b^t	counts_p^t	counts_q^t	$\hat{p}^t$	$\hat{q}^t$	$A\hat{q}^t$ (row losses)	$(\hat{p}^t)^\top A$ (col losses)	(a^{t+1}, b^{t+1})
0	n/a	n/a	(5,0,5)	(10,0,0)	(0.50,0.00,0.50)	(1.00,0.00,0.00)	(2.00,1.00,0.00)	(1.00,0.50,1.50)	(3,2)
1	3	2	(5,0,6)	(10,1,0)	(0.45,0.00,0.55)	(0.91,0.09,0.00)	(1.82,1.09,0.09)	(0.91,0.55,1.55)	(3,2)
2	3	2	(5,0,7)	(10,2,0)	(0.42,0.00,0.58)	(0.83,0.17,0.00)	(1.67,1.17,0.17)	(0.83,0.58,1.58)	(3,2)

Table 2.1: Early rounds of fictitious play in the 3×3 zero-sum game with initial counts $\text{counts}_p^0 = (5, 0, 5)$ and $\text{counts}_q^0 = (10, 0, 0)$. P1 (minimizer) chooses $a^{t+1} \in \arg \min_i (A\hat{q}^t)_i$, and P2 (maximizer) chooses $b^{t+1} \in \arg \max_j (-\hat{p}^t)^\top A_j$ (equivalently, $\arg \min_j (\hat{p}^t)^\top A_j$).

At time $t = 0$, the empirical frequencies derived from the initial counts are $\hat{p}^0 = (0.50, 0.00, 0.50)$ and $\hat{q}^0 = (1.00, 0.00, 0.00)$. P1's expected row losses are $A\hat{q}^0 = (2.00, 1.00, 0.00)$, and thus the minimizer chooses row 3. P2's expected column

losses are $(\hat{p}^0)^\top A = (1.00, 0.50, 1.50)$, and thus the maximizer chooses column 2. Thus, the next play is $(a^1, b^1) = (3, 2)$.

At time $t = 1$, the counts are updated to $(5, 0, 6)$ for P1 and $(10, 1, 0)$ for P2, yielding empirical distributions $\hat{p}^1 = (0.45, 0.00, 0.55)$ and $\hat{q}^1 = (0.91, 0.09, 0.00)$. The expected row losses are $A\hat{q}^1 = (1.82, 1.09, 0.09)$, which again make row 3 the best choice for the minimizer. The expected column losses are $(0.91, 0.55, 1.55)$, and thus column 2 remains the maximizer's best response. Hence the next play is $(a^2, b^2) = (3, 2)$.

At time $t = 2$, the cumulative counts become $(5, 0, 7)$ for P1 and $(10, 2, 0)$ for P2, with empirical distributions $\hat{p}^2 = (0.42, 0.00, 0.58)$ and $\hat{q}^2 = (0.83, 0.17, 0.00)$. The row losses $A\hat{q}^2 = (1.67, 1.17, 0.17)$ confirm that row 3 is still optimal for P1, while the column losses $(0.83, 0.58, 1.58)$ make column 2 continue as P2's best response. Thus, the next play is $(a^3, b^3) = (3, 2)$.

These initial rounds show a reinforcement pattern: P1 repeatedly chooses row 3, while P2 repeatedly chooses column 2. The empirical frequencies shift incrementally toward heavier weight on these actions, illustrating how fictitious play can persist in repeated best responses for many rounds before gradually converging toward the long-run mixed-strategy equilibrium.

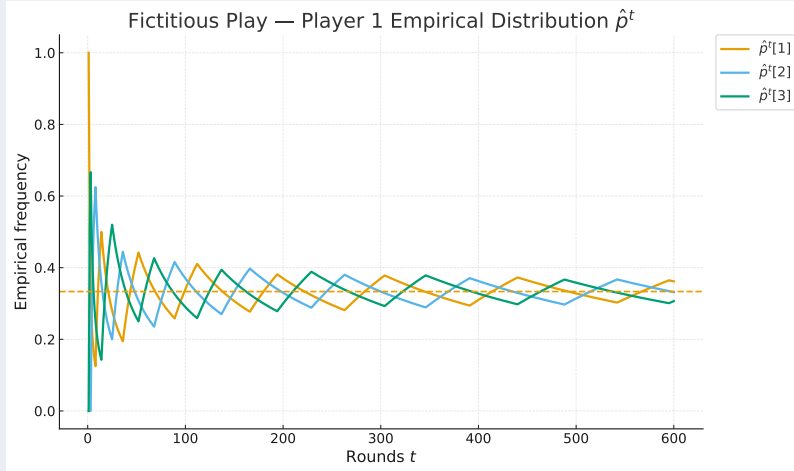


Figure 2.4: Fictitious play in the 3×3 zero-sum game: P1 empirical distribution $\hat{p}^t$. Dashed line marks $1/3$.

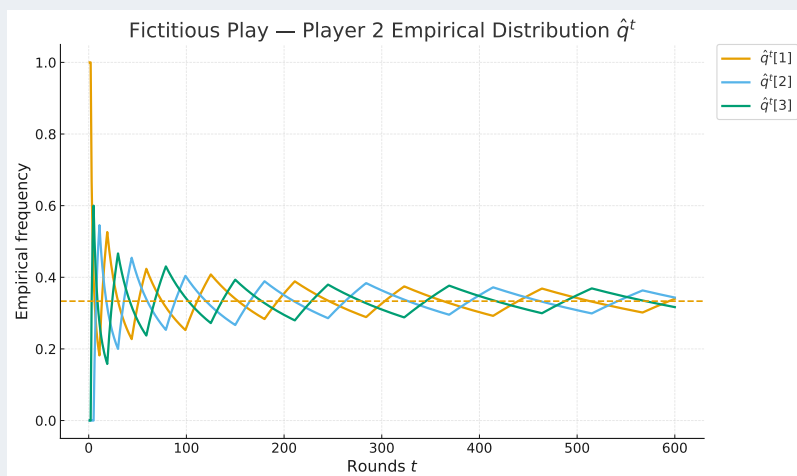


Figure 2.5: Fictitious play in the 3×3 zero-sum game: P2 empirical distribution $\hat{q}^t$. Dashed line marks $1/3$.

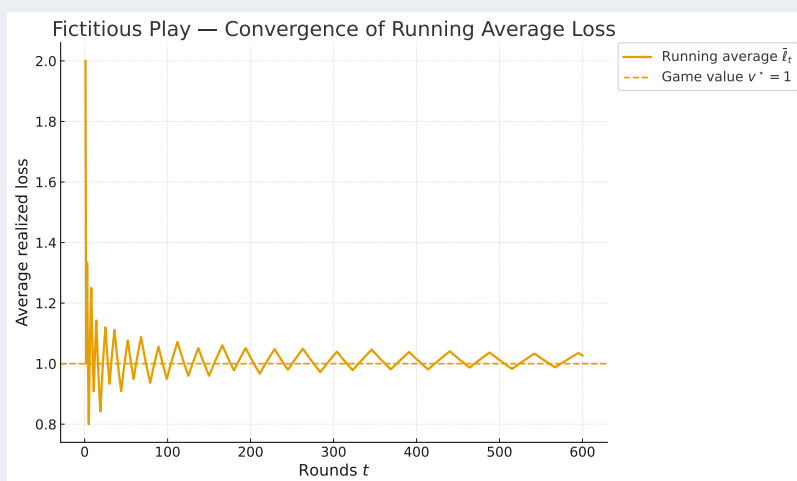


Figure 2.6: Running average of realized stage losses $\bar{\ell}_t$ converging to the game value $v^* = 1$.

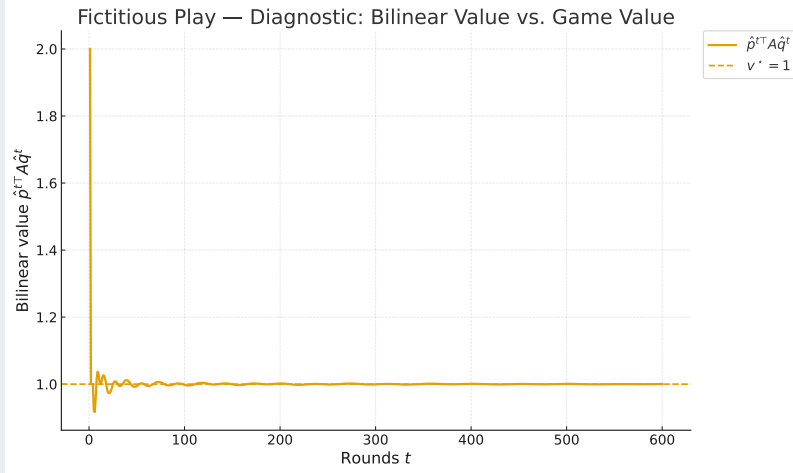


Figure 2.7: Diagnostic: bilinear value $\hat{p}^{t\top} A \hat{q}^t$ alongside $v^* = 1$.

Figures 2.4 to 2.7 complement Table 2.1 by illustrating the evolution of fictitious play in the 3×3 zero-sum game over hundreds of update rounds. The payoff structure in A is symmetric and cyclic, and neither player can secure a long-run advantage with a fixed pure strategy. Instead, equilibrium requires that both players randomize.

Figure 2.4 shows the empirical distribution $\hat{p}^t$ of P1's row choices. Although the initial rounds are marked by oscillations caused by alternating the best responses, the trajectory gradually settles around the uniform distribution $(1/3, 1/3, 1/3)$, indicated by the dashed line. A similar pattern is observed in Figure 2.5 for P2's empirical distribution $\hat{q}^t$. This symmetry reflects the structure of the game: any persistent deviation from uniform play would expose one player to exploitable weaknesses.

Figure 2.6 shows the running average of realized stage losses $\bar{\ell}_t$. Despite short-term fluctuations, the average rapidly converges to the game value $v^* = 1$, represented by the dashed line. This confirms that even if the sequence of pure strategies (a^t, b^t) remains erratic, the long-run empirical outcome stabilizes at the equilibrium value.

Finally, Figure 2.7 provides a diagnostic view of the bilinear product $\hat{p}^{t\top} A \hat{q}^t$, which measures the expected loss according to current empirical beliefs. The trajectory hovers around $v^* = 1$ with diminishing variation.

Taken together, the figures illustrate Robinson's classical result: in zero-sum finite games with two players, fictitious play converges to a mixed-strategy saddle-point equilibrium. In this specific case, the empirical strategies of both players approach the uniform distribution, and the average loss converges to the value $v^* = 1$. These observations reinforce the intuition that learning through iterative best responses, even without full knowledge of the opponent's strategy, suffices to achieve equilibrium play in repeated interactions.

□

The classic result of fictitious play is due to [101], who showed that in finite two-player zero-

sum games, the empirical strategies generated by fictitious play converge to a mixed-strategy saddle-point equilibrium.

While fictitious play is guaranteed to converge in finite two-player zero-sum games [21, 101], its behavior in more general settings is far less straightforward. In general-sum games, convergence to mixed-strategy Nash equilibria is not assured, and indeed there exist simple 2×2 examples where the process fails to converge [13]. In other instances, fictitious play can exhibit persistent cycling or converge to behavior that does not correspond to any equilibrium; a well-known illustration of this phenomenon is Shapley's three-action counterexample [110], in which the empirical distribution of play follows a limit cycle. To address these limitations, researchers have developed modified learning rules, such as smooth or regularized fictitious play [38, 108, 109], regret-minimization dynamics [52], and general no-regret algorithms [23], which often have better convergence properties in broader classes of games. Nevertheless, in the restricted but important setting of two-player zero-sum games, fictitious play remains an appealing approach: it is simple to implement, easy to interpret, and provably convergent to equilibrium strategies through repeated iterative play.

2.6 ■ Further Notes and Reading

This lecture has focused exclusively on zero-sum games in the setting of *finite matrix games*, where both players have finite action sets and the existence of saddle-point equilibria in mixed strategies is guaranteed by the classical minimax theorem. Within this discrete framework, equilibrium computation admits particularly transparent characterizations via security strategies, graphical methods, and linear programming. The zero-sum formulation, however, extends well beyond finite matrix games. When players choose from infinite action sets, as in *continuous-kernel games* or *dynamic zero-sum games*, the existence and characterization of equilibria become more delicate. In such settings, a general minimax theorem does not hold without additional structural assumptions, such as convexity and compactness of action spaces or continuity and regularity of payoff functions. These extensions, together with the corresponding analytical tools, will be taken up explicitly in later lectures, beginning with Lecture 9.

Readers interested in further analytical and computational aspects of zero-sum games beyond the finite matrix case are referred to [98] and [12]. Moreover, while not developed in this lecture, the zero-sum framework underlies a wide range of modern applications, including adversarial learning architectures such as generative adversarial networks, robust optimization and control formulations, and max-min fairness criteria in resource allocation.

Generative adversarial networks GANs are generative models that can be interpreted directly as a two-player zero-sum game between a *generator* G and a *discriminator* D . This formulation was first introduced in [43], marking a significant departure from earlier generative modeling approaches such as variational autoencoders and maximum-likelihood estimation. Instead of explicitly modeling data distributions, GANs leverage adversarial training: the generator produces synthetic samples in an attempt to fool the discriminator, while the discriminator improves by distinguishing real data from generated data. This adversarial setup drew immediate attention because it provided a natural game-theoretic framework for unsupervised learning, linked directly to the classical minimax principle, and led to striking empirical successes in image generation, synthesis, and beyond. The generator's objective is to produce samples that are indistinguishable from real data, while the discriminator's objective is to correctly classify whether a sample is real or generated. This adversarial setup naturally fits into the zero-sum framework: any improvement in the discriminator's classification accuracy corresponds to a degradation in the generator's performance, and conversely, if the generator succeeds in fooling the discrimina-

tor, the discriminator's payoff decreases.

Formally, the interaction is expressed as a minimax optimization problem:

$$\min_G \max_D V(G, D) = \mathbb{E}_{x \sim p_{\text{data}}} [\log D(x)] + \mathbb{E}_{z \sim p_z} [\log(1 - D(G(z)))].$$

Here, p_{data} denotes the true distribution of training data, and p_z is a prior distribution (e.g., Gaussian or uniform) from which the generator samples latent variables z . The generator G maps z into the data space, while the discriminator D outputs a probability in $[0, 1]$ representing its confidence that an input comes from the real distribution.

From a game-theoretic perspective, the discriminator's strategy space consists of all possible classifiers $D : \mathcal{X} \rightarrow [0, 1]$, and its objective is to maximize $V(G, D)$ by assigning high values $D(x) \approx 1$ to real samples and low values $D(G(z)) \approx 0$ to generated ones. The generator's strategy space consists of all conditional distributions p_G induced by mappings $G : \mathcal{Z} \rightarrow \mathcal{X}$, and its objective is to minimize $V(G, D)$ by producing synthetic samples that make the discriminator's classification difficult, ideally forcing $D(G(z)) \approx \frac{1}{2}$.

At equilibrium, this interaction corresponds to a *saddle point* of the game. In particular, if the generator succeeds in perfectly matching the real distribution ($p_G = p_{\text{data}}$), then for every discriminator D the best achievable payoff is

$$V(G, D) = \mathbb{E}_{x \sim p_{\text{data}}} [\log D(x)] + \mathbb{E}_{x \sim p_{\text{data}}} [\log(1 - D(x))],$$

which is maximized when $D(x) = \frac{1}{2}$ for all x . In this case, the discriminator cannot do better than random guessing, and the generator has achieved its objective. Formalizing GANs as zero-sum games is crucial because it clarifies that training is not a simple optimization problem but an equilibrium problem between two strategically opposed players. This perspective provides a rigorous mathematical framework, through minimax theorems, saddle points, and equilibrium concepts, to define success precisely as condition $p_G = p_{\text{data}}$. It also explains why GAN training is often unstable in practice: in high-dimensional, non-convex settings, equilibria can be hard to find or fail to exist, leading to oscillations or mode collapse. Beyond theoretical clarity, the game-theoretic formalism connects GANs to broader domains such as robustness, control, and fairness, where adversarial dynamics also play a central role, thereby guiding both the analysis of convergence and the design of improved algorithms.

Robust control and optimization. Robustness in control and optimization can also be framed as a zero-sum game. This perspective became particularly influential in the late 20th century, most notably in the development of H_∞ control theory [11]. Classical optimal control assumes a known system model and disturbance statistics, but in practice, uncertainties and unmodeled dynamics can severely degrade performance. By introducing a fictitious adversary that chooses disturbances to maximize loss, the control problem is reformulated as a two-player minimax game. The decision-maker (controller) seeks to minimize the worst-case cost, while the adversary (disturbance) seeks to maximize it. Formally, the interaction takes the form

$$\min_{u \in \mathcal{U}} \max_{w \in \mathcal{W}} J(u, w),$$

where $J(u, w)$ is the loss under control input u and disturbance w . Classical H_∞ control theory is built on this minimax formulation and ensures that performance is robust against worst-case perturbations.

The significance of this formalism is that it elevates robustness from a heuristic idea (“designing a system that works under uncertainty”) to a precise mathematical principle. By modeling disturbances as an intelligent adversary, the framework forces the controller to prepare for the

worst possible perturbations, thereby guaranteeing safety and performance margins even in extreme conditions. This worst-case perspective contrasts with stochastic optimization, which focuses on average-case behavior and may underestimate rare but catastrophic disturbances. Moreover, the zero-sum structure provides access to powerful tools from game theory and convex analysis: minimax duality, saddle-point conditions, and dynamic programming can all be leveraged to design controllers with provable guarantees. In modern applications such as cyber-physical systems, adversarial machine learning, and resilient network design, the adversary may represent not just random noise but a strategic attacker. The zero-sum game formalism is therefore essential for systematically addressing robustness in settings where uncertainty is structured, intelligent, and potentially malicious, making it a cornerstone for decision-making under uncertainties.

Max-min fairness. Fairness has long been a central theme in philosophy, economics, and political science, and has become increasingly important in modern applications such as network resource allocation, multi-agent systems, and machine learning [15]. Classical efficiency-driven optimization often seeks to maximize aggregate utility or average performance, but such objectives may mask significant inequalities, leaving disadvantaged groups systematically underserved. Without a precise mathematical framework, fairness risks remaining a vague ideal that is difficult to enforce or compare between scenarios. Formalizing fairness within optimization and game theory makes it possible to design mechanisms that guarantee equitable outcomes, analyze trade-offs between efficiency and equity, and protect vulnerable groups in a principled way.

A natural and powerful way to formalize fairness is through a minimax principle. Suppose that resources are allocated across groups indexed by i , with utility u_i assigned to each group. The fairness objective can be written as

$$\max_{x \in \mathcal{X}} \min_i u_i(x),$$

where x denotes the allocation decision. This max-min criterion guarantees that the welfare of the worst-off group is maximized, thereby providing a precise and principled notion of equity. An allocation that satisfies this criterion is called *max-min fair* [97, 75].

The significance of this formalism is that it elevates fairness from an aspirational goal to a rigorous design principle. Unlike objectives that maximize average utility while potentially ignoring inequality, the minimax perspective explicitly protects the most disadvantaged group. In practice, this prevents resource allocation policies from sacrificing the well-being of a minority in order to improve aggregate performance. The zero-sum game view underscores the adversarial nature of fairness: the decision-maker chooses an allocation to maximize utility, while an implicit adversary selects the worst-off group to reveal inequities. Solving the resulting saddle-point problem yields allocations that are robust against inequality and are equitable across all groups, providing a principled foundation for fairness in diverse applications.

As an illustration, consider an allocation problem with three groups. Suppose that the total resources are limited by $u_1 + u_2 + u_3 \leq 12$, and, due to external constraints, group 3 cannot receive more than 2 units: $u_3 \leq 2$. The fairness objective is to maximize the minimum utility across the groups: $\max_{(u_1, u_2, u_3)} \min\{u_1, u_2, u_3\}$, subject to the constraint imposed on u_3 .

Without the constraint on u_3 , the optimal max-min solution would allocate $u_1 = u_2 = u_3 = 4$, giving each group equal utility. However, the upper bound $u_3 \leq 2$ caps group 3's utility, and thus the best achievable fairness level is $\min\{u_1, u_2, u_3\} = 2$. The optimal allocation in this case is $u_1 = u_2 = 5$, $u_3 = 2$, which exhausts the total budget of 12 while maximizing the minimum utility across groups. This example illustrates how the minimax principle enforces protection of the worst-off group. Even though groups 1 and 2 could in principle receive more, the fairness objective prioritizes raising group 3's utility as much as possible and then only distributes the remainder among the others. Such solutions guarantee equity by design, and prevent systemic

disadvantage to any subgroup.

The significance of zero-sum games lies in their ability to expand the perspective from a one-agent, closed optimization problem to a two-agent adversarial system by "opening" the system. In the single-agent view, the world is passive and the agent minimizes a cost in isolation; in the zero-sum view, an adversary is introduced who actively resists by maximizing loss. This embedding transforms optimization into interaction and provides a principled way to address conflict and balance. In GANs, it casts generative modeling as a dialogue between generator and discriminator; in robustness, it compels controllers to design policies that withstand worst-case disturbances; and in fairness, it treats inequity as an implicit adversary whose presence forces the decision-maker to protect the worst-off group. Across these domains, the zero-sum framework uncovers hidden tensions, clarifies equilibrium structures, and provides formal solution concepts for emerging problems.

2.7 ■ Exercises

Exercise 2.1 (Colonel Blotto Game). Colonel Blotto has 4 regiments with which to occupy two posts, and Lieutenant Kije has 3 regiments with which to occupy the same posts. Each commander must decide, simultaneously and without communication, how to allocate his regiments between the two posts. At each post, the side sending the larger number of regiments captures the post and all regiments sent there by the other side, scoring one point for the captured post and one point for each captured regiment. If both sides send the same number of regiments to a post, both forces withdraw from that post and no points are scored there. The payoff to Colonel Blotto is the total number of points he scores minus the total number of points Lieutenant Kije scores, and hence the game is zero-sum. Colonel Blotto has five pure strategies,

$$X = \{(4, 0), (3, 1), (2, 2), (1, 3), (0, 4)\},$$

where (n_1, n_2) denotes sending n_1 regiments to post 1 and n_2 regiments to post 2, and Lieutenant Kije has four pure strategies,

$$Y = \{(3, 0), (2, 1), (1, 2), (0, 3)\}.$$

The resulting payoff matrix, whose entries are the payoffs to Colonel Blotto (the row player and maximizer), is

	(3, 0)	(2, 1)	(1, 2)	(0, 3)
(4, 0)	4	2	1	0
(3, 1)	1	3	0	-1
(2, 2)	-2	2	2	-2
(1, 3)	-1	0	3	1
(0, 4)	0	1	2	4

- (i) Verify, directly from the capture rules stated above, the entries of the payoff matrix in the row (3, 1).
- (ii) Compute the security levels of the two players in pure strategies and show that the game admits no saddle-point equilibrium in pure strategies.
- (iii) Compute a mixed-strategy saddle-point equilibrium and the value of the game. *Hint:* Interchanging the labels of the two posts leaves the game unchanged; restrict attention to mixed strategies that are invariant under this interchange, and use the support-based indifference conditions.

- (iv) Formulate the game as a primal-dual linear programming pair, as in the linear programming formulation of this lecture, after shifting all entries by a positive constant to make them strictly positive. Implement the formulation in Python or MATLAB, solve it numerically with a linear programming solver (for example, `scipy.optimize.linprog` or `linprog`), and report the value of the game and the optimal mixed strategies of both players to four decimal places. Compare the numerical output with the answer obtained in part (iii).

Exercise 2.2 (Value and Optimal Strategies via the Inverse-Matrix Criterion). The following theorem gives a closed-form solution for the value and optimal strategies of certain square matrix games.

Theorem (see Theorem II.5.6 of [91]). Let $A \in \mathbb{R}^{n \times n}$ be the payoff matrix of a two-person zero-sum game. Suppose that A is nonsingular and $\mathbf{1}^\top A^{-1} \mathbf{1} \neq 0$, where $\mathbf{1} \in \mathbb{R}^n$ is the all-ones vector. Define

$$V := \frac{1}{\mathbf{1}^\top A^{-1} \mathbf{1}}, \quad p^\top := V \mathbf{1}^\top A^{-1}, \quad q := V A^{-1} \mathbf{1}.$$

If $p \geq 0$ and $q \geq 0$ (componentwise), then p and q are optimal mixed strategies of the row and column players, respectively, and the value of the game is V .

- (i) Prove the theorem. *Hint:* Show that $p^\top A = V \mathbf{1}^\top$ and $Aq = V \mathbf{1}$, and conclude that $p^\top Aq' = V$ for every $q' \in \Delta_n$ and $(p')^\top Aq = V$ for every $p' \in \Delta_n$, so that (p, q) satisfies the saddle-point inequalities.

- (ii) Verify that the matrix

$$A = \begin{pmatrix} 1 & 0 \\ -1 & 2 \end{pmatrix}$$

satisfies the hypotheses of the theorem, namely that A is nonsingular and $\mathbf{1}^\top A^{-1} \mathbf{1} \neq 0$.

- (iii) Compute V , p , and q for this matrix, confirm that $p \geq 0$ and $q \geq 0$, and verify directly that (p, q) satisfies the saddle-point inequalities, so that V is the value of the game.

Exercise 2.3 (A Guessing Game). Consider the following guessing game. P2 secretly chooses an integer $j \in \{1, 2, \dots, n\}$ with $n \geq 2$, and P1 simultaneously announces a guess $i \in \{1, 2, \dots, n\}$. The payoff to P1 (the maximizer) is determined as follows: if $i = j$ (the guess is correct), P1 receives $+2$; if $|i - j| = 1$ (the guess is off by exactly one), P1 pays 1 (that is, receives -1); otherwise, the payoff is 0.

- (i) Show that the payoff matrix of the game is the $n \times n$ tridiagonal matrix A_n with entries 2 on the main diagonal, -1 on the first superdiagonal and subdiagonal, and 0 elsewhere.
- (ii) Prove that A_n is nonsingular with inverse $A_n^{-1} = B_n = (b_{ij})$, where

$$b_{ij} = \frac{i(n+1-j)}{n+1} \quad \text{for } i \leq j, \quad b_{ij} = b_{ji} \quad \text{for } i > j.$$

- (iii) Apply the inverse-matrix criterion of the preceding exercise to the case $n = 3$: compute the value of the game and the optimal mixed strategies of both players, and verify that the nonnegativity conditions of the criterion are satisfied.

Exercise 2.4 (Symmetric Matrix Games). A square matrix $A = (a_{ij})$ is said to be *skew-symmetric* if

$$a_{ij} = -a_{ji} \quad \text{for all } i, j.$$

A matrix game is called *symmetric* if its payoff matrix is skew-symmetric.

- (i) Prove that if a matrix game is symmetric, then its value is zero. In particular, prove that if x is an optimal mixed strategy for P1, then x is also an optimal mixed strategy for P2.
- (ii) Verify that the matrix

$$A = \begin{pmatrix} 0 & 1 & -2 \\ -1 & 0 & 3 \\ 2 & -3 & 0 \end{pmatrix}$$

is skew-symmetric, compute the value of the game and an optimal mixed strategy, and confirm that the optimal strategies of both players coincide.

Exercise 2.5 (Fictitious Play and Airport Security Screening). In many airports, security officers must decide how to allocate limited resources to screen passengers. Passengers, some of whom may be malicious, choose which checkpoint to go through. Both sides adapt over time: security observes where risks concentrate and adjusts staffing, while passengers observe delays and change their choices. This interaction can be modeled as a two-person zero-sum game. Suppose that there are two checkpoints, A and B. The defender (security) can either allocate additional staff to A (denoted SA) or allocate additional staff to B (denoted SB). The attacker (a malicious passenger) chooses which checkpoint to target: A or B. If the attacker is caught, the defender receives payoff +1 and the attacker receives -1; if the attacker passes through undetected, the defender receives payoff -1 and the attacker receives +1. The probability of detection is 0.8 at the checkpoint with additional staff and 0.3 at the other checkpoint.

- (i) Derive the 2×2 matrix of expected payoffs to the defender, whose rows correspond to the defender's actions SA and SB and whose columns correspond to the attacker's targets A and B.
- (ii) Compute the mixed-strategy saddle-point equilibrium and the value of this game, and interpret the equilibrium in terms of the defender's allocation policy and the attacker's targeting behavior.
- (iii) Suppose that the interaction repeats daily and that on day 1 the defender staffs checkpoint A while the attacker targets checkpoint A. From day 2 onward, both players update their choices by fictitious play: the defender best-responds to the empirical frequency of the attacker's past targets, and the attacker best-responds to the empirical frequency of the defender's past allocations, with ties broken in favor of checkpoint A. Compute the play by hand for days 1 to 5. Then implement the fictitious play recursion in Python or MATLAB, simulate it for $T = 500$ days, plot the empirical frequencies of both players over time together with the equilibrium probabilities obtained in part (ii), and report the empirical frequencies at $T = 500$. Compare the limiting behavior with the saddle-point equilibrium.
- (iv) Discuss why fictitious play is a reasonable model of adaptation in this airport security context, identify at least one assumption of the model that may be unrealistic, and propose one modification of the model that would better match real-world behavior.

Chapter 3

Nash Equilibrium: Concepts and Definition

This lecture extends the study of two-player zero-sum games to the broader class of nonzero-sum games. In zero-sum settings, the solution concept of saddle-point equilibrium captures stability, since one player's gain is exactly the other's loss. In contrast, nonzero-sum games allow for partially aligned or conflicting interests, and the central solution concept becomes the Nash equilibrium. The focus of this lecture is on the formal definition of Nash equilibrium and its existence properties. We will study both pure and mixed strategies, and explore fixed-point methods that establish the general existence of equilibria. The lecture that follows this one (Lecture 4) will focus on computational aspects and interpretations of Nash equilibria, including methods for finding them and understanding their implications in applied contexts.

3.1 ■ Finite N -Person Nonzero-Sum Static Games

In the previous lecture (Lecture 2), we introduced the mathematical foundations of two-player zero-sum games. We discussed the minimax theorem, saddle-point equilibria, and developed intuition through examples. These examples highlighted the interplay between strategic choices and payoff structures in settings where the interests of players are strictly opposed. This lecture generalizes the analysis to finite N -person nonzero-sum games. These games provide a broad modeling framework for analyzing interactions among multiple agents whose objectives may be cooperative, competitive, or partially aligned. The significance of this generalization lies in its applicability to real-world settings where numerous agents, such as firms, voters, or autonomous AI systems, interact with various incentives and limited coordination. We begin by formalizing the structure of such games and introducing the key solution concept: the Nash equilibrium. We then explore analytical methods for characterizing equilibria and conclude with a discussion on their existence.

A static non-cooperative game is formally specified by three ingredients: a set of players, a set of actions available to each player, and payoff functions that capture the consequences of joint decisions. Let $\mathcal{N} = \{1, 2, \dots, N\}$ denote the finite set of players. Each player $i \in \mathcal{N}$ is endowed with a finite set A_i of admissible actions, called the *pure-strategy space*. A pure strategy (in this “static” context) represents a deterministic choice of action by a player.

The space of all possible pure-strategy combinations is the Cartesian product $A = \prod_{i \in \mathcal{N}} A_i$, so that a pure-strategy profile is an element $a = (a_1, \dots, a_N) \in A$, where $a_i \in A_i$ is the action chosen by player i . In many games, playing a pure strategy deterministically may be predictable or exploitable. Players may instead adopt *mixed strategies*, which are probability distributions

over their pure-strategy sets. Formally, a mixed strategy for player i is an element

$$\sigma_i \in \Delta(A_i), \quad \Delta(A_i) = \left\{ \sigma_i : A_i \rightarrow [0, 1] \mid \sum_{a_i \in A_i} \sigma_i(a_i) = 1 \right\}.$$

The set $\Delta(A_i)$ is the $(|A_i| - 1)$ -dimensional probability simplex, whose elements represent all possible randomizations over A_i .

A *mixed-strategy profile* is a tuple

$$\sigma = (\sigma_1, \dots, \sigma_N) \in \prod_{i \in \mathcal{N}} \Delta(A_i),$$

which induces a joint probability distribution over the set of pure-strategy profiles A . Intuitively, this describes the likelihood of each action combination when players randomize independently according to their chosen mixed strategies.

Each player i is equipped with a *utility function* $u_i : A \rightarrow \mathbb{R}$, which assigns a real-valued payoff to every pure-strategy profile. This function represents player i 's preferences over outcomes. Throughout this lecture, we adopt the convention that each player seeks to maximize her own utility, in expectation when strategies are mixed. When players use mixed strategies, the outcome is stochastic, and each player evaluates outcomes in expectation. The expected utility of player i under profile σ is

$$U_i(\sigma) := \mathbb{E}_{a \sim \sigma} [u_i(a)] = \sum_{a \in A} \left(\prod_{j \in \mathcal{N}} \sigma_j(a_j) \right) u_i(a),$$

where the expectation is taken with respect to the product distribution induced by σ .

Example 3.1 Game of Chicken (Intersection Game).

Two players simultaneously choose whether to *Swerve* (S) or *Stay* (T). If one player stays while the other swerves, the player who stays earns a payoff of 2, while the player who swerves incurs a payoff of -1 . If both players swerve, they each receive a payoff of 0. If both players stay, they crash and each receives a payoff of -5 .

The sets of pure strategies are $A_1 = A_2 = \{S, T\}$. The payoff function for P1 is

$$u_1(S, S) = 0, \quad u_1(S, T) = -1, \quad u_1(T, S) = 2, \quad u_1(T, T) = -5,$$

and the payoff function for P2 is defined symmetrically, i.e.,

$$u_2(S, S) = 0, \quad u_2(S, T) = 2, \quad u_2(T, S) = -1, \quad u_2(T, T) = -5.$$

The resulting game can be represented by the following payoff matrix, where P1 chooses rows and P2 chooses columns:

	P2	
	S	T
P1		
S	(0, 0)	(-1, 2)
T	(2, -1)	(-5, -5)

Suppose that P2 plays the mixed strategy σ_2 with

$$\sigma_2(S) = q, \quad \sigma_2(T) = 1 - q.$$

To compute P1's expected payoff from choosing a pure action $a_1 \in \{S, T\}$, we evaluate the expected utility function $U_1(e_{a_1}, \sigma_2)$, where e_{a_1} denotes the degenerate mixed strategy that plays a_1 with probability 1. Thus, P1's expected payoff from choosing S is

$$U_1(e_S, \sigma_2) = u_1(S, S)q + u_1(S, T)(1 - q) = 0 \cdot q + (-1)(1 - q) = -1 + q,$$

and from choosing T is

$$U_1(e_T, \sigma_2) = u_1(T, S)q + u_1(T, T)(1 - q) = 2q + (-5)(1 - q) = 7q - 5.$$

□

Example 3.2 Prisoner's Dilemma.

Consider a two-player two-action game with $N = 2$, where each player i has the pure-strategy set $A_i = \{\text{Cooperate}, \text{Defect}\}$. The set of pure-strategy profiles is

$$A = A_1 \times A_2 = \{(C, C), (C, D), (D, C), (D, D)\}.$$

For instance, the profile $a = (C, D)$ means that P1 cooperates while P2 defects.

The payoff structure can be represented as:

P1	P2	
	C	D
C	(3, 3)	(0, 5)
D	(5, 0)	(1, 1)

To see why defection is dominant, consider P1's expectations about P2. If P2 cooperates, then P1 earns 3 from cooperating and 5 from defecting, and thus defection is strictly better. If P2 defects, then P1 earns 0 from cooperating and 1 from defecting, and thus again defection is strictly better. Thus, *Defect* is a dominant strategy for P1. By symmetry, the same reasoning shows that *Defect* is also a dominant strategy for P2.

Because both players have a dominant strategy to defect, the resulting outcome is (D, D) , with payoff $(1, 1)$. This outcome is the unique equilibrium of the game. It is individually rational, since no player can gain by unilaterally deviating, but it is inefficient, because both players would be strictly better off under mutual cooperation (C, C) with payoffs $(3, 3)$. As we will see shortly, this constitutes a pure-strategy Nash equilibrium. □

3.2 ■ Nash Equilibrium

We now turn to the central solution concept of non-cooperative game theory, the *Nash equilibrium*. Whereas dominant strategies and saddle points provide clear prescriptions in special cases, most strategic situations lack such straightforward solutions. In general, players must form expectations about the behavior of others and choose strategies that are mutually consistent with these expectations. The Nash equilibrium formalizes this idea. Each player's strategy is a best response to the strategies of the others. In other words, no player can improve her expected payoff by unilaterally deviating.

Definition 3.3 (Mixed-Strategy Nash Equilibrium). A mixed-strategy profile $\sigma^* = (\sigma_1^*, \dots, \sigma_N^*) \in \prod_{i \in \mathcal{N}} \Delta(A_i)$ is called a Nash equilibrium if for every player $i \in \mathcal{N}$,

$$U_i(\sigma_i^*, \sigma_{-i}^*) \geq U_i(\sigma_i, \sigma_{-i}^*) \quad \text{for all } \sigma_i \in \Delta(A_i),$$

where $\sigma_{-i}^* := (\sigma_1^*, \dots, \sigma_{i-1}^*, \sigma_{i+1}^*, \dots, \sigma_N^*)$ denotes the strategy profile of all players other than i .

Equivalently, each σ_i^* must satisfy the best-response condition

$$\sigma_i^* \in \arg \max_{\sigma_i \in \Delta(A_i)} U_i(\sigma_i, \sigma_{-i}^*).$$

A Nash equilibrium represents a situation of *mutual best responses*: each player's strategy is optimal given the strategies chosen by the others. In this state, no single player can increase their expected payoff by unilaterally deviating, and thus a player has no incentive to change behavior as long as others adhere to their equilibrium strategies.

Pure strategies can be viewed as a special case of mixed strategies. In particular, a pure strategy corresponds to a degenerate probability distribution that assigns probability one to a single action. Let A_i be the finite set of admissible actions available to player $i \in \mathcal{N}$. A *pure-strategy* for player i is a deterministic choice of a single action from A_i , i.e., an element $a_i \in A_i$.

In an N -player game, a *pure-strategy profile* is a tuple $a = (a_1, \dots, a_N) \in \prod_{i \in \mathcal{N}} A_i$ in which each player selects a single action without randomization. The utility functions $u_i : A \rightarrow \mathbb{R}$ assign a real-valued payoff to every such profile.

Definition 3.4 (Pure-Strategy Nash Equilibrium). A pure-strategy profile $a^* = (a_1^*, \dots, a_N^*) \in \prod_{i \in \mathcal{N}} A_i$ is a pure-strategy Nash equilibrium if for every player $i \in \mathcal{N}$,

$$u_i(a_i^*, a_{-i}^*) \geq u_i(a_i, a_{-i}^*) \quad \text{for all } a_i \in A_i,$$

where a_{-i}^* denotes the actions chosen by all players other than i .

A pure-strategy equilibrium represents a stable outcome where no player can benefit by changing her decision, assuming that others remain fixed. Such equilibria are often desirable for their interpretability and deterministic nature. However, pure-strategy equilibria do not always exist. In many competitive games, players have incentives to continually outguess or counter one another, leading to cycles of best responses. This failure motivates the broader concept of mixed strategies, already introduced.

3.3 ■ Examples Illustrating Nash Equilibria

To make the abstract definitions concrete, we discuss four well-known games, namely, the Prisoner's Dilemma, the Battle of the Sexes, a constant-sum game, and the Second-Price Auction.

In each case, we present the payoff matrix, compute expected utilities under mixed strategies, and explicitly verify Nash equilibrium conditions. These examples highlight different aspects of strategic reasoning: dominance, coordination, and risk.

Example 3.5 Prisoner's Dilemma (Continued).

In the context of the Prisoner's Dilemma, the outcome (D, D) is an example of a pure-strategy Nash equilibrium. At this profile, P1 earns 1 when both defect. If P1 were to deviate to C while P2 remains at D , her payoff would drop from 1 to 0, and thus deviation is unprofitable. Similarly, P2 earns 1 at (D, D) and would drop to 0 if deviating to C while P1 stays at D . Since neither player can gain from unilateral deviation, (D, D) satisfies the best-response condition and therefore constitutes a pure-strategy Nash equilibrium of the game. $\square$

Example 3.6 Battle of the Sexes.

In this coordination game, two players want to spend time together but have different preferences: P1 prefers Opera, while P2 prefers Football, with a bi-matrix game capturing this scenario being as follows:

P1	P2	
	Opera	Football
Opera	(2, 1)	(0, 0)
Football	(0, 0)	(1, 2)

Let P1 choose Opera with probability p and Football with probability $1 - p$, while P2 chooses Opera with probability q and Football with probability $1 - q$. The expected utilities are

$$U_1(p, q) = 2pq + (1 - p)(1 - q), \quad U_2(p, q) = pq + 2(1 - p)(1 - q).$$

Suppose that we are given the mixed-strategy profile $\sigma_1^* = (2/3, 1/3)$ and $\sigma_2^* = (1/3, 2/3)$. To verify that this is indeed a Nash equilibrium, we check that each player is indifferent between her two actions when facing the other's equilibrium mixture.

For P1, if P2 plays Opera with probability $q = 1/3$, the payoff from playing Opera is $2q = 2/3$, while the payoff from playing Football is $1 - q = 2/3$. Since both payoffs are equal, P1 is indifferent, and thus willing to mix between Opera and Football with, for example, probabilities $(2/3, 1/3)$.

For P2, if P1 plays Opera with probability $p = 2/3$, the payoff from playing Opera is $p = 2/3$, while the payoff from playing Football is $2(1 - p) = 2/3$. Again, the payoffs are the same, and thus P2 is indifferent and can mix between Opera and Football with, for example, probabilities $(1/3, 2/3)$.

Since each player's strategy makes the other one indifferent, neither would have an incentive to deviate. Therefore, the profile (σ_1^*, σ_2^*) is a valid mixed-strategy Nash equilibrium of the game. Note that this coordination game also has two pure-strategy equilibria (Opera, Opera) and (Football, Football), which shows that it is possible for a game to have both pure- and mixed-strategy equilibria. $\square$

Example 3.7 A Constant-Sum 2×2 Game.

Consider the constant-sum game

	P2	
	Left	Right
Top	(2, 0)	(0, 2)
Bottom	(0, 2)	(2, 0)

and let $\sigma_1 = (p, 1 - p)$ and $\sigma_2 = (q, 1 - q)$ denote the mixed strategies. Expected payoffs are $U_1(\sigma_1, \sigma_2) = 2pq + 2(1 - p)(1 - q)$ and $U_2(\sigma_1, \sigma_2) = 2p(1 - q) + 2(1 - p)q$. For a pure action a_i , write e_{a_i} for the degenerate mixed strategy.

We can verify that $\sigma_1^* = (\frac{1}{2}, \frac{1}{2})$ and $\sigma_2^* = (\frac{1}{2}, \frac{1}{2})$ form a Nash equilibrium. If P2 uses σ_2^* , then $U_1(e_{\text{Top}}, \sigma_2^*) = 1$ and $U_1(e_{\text{Bottom}}, \sigma_2^*) = 1$. Since $U_1(\cdot, \sigma_2^*)$ is affine in σ_1 , any deviation $\sigma'_1 = \lambda e_{\text{Top}} + (1 - \lambda)e_{\text{Bottom}}$ yields

$$U_1(\sigma'_1, \sigma_2^*) = \lambda \cdot 1 + (1 - \lambda) \cdot 1 = 1.$$

Thus no pure or mixed deviation gives P1 more than $U_1(\sigma_1^*, \sigma_2^*) = 1$. Checking pure deviations is therefore sufficient because a linear objective over a simplex attains its maximum at a vertex.

If P1 uses σ_1^* , then $U_2(e_{\text{Left}}, \sigma_1^*) = 1$ and $U_2(e_{\text{Right}}, \sigma_1^*) = 1$, and the same affine argument shows that every mixed deviation yields a payoff of 1. Thus σ_2^* is also a best response to σ_1^* . Hence (σ_1^*, σ_2^*) is a mixed-strategy Nash equilibrium. Since at every pure outcome one player strictly prefers to deviate, no pure-strategy equilibrium exists.

An affine transformation of payoffs converts this game into the standard *Matching Pennies* zero-sum game, and thus its equilibrium structure is identical: a unique fully mixed Nash (or equivalently saddle-point) equilibrium; recall the saddle-point analysis of zero-sum games in Lecture 2. Strategic equivalence to a zero-sum game, of which this is an instance, is treated systematically in Section 3.4. $\square$

Example 3.8 Second-Price Auction.

There are N bidders, each with a private valuation v_i for a single indivisible item. Each bidder simultaneously submits a sealed bid b_i . The bidder with the highest bid wins and pays the *second-highest* bid. The payoff to bidder i is

$$u_i(b_i, b_{-i}) = \begin{cases} v_i - m_{-i}, & \text{if } b_i \text{ is (weakly) the highest bid and } i \text{ wins,} \\ 0, & \text{otherwise,} \end{cases}$$

where $m_{-i} = \max_{j \neq i} b_j$ is the highest competing bid. We now show that truthful bidding $b_i = v_i$ is a pure-strategy Nash equilibrium.

Without loss of generality, order the valuations so that

$$v_1 \geq v_2 \geq \dots \geq v_k > v_{k+1} \geq \dots \geq v_N,$$

so the top k bidders have the common highest value $v_1 = \dots = v_k$ strictly exceeding the rest. We analyze $k = 1$ and $k > 1$ separately.

Case $k = 1$: Unique highest valuation. Bidder 1 has $v_1 > v_2$. If all others bid truthfully, then for bidder 1 the highest competing bid is $m_{-1} = v_2$. With $b_1 = v_1$, bidder 1 wins, pays v_2 , and earns $v_1 - v_2 > 0$. Any deviation with $b_1 < v_2$ loses (payoff $0 < v_1 - v_2$), while any $b_1 \geq v_2$ still wins and pays v_2 , yielding the same payoff $v_1 - v_2$. For any bidder $i \geq 2$, truthful bidding yields payoff 0. If $b_i > v_1$, the bidder wins and pays v_1 , obtaining $v_i - v_1 < 0$; if $b_i = v_1$, tie-breaking either yields a loss (0) or a win with negative payoff $v_i - v_1 < 0$. Thus no deviation is profitable.

Case $k > 1$: Multiple highest valuations. Suppose $v_1 = \dots = v_k > v_{k+1}$. With truthful bids, the winner is selected by a tie-break among bidders $1, \dots, k$, pays the second-highest bid v_1 , and earns 0. A top-valued bidder $i \leq k$ who bids $b_i > v_1$ still pays v_1 upon winning, and thus payoff remains 0; bidding $b_i < v_1$ forfeits winning and yields 0. A lower-valued bidder $i > k$ who bids $b_i \leq v_1$ loses (payoff 0), while bidding $b_i \geq v_1$ risks winning at price v_1 and yields $v_i - v_1 < 0$. Hence no deviation is profitable.

For any bidder i facing highest competing bid m_{-i} ,

$$u_i(b_i; v_i, m_{-i}) = \begin{cases} v_i - m_{-i}, & b_i > m_{-i}, \\ \pi_i(v_i - m_{-i}), & b_i = m_{-i}, \\ 0, & b_i < m_{-i}, \end{cases}$$

where $\pi_i \in [0, 1]$ is the tie-breaking win probability conditional on a tie. If $v_i < m_{-i}$, the payoff is never positive; if $v_i = m_{-i}$, it is 0; if $v_i > m_{-i}$, bidding $b_i = v_i$ attains the maximal payoff $v_i - m_{-i}$. Therefore $b_i = v_i$ is a best response for every bidder, and the truthful profile $(b_1, \dots, b_N) = (v_1, \dots, v_N)$ is a dominant-strategy Nash equilibrium of the second-price auction.

□

3.4 ■ Strategic Equivalence

It is often useful to transform a game into a *strategically equivalent* form that preserves the set of Nash equilibria while simplifying the payoff structure. Two games are said to be *strategically equivalent* if each player's utility function in one game can be obtained from that in the other by a positive affine transformation whose additive term does not depend on that player's own action. Formally, the game with utilities

$$\tilde{u}_i(a_i, a_{-i}) = \alpha_i u_i(a_i, a_{-i}) + \beta_i(a_{-i}), \quad \alpha_i > 0, \quad i \in \mathcal{N},$$

is strategically equivalent to the game with utilities u_i . The positive factor α_i preserves player i 's preference ordering over actions, while the term $\beta_i(a_{-i})$ is independent of a_i and hence does not affect player i 's best-response correspondence. Consequently the two games have identical best-response maps and therefore the same Nash equilibria, in both pure and mixed strategies. Strategic equivalence is routinely used to normalize payoffs, remove extraneous constants, and bring a game into a canonical form without altering its equilibrium structure. Two such canonical forms are of particular importance and mark the opposite extremes of strategic interaction.

Equivalence to a team. The cooperative extreme arises when the transformed utilities coincide for all players,

$$\tilde{u}_1 \equiv \tilde{u}_2 \equiv \cdots \equiv \tilde{u}_N =: W.$$

The game is then strategically equivalent to a *team problem* (a common-interest game), in which all players share the single objective W . Every Nash equilibrium of a team problem is *person-by-person optimal*: no player can increase W by deviating unilaterally. Person-by-person optimality is, however, weaker than *global* (team) optimality, which requires joint maximization,

$$W(a^*) = \max_{a \in A} W(a).$$

A team-optimal profile is always a Nash equilibrium, but the converse can fail, and hence some person-by-person optimal profiles need not be globally optimal. Games strategically equivalent to a team admit an exact potential (namely W itself) and thus form a subclass of the potential games taken up in Lecture 4. The inclusion is strict: an exact potential game need not have identical payoffs, as the congestion games of Lecture 4 illustrate.

Equivalence to a zero-sum game. The antagonistic extreme arises, for two players, when the transformed utilities are constant-sum,

$$\tilde{u}_1(a_1, a_2) + \tilde{u}_2(a_1, a_2) = \kappa \quad \text{for all } (a_1, a_2),$$

for some constant κ . The game is then strategically equivalent to a *zero-sum* (equivalently, constant-sum) game: because each $\alpha_i > 0$ and each β_i is independent of the corresponding own action, the equilibria of the original and transformed games coincide, and they are precisely the saddle points of $\tilde{u}_1$. The entire minimax structure of Lecture 2 (existence of a value, interchangeability of equilibria, and the linear-programming and learning-based algorithms) therefore applies to any game in this class, even when it is not presented in zero-sum form.

These two extremes, common interest and pure conflict, bracket the range of strategic interaction, with general nonzero-sum games lying between them. The two can also be combined: two teams opposing each other form a zero-sum game between two collective players, each of which is internally of common interest. A game that admits no strategic equivalence of either kind is a genuine nonzero-sum game. The same classification carries over to the continuous-kernel games of Lecture 9 and underlies the potential-game structure developed in Lecture 4.

3.5 ■ Properties of Nash Equilibria

The Nash equilibrium in mixed strategies exists under general conditions for finite games. This section introduces the key mathematical properties and tools needed to establish this foundational result. We develop the necessary topological and convex-analytic concepts to formalize and prove the existence of equilibrium using a fixed-point approach; see Appendix B (Mathematical Background) for further background on compactness, convexity, and fixed point theorems. The central technical object is the best response correspondence, and we focus on its structure, continuity properties, and fixed-point implications.

3.5.1 ■ Preliminaries: Convexity and Continuity

Definition 3.9 (Convex Set). A set $C \subseteq \mathbb{R}^n$ is convex if for all $x, y \in C$ and all $\lambda \in [0, 1]$, we have

$$\lambda x + (1 - \lambda)y \in C.$$

Definition 3.10 (Compact Set). A set $S \subseteq \mathbb{R}^n$ is compact if it is closed and bounded. Equivalently, every sequence in S has a convergent subsequence whose limit also lies in S .

Definition 3.11 (Correspondence). Let $S \subseteq \mathbb{R}^n$. A correspondence is a map $f : S \rightrightarrows \mathbb{R}^k$ that assigns to each $x \in S$ a subset $f(x) \subseteq \mathbb{R}^k$.

Definition 3.12 (Graph of a Correspondence). The graph of a correspondence $f : S \rightrightarrows Y$ is

$$\text{graph}(f) = \{(x, y) \in S \times Y : y \in f(x)\}.$$

Definition 3.13 (Closed Graph). A correspondence $f : S \rightrightarrows Y$ has a closed graph if for every sequence $x^m \rightarrow x$ and $y^m \rightarrow y$ with $y^m \in f(x^m)$, it follows that $y \in f(x)$.

In many applications in economics, optimization, and game theory, we encounter *correspondences*, i.e., set-valued maps that assign to each input x a set $f(x)$ of possible outputs. To ensure stability of equilibria or feasibility of solutions, we need continuity-like properties for such set-valued maps. One of the most important ones is *upper hemicontinuity* (UHC). Intuitively, UHC prevents new values from appearing suddenly in the image of a correspondence as the input changes slightly.

The definition can be given in several equivalent forms, depending on the underlying mathematical setting.

Definition 3.14 (Upper Hemicontinuity). Let $f : S \rightrightarrows Y$ be a correspondence. The following are equivalent ways of defining upper hemicontinuity.

(i) (Open set form) f is upper hemicontinuous at $x \in S$ if for every open set $V \subseteq Y$ with $f(x) \subseteq V$, there exists a neighborhood U of x in S such that $f(x') \subseteq V$ for all $x' \in U$. The correspondence f is upper hemicontinuous if this holds for all $x \in S$.

(ii) (Sequential form) Let S and Y be metric spaces. f is upper hemicontinuous at $x \in S$ if for every sequence $x^m \rightarrow x$ in S and for every sequence $y^m \in f(x^m)$ with $y^m \rightarrow y$, it follows that $y \in f(x)$. Equivalently, no new limit points appear outside $f(x)$.

Definition 3.15 (Lower Hemicontinuity). A correspondence $f : S \rightrightarrows Y$ is lower hemicontinuous if for every sequence $x^m \rightarrow x$ and every $y \in f(x)$, there exists a sequence $y^m \in f(x^m)$

with $y^m \rightarrow y$.

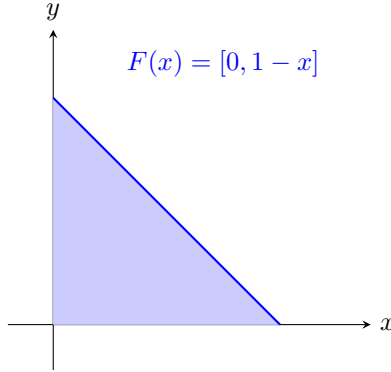


Figure 3.1: The correspondence $F(x) = [0, 1 - x]$ illustrated by shading the feasible region. As x increases, the vertical slice of possible y values shrinks, showing upper hemicontinuity.

Example 3.16 Shrinking interval.

Consider the correspondence $F : [0, 1] \rightrightarrows [0, 1]$ defined by $F(x) = [0, 1 - x]$, depicted in Figure 3.1. For each $x \in [0, 1]$, the image $F(x)$ is a closed interval that shrinks as x increases, with $F(0) = [0, 1]$ and $F(1) = [0, 0] = \{0\}$.

Upper hemicontinuity. Fix $x \in [0, 1]$ and let $V \subseteq [0, 1]$ be an open set with $F(x) \subseteq V$. Since $F(x)$ is compact, there exists $\varepsilon > 0$ such that $[0, 1 - x + \varepsilon) \subseteq V$. For x' sufficiently close to x with $|x' - x| < \varepsilon$, we have

$$F(x') = [0, 1 - x'] \subseteq [0, 1 - x + \varepsilon) \subseteq V,$$

which shows that F is upper hemicontinuous. The same conclusion can be reached using the sequential form. Suppose $x^m \rightarrow x$ and $y^m \in F(x^m)$ with $y^m \rightarrow y$. Since $0 \leq y^m \leq 1 - x^m$ for all m , taking limits yields $0 \leq y \leq 1 - x$, and hence $y \in F(x)$.

Lower hemicontinuity. Let $x^m \rightarrow x$ and fix any $y \in F(x) = [0, 1 - x]$. Define $y^m := \min\{y, 1 - x^m\}$. Then, $y^m \in F(x^m)$ for all m , and $y^m \rightarrow y$. Indeed, if $y < 1 - x$, then eventually $y^m = y$, while if $y = 1 - x$, then $y^m = 1 - x^m \rightarrow 1 - x$. Thus F is lower hemicontinuous.

The correspondence $F(x) = [0, 1 - x]$ is both upper and lower hemicontinuous. $\square$

Example 3.17 Jump correspondence.

Define the correspondence $F : \mathbb{R} \rightrightarrows \mathbb{R}$ by

$$F(x) = \begin{cases} 0, & x < 0, \\ \{0, 1\}, & x = 0, \\ 1, & x > 0. \end{cases}$$

Upper hemicontinuity at $x = 0$. Let V be an open set containing $F(0) = \{0, 1\}$. Since V is open, there exists $\delta \in (0, 1)$ such that $(-\delta, \delta) \subseteq V$ and $(1-\delta, 1+\delta) \subseteq V$. Choose $U := (-\delta, \delta)$. If $x' \in U$ with $x' < 0$, then $F(x') = \{0\} \subseteq V$; if $0 < x' < \delta$, then $F(x') = \{1\} \subseteq V$. Thus F is upper hemicontinuous at 0. The sequential form yields the same conclusion. If $x^m \rightarrow 0$ and $y^m \in F(x^m)$ with $y^m \rightarrow y$, then necessarily $y \in \{0, 1\} = F(0)$, which is precisely the requirement for upper hemicontinuity.

Upper hemicontinuity at $x \neq 0$. For $x \neq 0$, the correspondence is locally constant, taking only the singleton value $\{0\}$ for $x < 0$ or $\{1\}$ for $x > 0$. Hence F is trivially upper hemicontinuous at all $x \neq 0$.

Failure of lower hemicontinuity at $x = 0$. Take $y = 0 \in F(0)$ and a sequence $x^m \downarrow 0$ with $x^m > 0$. Then, $F(x^m) = \{1\}$, and thus any $y^m \in F(x^m)$ equals 1 and cannot converge to $y = 0$. Thus, the lower hemicontinuity condition fails at 0. Symmetrically, taking $y = 1 \in F(0)$ and $x^m \uparrow 0$ with $x^m < 0$ also shows failure.

Lower hemicontinuity at $x \neq 0$. If $x < 0$ (resp. $x > 0$), then $F(x) = \{0\}$ (resp. $F(x) = \{1\}$). For any sequence $x^m \rightarrow x$, choosing $y^m \equiv 0$ (resp. $y^m \equiv 1$) ensures $y^m \in F(x^m)$ and $y^m \rightarrow y$. Therefore, F is lower hemicontinuous at all $x \neq 0$.

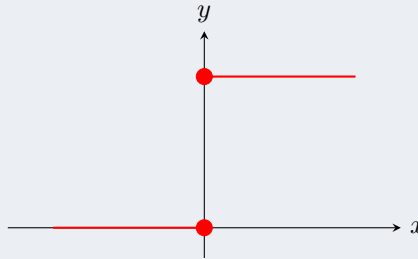


Figure 3.2: The jump correspondence $F(x) = \{0\}$ for $x < 0$, $F(0) = \{0, 1\}$, $F(x) = \{1\}$ for $x > 0$. *Upper hemicontinuity* holds at $x = 0$: any limit of $y^m \in F(x^m)$ with $x^m \rightarrow 0$ lies in $\{0, 1\} = F(0)$. *Lower hemicontinuity* fails at $x = 0$: taking $x^m \downarrow 0$ with $x^m > 0$ forces $y^m = 1$, which cannot approximate $y = 0 \in F(0)$.

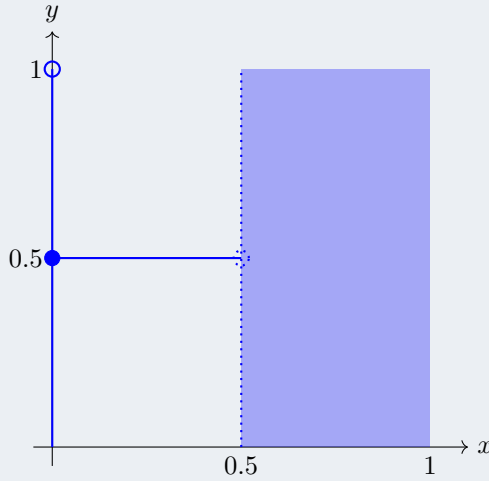


Figure 3.3: Graph of the correspondence F : vertical segment at $x = 0$ (excluding $(0, 1)$), a horizontal line at $y = 0.5$ for $x \in [0, 0.5]$, and a shaded region for $x \in (0.5, 1]$. Open boundaries are indicated with dotted lines or open circles.

□

Example 3.18

Define the correspondence $F : [0, 1] \rightrightarrows [0, 1]$ by

$$F(x) = \begin{cases} [0, 1), & x = 0, \\ \{0.5\}, & x \in (0, 0.5], \\ [0, 1], & x \in (0.5, 1]. \end{cases}$$

For every x and $y \in F(x)$, we can approximate y by nearby selections $y' \in F(x')$ whenever $x' \rightarrow x$. At $x = 0$, any $y \in [0, 1)$ can be preserved when approaching from $x' > 0.5$, since there $F(x') = [0, 1]$. At $x \in (0, 0.5]$, we have $F(x) = \{0.5\}$, which is trivially LHC. At $x \in (0.5, 1]$, $F(x) = [0, 1]$, which is closed under limits. Thus, F is lower hemicontinuous at every point.

At $x = 0$, we have $F(0) = [0, 1)$. Consider the open set $V = (-\varepsilon, 1)$ containing $F(0)$. If F were UHC at 0, there would exist a neighborhood U of 0 such that $F(x') \subset V$ for all $x' \in U$. But for any $x' \in (0.5, 1]$, we have $F(x') = [0, 1]$, which includes the point $1 \notin V$. Hence the UHC condition fails at $x = 0$.

The correspondence F is lower hemicontinuous everywhere, but not upper hemicontinuous at $x = 0$. This example illustrates how LHC allows feasible sets to “expand suddenly,” while UHC forbids such jumps.

□

Definition 3.19 (Best Response Correspondence). Let $\mathcal{N} = \{1, \dots, N\}$ be the set of players. Given the mixed-strategy profile $\sigma_{-i} \in \prod_{j \neq i} \Delta(A_j)$ of all players other than player i , the best response correspondence of player i is defined by

$$\text{BR}_i(\sigma_{-i}) := \arg \max_{\sigma_i \in \Delta(A_i)} U_i(\sigma_i, \sigma_{-i}),$$

where U_i denotes the expected utility function of player i .

We now establish key structural properties of the best response correspondence.

Lemma 3.20 (Properties of the Best-Response Correspondence). Let $\mathcal{G} = (\mathcal{N}, (A_i)_{i \in \mathcal{N}}, (u_i)_{i \in \mathcal{N}})$ be a finite N -player game, where each A_i is a finite set of pure strategies for player i and $u_i : \prod_{j \in \mathcal{N}} A_j \rightarrow \mathbb{R}$ is the payoff function for player i . Let $\Delta(A_i)$ denote the set of mixed strategies of player i , $\sigma_i \in \Delta(A_i)$, and $\sigma_{-i} \in \prod_{j \neq i} \Delta(A_j)$ denote the opponents' profile. The expected payoff to player i is

$$U_i(\sigma_i, \sigma_{-i}) = \sum_{a_i \in A_i} \sigma_i(a_i) \sum_{a_{-i} \in A_{-i}} \left(\prod_{j \neq i} \sigma_j(a_j) \right) u_i(a_i, a_{-i}).$$

The best-response correspondence is

$$\text{BR}_i(\sigma_{-i}) = \arg \max_{\sigma_i \in \Delta(A_i)} U_i(\sigma_i, \sigma_{-i}).$$

Then, for each $i \in \mathcal{N}$:

- (1) $\Delta(A_i)$ is non-empty, convex, and compact in the Euclidean topology;
- (2) $U_i(\sigma_i, \sigma_{-i})$ is linear in σ_i for fixed σ_{-i} ;
- (3) $\text{BR}_i(\sigma_{-i})$ is non-empty and convex-valued for all σ_{-i} ; and
- (4) BR_i is upper hemicontinuous as a correspondence from $\prod_{j \neq i} \Delta(A_j)$ to $\Delta(A_i)$.

Proof. Since A_i is finite with $|A_i| = m_i \geq 1$, the set $\Delta(A_i) = \{\sigma_i \in \mathbb{R}^{m_i} : \sigma_i(a_i) \geq 0, \sum_{a_i} \sigma_i(a_i) = 1\}$ is a standard simplex in $\mathbb{R}^{m_i}$. It is non-empty because it contains, for example, the uniform distribution. It is convex because convex combinations of probability distributions are again probability distributions, and it is compact in the Euclidean topology by the Heine-Borel Theorem [104], being closed and bounded.

For fixed σ_{-i} , the expected payoff $U_i(\sigma_i, \sigma_{-i})$ can be expressed as $\sum_{a_i \in A_i} \sigma_i(a_i) c_{a_i}$, where each coefficient $c_{a_i} = \sum_{a_{-i} \in A_{-i}} \left(\prod_{j \neq i} \sigma_j(a_j) \right) u_i(a_i, a_{-i})$ is constant with respect to σ_i . This shows that U_i is linear in σ_i .

Since $U_i(\cdot, \sigma_{-i})$ is continuous on the compact nonempty set $\Delta(A_i)$, Weierstrass's theorem guarantees that it has a maximum. The set of maximizers $\text{BR}_i(\sigma_{-i})$ is therefore non-empty. Moreover, since the objective function is linear in σ_i , the set of maximizers is a face of the simplex and hence convex.

Finally, the feasible set $\Delta(A_i)$ is constant in σ_{-i} and compact, and U_i is jointly continuous in (σ_i, σ_{-i}) because it is multilinear in its arguments. By Berge's Maximum Theorem [123], the argmax correspondence $\sigma_{-i} \mapsto \text{BR}_i(\sigma_{-i})$ is upper hemicontinuous with compact and nonempty values. Convex-valuedness has already been established, thus completing the proof. $\square$

3.5.2 ■ Existence of Nash Equilibrium via Kakutani's Fixed Point Theorem

A foundational proof for existence of mixed-strategy Nash equilibrium, especially suitable for set-valued best response correspondences, is based on Kakutani's fixed point theorem. This

approach is constructive in the sense of establishing sufficient conditions on correspondences that guarantee fixed points.

We begin with the formal statement of Kakutani's theorem and proceed to develop the necessary lemmas.

Lemma 3.21 (Kakutani's Fixed Point Theorem). *Let $S \subset \mathbb{R}^n$ be a non-empty, compact, and convex set, and let $F : S \rightrightarrows S$ be a correspondence such that:*

- (i) *For every $x \in S$, $F(x)$ is non-empty, convex, and closed.*
- (ii) *F is upper hemicontinuous.*

Then, F has at least one fixed point; that is, there exists $x^ \in S$ such that $x^* \in F(x^*)$.*

We now apply this result to the setting of N -player finite games. To do so, we must verify that the aggregated best response correspondence $\text{BR}(\cdot)$ satisfies the conditions of Kakutani's Fixed-Point Theorem (FPT).

Lemma 3.22 (Strategy Space is Compact and Convex). *Let A_i be a finite action set for player $i \in \mathcal{N}$. Then, the mixed-strategy set $\Delta(A_i)$ is a non-empty, compact, and convex subset of $\mathbb{R}^{|A_i|}$, and the Cartesian product $\Delta := \prod_{i \in \mathcal{N}} \Delta(A_i)$ is compact and convex in $\mathbb{R}^{\sum_i |A_i|}$.*

Lemma 3.23 (Best-Response Correspondence Properties). *For every player $i \in \mathcal{N}$, the best-response correspondence*

$$\text{BR}_i : \prod_{j \neq i} \Delta(A_j) \rightrightarrows \Delta(A_i)$$

is non-empty-valued, convex-valued, and upper hemicontinuous.

Proof. Non-emptiness follows from the Weierstrass Theorem: for each fixed σ_{-i} , the expected payoff function $U_i(\cdot, \sigma_{-i})$ is continuous on the non-empty, compact set $\Delta(A_i)$, and thus a maximizer exists. Convexity of $\text{BR}_i(\sigma_{-i})$ holds because U_i is linear in σ_i ; the maximization of a linear function over a convex set yields a convex set of maximizers. Upper hemicontinuity follows from Berge's Maximum Theorem: the set $\Delta(A_i)$ is compact and independent of σ_{-i} , and U_i is jointly continuous in (σ_i, σ_{-i}) , and thus the argmax correspondence is upper hemicontinuous with non-empty, compact values. $\square$

Lemma 3.24 (Aggregated Best Response Correspondence). *The aggregated best response correspondence*

$$\text{BR}(\sigma) := \text{BR}_1(\sigma_{-1}) \times \text{BR}_2(\sigma_{-2}) \times \cdots \times \text{BR}_N(\sigma_{-N})$$

is upper hemicontinuous and has non-empty, convex values for all $\sigma \in \Delta$.

Proof. The product of upper hemicontinuous correspondences with non-empty convex values remains upper hemicontinuous and convex-valued. Hence, $\text{BR}(\cdot)$ inherits these properties from the individual BR_i . $\square$

We are now in a position to state and prove the existence theorem via Kakutani's FPT.

Theorem 3.25 (Existence of Nash Equilibrium via Kakutani's FPT). *Every finite N -player non-cooperative static game admits at least one Nash equilibrium in mixed strategies.*

Proof. Let $\Delta = \prod_{i \in N} \Delta(A_i)$ denote the mixed-strategy profile space. From the previous lemmas:

- Δ is compact and convex.
- $\text{BR} : \Delta \rightrightarrows \Delta$ is upper hemicontinuous.
- $\text{BR}(\sigma)$ is non-empty, convex, and closed for each $\sigma \in \Delta$.

Therefore, by Kakutani's fixed point theorem, there exists $\sigma^* \in \Delta$ such that $\sigma^* \in \text{BR}(\sigma^*)$. This σ^* is a mixed-strategy Nash equilibrium. $\square$

Kakutani's approach is especially useful for proving existence in general games where the best response sets are correspondences rather than functions. This formulation highlights the set-valued nature of strategic interdependence in equilibrium.

3.5.3 ■ Existence of Mixed-Strategy Nash Equilibrium via Brouwer's Fixed Point Theorem

In this section, we present an alternative constructive proof for existence of mixed-strategy Nash equilibrium for finite games using Brouwer's fixed-point theorem. Unlike the Kakutani-based approach, which relies on set-valued correspondences, the Brouwer method employs a continuous single-valued function constructed to preserve best-response behavior. We first state the necessary fixed-point result and supporting lemmas before presenting the theorem and its proof.

Lemma 3.26 (Brouwer's Fixed Point Theorem). *Let $S \subset \mathbb{R}^n$ be a non-empty, compact, and convex set, and let $f : S \rightarrow S$ be a continuous function. Then, there exists $x^* \in S$ such that $f(x^*) = x^*$.*

This foundational result ensures the existence of fixed points for continuous self-maps over convex, compact domains in a Euclidean space. Intuitively, a fixed point is a location that remains unchanged under the mapping f . The assumptions are crucial: compactness rules out “escaping to infinity,” convexity prevents “holes” that could break the averaging arguments behind fixed-point existence, and continuity ensures there are no jumps that could avoid the intersection argument. To build geometric intuition, we first treat low-dimensional cases, and then connect the result to the mixed-strategy space in game theory.

In one dimension, take $S = [0, 1]$. Consider $g(x) = f(x) - x$. Since f maps $[0, 1]$ into itself, we have $g(0) = f(0) - 0 \geq 0$ and $g(1) = f(1) - 1 \leq 0$. Continuity of g and the Intermediate Value Theorem imply that there is $x^* \in [0, 1]$ with $g(x^*) = 0$, equivalently $f(x^*) = x^*$. Figure 3.4 depicts this as the intersection of the graph $y = f(x)$ with the diagonal $y = x$, which must occur within the unit square.

In two dimensions, let S denote the closed unit disk

$$D^2 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}.$$

A continuous mapping $f : D^2 \rightarrow D^2$ may twist, stretch, or compress the disk, yet *Brouwer's Fixed Point Theorem* guarantees that at least one point must remain stationary, $x^* = f(x^*)$. To see why, assume that to the contrary every point of the disk moves, and thus $f(x) \neq x$ for all $x \in D^2$. For each point $z \in D^2$, draw a ray starting at $f(z)$ and passing through z , and let $r(z)$ denote the intersection of this ray with the boundary circle $S^1 = \partial D^2$. This construction defines a continuous map $r : D^2 \rightarrow S^1$ that fixes every boundary point ($r(z) = z$ for all $z \in S^1$), thus constituting a *continuous retraction* of the disk onto its boundary.

However, such a retraction cannot exist. The *No Retraction Theorem* (e.g., see [80, Theorem 17.4] or [53, Section 0.2]) states that there is no continuous map $r : D^2 \rightarrow S^1$ satisfying $r(x) = x$ for all $x \in S^1$. The reason is topological: the disk D^2 is contractible ($\pi_1(D^2) = 0$),

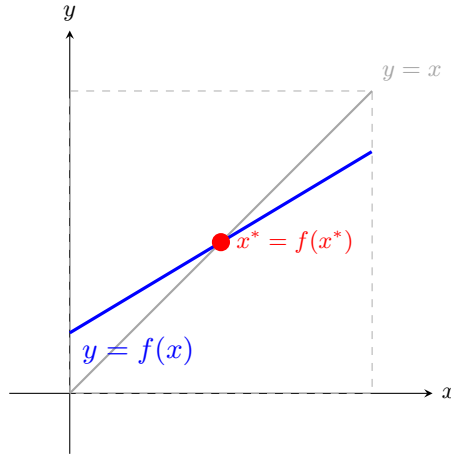


Figure 3.4: One-dimensional intuition. A continuous $f : [0, 1] \rightarrow [0, 1]$ yields $g(x) = f(x) - x$ with $g(0) \geq 0$ and $g(1) \leq 0$, and thus some x^* satisfies $f(x^*) = x^*$. Geometrically, $y = f(x)$ intersects $y = x$ within the unit square.

whereas the circle S^1 has a nontrivial fundamental group ($\pi_1(S^1) = \mathbb{Z}$). The existence of r would imply an impossible embedding of these distinct topological structures. This contradiction shows that the assumption must be false; therefore, at least one fixed point $x^* = f(x^*)$ must exist. Figure 3.5 illustrates this reasoning: each point is mapped to a new position within the disk, yet one point necessarily coincides with its image, producing a zero-length arrow that marks the fixed point.

The same principle extends to higher dimensions. In game theory, the relevant space is the mixed-strategy simplex $\Delta(A) = \{\sigma \in \mathbb{R}_{\geq 0}^m : \sum_{k=1}^m \sigma_k = 1\}$ for a finite set of m pure strategies. The simplex is the convex hull of the pure strategies, hence convex; it is also closed and bounded, hence compact; and it is non-empty. Many equilibrium maps of interest, such as best-response correspondences that admit a continuous selection, smooth regularizations of best-response, or payoff-monotone dynamics with continuous time-1 maps, define continuous self-maps $\beta : \Delta(A) \rightarrow \Delta(A)$. Brouwer's theorem then guarantees a fixed point σ^* with $\beta(\sigma^*) = \sigma^*$, which corresponds to a mixed-strategy equilibrium for the chosen continuous selection.

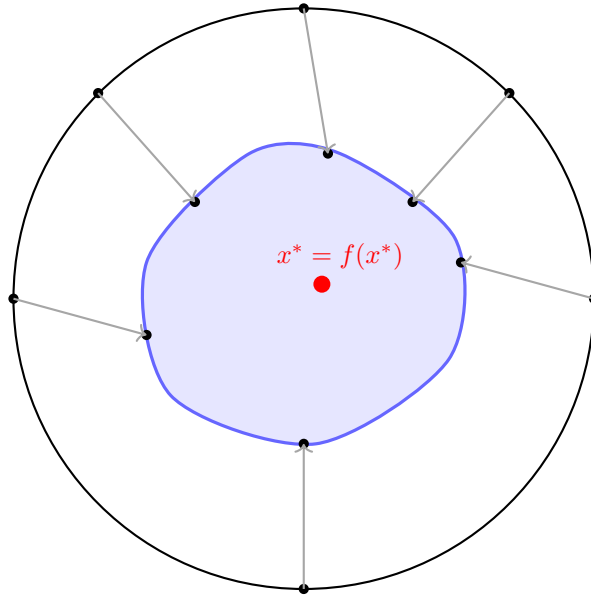
The following lemma establishes the specific convexity and compactness properties of the mixed-strategy space needed for the application of Brouwer's theorem to the existence of an equilibrium.

Lemma 3.27 (Compactness and Convexity of Strategy Space). *Let A_i be a finite set of pure strategies for player i . Then, the mixed-strategy set $\Delta(A_i)$ is a non-empty, compact, and convex subset of $\mathbb{R}^{|A_i|}$. Moreover, the product space $\prod_{i \in \mathcal{N}} \Delta(A_i)$ is compact and convex in the product topology.*

Lemma 3.28 (Continuity of Expected Utility Functions). *For each player $i \in \mathcal{N}$, the expected utility function*

$$U_i(\sigma_1, \dots, \sigma_N) = \mathbb{E}[u_i(a)] = \sum_{a \in A} \left(\prod_{j \in \mathcal{N}} \sigma_j(a_j) \right) u_i(a)$$

is continuous in all its arguments $\sigma = (\sigma_1, \dots, \sigma_N)$.



$$S = D^2 \subset \mathbb{R}^2$$

Figure 3.5: Two-dimensional intuition for Brouwer's Fixed Point Theorem. The figure shows a continuous self-map $f : D^2 \rightarrow D^2$. Each arrow connects a point z to its image $f(z)$. If every point were displaced ($f(z) \neq z$), one could extend these arrows to define a continuous retraction of the disk onto its boundary S^1 , which is topologically impossible by the No Retraction Theorem. Hence, at least one point x^* must satisfy $f(x^*) = x^*$.

We are now ready to prove the existence of equilibrium using Brouwer's theorem.

Theorem 3.29 (Existence of Mixed-Strategy Nash Equilibrium in Finite Games via Brouwer's Fixed Point Theorem). *Every finite N -player non-cooperative static game admits at least one Nash equilibrium in mixed strategies.*

Proof. Let $Y^i = \Delta(A_i)$ be the mixed-strategy simplex for player i , and let $Y = \prod_{i \in \mathcal{N}} Y^i$ be the Cartesian product of all players' strategy spaces. For each player i and pure action $a_i \in A_i$, define the deviation function:

$$\psi_{a_i}^i(y^1, \dots, y^N) := u_i(a_i, y^{-i}) - u_i(y^i, y^{-i}),$$

where $y^{-i} = (y^1, \dots, y^{i-1}, y^{i+1}, \dots, y^N)$ denotes the strategies of all players other than i . The function $\psi_{a_i}^i$ measures the gain in payoff for player i by unilaterally switching to pure strategy a_i .

Define the correction term:

$$c_{a_i}^i := \max\{\psi_{a_i}^i, 0\}, \quad \forall a_i \in A_i, i \in \mathcal{N}.$$

This correction captures the incentive to deviate. We now define a continuous transformation

$T : Y \rightarrow Y$ by modifying the original strategy profile $y = (y^1, \dots, y^N)$ as follows:

$$\bar{y}_{a_i}^i := \frac{y_{a_i}^i + c_{a_i}^i}{1 + \sum_{a'_i \in A_i} c_{a'_i}^i}, \quad \forall a_i \in A_i, i \in \mathcal{N}.$$

That is, we adjust the probability of each action by increasing it in proportion to the potential payoff gain, and then renormalize to ensure that the result is a valid probability distribution.

The mapping $T = (T^1, \dots, T^N)$ is continuous (as a composition of continuous operations on a compact convex domain), and maps the compact convex set Y into itself.

By Brouwer's fixed point theorem, T has a fixed point $y^* \in Y$, and thus $T(y^*) = y^*$. It remains to show that fixed points are exactly mixed-strategy Nash equilibria.

(1) Fixed point $\Rightarrow$ Nash equilibrium. Fix $i \in \mathcal{N}$. The fixed-point condition states that for every $a_i \in A_i$,

$$y_{a_i}^{i*} = \frac{y_{a_i}^{i*} + c_{a_i}^i}{1 + \sum_{a'_i \in A_i} c_{a'_i}^i}.$$

Rearranging,

$$c_{a_i}^i = \left(\sum_{a'_i} c_{a'_i}^i \right) y_{a_i}^{i*} \quad \forall a_i \in A_i.$$

Summing this equality over a_i leads to

$$\sum_{a_i} c_{a_i}^i = \left(\sum_{a'_i} c_{a'_i}^i \right) \sum_{a_i} y_{a_i}^{i*} = \sum_{a'_i} c_{a'_i}^i.$$

Thus the identity is consistent for any nonnegative vector $c^i = (c_{a_i}^i)_{a_i}$ proportional to the probability vector $y^{i*} = (y_{a_i}^{i*})_{a_i}$. However, by definition, each $c_{a_i}^i \geq 0$ and it represents the incentive to deviate from the current profile. If $\sum_{a_i} c_{a_i}^i > 0$, then $c_{a_i}^i > 0$ for every a_i in the support of y^{i*} , which means that all actions played with positive probability would yield a strictly positive gain from deviation. This contradicts the fixed-point condition, since an equilibrium cannot assign positive weight to actions that all admit profitable deviations. Therefore the only consistent possibility is

$$\sum_{a_i} c_{a_i}^i = 0 \implies c_{a_i}^i = 0 \text{ for all } a_i.$$

Consequently, $c_{a_i}^i = 0$ for all i and a_i , which implies $\psi_{a_i}^i \leq 0$ for all pure deviations. Hence no player can gain by deviating, and y^* is a Nash equilibrium.

(2) Nash equilibrium $\Rightarrow$ fixed point. Conversely, if y^* is a mixed-strategy Nash equilibrium, then for every i and a_i we have $\psi_{a_i}^i \leq 0$, and thus by construction $c_{a_i}^i = \max\{\psi_{a_i}^i, 0\} = 0$. Substituting into the definition of T gives $\bar{y}_{a_i}^i = y_{a_i}^i$ for all i and a_i , and hence $T(y^*) = y^*$ and y^* is a fixed point.

Combining (1) and (2), the fixed points of T coincide with the mixed-strategy Nash equilibria of the game, which completes the proof.

□

Remark 3.30. The above proof offers a constructive transformation whose fixed points coincide exactly with the set of mixed-strategy Nash equilibria. Unlike the Kakutani-based method, which

relies on set-valued upper hemicontinuous correspondences, the Brouwer approach constructs a continuous function on a convex, compact set. This perspective is often used in algorithmic game theory to motivate computational procedures. Computational methods for Nash equilibria are the subject of Lecture 4, and learning dynamics that converge to equilibrium are studied in Lecture 30.

The above result applies only to finite games, i.e., games where each player has a finite set of actions. For games with infinite action sets, further conditions are needed to ensure existence, such as compactness of action sets and continuity of utility functions. In such settings, Glicksberg's generalization of Kakutani's theorem provides the appropriate fixed-point machinery [42].

Example 3.31 A nonzero-sum $2 \times \infty$ game with no equilibrium.

We construct a nonzero-sum game with two row choices for P1 and countably many column choices for P2. Let P1's pure strategies be T (Top) and B (Bottom). P2's pure strategies are columns $j \in \mathbb{N} = \{1, 2, \dots\}$. Payoffs (to P1, P2) are

$$u(T, j) = \left(\frac{1}{2}, 1 - \frac{1}{j}\right), \quad u(B, j) = (0, 0) \quad (j = 1, 2, \dots).$$

This game is nonzero-sum: the sum depends on both the row and the column.

Finite truncations admit equilibria. Fix $k \in \mathbb{N}$ and truncate P2's actions to $\{1, \dots, k\}$. For any column $j \leq k$, P1 strictly prefers T to B because $\frac{1}{2} > 0$, and hence T is a strict best response to every column. Given P1 plays T , P2's payoff on column j is $1 - \frac{1}{j}$, which is strictly increasing in j on $\{1, \dots, k\}$, and thus P2's best response is $j = k$. Therefore (T, k) is a pure-strategy Nash equilibrium of the truncated game, with equilibrium payoffs

$$u(T, k) = \left(\frac{1}{2}, 1 - \frac{1}{k}\right) \xrightarrow{k \rightarrow \infty} \left(\frac{1}{2}, 1\right).$$

P1	P2				
	$j=1$	2	3	4	5
T	$(\frac{1}{2}, 0)$	$(\frac{1}{2}, \frac{1}{2})$	$(\frac{1}{2}, \frac{2}{3})$	$(\frac{1}{2}, \frac{3}{4})$	$(\frac{1}{2}, \frac{4}{5})$
B	$(0, 0)$	$(0, 0)$	$(0, 0)$	$(0, 0)$	$(0, 0)$

Figure 3.6: Payoff table for the first few columns. For every j , P1 strictly prefers T over B . For fixed k , the unique best response of P2 is $j = k$.

The countably infinite game has no equilibrium. Now allow all columns $j \in \mathbb{N}$ as possible choices for P2. Since $u_1(T, j) = \frac{1}{2} > 0 = u_1(B, j)$ for every j , P1's unique best response to any strategy of P2 is T ; hence in any equilibrium P1 must play T with probability one.

Conditional on P1 playing T , P2's payoff from pure column j is $1 - \frac{1}{j}$. The sequence

$$0, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \dots$$

is strictly increasing and approaches $\sup_j(1 - \frac{1}{j}) = 1$. However, no finite j attains the value 1.

For any mixed strategy $z = (z_j)_{j \geq 1}$ of P2, the expected payoff is

$$\mathbb{E}_z \left[1 - \frac{1}{j} \right] = 1 - \sum_{j=1}^{\infty} \frac{z_j}{j}.$$

Since $\sum_{j=1}^{\infty} \frac{z_j}{j} > 0$ for any probability distribution on $\mathbb{N}$, this expectation is always strictly less than 1. Moreover, for any z , P2 can shift a small positive probability mass from finitely many low indices to sufficiently large j and strictly increase her payoff (because $1/j$ decreases as j grows). Thus P2 has *no best response* to P1's strategy T . Therefore, the semi-infinite game admits no Nash equilibrium.

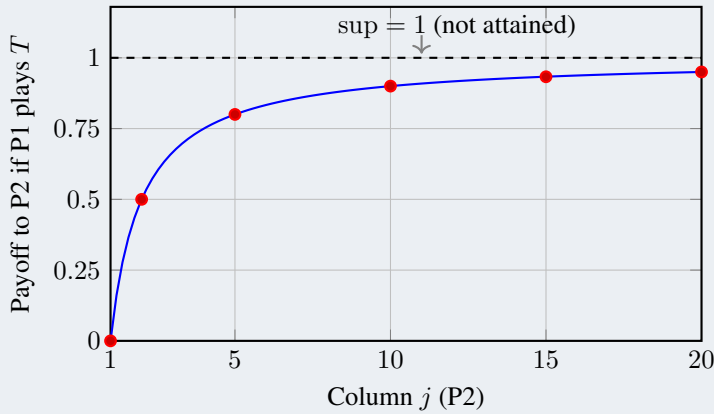


Figure 3.7: P2's payoff increases with column j according to $(j - 1)/j$ and approaches 1 as $j \rightarrow \infty$, but never reaches it. Thus no maximizing column or mixed strategy exists, and P2 has no best response in the semi-infinite game.

Why existence fails. In every finite truncation, P2's mixed-strategy set is a simplex in $\mathbb{R}^k$, which is compact. Compactness guarantees the maximum is attained, and equilibrium follows. In the countably infinite game, however, P2's strategy set is the probability simplex over $\mathbb{N}$, which is *not compact*. This lack of compactness invalidates the maximization argument: although P2's payoff is continuous and bounded above by 1, the supremum is not achieved, and thus her best-response correspondence is empty. Standard existence theorems (e.g. Kakutani [61], Glicksberg [42]) rely, as discussed, on compactness and upper hemicontinuity of best responses. Here compactness fails, and thus equilibrium existence fails as well.

□

Example 3.32 From Discretization to Nonexistence by Discontinuity.

A simple yet illuminating example of the nonexistence of Nash equilibrium in continuous games arises from a “minimum-choice” game. Two players simultaneously select real-valued actions from the interval $(0, 1]$. The payoff to P1 is

$$g(a_1, a_2) = \begin{cases} 1, & a_1 < a_2, \\ \frac{1}{2}, & a_1 = a_2, \\ 0, & a_1 > a_2, \end{cases}$$

and P2 receives $1 - g(a_1, a_2)$. Thus, the player choosing the smaller action wins, ties split the payoff, and choosing the larger action yields zero.

To build intuition, consider a discretized version of the game in which each player chooses from the finite grid

$$a_i = \frac{i}{n}, \quad i = 1, \dots, n.$$

P1’s payoff matrix becomes

$$g_{ij}^{(n)} = \begin{cases} 1, & i < j, \\ \frac{1}{2}, & i = j, \\ 0, & i > j, \end{cases}$$

with P2 earning $1 - g_{ij}^{(n)}$.

For $n = 3$, P1’s payoff matrix is

$$G^{(3)} = \begin{pmatrix} \frac{1}{2} & 1 & 1 \\ 0 & \frac{1}{2} & 1 \\ 0 & 0 & \frac{1}{2} \end{pmatrix},$$

shown in Table 3.1.

$G^{(3)}$	$a_2 = 1/3$	$a_2 = 2/3$	$a_2 = 1$
$a_1 = 1/3$	0.5	1	1
$a_1 = 2/3$	0	0.5	1
$a_1 = 1$	0	0	0.5

Table 3.1: P1’s payoff matrix in the 3×3 discretized game.

In every finite discretization, the action pair $(1/n, 1/n)$ is a Nash equilibrium: both players choose the smallest point in the grid and obtain $1/2$. Any upward deviation loses to the opponent’s lowest action.

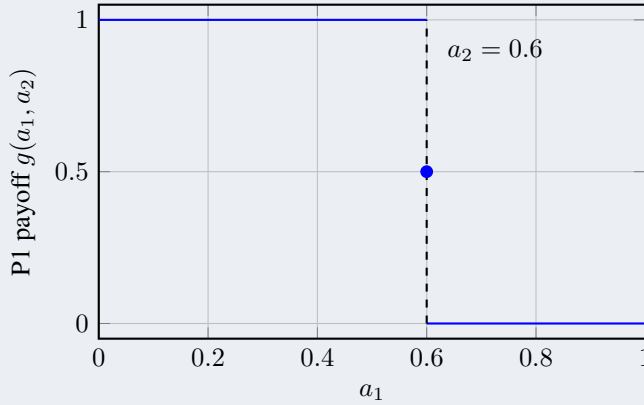


Figure 3.8: P1's payoff as a function of a_1 when $a_2 = 0.6$.

As $n \rightarrow \infty$, the grid approaches the continuum $(0, 1]$, and the discretized games converge to the continuous payoff function $g(a_1, a_2)$. This function is discontinuous along the diagonal $a_1 = a_2$, since the payoff jumps between 1, $1/2$, and 0 at the tie point. Figure 3.8 illustrates P1's payoff when P2 fixes $a_2 = 0.6$.

To see that no pure-strategy equilibrium exists, consider any $(a_1, a_2) \in (0, 1]^2$. If $a_1 < a_2$, then P2 can deviate to some a'_2 strictly between 0 and a_1 and thus change a payoff of 0 to a payoff of 1. If $a_1 > a_2$, the situation is symmetric, since P1 can profitably deviate to some $a'_1 < a_2$ and obtain payoff 1. Finally, if $a_1 = a_2$, each player receives only $1/2$, and either one can obtain 1 by choosing a slightly smaller action. In each case, at least one player has a profitable deviation, and thus no pure action pair is stable.

Let σ_2 be any distribution over $(0, 1]$ with cumulative distribution function F_2 . If P1 plays a pure action a , her expected payoff is

$$u_1(a \mid \sigma_2) = 1 - F_2(a) + \frac{1}{2} \Pr(a_2 = a).$$

Because 0 is excluded from the action set, we have $\lim_{a \downarrow 0} F_2(a) = 0$, and thus

$$\sup_{a > 0} u_1(a \mid \sigma_2) = 1.$$

P1 can therefore guarantee a payoff arbitrarily close to 1 by choosing a sufficiently close to 0, but she can never reach exactly 1 unless P2 leaves a positive gap above 0, which she would immediately exploit. The same reasoning applies to P2. The game therefore admits no Nash equilibrium. If the action sets were closed to include 0, then the profile $(0, 0)$ would constitute an equilibrium.

□

3.6 ■ Further Notes and Reading

In our discussion of Nash equilibrium, we have assumed the existence of payoff functions that assign real numbers to outcomes. This assumption can be justified by appealing to the axiomatic foundations of *utility theory*. Starting from primitive preference relations, defined by axioms such as completeness, transitivity, and continuity, it can be shown that there exists a utility function that faithfully represents these preferences in the sense that one outcome is preferred over another if and only if it has a higher utility value. Thus, the payoff functions used in equilibrium analysis are not arbitrary, but arise as consistent numerical representations of underlying preferences. For further details, we refer the reader to standard treatments of utility theory, such as the ones by von Neumann and Morgenstern [138], Debreu [30], and Fishburn [35], which establish both the existence and uniqueness (up to positive affine transformations) of such utility functions.

This lecture has focused on existence and characterization of equilibrium in *static (normal-form) games*. Existence has been established using fixed-point theorems such as those of Brouwer and Kakutani, which ensure that every finite static game admits at least one Nash equilibrium in mixed strategies.

Beyond existence, subsequent research has investigated the robustness and refinement of equilibria. Within the class of static games, Selten's trembling-hand perfection [107] refines Nash equilibrium by requiring robustness to small, unintended perturbations of players' strategies, thereby ruling out equilibria that rely on non-credible behavior. Other influential refinement concepts arise naturally in *dynamic games* and therefore lie beyond the scope of this lecture. Harsanyi's purification theorem [51] provides an interpretation of mixed-strategy equilibria as limits of pure-strategy equilibria in games with small payoff perturbations and is most naturally formulated in dynamic or informationally extended settings. Similarly, Kreps and Wilson's sequential equilibrium [68] generalizes Nash equilibrium to dynamic games with incomplete information by explicitly incorporating beliefs and Bayesian updating at every information set. These dynamic refinements will be revisited in later lectures, where extensive-form games and dynamic strategic interaction are introduced explicitly.

3.7 ■ Exercises

Exercise 3.1 (Fixed Points and Iterative Best-Response Dynamics). Consider a two-player two-action game in which each player chooses a mixed strategy in $[0, 1]$. P1's strategy at time t is denoted by $p_t \in [0, 1]$, representing the probability of playing action A , and P2's strategy is $q_t \in [0, 1]$, representing the probability of playing action A . The players update their strategies over time by the *best-response smoothing rule*

$$p_{t+1} = \frac{1}{2}p_t + \frac{1}{2}\text{BR}_1(q_t), \quad q_{t+1} = \frac{1}{2}q_t + \frac{1}{2}\text{BR}_2(p_t),$$

where $\text{BR}_i(\cdot)$ denotes the best response of player i . In parts (i) and (ii), assume that each best-response mapping $\text{BR}_i : [0, 1] \rightarrow [0, 1]$ is single-valued and continuous.

- (i) Show that the update rule defines a continuous self-map

$$F : [0, 1]^2 \rightarrow [0, 1]^2, \quad F(p, q) = \left(\frac{1}{2}p + \frac{1}{2}\text{BR}_1(q), \frac{1}{2}q + \frac{1}{2}\text{BR}_2(p) \right),$$

and explain why Brouwer's Fixed Point Theorem guarantees the existence of at least one fixed point $(p^*, q^*) \in [0, 1]^2$.

- (ii) Prove that (p^*, q^*) is a fixed point of F if and only if $p^* = \text{BR}_1(q^*)$ and $q^* = \text{BR}_2(p^*)$, and conclude that every fixed point of the smoothed dynamics is a profile of mutual best responses, that is, a Nash equilibrium of the underlying game.

- (iii) Derive the best-response correspondences $BR_1(q)$ and $BR_2(p)$ for the *Battle of the Sexes* game with payoff matrix

P1	P2	
	Opera	Football
Opera	(2, 1)	(0, 0)
Football	(0, 0)	(1, 2)

where p and q denote the probabilities that P1 and P2, respectively, play Opera. Determine all profiles (p^*, q^*) that satisfy the mutual best-response condition of part (ii), allowing BR_i to be set-valued at points of indifference, and compare them with the mixed-strategy Nash equilibrium

$$\sigma_1^* = (2/3, 1/3), \quad \sigma_2^* = (1/3, 2/3).$$

Discuss whether the fixed points of the dynamics coincide with the Nash equilibria of the game.

- (iv) Implement the smoothed best-response iteration for the Battle of the Sexes game in Python or MATLAB, using the single-valued selections

$$BR_1(q) = \begin{cases} 1, & q > 1/3, \\ 0, & q \leq 1/3, \end{cases} \quad BR_2(p) = \begin{cases} 1, & p > 2/3, \\ 0, & p \leq 2/3. \end{cases}$$

Run $T = 200$ iterations from each of the initial conditions $(p_0, q_0) \in \{(0.9, 0.9), (0.1, 0.1), (0.5, 0.5)\}$. Report the final iterates (p_T, q_T) in a table, and plot p_t and q_t against t for each initial condition. Compare the observed limiting behavior with the equilibria found in part (iii); in particular, verify that the first two trajectories converge to the pure-strategy equilibria (1, 1) and (0, 0), respectively, while the third oscillates in a neighborhood of the mixed-strategy equilibrium $(2/3, 1/3)$ without converging to it.

Exercise 3.2 (Closedness of the Feasible-Set Mapping). A point-to-set mapping $\Gamma : \mathbb{R}^p \rightrightarrows \mathbb{R}^n$ is said to be closed, or upper hemicontinuous, at $\theta^* \in \mathbb{R}^p$ if, given any sequence $\theta^k \rightarrow \theta^*$ and any sequence $x^k \in \Gamma(\theta^k)$ such that $x^k \rightarrow x^*$, it follows that $x^* \in \Gamma(\theta^*)$.

- (i) Determine whether the following mapping is upper hemicontinuous:

$$\Gamma(x) = \begin{cases} [0, 1], & \text{if } x = \frac{1}{3}, \\ \{0\}, & \text{if } x \in [0, \frac{1}{3}), \\ \{1\}, & \text{if } x \in (\frac{1}{3}, 1]. \end{cases}$$

If it is upper hemicontinuous, construct an example of a mapping that is not upper hemicontinuous; if it is not, construct an example of a mapping that is upper hemicontinuous. Justify your answers.

- (ii) Consider the following parametric mathematical programming problem:

$$\begin{aligned} \min_x \quad & f(x, \theta) \\ \text{s.t.} \quad & f^i(x, \theta) \leq 0, \quad i \in \mathcal{P} = \{1, \dots, m\}. \end{aligned}$$

Here, there are two types of variables: the parameter $\theta \in \mathbb{R}^p$ and the decision variable $x \in \mathbb{R}^n$. Assume that the functions $f, f^i : \mathbb{R}^{n+p} \rightarrow \mathbb{R}, i \in \mathcal{P}$, are defined and continuous on the entire space $\mathbb{R}^{n+p}$. The feasible set of this optimization problem is

$$\mathcal{F}(\theta) = \{x \in \mathbb{R}^n : f^i(x, \theta) \leq 0, i \in \mathcal{P}\}.$$

A global optimum of the problem is denoted by $x^*(\theta)$, the set of all global optima is denoted by $\mathcal{F}^*(\theta)$, and the corresponding optimal value is

$$f(x^*(\theta), \theta) =: f^*(\theta).$$

Show that the point-to-set mapping F that maps θ to the feasible set $\mathcal{F}(\theta)$ is closed at every $\theta^* \in \mathbb{R}^p$.

Exercise 3.3 (Lower Hemicontinuity of the Feasible-Set Mapping). A point-to-set mapping $\Gamma : \mathbb{R}^p \rightrightarrows \mathbb{R}^n$ is open, or lower hemicontinuous, at $\theta^* \in \mathbb{R}^p$ if, given any sequence $\theta^k \rightarrow \theta^*$ and any point $x^* \in \Gamma(\theta^*)$, there is a sequence $x^k \in \Gamma(\theta^k)$ such that $x^k \rightarrow x^*$. A point-to-set mapping Γ is continuous at $\theta^* \in \mathbb{R}^p$ if it is both closed and open at θ^* . Consider the parametric programming problem of the preceding exercise, and assume in addition that the functions $f^i(\cdot, \theta)$, $i \in \mathcal{P}$, are convex in x for each fixed θ , and that Slater's condition holds at θ^* ; that is, there exists x' such that $f^i(x', \theta^*) < 0$ for all $i \in \mathcal{P}$.

- (i) Show that for every $x^* \in \mathcal{F}(\theta^*)$ and every $\lambda \in (0, 1]$, the point $x_\lambda := \lambda x' + (1 - \lambda)x^*$ satisfies $f^i(x_\lambda, \theta^*) < 0$ for all $i \in \mathcal{P}$.
- (ii) Prove, using part (i), that the point-to-set mapping F from θ to $\mathcal{F}(\theta)$ is lower hemicontinuous at θ^* under the stated assumptions.

Exercise 3.4 (Failure of Fixed Points without Convexity). Brouwer's and Kakutani's fixed point theorems require the domain to be non-empty, compact, and convex, and Kakutani's theorem additionally requires the correspondence to have convex values. Compactness alone is not sufficient. This exercise examines two instances of failure: a compact interval with a non-convex-valued correspondence, and a compact but non-convex planar domain. Consider first the correspondence $F : [0, 1] \rightrightarrows [0, 1]$ defined by

$$F(x) = \begin{cases} \left\{ \frac{1}{3} + \frac{1}{2}x \right\}, & x \in [0, \frac{1}{2}), \\ \left[\frac{7}{12} - 0.08, \frac{7}{12} + 0.08 \right] \cup \left[\frac{1}{3} - 0.10, \frac{1}{3} + 0.10 \right], & x = \frac{1}{2}, \\ \left\{ \frac{1}{3}x + \frac{1}{6} \right\}, & x \in (\frac{1}{2}, 1]. \end{cases}$$

- (i) Show that F has no fixed point; that is, there is no $x \in [0, 1]$ such that $x \in F(x)$. Determine whether F is upper hemicontinuous and whether it is lower hemicontinuous, justifying each answer using either the sequential or the open-set definition. Explain why the lack of convexity of $F(\frac{1}{2})$ is essential to the failure of Kakutani's fixed point theorem, and compare with a convex-valued modification of F (for example, replacing the two disjoint intervals at $x = \frac{1}{2}$ by their convex hull).
- (ii) Implement, in Python or MATLAB, a computation of the distance function

$$d(x) = \min_{y \in F(x)} |y - x|$$

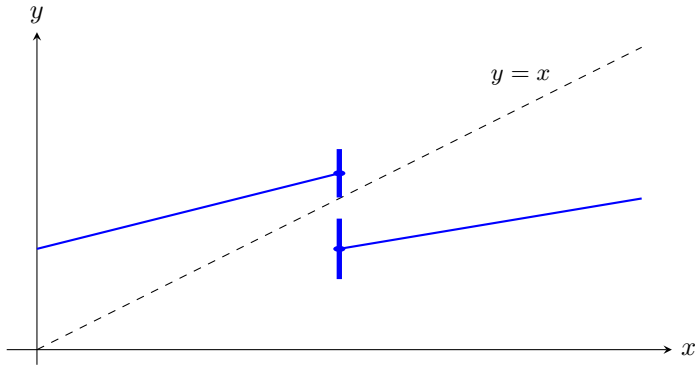


Figure 3.9: The correspondence $F : [0, 1] \rightrightarrows [0, 1]$. For $x \in [0, \frac{1}{2})$ the branch is $y = \frac{1}{3} + \frac{1}{2}x$; for $x \in (\frac{1}{2}, 1]$ the branch is $y = \frac{1}{3} + \frac{1}{3}(x - \frac{1}{2})$. At $x = \frac{1}{2}$, the image is the non-convex union $[\frac{7}{12} - 0.08, \frac{7}{12} + 0.08] \cup [\frac{1}{3} - 0.10, \frac{1}{3} + 0.10]$. The dashed line shows $y = x$. The point $(\frac{1}{2}, \frac{1}{2})$ lies on the diagonal but not in $F(\frac{1}{2})$, illustrating how non-convexity can destroy the existence of a fixed point.

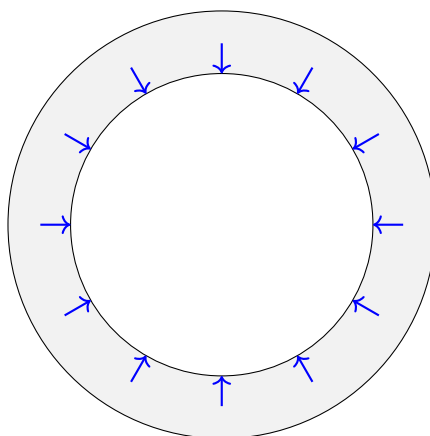
on the grid $x \in \{0, 0.001, 0.002, \dots, 1\}$ (evaluating d exactly at the grid point $x = 0.5$ using the two intervals in $F(\frac{1}{2})$). Plot $d(x)$ against x , and report the minimum value of d over the grid and the point at which it is attained. Verify numerically that the minimum is strictly positive, approximately 3.3×10^{-3} , attained at $x = \frac{1}{2}$, which confirms that the graph of F remains bounded away from the diagonal $y = x$.

- (iii) Let $S = \{(x, y) \in \mathbb{R}^2 : 1 \leq x^2 + y^2 \leq 2\}$ be the closed annulus, and define $f : S \rightarrow S$ by

$$f(x, y) = \frac{(x, y)}{\sqrt{x^2 + y^2}} \quad (\text{radial projection to the inner circle}).$$

Prove that f is continuous, maps S into itself, and has no fixed point. Figure 3.10 shows how every point in the annulus is mapped inward to the inner boundary.

- (iv) Explain why convexity of the domain rules out such “swapping” or “radial collapse” behaviors, thereby ensuring the existence of fixed points in convex domains such as intervals, disks, and simplices.



Radial collapse $f(x, y) = (x, y) / \sqrt{x^2 + y^2}$

Figure 3.10: Annulus $1 \leq x^2 + y^2 \leq 2$ with the mapping f that projects every point to the inner circle. No point remains fixed.

Chapter 4

Nash Equilibrium: Properties and Computation

In the previous lecture, we introduced the concept of Nash equilibrium and developed the theoretical foundations for its existence and characterization in finite games. We now turn to the practical side: how mixed-strategy equilibria can be identified, computed, and understood through explicit examples. The computational aspects of mixed-strategy Nash equilibria are crucial, not only for illustrating abstract theory, but also for applications in diverse fields, such as economics, engineering, and computer science, where agents interact strategically, and equilibrium analysis provides predictive power.

In this lecture, we begin with a classical example, the *Battle of the Sexes* game, to illustrate the structure of best responses, the logic of mixed strategies, and the derivation of equilibria. This example will allow us to highlight both the intuitive and formal properties of Nash equilibrium and to motivate general computational approaches. We then discuss iterative methods, such as best-response dynamics, which provide algorithmic procedures for approximating equilibria in more complex settings.

4.1 ■ Computational Methods for Mixed-Strategy Nash Equilibrium

4.1.1 ■ Battle of the Sexes: A Canonical Example

The *Battle of the Sexes* is a classical coordination game that illustrates situations with both common interests and conflicting preferences. As introduced earlier in Lecture 3, two players wish to spend time together but disagree on which activity to attend. P1 prefers Opera, while P2 prefers Football. The game is represented as a bi-matrix game, with P1 choosing rows and P2 choosing columns, and with the ordering of actions consistent with the earlier presentation: Opera as the first action and Football as the second. The payoff matrix is

P1	P2	
	Opera	Football
Opera	(2, 1)	(0, 0)
Football	(0, 0)	(1, 2)

Let $A_1 = A_2 = \{\text{Opera}, \text{Football}\}$ denote the action sets. A mixed strategy for P1 is $\sigma_1 = (x, 1 - x) \in \Delta(A_1)$, where $x \in [0, 1]$ is the probability of choosing Opera. Similarly, a mixed

strategy for P2 is $\sigma_2 = (y, 1 - y) \in \Delta(A_2)$, where $y \in [0, 1]$ is the probability of choosing Opera.

The expected utilities under the mixed-strategy profile (σ_1, σ_2) are

$$U_1(x, y) = 2xy + (1 - x)(1 - y) = 3xy - x - y + 1,$$

$$U_2(x, y) = xy + 2(1 - x)(1 - y) = 3xy - 2x - 2y + 2.$$

Each player's best response maximizes her expected utility given the other player's strategy. Fixing $y \in [0, 1]$, the function $U_1(x, y)$ is linear in x with slope $3y - 1$, yielding

$$\text{BR}_1(y) = \begin{cases} \{0\}, & \text{if } y < \frac{1}{3}, \\ [0, 1], & \text{if } y = \frac{1}{3}, \\ \{1\}, & \text{if } y > \frac{1}{3}, \end{cases}$$

where $x = 1$ corresponds to choosing Opera and $x = 0$ to choosing Football.

Similarly, fixing $x \in [0, 1]$, the function $U_2(x, y)$ is linear in y with slope $3x - 2$, and thus

$$\text{BR}_2(x) = \begin{cases} \{0\}, & \text{if } x < \frac{2}{3}, \\ [0, 1], & \text{if } x = \frac{2}{3}, \\ \{1\}, & \text{if } x > \frac{2}{3}. \end{cases}$$

Figure 4.1 provides a graphical representation of these best-response correspondences in the unit square $[0, 1]^2$. The blue curves depict P1's best responses BR_1 , while the red curves depict P2's best responses BR_2 . The dashed horizontal and vertical segments indicate indifference sets, where a player is indifferent between her two pure strategies. Nash equilibria correspond to intersections of the two best-response correspondences. As shown in the figure, there are three such intersections: two pure-strategy equilibria at (Opera, Opera), i.e., $(1, 1)$, and (Football, Football), i.e., $(0, 0)$, and a unique mixed-strategy equilibrium at $(x^*, y^*) = (\frac{2}{3}, \frac{1}{3})$.

As discussed extensively in Lecture 3, these best-response correspondences are upper hemi-continuous with nonempty, convex values, ensuring the applicability of Kakutani's fixed-point theorem (see Appendix B (Mathematical Background)) and hence the existence of at least one mixed-strategy Nash equilibrium.

The mixed-strategy equilibrium satisfies the *indifference principle*: each player chooses her mixing probability so as to make the other player indifferent between their pure strategies. This principle is restated formally in Subsection 4.1.3.

4.1.2 ■ Best Response Approach

The *Battle of the Sexes* game provides a simple but illuminating illustration of how the best response approach can be used to identify Nash equilibria. In that game, we first expressed each player's expected payoff as a function of the opponent's strategy. We then derived the best response correspondence for each player, showing which strategies were optimal given the strategy of the other. Finally, by examining where these best response functions intersected, we were able to identify the set of Nash equilibria.

This case highlights three essential ingredients of the method. The first is to compute the expected payoff for each player as a function of the opponent's strategies. The second is the derivation of the best response correspondence, which translates payoffs into optimal reactions. The third is the identification of fixed points, namely strategy profiles in which each player's strategy is a best response to the others. In small-size two-player games, such as the Battle of the Sexes, this process can be carried out explicitly, even graphically, by plotting best responses

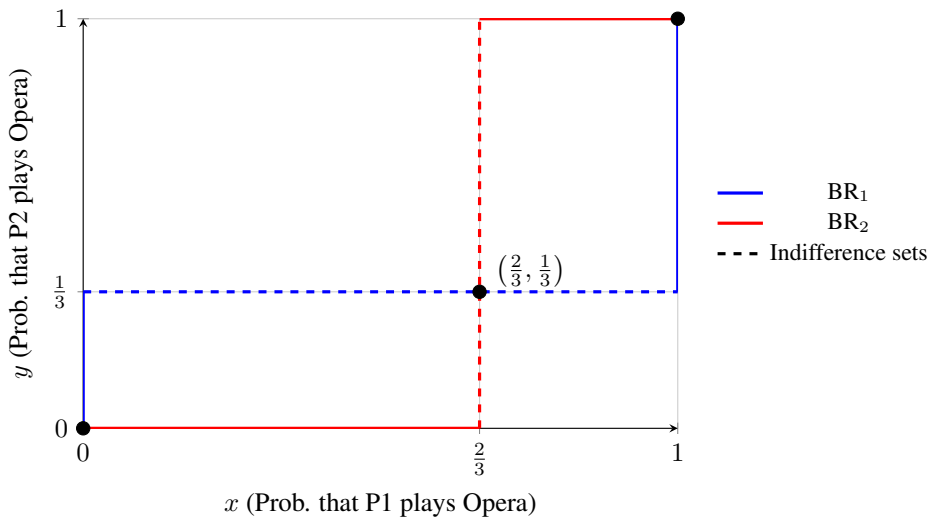


Figure 4.1: Best-response correspondences for the Battle of the Sexes. Blue lines represent BR_1 , red lines represent BR_2 , and dashed segments indicate indifference sets. Intersections yield the two pure-strategy equilibria and the mixed-strategy equilibrium $(\frac{2}{3}, \frac{1}{3})$.

and observing their intersections as in Figure 4.1. In larger games, however, even with only two players, direct graphical methods are not always feasible, and one must resort to systematic and algorithmic procedures.

The same reasoning extends beyond the Battle of the Sexes to the general case of finite games. Let $\mathcal{N} = \{1, 2, \dots, N\}$ denote the set of players. As in Lecture 3, each player i is endowed with a finite set of actions, A_i , and their mixed-strategy space is given by the probability simplex $\Delta(A_i)$. Throughout this lecture, we retain the convention that each player maximizes a utility (payoff) function; whenever costs are used instead, as in some examples and exercises, the players are explicitly designated as minimizers. The set of all mixed-strategy profiles is the Cartesian product

$$\Sigma := \prod_{i \in \mathcal{N}} \Delta(A_i).$$

As discussed in Lecture 3, for any profile $\sigma \in \Sigma$, the expected payoff of player i is computed as

$$U_i(\sigma) = \sum_{a \in A} \left(\prod_{j \in \mathcal{N}} \sigma_j(a_j) \right) u_i(a), \quad A = \prod_{i \in \mathcal{N}} A_i,$$

which represents the average payoff across all pure profiles, weighted by the probabilities induced by the players' mixed strategies. Given this representation, the best response correspondence of player i is

$$BR_i(\sigma_{-i}) = \arg \max_{\sigma_i \in \Delta(A_i)} U_i(\sigma_i, \sigma_{-i}).$$

By construction, a Nash equilibrium $\sigma^* \in \Sigma$ is precisely a fixed point of the best response mapping: for every player $i \in \mathcal{N}$, the strategy σ_i^* must belong to the set of best responses against σ_{-i}^* .

Iterative best response dynamics. The fixed-point characterization of Nash equilibrium naturally motivates an iterative procedure known as *best response dynamics*. Beginning with an

initial mixed-strategy profile $\sigma^{(0)} \in \Sigma$, players repeatedly update their strategies in response to those played by their opponents in the previous round:

$$\sigma_i^{(n+1)} \in \text{BR}_i(\sigma_{-i}^{(n)}), \quad i \in \mathcal{N}.$$

Since best responses are generally *not* unique, this update rule defines a correspondence rather than a single trajectory: different selections from $\text{BR}_i(\cdot)$ may generate different sequences $\{\sigma^{(n)}\}_{n \geq 0}$, and these trajectories need not converge. Even in two-player games, best response dynamics can cycle or diverge, and convergence to a mixed-strategy Nash equilibrium is not guaranteed. Likewise, convergence to a pure-strategy equilibrium cannot be assumed in general, since many finite normal-form games admit no pure-strategy Nash equilibrium.

In games with $N > 2$ players, the *order* in which players update matters significantly. Parallel (simultaneous) updates, sequential (Gauss-Seidel) updates, and asynchronous update schedules can generate qualitatively different trajectories and may differ in their convergence behavior. Thus, best response dynamics should be understood as a family of adaptive processes rather than a single algorithm. Despite these limitations, best response dynamics remain conceptually and practically valuable.

From a computational perspective, they provide constructive local search procedures for equilibria in many games of moderate size. From a behavioral perspective, they model a natural form of adaptive play in which agents myopically best respond to observed behavior. This interpretation is widely used in experimental and applied settings, where players are typically not assumed to know the equilibrium a priori, but instead adjust their play incrementally in response to others.

Example 4.1 A 3×3 Non-Zero-Sum Game.

Consider two players with three actions each: P1 chooses T, M, B and P2 chooses L, C, R . Payoffs are represented by the bi-matrix:

P1	P2		
	L	C	R
T	(2, 1)	(0, 3)	(3, 0)
M	(1, 3)	(2, 1)	(0, 2)
B	(0, 0)	(3, 2)	(1, 3)

To compute best responses, let P2 randomize with probabilities (q_L, q_C, q_R) . P1's expected payoff from T is $2q_L + 0 \cdot q_C + 3q_R = 2q_L + 3q_R$, from M it is $1 \cdot q_L + 2q_C + 0 \cdot q_R = q_L + 2q_C$, and from B it is $0 \cdot q_L + 3q_C + 1 \cdot q_R = 3q_C + q_R$. A natural candidate equilibrium has P1 mixing between T and M while placing zero probability on B ; in that case, P1's indifference between T and M requires

$$2q_L + 3q_R = q_L + 2q_C,$$

and off-support optimality requires $3q_C + q_R \leq 2q_L + 3q_R$ (so that B does not exceed the on-support value). Now let P1 randomize with probabilities (p_T, p_M, p_B) . P2's expected payoff from L is $1 \cdot p_T + 3p_M + 0 \cdot p_B = p_T + 3p_M$; from C it is $3p_T + 1 \cdot p_M + 2p_B = 3p_T + p_M + 2p_B$; and from R it is $0 \cdot p_T + 2p_M + 3p_B = 2p_M + 3p_B$. Seeking a tractable interior support, suppose that P2 mixes between L

and C while placing zero probability on R . Then, P2's indifference between L and C reduces to

$$p_T + 3p_M = 3p_T + p_M + 2p_B.$$

Imposing $p_B = 0$ (consistent with our candidate support for P1) simplifies this to $p_T + 3p_M = 3p_T + p_M$, yielding $p_T = \frac{1}{2}$ and $p_M = \frac{1}{2}$. With $(p_T, p_M, p_B) = (\frac{1}{2}, \frac{1}{2}, 0)$, P2's payoff from R is $2p_M + 3p_B = 1$, while the indifference value at L and C is $p_T + 3p_M = 2$, and thus R is indeed off-support as desired.

Returning to P1's indifference, set $q_R = 0$ to match the candidate support for P2. The condition $2q_L + 3q_R = q_L + 2q_C$ becomes $2q_L = q_L + 2q_C$, hence $q_L = 2q_C$. Enforcing $q_L + q_C = 1$ gives $q_C = \frac{1}{3}$ and $q_L = \frac{2}{3}$. With $(q_L, q_C, q_R) = (\frac{2}{3}, \frac{1}{3}, 0)$, P1's value on the support equals $2q_L = \frac{4}{3}$ for T and $q_L + 2q_C = \frac{4}{3}$ for M . The off-support action B yields $3q_C + q_R = 1$, which is strictly lower, confirming that B is not a best response.

We have thus identified a mixed-strategy Nash equilibrium:

$$\sigma_1^* = (\frac{1}{2}, \frac{1}{2}, 0), \quad \sigma_2^* = (\frac{2}{3}, \frac{1}{3}, 0).$$

This profile is a fixed point of the best response correspondence: given σ_2^* , P1 is exactly indifferent between T and M and strictly prefers them to B ; given σ_1^* , P2 is exactly indifferent between L and C and strictly prefers them to R . The example underscores how the best response approach scales in larger nonzero-sum games: rather than plotting reaction curves, one solves the indifference equations on the candidate supports and verifies off-support optimality to certify equilibrium. $\square$

4.1.3 ■ Indifference Principle

Another key observation from the Battle of the Sexes game is that in a mixed-strategy equilibrium, a player must be *indifferent* among all the pure actions over which they randomize. If one action in the support of the strategy yielded a strictly higher expected payoff than another, then the player would strictly prefer that action and would not assign positive probability to the inferior one, as we have seen in Example 4.1 above. This observation provides a practical way to compute mixed-strategy equilibria: one equates the expected payoffs of the actions in the support of the mixed strategy.

Theorem 4.2 (Indifference Principle for Mixed-Strategy Equilibria in N -Player Games).

Let G be a finite N -player static game. A mixed-strategy profile $\sigma^* = (\sigma_1^*, \dots, \sigma_N^*)$ is a Nash equilibrium if, and only if, for every player $i \in \mathcal{N} = \{1, \dots, N\}$, every pure action in the support of σ_i^* yields the same expected payoff against the opponents' mixed strategies, and every pure action outside the support yields no higher payoff.

Equivalently, for each player i there exists a payoff level v_i such that

$$a_i \in \text{supp}(\sigma_i^*) \Rightarrow U_i(a_i, \sigma_{-i}^*) = v_i, \quad a_i \notin \text{supp}(\sigma_i^*) \Rightarrow U_i(a_i, \sigma_{-i}^*) \leq v_i,$$

where U_i denotes the expected payoff against the mixed strategies $\sigma_{-i}^* = (\sigma_j^*)_{j \neq i}$.

Proof. Suppose that σ^* is a mixed-strategy Nash equilibrium. Then for each player i , the strategy σ_i^* is a best response to σ_{-i}^* . If two pure actions in $\text{supp}(\sigma_i^*)$ yielded different expected payoffs against σ_{-i}^* , shifting probability mass from a lower-payoff to a higher-payoff support

action would strictly increase player i 's expected payoff, contradicting optimality of σ_i^* . Thus all support actions yield the same payoff v_i . Any off-support action must yield payoff at most v_i , otherwise σ_i^* would not be a best response. This proves the “only if” direction.

Conversely, suppose that for each player i , all pure actions in $\text{supp}(\sigma_i^*)$ yield the common payoff v_i against σ_{-i}^* , and any off-support action yields payoff at most v_i . Then no unilateral deviation, whether by reallocating probability among support actions or by assigning positive probability to an off-support action, can increase the expected payoff above v_i . Thus σ_i^* is a best response to σ_{-i}^* for every i , and so σ^* is a Nash equilibrium. $\square$

If σ_i^* is pure (that is, a one-point probability distribution), then its support is a singleton and the indifference condition is vacuously satisfied; in this case, the opponent needs only to respond best to that pure action. Off-support actions are allowed to tie at the value v_i ; the weak inequality $\leq v_i$ covers this possibility, and best-response sets may be multi-valued at equilibrium. Finally, the indifference requirement applies only to actions played with strictly positive probability, not to all available actions.

Example 4.3 A 2×3 Entry Game.

To illustrate the Indifference Principle in a richer setting, consider a game between an entrant (P1) and an incumbent (P2). P1 chooses either to *Enter* (E) the market or to *Stay Out* (O). P2 simultaneously selects one of three possible responses, labeled A , B , and C . The payoff matrix, with entries (u_1, u_2) , is given by

P1	P2		
	A	B	C
E	(2, 1)	(0, 0)	(3, 2)
O	(1, 0)	(1, 1)	(1, 0)

Suppose P1 enters with probability p and stays out with probability $1 - p$, while P2 chooses a mixed strategy (q_A, q_B, q_C) over the three actions. The expected payoff to P1 from playing E is $2q_A + 3q_C$, while the expected payoff from playing O is always 1. For P1 to be willing to randomize, she must be indifferent between E and O , which imposes the condition $2q_A + 3q_C = 1$.

Turning to P2, the expected payoff from choosing A is p , from choosing B is $1 - p$, and from choosing C is $2p$. If P2 were to mix between B and C , indifference requires $1 - p = 2p$, which yields $p = \frac{1}{3}$. At this probability, P2 obtains $\frac{2}{3}$ from either B or C , while A yields only $\frac{1}{3}$, and thus A can be excluded from the support. Thus, P2's equilibrium mixture assigns positive probabilities only to B and C .

The indifference condition for P1 then requires $2q_A + 3q_C = 1$. Since $q_A = 0$ at equilibrium, this reduces to $3q_C = 1$, and thus $q_C = \frac{1}{3}$ and $q_B = \frac{2}{3}$. At equilibrium, therefore, P1 randomizes by entering with probability $\frac{1}{3}$ and staying out with probability $\frac{2}{3}$, while P2 responds with the mixture $(0, \frac{2}{3}, \frac{1}{3})$.

This solution illustrates the full power of the indifference principle. P1 mixes in such a way that P2 is left indifferent between B and C , while P2 in turn mixes so that P1 is exactly indifferent between entering and staying out. The equilibrium mixture

is pinned down entirely by these indifference equations, showing how the principle generalizes to more complex asymmetric games. $\square$

4.1.4 ■ Bilinear Programming Methods for Computing Nash Equilibria in Finite Bi-matrix Games

In Lecture 2, we used linear programming to compute saddle-point equilibria in zero-sum games. This approach can be generalized to bi-matrix games, where the payoff structure is nonzero-sum and both players' objectives must be satisfied simultaneously. In such settings, the problem can be cast as a bilinear program.

We now present a variational characterization of mixed-strategy Nash equilibria in bi-matrix games using bilinear programming. In what follows, $A, B \in \mathbb{R}^{m \times n}$ denote the payoff matrices of P1 (the row player) and P2 (the column player), respectively, and

$$\Delta_m := \{y \in \mathbb{R}^m \mid y \geq 0, y^\top \mathbf{1}_m = 1\}, \quad \Delta_n := \{z \in \mathbb{R}^n \mid z \geq 0, z^\top \mathbf{1}_n = 1\}$$

denote the probability simplices of mixed strategies for the two players.

Proposition 4.4 (Bilinear Program for Mixed-Strategy Nash Equilibria). *A strategy pair $(y^*, z^*) \in \Delta_m \times \Delta_n$ is a mixed-strategy Nash equilibrium of the bi-matrix game (A, B) if, and only if, there exists a pair $(p^*, q^*) \in \mathbb{R}^2$ such that (y^*, z^*, p^*, q^*) solves*

$$\begin{aligned} \min_{y, z, p, q} \quad & p + q - y^\top A z - y^\top B z \\ \text{such that} \quad & A z \leq p \mathbf{1}_m, \\ & B^\top y \leq q \mathbf{1}_n, \\ & y \in \Delta_m, \quad z \in \Delta_n. \end{aligned} \tag{4.1}$$

Moreover, the optimal value equals 0.

Proof. ($\Rightarrow$) Let (y^*, z^*) be a mixed-strategy Nash equilibrium. Define the equilibrium payoffs $p^* := y^{*\top} A z^*$ and $q^* := y^{*\top} B z^*$. Best-response optimality of y^* against z^* means that no pure row i yields more than p^* , i.e., $(A z^*)_i \leq p^*$ for all i , equivalently $A z^* \leq p^* \mathbf{1}_m$. Similarly, optimality of z^* against y^* leads to $B^\top y^* \leq q^* \mathbf{1}_n$. Thus (y^*, z^*, p^*, q^*) is feasible for (4.1). By feasibility and the simplex constraints,

$$y^{*\top} A z^* \leq p^*, \quad y^{*\top} B z^* \leq q^*,$$

and thus the objective at (y^*, z^*, p^*, q^*) is

$$p^* + q^* - y^{*\top} A z^* - y^{*\top} B z^* = 0.$$

Hence the optimal value is at most 0.

($\Leftarrow$) Conversely, let (y^*, z^*, p^*, q^*) be an optimal solution of (4.1). From the constraints and $y^*, z^* \in \Delta$,

$$y^{*\top} A z^* \leq p^*, \quad y^{*\top} B z^* \leq q^*,$$

and thus the objective function is always nonnegative: $p^* + q^* - y^{*\top} A z^* - y^{*\top} B z^* \geq 0$. Optimality therefore forces equality,

$$p^* = y^{*\top} A z^*, \quad q^* = y^{*\top} B z^*,$$

and the deviation constraints become

$$Az^* \leq (y^{*\top} Az^*) \mathbf{1}_m, \quad B^\top y^* \leq (y^{*\top} Bz^*) \mathbf{1}_n.$$

These are exactly the no-profitable-deviation conditions for each player: no pure row (column) yields a payoff above the mixed payoff against z^* (resp. y^*). Hence y^* is a best response to z^* and z^* is a best response to y^* ; thus (y^*, z^*) is a mixed-strategy Nash equilibrium.

Finally, note that if some action in the support of y^* (resp. z^*) had strict inequality, then

$$y^{*\top} Az^* = \sum_i y_i^* (Az^*)_i < \sum_i y_i^* (y^{*\top} Az^*) = y^{*\top} Az^*,$$

a contradiction. Therefore, inequalities bind on supports, recovering the usual indifference property at equilibrium. This completes the proof. $\square$

Example 4.5 3×3 bi-matrix.

Consider the following two-player game, where P1 chooses rows and P2 chooses columns. The payoff pairs are written in the form (u_1, u_2) , with P1's payoff listed first:

P1	P2		
	1	2	3
1	(2, -1)	(-1, 3)	(0, 0)
2	(-3, 2)	(1, -2)	(4, 1)
3	(0, 0)	(2, 1)	(-2, 2)

Equivalently, the game can be represented as a pair of payoff matrices (A, B) , where A records P1's payoffs and B records P2's:

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -3 & 1 & 4 \\ 0 & 2 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} -1 & 3 & 0 \\ 2 & -2 & 1 \\ 0 & 1 & 2 \end{bmatrix}.$$

Here, the entries are picked as arbitrary real numbers and may include negative payoffs. Let P1's mixed strategy be $y = (y_1, y_2, y_3)^\top \in \Delta_3$ and P2's be $z = (z_1, z_2, z_3)^\top \in \Delta_3$, with $\Delta_3 = \{w \in \mathbb{R}_{\geq 0}^3 : \mathbf{1}^\top w = 1\}$. Define the equilibrium value variables as $p, q \in \mathbb{R}$.

Using the adjusted formulation,

$$\begin{aligned} \min_{y, z, p, q} \quad & p + q - y^\top Az - y^\top Bz \\ \text{such that} \quad & Az \leq p \mathbf{1}_3, \\ & B^\top y \leq q \mathbf{1}_3, \\ & y \in \Delta_3, z \in \Delta_3. \end{aligned} \tag{BP}$$

Writing the deviation constraints component-wise makes the role of negative entries explicit. First,

$$Az = \begin{bmatrix} 2z_1 - z_2 + 0 \cdot z_3 \\ -3z_1 + z_2 + 4z_3 \\ 0 \cdot z_1 + 2z_2 - 2z_3 \end{bmatrix} \leq \begin{bmatrix} p \\ p \\ p \end{bmatrix}, \quad \text{i.e.,} \quad \begin{cases} 2z_1 - z_2 \leq p, \\ -3z_1 + z_2 + 4z_3 \leq p, \\ 2z_2 - 2z_3 \leq p, \end{cases}$$

and

$$B^\top y = \begin{bmatrix} -y_1 + 2y_2 + 0 \cdot y_3 \\ 3y_1 - 2y_2 + y_3 \\ 0 \cdot y_1 + y_2 + 2y_3 \end{bmatrix} \leq \begin{bmatrix} q \\ q \\ q \end{bmatrix}, \quad \text{i.e.,} \quad \begin{cases} -y_1 + 2y_2 \leq q, \\ 3y_1 - 2y_2 + y_3 \leq q, \\ y_2 + 2y_3 \leq q. \end{cases}$$

The simplex constraints are

$$y_i \geq 0, \quad \sum_{i=1}^3 y_i = 1, \quad z_j \geq 0, \quad \sum_{j=1}^3 z_j = 1.$$

The vectors Az and $B^\top y$ list each pure action's expected payoff against the opponent's mix. The inequalities $Az \leq p\mathbf{1}$ and $B^\top y \leq q\mathbf{1}$ enforce “no profitable deviation.” Negativity of entries in A, B does not affect the logic: equilibrium depends on *comparisons* relative to p and q , and not on absolute signs. If strictly positive payoff matrices are preferred, one may add a positive constant to every entry of A and another to every entry of B , here 4 and 3 respectively, which makes both matrices positive without changing the equilibria of the underlying game: the shift raises p and q by the same constants and leaves every comparison intact. At optimality, the objective function attains 0 and $p = y^\top Az$, $q = y^\top Bz$, with equalities binding on the supports, recovering indifference.

□

The bilinear formulation highlights the central computational difficulty: the objective contains bilinear terms $y^\top Az$ and $y^\top Bz$, making the program nonconvex. For small-size games, one can solve the problem by enumerating supports or by applying iterative descent methods. However, as the numbers of actions grow, the feasible sets expand combinatorially. In an $m \times n$ bi-matrix game, the support-enumeration approach may require checking $\binom{m}{k} \binom{n}{\ell}$ combinations for supports of size k and ℓ , which becomes computationally prohibitive for large m and n . Indeed, computing mixed-strategy Nash equilibria in bi-matrix games is known to be PPAD-complete, and thus no efficient algorithm is expected in general. The bilinear programming viewpoint thus provides a unifying mathematical formulation, but in practice its direct use is limited to small or specially structured games. For larger systems, one typically resorts to approximation schemes, heuristics, or specialized algorithms that exploit additional structure.

4.1.5 ■ Robustness of Nash Equilibrium and the Need for Refinement

The Nash equilibrium concept guarantees internal consistency: no player has an incentive to deviate unilaterally given the strategies of the others. However, this notion alone does not address

robustness. In many practical situations, players may make mistakes with small probability, or they may face slight uncertainty about others' actions. If an equilibrium ceases to be optimal under such minimal perturbations, then it may be behaviorally implausible or fragile as a predictive tool.

The *Battle of the Sexes* game provides a simple illustration of this distinction. As shown earlier, the game admits two pure-strategy Nash equilibria, (Opera, Opera) and (Football, Football), as well as a mixed-strategy equilibrium in which each player randomizes between Opera and Football. At the pure equilibria, each player's action is a *strict* best response: any unilateral deviation reduces that player's payoff, even if beliefs about the opponent's behavior are slightly perturbed. By contrast, at the mixed-strategy equilibrium each player is exactly indifferent between the two actions. Even an arbitrarily small perturbation, such as assigning a slightly higher probability to one action, breaks this indifference and makes one action strictly preferable. In this sense, the mixed-strategy equilibrium is comparatively fragile, while the pure equilibria are robust.

Refinement concepts formalize this intuition by strengthening the equilibrium requirement to account for small perturbations. One of the most influential such refinements is *trembling-hand perfection*. The basic idea is that players may occasionally make mistakes (or *trembles*), so that every pure action is played with a small but positive probability. A strategy profile is said to be trembling-hand perfect if it can be obtained as the limit of a sequence of Nash equilibria of perturbed games in which all players are required to use fully mixed strategies. Equivalently, a trembling-hand perfect equilibrium is a Nash equilibrium that remains a best response when opponents' strategies are subject to arbitrarily small perturbations.

Operationally, the refinement is carried out by considering sequences of strategy profiles in which players mostly follow the equilibrium strategies but occasionally deviate with vanishingly small probability. An equilibrium is deemed behaviorally credible only if no player has an incentive to deviate even in the presence of such trembles. In the Battle of the Sexes game, the pure equilibria (Opera, Opera) and (Football, Football) satisfy this requirement, since they are strict Nash equilibria. The mixed-strategy equilibrium does not: its reliance on exact indifference causes it to break down under arbitrarily small perturbations. From the perspective of refinement theory, only the strict pure equilibria emerge as robust predictions of play.

A detailed treatment of trembling-hand perfection, along with other equilibrium refinements and their relationships to dynamic and learning-based models of play, will be discussed in Lecture 7.

4.2 ■ Special Classes of Games

In this section, we examine two fundamental classes of games, *potential games* and *supermodular games*, which possess rich mathematical structures that guarantee desirable equilibrium properties, including convergence of learning dynamics and the existence of pure-strategy Nash equilibria.

4.2.1 ■ Potential Games

Potential games are characterized by the existence of a global function that captures the local incentives of all players [78]. In Lecture 3, we introduced the notion of *strategic equivalence*, where bi-matrix games can be transformed into canonical forms such as zero-sum or team games. Potential games constitute an important class within this broader framework, because the existence of a potential function effectively aligns players' marginal incentives with a single global objective.

A particularly important subclass of potential games consists of games that are *strategically*

equivalent to a team game: these are games in which all players' incentives can be made perfectly aligned via suitable positive affine transformations of their payoff functions. Every finite game that is strategically equivalent to a team game admits a pure-strategy Nash equilibrium, since all players share (up to affine transformations) a common objective, and this team objective attains a maximum on the finite strategy space.

Definition 4.6 (Exact Potential Game). Let $\mathcal{G} = (\mathcal{N}, (A_i)_{i \in \mathcal{N}}, (u_i)_{i \in \mathcal{N}})$ be a finite normal-form game. It is an exact potential game if there exists a function $\Phi : \prod_{i \in \mathcal{N}} A_i \rightarrow \mathbb{R}$ such that for every player $i \in \mathcal{N}$, for all $a_{-i} \in A_{-i}$, and for all $a_i, a'_i \in A_i$,

$$u_i(a_i, a_{-i}) - u_i(a'_i, a_{-i}) = \Phi(a_i, a_{-i}) - \Phi(a'_i, a_{-i}).$$

The function Φ is called a *potential function* for the game. It records the marginal change in players' incentives induced by any unilateral deviation. In an exact potential game, every player's payoff change from a unilateral deviation coincides with the corresponding change in the potential function.

Proposition 4.7. Every finite exact potential game admits at least one pure-strategy Nash equilibrium.

Proof. Since the strategy space $\prod_{i \in \mathcal{N}} A_i$ is finite, the potential function Φ attains a maximum. Any maximizer of Φ is a pure-strategy Nash equilibrium, because a unilateral deviation by any player would increase her utility if, and only if, it increased Φ , which contradicts maximality. Hence no player has a profitable deviation at such a maximizer. $\square$

Moreover, in exact potential games, *better-response dynamics*, in which at each iteration a player chooses a strictly payoff-improving deviation, converge to a pure-strategy Nash equilibrium. This follows from the fact that each better response increases Φ strictly, and hence the dynamics cannot cycle.

Example 4.8 Congestion Game as an Exact Potential Game.

Consider a two-player congestion game with facilities F_1 and F_2 . If n_f players use facility $f \in \{1, 2\}$, then a user of f receives utility $v_f(n_f)$, where utilities decrease with congestion:

$$v_1(1) = 4, \quad v_1(2) = 3, \quad v_2(1) = 2, \quad v_2(2) = 1.$$

The induced bi-matrix is

	P2	
	F_1	F_2
P1 F_1	(3, 3)	(4, 2)
F_2	(2, 4)	(1, 1)

A standard Rosenthal potential for congestion games is

$$\Phi(a_1, a_2) = \sum_{f \in \{1, 2\}} \sum_{k=1}^{n_f(a_1, a_2)} v_f(k),$$

where $n_f(a_1, a_2)$ is the number of players choosing facility f . Evaluating Φ on the four action profiles gives:

$$\Phi(F_1, F_1) = 4 + 3 = 7,$$

$$\Phi(F_2, F_2) = 2 + 1 = 3,$$

$$\Phi(F_1, F_2) = 4 + 2 = 6,$$

$$\Phi(F_2, F_1) = 2 + 4 = 6.$$

Thus Φ is uniquely maximized at (F_1, F_1) .

As discussed earlier in the context of strategic equivalence (also in Lecture 3), adding positive constants to individual players' payoffs does not affect best responses or Nash equilibria. By adding 2 to the first column of P1's payoff matrix and 2 to the first row of P2's payoff matrix, the game becomes strategically equivalent to a team game with the single payoff matrix

$$T = \begin{pmatrix} 5 & 4 \\ 4 & 1 \end{pmatrix}.$$

The unique maximizer of T is the entry 5, corresponding to the joint action (F_1, F_1) . Since affine transformations preserve best responses, this team-optimal solution is also the unique Nash equilibrium of the original congestion game. (Any positive scaling or uniform shifting of T would yield the same conclusion.)

Equilibria and better-response dynamics. The potential function is strictly increasing along any better-response path, and the strategy space is finite, and thus every sequence of better responses converges in finitely many steps. Because both the potential Φ and the strategically equivalent team function T attain their unique maximum at (F_1, F_1) , the game has a *unique* pure-strategy Nash equilibrium: (F_1, F_1) . Indeed, each player strictly prefers F_1 to F_2 regardless of the opponent's choice, and thus the equilibrium is also a dominant-strategy equilibrium. $\square$

We can generalize the two-player congestion game into the framework of *atomic congestion games*. Let $\mathcal{N} = \{1, \dots, N\}$ be a finite set of players, and let $\mathcal{F}$ denote a finite set of facilities. Each player $i \in \mathcal{N}$ selects an action a_i from a nonempty action set $A_i \subseteq 2^{\mathcal{F}}$. In particular, for *singleton congestion games*, one has $A_i \subseteq \mathcal{F}$, so that each player chooses exactly one facility. Given an action profile $a = (a_1, \dots, a_N) \in \prod_{i \in \mathcal{N}} A_i$, define the *load* on facility $f \in \mathcal{F}$ as

$$\ell_f(a) := |\{i \in \mathcal{N} : f \in a_i\}|,$$

i.e., the number of players making use of facility f under profile a . Each facility f is associated with a non-decreasing (discrete) *latency function* $c_f : \{0, 1, \dots, N\} \rightarrow \mathbb{R}_{\geq 0}$, which specifies the cost per player as a function of congestion. The total cost incurred by player i under profile a is

$$C_i(a) = \sum_{f \in a_i} c_f(\ell_f(a)).$$

Players are cost-minimizers, and thus utilities are taken as $u_i(a) = -C_i(a)$.

Define the function $\Phi : \prod_{i \in \mathcal{N}} A_i \rightarrow \mathbb{R}$ by

$$\Phi(a) = \sum_{f \in \mathcal{F}} \sum_{k=1}^{\ell_f(a)} c_f(k).$$

This function is an *exact potential* in the sense of Rosenthal (1973) [102]: for any player $i \in \mathcal{N}$ and any unilateral deviation from a_i to a'_i ,

$$\Phi(a'_i, a_{-i}) - \Phi(a_i, a_{-i}) = C_i(a'_i, a_{-i}) - C_i(a_i, a_{-i}).$$

Hence changes in an individual player's cost are exactly reflected by changes in the potential.

Since the action space $\prod_{i \in \mathcal{N}} A_i$ is finite, the potential function Φ attains a minimum. Therefore, at least one pure-strategy Nash equilibrium exists, corresponding to a local (indeed global) minimum of Φ . Moreover, any sequence of unilateral better-responses strictly decreases Φ ; because Φ is bounded below, such sequences must terminate in finitely many steps. This property, known as the *finite improvement property*, guarantees convergence to a pure-strategy Nash equilibrium under better-response dynamics.

The notion of potential games extends beyond exact potential games. In particular, a game may admit an *ordinal potential* or a *weighted potential* function that preserves only the *direction* (not the magnitude) of unilateral improvements. That is, while exact potential games require equality between changes in utilities and changes in potential values, ordinal and weighted variants only require that these changes share the same sign or are scaled by constant weights. Such relaxations still ensure the existence of at least one pure-strategy Nash equilibrium and capture a broader class of coordination and congestion phenomena [78].

Remark 4.9. Games that are strategically equivalent to team games form a *strict subclass* of exact potential games. To see this, suppose that a game admits a representation

$$u_i(a) = W(a) + f_i(a_{-i}), \quad i \in \mathcal{N},$$

for some function $W : \prod_{i \in \mathcal{N}} A_i \rightarrow \mathbb{R}$ and functions $f_i : A_{-i} \rightarrow \mathbb{R}$. Since the terms f_i do not depend on player i 's own action, they are strategically irrelevant and do not affect best responses. Consequently, for any unilateral deviation by player i , the change in that player's payoff coincides exactly with the change in the common function W . It follows that such games are exact potential games, with potential function $\Phi = W$ (up to an additive constant).

The inclusion is strict. Exact potential games require only that players' *marginal incentives* be aligned through a potential function; they do not require players' payoffs to be globally aligned with a single objective. As a result, there exist exact potential games that are not strategically equivalent to team games. For example, congestion games with player-specific cost functions admit exact (Rosenthal) potentials but cannot, in general, be transformed into team games. See [78] for explicit constructions and a general characterization.

4.2.2 ■ Supermodular Games

Supermodular games constitute an important class of games in which players' strategies act as *strategic complements* [135]. This means that when one player increases the level of her action, it makes other players more willing to increase theirs. Such positive feedback structures are common in economics, networks, and technology adoption, and they guarantee not only that pure-strategy Nash equilibria exist, but also that these equilibria can be found through simple iterative processes, as further discussed below.

Background: Lattices, Sublattices, and Increasing Differences. A *lattice* is a partially ordered set $(X, \leq)$ in which every pair of elements $x, y \in X$ has a greatest lower bound (the *meet*, written $x \wedge y$) and a least upper bound (the *join*, written $x \vee y$). For totally ordered sets such as subsets of $\mathbb{R}$ with the usual order, the meet and join of $\{x, y\}$ are simply $\min\{x, y\}$ and $\max\{x, y\}$, respectively. Thus, sets like $\{0, 1\}$ and $\{0, 1, 2\}$ are lattices because the min and max of any two elements always belong to the set. Even $\{0, 2\}$ is a lattice in this sense: the meet of 0 and 2 is 0, their join is 2, and both are in the set. The presence or absence of “in-between” values is irrelevant for the lattice property itself.

A *sublattice* is simply a subset of a lattice that is itself a lattice under the same order. This means that for any x, y in the subset, both $x \wedge y$ and $x \vee y$ also belong to the subset. For example, $\{0, 2\}$ is a sublattice of $\{0, 1, 2\}$, since the min and max of any two of its elements remain inside $\{0, 2\}$. By contrast, the restricted profile set $S = \{(0, 1), (1, 0)\} \subset \{0, 1\}^2$ is *not* a sublattice: the meet of $(0, 1)$ and $(1, 0)$ is $(0, 0)$ and the join is $(1, 1)$, but neither of these is contained in S .

When there are multiple players, we order action *profiles* componentwise (the *product order*):

$$(a_1, \dots, a_N) \leq (a'_1, \dots, a'_N) \iff a_i \leq a'_i \text{ for every } i.$$

If each A_i is a lattice, then the product $A = \prod_i A_i$ is also a lattice under this product order, with

$$(a \wedge a')_i = \min\{a_i, a'_i\}, \quad (a \vee a')_i = \max\{a_i, a'_i\}.$$

In our running examples, $A_i = \{0, 1\}$ (binary coordination) and $A_i = \{0, 1, 2\}$ (three-level effort) are lattices, and thus the profile sets $\{0, 1\}^2$ and $\{0, 1, 2\}^2$ are lattices under the product order. The lattice structure ensures that equilibria can be compared component by component.

A payoff $u_i(a_i, a_{-i})$ has *increasing differences* if the marginal gain from raising a_i is (weakly) greater when a_{-i} is greater. In continuous variables, this is often checked by a nonnegative cross-partial $\frac{\partial^2 u_i}{\partial a_i \partial a_j} \geq 0$; in discrete settings (as in our examples), the equivalent check is non-negativity of the cross-difference

$$\Delta_i \Delta_j u_i(a_i, a_j) = u_i(a_i+1, a_j+1) - u_i(a_i+1, a_j) - u_i(a_i, a_j+1) + u_i(a_i, a_j) \geq 0,$$

whenever the four terms are defined. For the three-level effort payoff $u_i(a_i, a_j) = 2a_i + a_i a_j - a_i^2$, we have, on the grid $\{0, 1, 2\}^2$, $\Delta_i \Delta_j u_i \equiv 1 \geq 0$, and thus increasing differences hold.

The second key concept is *increasing differences*. A payoff function $u_i(a_i, a_{-i})$ has increasing differences if the marginal benefit of raising a_i is larger when the opponents' strategies a_{-i} are higher. Formally, for all $a'_i > a_i$ and $a'_{-i} > a_{-i}$,

$$u_i(a'_i, a'_{-i}) - u_i(a_i, a'_{-i}) \geq u_i(a'_i, a_{-i}) - u_i(a_i, a_{-i}).$$

This condition captures strategic complementarities.

Definition 4.10 (Supermodular Game). A finite game $\mathcal{G} = (\mathcal{N}, (A_i)_{i \in \mathcal{N}}, (u_i)_{i \in \mathcal{N}})$ is called a *supermodular game* if the following conditions hold:

- (1) Each action set A_i is a nonempty finite sublattice of $\mathbb{R}$, i.e. for any $a_i, a'_i \in A_i$, both $\min\{a_i, a'_i\}$ and $\max\{a_i, a'_i\}$ are also in A_i .
- (2) Each payoff function $u_i : A \rightarrow \mathbb{R}$ has increasing differences in (a_i, a_{-i}) : for all $a'_i > a_i$ and $a'_{-i} > a_{-i}$,

$$u_i(a'_i, a'_{-i}) - u_i(a_i, a'_{-i}) \geq u_i(a'_i, a_{-i}) - u_i(a_i, a_{-i}).$$

The increasing differences condition captures strategic complementarities: the incentive for a player to raise her action is stronger when the other players are also choosing higher actions.

Example 4.11 Three-Level Joint-Effort Game.

Two firms choose effort levels $a_i \in \{0, 1, 2\}$ (low, medium, high). Player i 's payoff is

$$u_i(a_i, a_j) = 2a_i + a_i a_j - a_i^2,$$

which combines a direct benefit $2a_i$, a positive interaction $a_i a_j$ (strategic complementarity), and a convex cost a_i^2 . The action set $A_i = \{0, 1, 2\}$ is a finite sublattice of $\mathbb{R}$ (closed under min and max), which will be one of the standing assumptions in the supermodular-games theorem stated later (see Theorem 4.12).

For P1 (rows) and P2 (columns), the payoff table is shown in Table 4.1.

Table 4.1: P1's payoff table in the three-level joint-effort game.

	$a_2 = 0$	$a_2 = 1$	$a_2 = 2$
$a_1 = 0$	0	0	0
$a_1 = 1$	1	2	3
$a_1 = 2$	0	2	4

Best replies are obtained by identifying the largest entries in each column:

$$\text{BR}_1(0) = \{1\}, \quad \text{BR}_1(1) = \{1, 2\}, \quad \text{BR}_1(2) = \{2\}.$$

By symmetry, P2 has the same correspondence.

On a discrete lattice, increasing differences are verified by the cross-difference operator:

$$\Delta_i \Delta_j u_i(a_i, a_j) := u_i(a_i+1, a_j+1) - u_i(a_i+1, a_j) - u_i(a_i, a_j+1) + u_i(a_i, a_j).$$

For all $(a_i, a_j) \in \{0, 1\}^2$, direct calculation yields $\Delta_i \Delta_j u_i(a_i, a_j) = 1 \geq 0$. Thus u_i has increasing differences, and the game is supermodular.

Supermodularity implies that best replies move (weakly) upward as the opponent's action increases. Here this is explicit: as a_j increases $0 \rightarrow 1 \rightarrow 2$, the minimal best reply moves $1 \rightarrow 1 \rightarrow 2$, and the maximal best reply moves $1 \rightarrow 2 \rightarrow 2$. Both selections are nondecreasing, and thus BR_i is isotone in the strong set order. These properties are exactly those required by the theorem stated later (see Theorem 4.12), which guarantees the existence of least and greatest Nash equilibria.

A profile is a Nash equilibrium if each action is a best reply to the other's. From Table 4.1,

$$(1, 1) \text{ is an equilibrium,} \quad (2, 2) \text{ is an equilibrium.}$$

No other equilibrium exists: if $a_j = 0$, the best reply is $a_i = 1$, ruling out $(0, 0)$ and all profiles containing a 0. In the product order, $(1, 1) \leq (2, 2)$, and thus $(1, 1)$ is the *least* equilibrium and $(2, 2)$ the *greatest* equilibrium. This is exactly the lattice structure predicted by Theorem 4.12.

Monotone best-response dynamics. This example also illustrates the monotone dynamics guaranteed by the theorem. Starting from the bottom element $(0, 0)$, each player's best reply is 1, giving $(0, 0) \mapsto (1, 1)$; if minimal best replies are used at $a_j = 1$, the process stabilizes at $(1, 1)$. Conversely, starting from the top $(2, 2)$ keeps the profile fixed, and if maximal best replies are used at $a_j = 1$, the process moves $(1, 1) \mapsto (2, 2)$ in one step. Thus minimal (resp. maximal) selections converge to the least (resp. greatest) equilibrium, which is exactly the behavior described by Theorem 4.12 for finite supermodular games in the following. $\square$

Theorem 4.12 (Tarski's Fixed Point Theorem for Supermodular Games [135]). *Let $\mathcal{G} = (\mathcal{N}, (A_i)_{i \in \mathcal{N}}, (u_i)_{i \in \mathcal{N}})$ be a finite-player normal-form game satisfying the following conditions:*

- (1) *Each action set A_i is a nonempty finite sublattice of $\mathbb{Z}$.*
- (2) *For each player i , the payoff function u_i has increasing differences in (a_i, a_{-i}) .*

Then, the following statements hold:

- (i) *For each player i , the best-response correspondence $\text{BR}_i(a_{-i})$ is nonempty and isotone (nondecreasing) in a_{-i} .*
- (ii) *The set of pure-strategy Nash equilibria is nonempty and forms a complete lattice under the product order.*
- (iii) *In particular, there exist a least equilibrium $a^{\min}$ and a greatest equilibrium $a^{\max}$ such that for every Nash equilibrium a^* ,*

$$a^{\min} \leq a^* \leq a^{\max}.$$

This theorem shows that equilibria in supermodular games are ordered: there is always a bottom equilibrium, a top equilibrium, and possibly others in between. In the *binary coordination game*, each player chooses 0 (not adopt) or 1 (adopt). The two equilibria are $(0, 0)$ and $(1, 1)$. Since $(0, 0) \leq (1, 1)$ componentwise, the first is the least equilibrium and the second is the greatest. Intuitively, one equilibrium corresponds to no adoption, the other to full adoption.

In the *three-level effort game*, each player chooses $a_i \in \{0, 1, 2\}$ with payoff $u_i(a_i, a_j) = 2a_i + a_i a_j - a_i^2$. The payoff function has increasing differences since $\Delta_i \Delta_j u_i(a_i, a_j) = 1 > 0$ everywhere, and thus the game is supermodular. Best responses show two equilibria: $(1, 1)$ and $(2, 2)$. Here, $(1, 1) \leq (2, 2)$, and thus $(1, 1)$ is the least equilibrium and $(2, 2)$ the greatest. If an additional equilibrium such as $(1, 2)$ had existed, it would necessarily have lain between them: $(1, 1) \leq (1, 2) \leq (2, 2)$. This is what it means for the set of equilibria to form a lattice.

The order structure is also reflected in adaptive dynamics. In the binary game, starting from $(0, 0)$, best responding keeps the system at the least equilibrium; starting from $(1, 1)$ keeps it at the greatest. In the three-level game, starting from $(0, 0)$ pushes players upward until they reach $(1, 1)$, while starting from $(2, 2)$ keeps them at the high-effort equilibrium. Thus, monotone best-response dynamics converge to the least or greatest equilibrium depending on the starting point. Other adaptive processes, such as fictitious play, introduced earlier in Lecture 2 and studied further in Lecture 30, also exhibit convergence properties in supermodular games under additional assumptions, though their behavior is generally more subtle than that of monotone best replies.

4.3 ■ From Dominance to Rationalizability to Nash Equilibrium: Motivation and Interpretation

The purpose of this section is to *motivate* and *interpret* the concept of Nash equilibrium by placing it in the broader hierarchy of solution concepts for strategic-form games. We begin with the most restrictive notion of rational behavior, dominant strategies, then move to the broader concept of rationalizability, and finally arrive at Nash equilibrium, which refines rationalizability by imposing belief consistency.

4.3.1 ■ Dominant Strategies

In the Prisoner's Dilemma, we have seen that (D, D) arises because *Defect* is a strictly dominant strategy for both players. Regardless of whether the opponent cooperates or defects, defection yields a strictly higher payoff. Thus, rational players will each defect, and the unique outcome is (D, D) . More formally, a pure strategy $a_i \in A_i$ for player i is a *strictly dominant strategy* if it yields a strictly higher payoff than any other strategy, for every possible profile of the opponents:

$$u_i(a_i, a_{-i}) > u_i(a'_i, a_{-i}), \quad \forall a'_i \neq a_i, \forall a_{-i} \in A_{-i}.$$

If each player has a strictly dominant strategy, the game has a unique and compelling outcome: each player plays her dominant strategy. The *iterated elimination of strictly dominated strategies* is a natural extension that can simplify the analysis of many games. However, many games do not have strictly dominant strategies, and in such cases we must turn to weaker concepts.

4.3.2 ■ Rationalizability: A Precursor to Nash

Rationalizability generalizes the idea of eliminating unreasonable strategies without requiring the strong dominance condition. It begins with the premise of *common knowledge of rationality*: all players are rational, everyone knows they are rational, everyone knows that everyone knows they are rational, and so on.

Formally, let A_i be the action set of player i and $A_{-i} = \prod_{j \neq i} A_j$. A pure strategy $a_i \in A_i$ is *rationalizable* if there exists a belief $\mu_i \in \Delta(A_{-i})$ such that:

- (i) (*Best response*) $a_i \in \arg \max_{a'_i \in A_i} u_i(a'_i, \mu_i)$.
- (ii) (*Belief restriction*) μ_i assigns probability only to strategies of the opponents that are themselves rationalizable.

The set of rationalizable strategies is obtained through *iterated elimination of never-best responses*:

- (i) Eliminate any strategy that is never a best response to any belief over the opponents' strategies.
- (ii) Restrict beliefs to the surviving strategies and repeat the elimination.
- (iii) Stop when no further eliminations are possible.

Since rationalizability requires only that each strategy be optimal for *some* belief consistent with rationality, the resulting set can be large and may include profiles whose supporting beliefs would be *incorrect* if the game were actually played.

Example 4.13

Consider the following bi-matrix game, where P1 chooses rows $A_1 = \{U, M, D\}$ and P2 chooses columns $A_2 = \{L, C, R\}$:

P1	P2		
	L	C	R
U	(1, 1)	(0, 0)	(0, 0)
M	(0, 0)	(-1, 0)	(0, 0)
D	(0, 0)	(-2, 0)	(1, 1)

We determine the set of rationalizable strategies via iterated elimination of never-best responses. For P1, strategy M is never a best response. Against L and C , strategy U yields a weakly higher payoff, and against R , strategy D yields a strictly higher payoff. Hence M is eliminated. For P2, strategy C is never a best response. Against every action of P1, either L or R yields a weakly higher payoff, and sometimes a strictly higher payoff. Hence C is eliminated.

After this first round, the surviving strategy sets are $R_1^{(1)} = \{U, D\}$, $R_2^{(1)} = \{L, R\}$. Restricting attention to the reduced game on $R_1^{(1)} \times R_2^{(1)}$, each remaining strategy is a best response to some belief over the opponent's surviving strategies:

- U is a best response to beliefs placing sufficiently high probability on L ;
- D is a best response to beliefs placing sufficiently high probability on R ;
- L is a best response to beliefs favoring U ;
- R is a best response to beliefs favoring D .

No further eliminations are therefore possible.

The final set of rationalizable strategies is $R_1 = \{U, D\}$, $R_2 = \{L, R\}$, and the set of rationalizable strategy profiles is $R_1 \times R_2$. Profiles such as (U, R) or (D, L) are rationalizable even though they are not Nash equilibria. The action of each player can be justified as optimal for some belief about the opponent that is consistent with common knowledge of rationality, even if those beliefs would be incorrect if the game were actually played.

This example illustrates the key distinction between rationalizability and Nash equilibrium. Rationalizability requires only that each strategy be optimal for *some* belief over rational opponent behavior. Nash equilibrium adds the additional requirement of *mutual correctness of beliefs*: players' beliefs must be confirmed by the strategies actually played. $\square$

4.3.3 ■ Nash Equilibrium: Adding Belief Consistency

The *Nash equilibrium* refines rationalizability by requiring that the beliefs justifying each player's strategy be *correct* in equilibrium. Formally, a strategy profile $a^* = (a_1^*, \dots, a_N^*)$ is a Nash equilibrium if, for each player i :

- (1) (*Best response condition*) $a_i^* \in \arg \max_{a_i \in A_i} u_i(a_i, a_{-i}^*)$.

(2) (*Belief consistency*) The belief supporting a_i^* assigns probability one to a_{-i}^* .

Example 4.14

In the 3×3 example above (Example 4.13), the rationalizable set $\{U, D\} \times \{L, R\}$ is reduced by the Nash conditions:

- Pure-Strategy Nash Equilibrium: (U, L) and (D, R) ; each is a mutual best response with correct beliefs.
- Mixed-Strategy Nash Equilibrium: P1 mixes $(\frac{1}{2}U, \frac{1}{2}D)$, P2 mixes $(\frac{1}{2}L, \frac{1}{2}R)$. This makes each player indifferent between her two strategies in the reduced game.

Outcomes like (U, R) are excluded because they require P1 to believe P2 will play L with high probability, which contradicts the actual play in that profile. $\square$

Every (pure or mixed) Nash equilibrium strategy is rationalizable, and every dominant strategy is part of some Nash equilibrium. Thus, we have the inclusions:

$$\{\text{Dominant Strategies}\} \subseteq \{\text{Nash Equilibria}\} \subseteq \{\text{Rationalizable Strategies}\}.$$

Dominant strategies are the most restrictive, rationalizability is more permissive, and Nash equilibrium lies between them in terms of predictive power, while ensuring self-consistent expectations.

4.3.4 ■ Interpretations of Nash Equilibrium

Based on the prior discussions in this lecture, the Nash equilibrium concept admits several complementary interpretations. Each highlights a different aspect of strategic stability. These perspectives provide a richer understanding of why Nash equilibrium is a central solution concept.

Mutual Best Response (Static Optimization View). A mixed-strategy Nash equilibrium is a fixed point of the best-response correspondence: $a_i^* \in \text{BR}_i(a_{-i}^*)$, $\forall i \in \mathcal{N}$. The strategy of each player is the optimal reaction to the strategies actually chosen by the others. From this perspective, the NE condition is simply a set of simultaneous optimality conditions, one for each player. This is the most immediate and widely used definition in static game theory: no player can improve her payoff by unilaterally deviating, given the strategies of the others.

Self-Enforcing Agreement. A mixed-strategy Nash equilibrium (NE) may be interpreted as an implicit *self-enforcing agreement*. Once the players adopt the equilibrium mixed strategies, no one can gain by changing their behavior unilaterally. Thus the equilibrium profile is stable without any external enforcement mechanism: the incentives of each player, taken together, sustain the outcome from within. This interpretation is especially natural in situations where equilibrium behavior emerges after experimentation or adjustment, so that players arrive at a point from which no further improvement is possible through unilateral changes.

A complementary way to view a mixed-strategy NE is as a *behavioral rule* that each player can follow independently. In this interpretation, no external mediator is involved; instead, each player privately randomizes according to her own equilibrium distribution. Because the mixed-strategy profile is a Nash equilibrium, each player's expected payoff from following her prescribed randomization is at least as high as any payoff she could obtain by deviating to a different

distribution. In this sense, the equilibrium mixed strategy serves as a personal recommendation that is inherently credible: each player finds it optimal to adhere to the rule, given that all others are doing the same.

Correct Beliefs and Consistent Play (Epistemic View). From the standpoint of epistemic game theory, a mixed-strategy Nash equilibrium corresponds to a situation in which: (i) the players are rational: each one chooses the best response to their beliefs about others; (ii) the players' beliefs are *correct*: each player's belief assigns probability 1 to the strategies that others actually choose. This formalizes the *belief consistency* idea discussed earlier: in equilibrium, expectations match reality. NE can therefore be interpreted as the result of mutual rationality combined with common knowledge of strategies.

Steady State of Learning or Adaptation (Dynamic/Repeated Play View). In many adaptive processes, such as fictitious play [21], no-regret learning [52] (to be discussed in Lecture 8), or certain evolutionary dynamics [55] (to be discussed in Lectures 28 and 29), a mixed-strategy Nash equilibrium corresponds to a steady state in which no player has an incentive to change her strategy, given the empirical distributions of other players' plays. In repeated or evolutionary settings, a mixed-strategy NE can be seen as a population state resistant to invasion: if the game is played once, no mutant strategy can achieve a higher payoff against the incumbent population. This dynamic perspective connects NE to long-run outcomes of decentralized adjustment processes, a theme developed further in Lecture 30.

4.4 ■ Further Notes and Reading

This lecture has discussed several special classes of static finite games, including potential games and supermodular games, for which structural properties enable insight into equilibrium existence, efficiency, and computation. Another important class of static games is formed by *Stackelberg games*, which model hierarchical decision making between leaders and followers. From a modeling perspective, Stackelberg games can be formulated as static games, since players choose actions from fixed strategy sets, and payoffs are defined over action profiles. What distinguishes Stackelberg games is not their static description, but the equilibrium interpretation, which presumes commitment by the leader, observability by the follower, and sequential rationality of responses. These features are most naturally captured using the extensive-form framework and backward induction. Readers may refer to Lecture 5 for a formal development of Stackelberg equilibrium and its connection to subgame-perfect equilibrium.

In addition to the computational methods discussed in this lecture, we can also leverage fictitious play for nonzero-sum games, as introduced in Lecture 2 for zero-sum games. However, for nonzero-sum games, convergence of fictitious play is not always guaranteed. A classical counterexample is Shapley's (1964) rock-paper-scissors game [110], where fictitious play cycles indefinitely and fails to converge. To address this, regularization techniques are often required to ensure convergence (see, e.g., [39, 79]). Beyond fictitious play, there are other important learning methods, which will be covered in the later lectures on evolutionary dynamics and learning in games, in particular Lecture 30.

For a broader perspective on the computational complexity of mixed-strategy Nash equilibria, the survey by Daskalakis, Goldberg, and Papadimitriou [28] provides foundational results on PPAD-completeness of equilibrium computation. Evolutionary perspectives are synthesized in Weibull [143], which highlights the stability of Nash equilibria as rest points of population dynamics. For algorithmic viewpoints, Nisan et al.'s handbook [86] surveys methods for analyzing structured games, including network games and auction-theoretic models.

The interpretations of mixed-strategy Nash equilibrium presented in Section 3.4, as mutual best response, self-enforcing agreement, epistemic consistency, and steady states of adaptive processes, have been extensively developed across game theory, economics, and evolutionary biology. For a foundational account, the original work by Nash [60] introduces the fixed-point interpretation that remains the bedrock of modern game theory. The epistemic interpretation, emphasizing rationality and common knowledge of beliefs, has been systematically studied in Aumann [6] and further elaborated in Brandenburger's interactive epistemology [19].

The view of Nash equilibrium as a self-enforcing agreement is captured in the broader literature on bargaining and conflict, particularly in Binmore's text [14] and Myerson's comprehensive treatment of game theory [84]. The interpretation of equilibrium as a steady state of learning or evolutionary dynamics has been developed in Brown's original formulation of fictitious play [21], in the learning framework of Fudenberg and Levine [39], and in the evolutionary perspective of Hofbauer and Sigmund [55].

Finally, the recognition that some equilibria may lack robustness motivates refinements of the concept. Selten's notion of trembling-hand perfection [107] provides a sharper characterization by requiring equilibria to be stable under small perturbations. Together, these works provide complementary perspectives on why Nash equilibrium has become a central and enduring solution concept in game theory.

4.5 - Exercises

Exercise 4.1 (Pure-Strategy Nash Equilibria in a Bi-matrix Game). Consider the following bi-matrix game, in which both players are minimizers. P1 chooses one of the five rows and P2 chooses one of the three columns, and each cell lists the cost pair (J_1, J_2) incurred by P1 and P2, respectively:

P1	P2		
	Col 1	Col 2	Col 3
Row 1	(8, 0)	(−4, −4)	(8, −4)
Row 2	(6, −4)	(1, −4)	(6, 0)
Row 3	(12, −4)	(0, 0)	(0, −4)
Row 4	(0, −4)	(0, 0)	(12, −4)
Row 5	(0, −4)	(4, −3)	(0, −4)

- (i) Compute the pure-strategy best-response correspondence of each player: for every column, list the cost-minimizing rows for P1, and for every row, list the cost-minimizing columns for P2.
- (ii) Determine all pure-strategy Nash equilibria of the game, and verify for each equilibrium that neither player can strictly reduce their cost by a unilateral deviation.

Exercise 4.2 (Pure- and Mixed-Strategy Equilibria in a 2×2 Game). Consider the following bi-matrix game, in which both players are minimizers and each cell lists the cost pair (J_1, J_2) :

P1	P2	
	Col 1	Col 2
Row 1	(5, −2)	(−7, 5)
Row 2	(−6, 4)	(−4, 2)

- (i) Show that the game admits no pure-strategy Nash equilibrium.
- (ii) Compute all mixed-strategy Nash equilibria by applying the indifference principle: determine the probability with which P1 chooses Row 1 and the probability with which P2 chooses Col 1, and state the resulting expected cost of each player at equilibrium.

Exercise 4.3 (Equilibria under Sign Reversal of the Payoff Matrices). Consider the bi-matrix game in which both players are minimizers, P1 chooses rows, P2 chooses columns, and the costs are

P1	P2	
	Col 1	Col 2
Row 1	(10, -5)	(-2, 2)
Row 2	(-1, 1)	(1, -1)

corresponding to the cost matrices

$$A = \begin{pmatrix} 10 & -2 \\ -1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} -5 & 2 \\ 1 & -1 \end{pmatrix}.$$

Let p denote the probability with which P1 chooses Row 1 and q the probability with which P2 chooses Col 1.

- (i) Compute the mixed-strategy Nash equilibrium (p^*, q^*) of the game (A, B) by applying the indifference principle.
- (ii) Compute the mixed-strategy Nash equilibrium of the game with cost matrices $(-A, -B)$, where both players are again minimizers.
- (iii) Determine whether the equilibria obtained in (i) and (ii) would change if the players were instead maximizers of the corresponding entries, and justify your answer by relating the maximization problems to the minimization problems above.
- (iv) Implement, in Python or MATLAB, a numerical verification of the equilibrium computed in (i). For each pair (p, q) on the grid $\{0, 0.01, 0.02, \dots, 1\}^2$, evaluate the exploitability

$$\varphi(p, q) = \left[J_1(p, q) - \min_{p' \in [0,1]} J_1(p', q) \right] + \left[J_2(p, q) - \min_{q' \in [0,1]} J_2(p, q') \right],$$

where $J_1(p, q) = y^\top A z$ and $J_2(p, q) = y^\top B z$ with $y = (p, 1-p)^\top$ and $z = (q, 1-q)^\top$. Report the grid point that minimizes φ , and confirm that it lies within 0.01 of (p^*, q^*) in each coordinate.

Exercise 4.4 (Equilibria of a Three-Player Game). Consider a three-player nonzero-sum game in which each player has two actions, labeled 1 and 2, and all players are minimizers. The triple $(a_{j,k,\ell}^1, a_{j,k,\ell}^2, a_{j,k,\ell}^3)$ denotes the costs incurred by P1, P2, and P3, respectively, when P1 chooses action j , P2 chooses action k , and P3 chooses action ℓ . The outcomes are

$$\begin{aligned} (a_{1,1,1}^1, a_{1,1,1}^2, a_{1,1,1}^3) &= (0, 0, 1), \\ (a_{1,2,1}^1, a_{1,2,1}^2, a_{1,2,1}^3) &= (-2, 1, 2), \\ (a_{2,1,1}^1, a_{2,1,1}^2, a_{2,1,1}^3) &= (0, -3, 1), \\ (a_{2,2,1}^1, a_{2,2,1}^2, a_{2,2,1}^3) &= (-1, 1, -2), \end{aligned}$$

$$(a_{1,1,2}^1, a_{1,1,2}^2, a_{1,1,2}^3) = (-2, 1, 1),$$

$$(a_{1,2,2}^1, a_{1,2,2}^2, a_{1,2,2}^3) = (-2, 0, 1),$$

$$(a_{2,1,2}^1, a_{2,1,2}^2, a_{2,1,2}^3) = (-1, -2, 2),$$

$$(a_{2,2,2}^1, a_{2,2,2}^2, a_{2,2,2}^3) = (1, 0, 0).$$

- (i) Formulate the game in strategic form by writing, for each action $\ell \in \{1, 2\}$ of P3, the induced 2×2 cost bi-matrix faced by P1 and P2.
- (ii) Determine all pure-strategy Nash equilibria of the game, and verify for each equilibrium that no player can strictly reduce her cost by a unilateral deviation.

Exercise 4.5 (Nonconvergence of Fictitious Play in the Shapley Game). Consider the bi-matrix game (A, B) in which both players are maximizers, P1 chooses rows, P2 chooses columns, and the payoff pairs (u_1, u_2) are

P1	P2		
	Col 1	Col 2	Col 3
Row 1	(2, 1)	(0, 0)	(1, 2)
Row 2	(1, 2)	(2, 1)	(0, 0)
Row 3	(0, 0)	(1, 2)	(2, 1)

This game, due to Shapley, has the uniform profile $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ for each player as its unique Nash equilibrium. Both players update their actions according to fictitious play, as introduced in Lecture 2, each choosing at every round a best response to the empirical frequency of the opponent's past play.

- (i) Show that fictitious play initialized at the action profile $(1, 1)$ generates a run of $(1, 1)$, followed by a run of $(1, 3)$, then runs of $(3, 3)$, $(3, 2)$, $(2, 2)$, $(2, 1)$, and so on, forming a repeating cycle, and explain why the sequence cycles rather than converging to a fixed profile.
- (ii) Prove that each run in the cycle is at least twice as long as the previous run, and state what this implies about the long-run behavior of fictitious play in this game.
- (iii) Show that the empirical mixed strategies (the time averages of play) do not converge but instead move along the cycle described in (i), and verify that the cycle does not pass through the uniform mixed strategy $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, the unique Nash equilibrium of the game.
- (iv) Implement fictitious play for this game in Python or MATLAB. Initialize at the action profile $(1, 1)$, break ties in favor of the action with the smallest index, and run $T = 10^4$ rounds. Plot each player's empirical frequencies of the three actions as functions of the round t (a logarithmic scale on the t -axis is recommended), and report the lengths of the successive runs observed. Confirm numerically the run-doubling behavior established in (ii) and the nonconvergence established in (iii).
- (v) Compare this behavior with the convergence of fictitious play in zero-sum matrix games established in Lecture 2, and explain why the convergence argument for zero-sum games does not extend to this bi-matrix game.

Chapter 5

Extensive-Form Games

Up to this point, our discussion has focused exclusively on *static games*, where all players move simultaneously and no temporal or informational structure is required. In such settings, specifying the players, their action sets, and their payoff functions suffices to describe the game, and concepts such as information sets do not arise. In this lecture, we move beyond the static framework and study *dynamic games with finite action sets*, in which players may act sequentially and may possess only partial information about earlier moves. To analyze such environments, we introduce the central notions of *information sets* and the *extensive form* of a game.

In a finite-action dynamic game, the extensive form represents the game as a decision tree: decision nodes correspond to points at which a player takes an action; edges represent available actions; and terminal nodes specify the resulting payoffs. Information sets group together decision nodes that a player cannot distinguish among when choosing an action; thus they encode the game's informational structure and distinguish perfect-information games from those with imperfect information. This explicit representation allows us to define *behavioral strategies* and the associated *concept of Nash equilibrium in behavioral strategies*.

Once the extensive form has been specified, we may construct the game's *normal form* (or strategic form) by listing each player's complete contingent plan, i.e., a pure strategy specifying an action at every information set, and the corresponding payoffs for every strategy profile. In finite extensive-form games with perfect recall, *Kuhn's Theorem* establishes the equivalence between mixed and behavioral strategies. Thus, for dynamic games with finite action sets, the extensive form provides the foundational description of the game's temporal and informational structure, while the normal form offers a compact representation derived from it.

5.1 ■ Introduction to Extensive-Form Games

To motivate the discussion, consider the classic *Matching Pennies* game, a two-player zero-sum game between Player 1 (P1) and Player 2 (P2). Its normal-form (matrix) representation is shown in Table 5.1.

While the matrix form fully specifies the players' payoffs, it does not encode additional structural features of the interaction, such as the order of moves or the information available to each player at the time of decision. Two extensive-form representations that correspond to the same strategic interaction are shown in Figures 5.1 and 5.2.

In both extensive-form representations, the player who moves second cannot distinguish which action was chosen by the first mover. This informational constraint is captured by the dashed line connecting the two decision nodes, which together form a single *information set*. As

P1	P2	
	H	T
H	(1, -1)	(-1, 1)
T	(-1, 1)	(1, -1)

Table 5.1: Matching Pennies: normal-form representation.

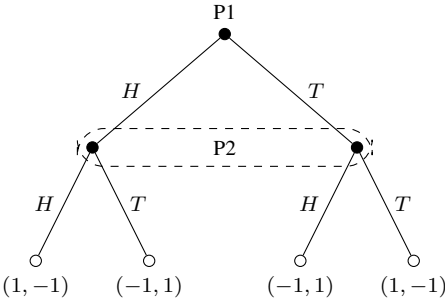


Figure 5.1: Extensive-form representation in which P1 moves first and P2 does not observe Player 1’s action.

a result, the second player must adopt the same decision rule at both nodes.
Although the two trees differ in their move order, they are *strategically equivalent* to each other and to the normal-form game in Table 5.1.

5.1.1 ■ Chance Moves and Nature’s Role

In many strategic interactions, outcomes are shaped not only by the players’ decisions but also by random events beyond anyone’s control. To model such uncertainty, extensive-form games introduce a special decision-maker called *Nature* (or the *chance player*), whose moves represent the realization of exogenous states, such as a coin flip, a card draw, or an environmental condition, that may affect payoffs, the order of moves, and the information available to the players.

The role played by chance moves depends critically on whether Nature’s realization is observed by the players or not. The following two examples illustrate two qualitatively different cases.

Example 5.1 Unobserved Chance Move: Payoff- and Information-Relevant Nature.

Consider the game depicted in Fig. 5.3. Nature first selects a state of the world, either ω_1 or ω_2 , each with probability $1/2$. The realized state determines both the payoff structure and the order of moves in the subsequent interaction. Crucially, *Nature’s realization is not fully observed by the players*, and hence the interaction following Nature’s move does not decompose into separate subgames.

Under ω_1 , play corresponds to a zero-sum Matching Pennies interaction. P1 moves first, and P2 responds without observing P1’s action. Under ω_2 , play instead corresponds to a *team (common-interest) problem*. P2 moves first, and P1 responds

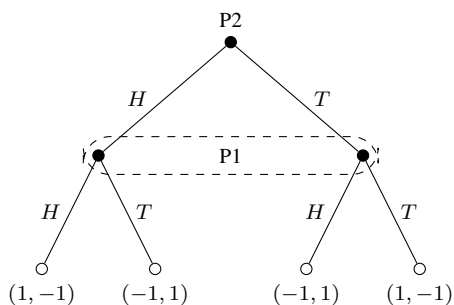


Figure 5.2: Strategically equivalent extensive-form representation in which P2 moves first and P1 does not observe Player 2's action.

without observing P2's action. Thus, Nature's move affects not only payoffs but also the strategic structure of play.

Figure 5.3 depicts the resulting extensive form. Player 1's post-Nature decision nodes x^{ω_1} and x^{ω_2} belong to the same information set, and hence P1 does not know which state Nature has selected when choosing an action. Similarly, Player 2's second-stage decision nodes $\{x^{\omega_1, H}, x^{\omega_1, T}, x^{\omega_2, H}, x^{\omega_2, T}\}$ form a single information set, and hence P2 cannot condition her action on either Nature's realization or the first mover's choice.

Because these information sets pool decision nodes across different actions of Nature, the interactions induced by ω_1 and ω_2 are not separate games. Instead, each player's optimal strategy depends on beliefs about Nature's realization, and therefore on the probability distribution with which Nature selects ω_1 and ω_2 .

□

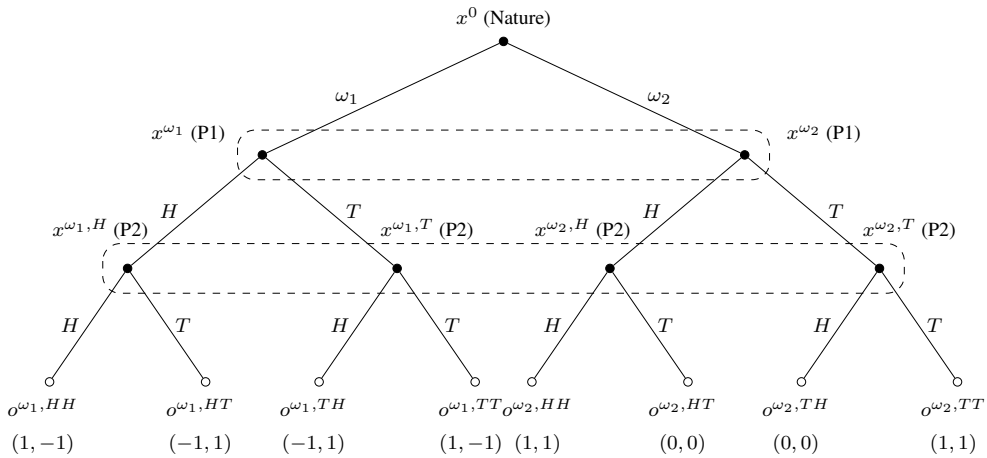


Figure 5.3: An extensive-form game with a payoff-relevant and information-relevant move by Nature. Nature selects ω_1 or ω_2 with given probabilities, but the players do not fully observe Nature's realization. P1 cannot condition her action on the realized state, and P2 cannot condition her action on either the state or P1's choice. Under ω_1 , play corresponds to a zero-sum Matching Pennies game, whereas under ω_2 it corresponds to a team (common-interest) problem. Because these induced strategic interactions are linked by the information structure, Nature's probabilities affect equilibrium behavior.

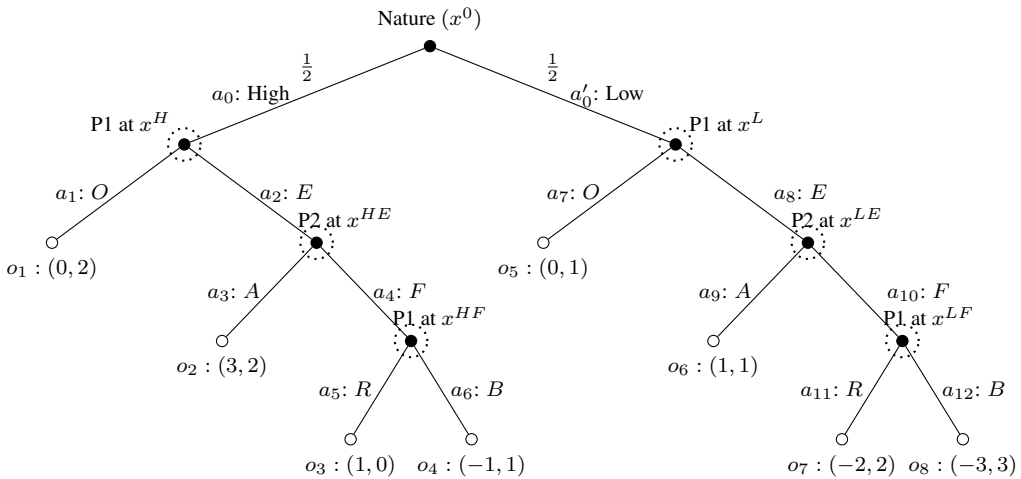


Figure 5.4: A complete-information extensive-form game with a chance move. Each decision node is enclosed by a dotted information-set circle (singleton information sets).

The first example illustrates the most subtle role of chance moves: Nature's action is *both payoff-relevant and information-relevant*. Because Nature's realization is unobserved, beliefs about the underlying state become a strategic object, and Nature's probability distribution directly affects equilibrium behavior.

Example 5.2 Observed Chance Move: Complete-Information Game with Nature.

Consider the game illustrated in Figure 5.4. Nature first draws the market demand, $\text{State} \in \{\text{High}, \text{Low}\}$, each with probability $\frac{1}{2}$, and this realization is publicly observed. The entrant (P1) then decides whether to *Enter* (E) or *Stay Out* (O). If P1 enters, the incumbent (P2) decides whether to *Accommodate* (A) or *Fight* (F). If P2 chooses to fight, P1 then decides whether to *Retaliate* (R) or *Back Down* (B).

All actions and Nature's realization are publicly observed. Consequently, every information set is a singleton, and the game has complete and perfect information. In this case, Nature's move selects the realized state but does not generate informational uncertainty for the players.

□

The contrast between the two examples highlights a fundamental distinction. When Nature's action is fully observed, each realization simply selects a subsequent interaction, and the analysis decomposes into independent games, one for each state. In such settings, Nature's probability distribution plays no strategic role. By contrast, when Nature's action is unobserved, as in Example 5.1, chance moves interact directly with the information structure of the game. Beliefs about Nature's realization become an integral part of strategic reasoning, and Nature's probabilities influence equilibrium behavior.

Example 5.1 illustrates two essential features of extensive-form games with chance moves. First, Nature's action can be *payoff-relevant* in that different realizations induce different payoff structures or strategic objectives. Second, Nature's action may be *information-relevant*, in that players may be unable to observe Nature's realization and must therefore form beliefs about the underlying state when choosing their actions. When these two features interact, the probability distribution of Nature becomes a fundamental component of strategic reasoning.

To analyze such settings in a systematic and general way, we now extend the extensive-form framework to explicitly incorporate *chance moves*. This requires treating Nature as a special player whose actions are governed by exogenous probability distributions rather than strategic choice. The resulting model, an *extensive-form game with chance moves*, provides a common language to represent random events, stochastic environments, and uncertainty about the state of the world within dynamic games.

Definition 5.3 (Extensive-Form Game with Perfect Information and Chance Moves). An extensive-form game is a tuple

$$G = \left(\mathcal{N}, \mathcal{V}, \mathcal{E}, x^0, (\mathcal{V}_i)_{i \in \mathcal{N} \cup \{0\}}, (p_x)_{x \in \mathcal{V}_0}, O, u \right),$$

where:

- $\mathcal{N}$ is a finite set of strategic players, and player 0 (Nature) represents chance;
- $(\mathcal{V}, \mathcal{E}, x^0)$ is a rooted, directed tree (the game tree) with:
 - $\mathcal{V}$ the set of nodes (decision and terminal points);
 - $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ the set of directed edges;
 - x^0 the unique root node.

Each non-terminal node $x \in \mathcal{V} \setminus O$ has one or more outgoing edges, and each edge $(x, y) \in \mathcal{E}$ is indicated by an action available at x . If $(x, y) \in \mathcal{E}$, we say that x is the parent of y and y is a child of x .

- $(\mathcal{V}_i)_{i \in \mathcal{N} \cup \{0\}}$ is a partition of the non-terminal nodes $\mathcal{V} \setminus O$, where:
 - $\mathcal{V}_i$ is the set of nodes where player i chooses an action;
 - $\mathcal{V}_0$ is the set of chance nodes, controlled by Nature.
- For each $x \in \mathcal{V}_0$, p_x is a probability distribution over the outgoing edges from x , specifying Nature's move probabilities.
- $O \subset \mathcal{V}$ is the set of terminal nodes (outcomes), which have no outgoing edges.
- $u : O \rightarrow \mathbb{R}^N$ assigns to each outcome $x \in O$ a payoff vector $u(x)$, where $u_i(x)$ is the payoff to player i .

Each node $x \in \mathcal{V}$ determines a unique sequence of edges leading from the root, called the history of play up to x , and denoted by $h(x)$. If x is a child of y , then $h(x) = (h(y), a)$, where a is the action labeling the edge (y, x) .

Throughout this lecture, the payoff functions u_i are utilities that the players seek to maximize, in expectation whenever chance moves or randomized strategies are involved. In two-player zero-sum examples we occasionally designate one player as the minimizer of the other's payoff; whenever we do so, this is stated explicitly.

5.1.2 ■ Information Sets in Extensive-Form Games

In Example 5.2, we saw a setting in which each player knows exactly where she is in the game tree and observes the entire history of play before making a move. In particular, even though the game includes chance moves by Nature, the realizations of Nature's randomizations are fully observed by all players. As a result, uncertainty plays no informational role in players' decision making. This assumption yields a game of *perfect information*, meaning that every past action and chance outcome is observable at the time of decision, and of *perfect recall*, meaning that players never forget what they have previously observed or the actions they themselves have taken.

By contrast, in Example 5.1, players cannot fully observe earlier actions or the realization of Nature's move, and therefore cannot distinguish among certain histories of play. In many strategic environments of this kind, players face uncertainty not about the structure of the game itself, but about which particular history has occurred when they are called upon to act. To model such situations, we introduce *information sets*, which specify the set of decision nodes that are observationally indistinguishable to a player at the time of action.

Information sets provide a flexible tool for representing *imperfect information*, which arises when some actions are hidden or when moves are made simultaneously. They also accommodate settings with *incomplete information*, in which Nature draws private types or payoff-relevant parameters that are observed by some players but not by others. The introduction of information sets thus provides a unified framework for modeling a wide range of informational environments within the extensive-form representation.

Definition 5.4 (Information Structure). Let $G = (\mathcal{N}, \mathcal{V}, \mathcal{E}, x^0, (\mathcal{V}_i)_{i \in \mathcal{N} \cup \{0\}}, (p_x)_{x \in \mathcal{V}_0}, O, u)$ be an extensive-form game as in Definition 5.3. For each player $i \in \mathcal{N}$, let $\mathcal{V}_i \subseteq \mathcal{V}$ denote the set of her decision nodes. The set $\mathcal{V}_i$ is partitioned into M_i information sets:

$$\mathcal{I}_i = \{\eta_i^1, \eta_i^2, \dots, \eta_i^{M_i}\}, \quad \bigcup_{k=1}^{M_i} \eta_i^k = \mathcal{V}_i, \quad \eta_i^k \cap \eta_i^{k'} = \emptyset \text{ for } k \neq k'.$$

Each information set $\eta_i^k \subseteq \mathcal{V}_i$ satisfies the following conditions:

- (i) (Indistinguishability) For all $x, x' \in \eta_i^k$, player i cannot distinguish whether node x or x'

has been reached, given her available information at that point in the game. Formally, x and x' are observationally equivalent for player i .

- (ii) (Consistency of Actions) Every node $x \in \eta_i^k$ admits the same number of feasible actions, $|A(x)| = \ell_i^k$, and these actions are labeled consistently across nodes in η_i^k . We denote this common action set by $A(\eta_i^k) = \{a_1^i, a_2^i, \dots, a_{\ell_i^k}^i\}$, where each label a_ℓ^i represents an equivalence class of edges leaving the nodes in η_i^k that correspond to the same decision from the player's perspective.

The pair $(\eta_i^k, A(\eta_i^k))$ thus defines both the informational state of player i , i.e., what she knows, and their strategic possibilities, i.e., what they can do, at that stage of the game.

An information set η_i^k formally represents player i 's *informational uncertainty* about the exact history of play at the time of decision. If each information set is a singleton, $|\eta_i^k| = 1$ for all i and k , the player knows precisely which node has been reached. If some information sets contain multiple nodes, $|\eta_i^k| > 1$, the player moves under uncertainty about previous actions or chance realizations.

Definition 5.5 (Extensive Form Game with Chance Moves and Information Structure). An extensive form game with chance moves and information structure is a tuple

$$G = (\mathcal{N}, \mathcal{V}, \mathcal{E}, x^0, (\mathcal{V}_i)_{i \in \mathcal{N} \cup \{0\}}, (p_x)_{x \in \mathcal{V}_0}, (\mathcal{I}_i)_{i \in \mathcal{N}}, O, u),$$

where:

- $\mathcal{N}$ is a finite set of players, and player 0 represents Nature;
- $(\mathcal{V}, \mathcal{E}, x^0)$ is a rooted directed tree with vertex set $\mathcal{V}$, edge set $\mathcal{E}$, and root node x^0 ;
- $(\mathcal{V}_i)_{i \in \mathcal{N} \cup \{0\}}$ partitions the nonterminal nodes $\mathcal{V} \setminus O$ among the players and Nature;
- For each $x \in \mathcal{V}_0$, p_x is a probability distribution over the outgoing edges from x , specifying Nature's move probabilities;
- For each player $i \in \mathcal{N}$, $\mathcal{I}_i$ is a partition of $\mathcal{V}_i$ into information sets, where each information set contains the nodes that player i cannot distinguish at the time of decision;
- $O \subset \mathcal{V}$ is the set of terminal nodes (outcomes);
- $u : O \rightarrow \mathbb{R}^N$ assigns to each terminal node $x \in O$ a payoff vector $u(x) = (u_1(x), \dots, u_N(x))$.

Definition 5.5 provides a unified framework for representing different informational assumptions through the specification of the information partitions $(\mathcal{I}_i)_{i \in \mathcal{N}}$. These partitions determine what each player observes about the history of play at the time of decision and thus encode the game's information structure. Every extensive-form game therefore implicitly specifies an underlying information structure, which determines what each player knows at every point where she may be required to act.

When every information set in $\mathcal{I}_i$ is a singleton, a player knows exactly which history has been reached, and the game has perfect information. When some information sets contain multiple decision nodes, the player cannot distinguish among the corresponding histories, allowing the representation of imperfect information. Furthermore, by allowing Nature to assign privately observed types at the beginning of the game, the same framework accommodates incomplete information. To clarify these distinctions, we now introduce several standard categories of information structure.

Perfect vs. Imperfect Information The distinction between *perfect* and *imperfect* information concerns what players observe about the *history of play* at the time they make decisions. It is

conceptually separate from the distinction between *complete* and *incomplete* information, which concerns players' knowledge of the game's primitives, such as payoff functions, action sets, or player types. A game has *perfect information* if, whenever a player is called upon to move, the player observes the entire sequence of past actions and outcomes. Formally, every information set in the extensive form is a singleton. In such games, players never face uncertainty about how the current decision node was reached; strategic uncertainty arises only from anticipating the future actions of other players.

A game has *imperfect information* if at least one player must move without observing some aspect of the past history. In the extensive form, this corresponds to the presence of information sets containing multiple decision nodes. Such situations arise in simultaneous-move games, games with hidden actions, or settings with delayed or noisy observations. In an imperfect-information game, players know the structure of the game and the payoff functions, but cannot determine which particular history within an information set has occurred at the time of their move. These issues will be developed further in Part IV on dynamic and repeated games and Part V on differential and stochastic games.

Public vs. Private Monitoring. A related but separate distinction concerns how information about past actions or outcomes is revealed to players over time. In games with *public monitoring*, all players observe a common signal about past actions or outcomes. Repeated games with publicly observed but noisy payoffs, or quality inspection results announced to all firms in a market, fall into this category. In such settings, players hold aligned beliefs about the history of play, even when observations are imperfect.

By contrast, under *private monitoring*, each player receives an idiosyncratic signal about past actions or outcomes. Repeated games in which players privately observe noisy signals of others' actions, or security games in which defenders and attackers observe different sensor readings, provide canonical examples. Here, players may hold different beliefs about the same underlying events, leading to persistent disagreement and more complex strategic behavior. These information structures will be discussed in Lecture 18.

Symmetric vs. Asymmetric Information. Information structures may also be classified as *symmetric* or *asymmetric*. In symmetric information structures, all players observe the same type of information at corresponding decision points, as in standard perfect-information games or repeated games with public signals. In asymmetric information structures, some players observe payoff-relevant information that others do not. Examples include signaling games, in which a sender privately observes a type assigned by Nature (the subject of Lecture 17); principal-agent problems with hidden effort; and Bayesian security games with private vulnerability or capability information. These settings will be developed further in Part IV on dynamic and repeated games and Part V on differential and stochastic games.

Complete vs. Incomplete Information. A game has *complete information* when all players understand the full structure of the game before it begins. This includes knowing who the players are, what actions are available to each player, when decisions are made, and how outcomes translate into payoffs for everyone. Complete information therefore concerns what players know about the game itself at the outset. Importantly, complete information does *not* require that players observe everything that happens during play. Players may be unaware of some actions taken by others, or may move simultaneously without seeing each other's choices. In such cases, the game has complete information while still involving *imperfect information* about the course of play.

A game has *incomplete information* when at least one player lacks knowledge of some

payoff-relevant aspect of the game. This may include uncertainty about other players' preferences, objectives, or constraints. Here, the uncertainty is not about what has already happened, but about the underlying features that determine how outcomes are valued. A standard way to represent incomplete information is to imagine that, before play begins, an external random process assigns each player a private characteristic that only that player observes. Players know the possible characteristics and how likely they are, but do not know which ones others have received. As a result, each player initially considers several possible versions of the game that differ only in these unobserved characteristics.

This construction, commonly known as the *Harsanyi transformation*, allows games with incomplete information to be analyzed within the same formal framework as games with perfect or imperfect information. The full framework of Bayesian games based on this construction will be introduced in Lecture 12. This distinction separates uncertainty about *the game itself*, which defines incomplete information, from uncertainty about *the history of play*, which defines imperfect information.

Example 5.6 Unobserved Chance Move: Payoff- and Information-Relevant Nature (Continued).

We revisit Example 5.1 to illustrate an extensive-form game with chance moves and information structure, as formalized in Definition 5.5. The game is given by

$$G = (\mathcal{N}, \mathcal{V}, \mathcal{E}, x^0, (\mathcal{V}_i)_{i \in \mathcal{N} \cup \{0\}}, (p_x)_{x \in \mathcal{V}_0}, (\mathcal{I}_i)_{i \in \mathcal{N}}, O, u).$$

The player set is $\mathcal{N} = \{1, 2\}$, and player 0 represents Nature. The game tree $(\mathcal{V}, \mathcal{E}, x^0)$ consists of a root node x^0 controlled by Nature, followed by first-stage decision nodes x^{ω_1} and x^{ω_2} , and second-stage decision nodes $x^{\omega_1, H}$, $x^{\omega_1, T}$, $x^{\omega_2, H}$, and $x^{\omega_2, T}$. The set of terminal nodes is denoted by O .

The partition of nonterminal nodes among the players is given by

$$\mathcal{V}_0 = \{x^0\}, \quad \mathcal{V}_1 = \{x^{\omega_1}, x^{\omega_2}\}, \quad \mathcal{V}_2 = \{x^{\omega_1, H}, x^{\omega_1, T}, x^{\omega_2, H}, x^{\omega_2, T}\}.$$

At the chance node x^0 , Nature selects ω_1 or ω_2 with probability $\frac{1}{2}$ each. The information structure is specified by partitions of each player's decision nodes. Let $\mathcal{I}_i$ denote the collection of information sets of player i .

For P1, $\mathcal{I}_1 = \{\eta_1\}$, $\eta_1 = \{x^{\omega_1}, x^{\omega_2}\}$, so P1 cannot distinguish between the two possible histories generated by Nature's initial move. For P2, $\mathcal{I}_2 = \{\eta_2\}$, $\eta_2 = \{x^{\omega_1, H}, x^{\omega_1, T}, x^{\omega_2, H}, x^{\omega_2, T}\}$. Thus, P2 observes neither Nature's realization nor P1's action, and must select the same action at all four of her decision nodes.

Each terminal node $o \in O$ corresponds to a unique history of play and is assigned a payoff vector $u(o) \in \mathbb{R}^2$, as shown in Fig. 5.3. The game therefore has complete information with respect to the structure of the game tree and the payoff functions, but imperfect information arising from the unobserved move of Nature. This informational uncertainty is captured by the information structure $(\mathcal{I}_1, \mathcal{I}_2)$, which specifies, for each player, the set of histories that are observationally indistinguishable at the time of decision and thereby constrains the set of feasible strategies.

Since each player has a single information set at her decision stage, both players must commit *ex ante* to actions that cannot be conditioned on either Nature's realization or the opponent's behavior. Consequently, although the game is represented in extensive form with sequential moves, the induced strategic interaction is equivalent to a simultaneous-move game under uncertainty, with informational constraints encoded explicitly by the partitions $\mathcal{I}_1$ and $\mathcal{I}_2$.

□

Example 5.7 Observed Chance Move: Complete-Information Game with Nature (Continued).

We continue Example 5.2, depicted in Figure 5.4, to further illustrate the extensive-form game with chance moves and information structure formalized in Definition 5.5. The game is given by the tuple $G = (\mathcal{N}, \mathcal{V}, \mathcal{E}, x^0, (\mathcal{V}_i)_{i \in \mathcal{N} \cup \{0\}}, (p_x)_{x \in \mathcal{V}_0}, (\mathcal{I}_i)_{i \in \mathcal{N}}, O, u)$.

The player set is $\mathcal{N} = \{1, 2\}$, and Nature (player 0) controls the root node x^0 . The vertex set is

$$\mathcal{V} = \{x^0, x^H, x^L, x^{HE}, x^{LE}, x^{HF}, x^{LF}\} \cup O,$$

where $O = \{o_1, \dots, o_8\}$ denotes the set of terminal nodes. The edge set $\mathcal{E} = \{a_0, a'_0, a_1, \dots, a_{12}\}$ consists of the labeled actions connecting each node to its children.

The partition of nonterminal nodes among the players is given by

$$\mathcal{V}_0 = \{x^0\}, \quad \mathcal{V}_1 = \{x^H, x^L, x^{HF}, x^{LF}\}, \quad \mathcal{V}_2 = \{x^{HE}, x^{LE}\}.$$

At the chance node x^0 , Nature's probability distribution is $p_{x^0}(a_0) = p_{x^0}(a'_0) = \frac{1}{2}$.

The information structure of the game is trivial. Since Nature's realization and all subsequent actions are publicly observed, every information set is a singleton. Formally, for each player $i \in \mathcal{N}$, $\mathcal{I}_i = \{\{x\} : x \in \mathcal{V}_i\}$. Thus, whenever a player is called upon to act, she observes the full history of play up to that point.

Each node $x \in \mathcal{V}$, except for terminal nodes, has a unique parent, and each edge represents a labeled action leading to a child node. For example, the history leading to terminal node o_4 is

$$h(o_4) = (a_0, a_2, a_4, a_6) = (\text{High}, E, F, B),$$

corresponding to the sequence $x^0 \xrightarrow{a_0} x^H \xrightarrow{a_2} x^{HE} \xrightarrow{a_4} x^{HF} \xrightarrow{a_6} o_4$. Each terminal node $o_k \in O$ is assigned a payoff vector $u(o_k) = (u_1(o_k), u_2(o_k))$, as labeled in the figure. This explicit labeling of nodes, edges, and outcomes makes the correspondence between the formal tuple $(\mathcal{N}, \mathcal{V}, \mathcal{E}, x^0, (\mathcal{V}_i), (p_x), (\mathcal{I}_i), O, u)$ and the graphical representation transparent.

The game has *complete information*, since the structure of the game, Nature's probability distribution, and all payoff functions are common knowledge. Moreover,

the game has *perfect information*, because all players observe Nature's realization and all prior actions at every decision point. In this setting, Nature's move selects the realized state of the world but does not generate informational uncertainty, and equilibrium analysis can therefore be conducted separately for each realization of Nature.

□

5.2 ■ From Extensive-Form to Normal-Form Games

The extensive-form representation of a game captures its *dynamic* and *informational* structure, including who moves when, what they know, and what payoffs result. The *normal form*, by contrast, summarizes the interaction as a simultaneous-move game in which each player chooses a complete plan of action at the outset based on the information she will receive. This section illustrates the correspondence between these two representations, beginning with a simple sequential example and then formalizing the concepts of pure and behavioral strategies.

Example 5.8 Extensive-Form Matching Pennies Game (Continued).

Consider again the *Matching Pennies* game. In the earlier version in Figure 5.1, P2 did not observe Player 1's move, and thus both of Player 2's decision nodes belonged to the same information set $\eta_2 = \{x_H, x_T\}$. This made the game one of *imperfect information*, since P2 could not distinguish whether P1 had played H or T . Now suppose that we modify the information structure so that P2 perfectly observes Player 1's move before acting, as depicted in Figure 5.5.

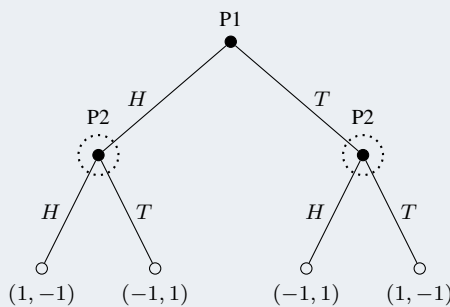


Figure 5.5: Modified Matching Pennies: P2 observes Player 1's move (perfect information).

Here, P2 has two distinct information sets: $\eta_2^{(H)} = \{x_H\}$, $\eta_2^{(T)} = \{x_T\}$. We may think of a public signal $y \in \{H, T\}$ being generated immediately after Player 1's move, revealing the action perfectly to Player 2. This new information structure expands Player 2's strategy space, since she can now condition her choice on the observed signal y .

□

5.2.1 ■ Pure Strategies in Extensive Form

A *pure strategy* for player i in an extensive-form game is a complete *contingency plan* that specifies a definite action at every information set belonging to that player. Formally, let $\mathcal{I}_i = \{\eta_i^1, \eta_i^2, \dots, \eta_i^{M_i}\}$ denote the collection of player i 's information sets, and let $A(\eta_i^k)$ be the set of feasible actions available at information set η_i^k . A pure strategy for player i is a function

$$s_i : \mathcal{I}_i \longrightarrow \bigcup_{k=1}^{M_i} A(\eta_i^k), \quad s_i(\eta_i^k) \in A(\eta_i^k) \quad \text{for all } k = 1, \dots, M_i.$$

That is, s_i prescribes a single action $s_i(\eta_i^k)$ from the available set $A(\eta_i^k)$ at each information set η_i^k . The collection of all such mappings defines the player's pure strategy set:

$$S_i = \prod_{k=1}^{M_i} A(\eta_i^k).$$

Intuitively, a pure strategy describes what player i would do at every point in the game where she might be called to move, even at information sets that may never be reached along the equilibrium path.

In the modified Matching Pennies game of Figure 5.5, P1 moves once at the root node and therefore has a single information set $\mathcal{I}_1 = \{\eta_1^1\}$, $A(\eta_1^1) = \{H, T\}$. Hence, Player 1's pure strategy set consists of choosing either H or T at that sole decision point: $S_1 = \{s_1^1, s_1^2\}$, where each pure strategy is a single rule mapping the information set to an action:

$$s_1^1 = H \quad \text{if at } \eta_1^1, \quad s_1^2 = T \quad \text{if at } \eta_1^1.$$

P2, on the other hand, has two information sets corresponding to the two possible observations of Player 1's move:

$$\mathcal{I}_2 = \{\eta_2^{(H)}, \eta_2^{(T)}\}, \quad A(\eta_2^{(H)}) = A(\eta_2^{(T)}) = \{H, T\}.$$

A pure strategy for P2 specifies an action for each information set:

$$s_2 = \begin{cases} a_2(\eta_2^{(H)}) & \text{if at } \eta_2^{(H)}, \\ a_2(\eta_2^{(T)}) & \text{if at } \eta_2^{(T)}, \end{cases} \quad a_2(\eta_2^{(H)}), a_2(\eta_2^{(T)}) \in \{H, T\}.$$

Explicitly, Player 2's four pure strategies can be written as:

$$s_2^1 = \begin{cases} H & \text{if at } \eta_2^{(H)}, \\ H & \text{if at } \eta_2^{(T)}, \end{cases} \quad s_2^2 = \begin{cases} T & \text{if at } \eta_2^{(H)}, \\ T & \text{if at } \eta_2^{(T)}, \end{cases}$$

$$s_2^3 = \begin{cases} H & \text{if at } \eta_2^{(H)}, \\ T & \text{if at } \eta_2^{(T)}, \end{cases} \quad s_2^4 = \begin{cases} T & \text{if at } \eta_2^{(H)}, \\ H & \text{if at } \eta_2^{(T)}. \end{cases}$$

Thus, Player 2's pure strategy set is $S_2 = \{s_2^1, s_2^2, s_2^3, s_2^4\}$. Each $s_2^m \in S_2$ represents a complete plan of action specifying how P2 will respond under every possible information condition. For example, s_2^3 corresponds to the intuitive "matching" rule: play the same action as P1 once her move is observed.

In shorthand notation, the pure strategies of both players can be expressed compactly as strings of letters representing their prescribed actions. For P1, who moves only once, we have:

$$s_1^1 := H, \quad s_1^2 := T.$$

For P2, whose strategy specifies an action at each of her two information sets, we have:

$$s_2^1 := HH, \quad s_2^2 := TT, \quad s_2^3 := HT, \quad s_2^4 := TH,$$

where, for P2, the first letter indicates the action taken if P1 played H , and the second letter indicates the action taken if P1 played T . This compact representation corresponds exactly to the row and column labels in the normal-form game of Table 5.2.

We can now construct the normal form of the modified game by enumerating each player's pure strategies. Player 1's pure strategy set is $S_1 = \{H, T\}$, and Player 2's is $S_2 = \{HH, HT, TH, TT\}$. The corresponding payoff table is given in Table 5.2.

P1	P2			
	HH	HT	TH	TT
H	(1, -1)	(1, -1)	(-1, 1)	(-1, 1)
T	(-1, 1)	(1, -1)	(-1, 1)	(1, -1)

Table 5.2: Normal form for the modified Matching Pennies game with perfect information for Player 2.

The expansion of Player 2's information partitions increases the number of her pure strategies from two (in the imperfect-information version) to four (in the perfect-information version). With full observability, P2 can now condition her choice on Player 1's action.

5.2.2 ■ General Correspondence between Extensive and Normal Forms

Given an extensive-form game with chance and information structure

$$G = (\mathcal{N}, \mathcal{V}, \mathcal{E}, x^0, (\mathcal{V}_i)_{i \in \mathcal{N} \cup \{0\}}, (p_x)_{x \in \mathcal{V}_0}, (\mathcal{I}_i)_{i \in \mathcal{N}}, O, u),$$

we can formally construct its strategically equivalent *normal-form representation*

$$\Gamma(G) = (\mathcal{N}, (S_i)_{i \in \mathcal{N}}, (u_i^\Gamma)_{i \in \mathcal{N}})$$

by specifying its components as follows.

Players. The set of players in the normal form coincides with that of the extensive form, $\mathcal{N}$.

Strategy sets. For each player $i \in \mathcal{N}$, let $\mathcal{I}_i = \{\eta_i^1, \eta_i^2, \dots, \eta_i^{M_i}\}$ denote the collection of information sets, and let $A(\eta_i^k)$ be the set of feasible actions at information set η_i^k . A *pure strategy* for player i is a complete contingent plan that assigns one feasible action to each of the player's information sets, subject to the indistinguishability and action-consistency requirements stated in Definition 5.4.

$$s_i : \mathcal{I}_i \rightarrow \bigcup_{k=1}^{M_i} A(\eta_i^k), \quad s_i(\eta_i^k) \in A(\eta_i^k) \text{ for all } k = 1, \dots, M_i.$$

Equivalently, the pure-strategy space of player i is the Cartesian product $S_i = \prod_{\eta_i \in \mathcal{I}_i} A(\eta_i)$, and each element $s_i \in S_i$ can be represented as

$$s_i = \begin{cases} a_{i,\eta_i^1} & \text{if at information set } \eta_i^1, \\ a_{i,\eta_i^2} & \text{if at information set } \eta_i^2, \\ \vdots & \vdots \\ a_{i,\eta_i^{M_i}} & \text{if at information set } \eta_i^{M_i}. \end{cases}$$

Here, each $a_{i,\eta_i^k} \in A(\eta_i^k)$ denotes the action that player i commits to take whenever the play of the game reaches some node $x \in \eta_i^k$. Because all nodes within an information set are indistinguishable to the player, the same action a_{i,η_i^k} applies uniformly across them. Thus, s_i constitutes a fully specified contingent plan, prescribing what player i will do at every decision point they might encounter in the game.

Induced outcomes and normal-form representation. Let H denote the set of all finite histories (paths of actions from the root x^0), and let $H_T \subset H$ be the set of terminal histories ending at terminal nodes O . Given a pure strategy profile $s = (s_i)_{i \in \mathcal{N}} \in \prod_{i \in \mathcal{N}} S_i$, the sequence of actions prescribed by s , together with Nature's randomizations $(p_x)_{x \in \mathcal{V}_0}$, induces a unique (possibly stochastic) terminal history $h(s) \in H_T$.

Formally, $h(s)$ is generated recursively starting from the root node x^0 :

- if $x \in \mathcal{V}_i$, the play follows the edge corresponding to the action $s_i(\eta_i^k)$, where $x \in \eta_i^k$;
 - if $x \in \mathcal{V}_0$, the play follows the probabilistic transition determined by Nature's distribution p_x .
- The resulting path defines the realized or expected terminal history $h(s)$.

The expected payoff to player i in the corresponding normal form is defined as

$$u_i^\Gamma(s) := \mathbb{E}_p[u_i(h(s))] , \quad \forall i \in \mathcal{N},$$

where the expectation is taken with respect to the probability measure over terminal histories induced by Nature's moves. If the game has no chance nodes ($\mathcal{V}_0 = \emptyset$), then the expectation is degenerate and $u_i^\Gamma(s) = u_i(h(s))$.

Collecting these components, the *normal-form game* associated with G is defined as

$$\Gamma(G) = (\mathcal{N}, (S_i)_{i \in \mathcal{N}}, (u_i^\Gamma)_{i \in \mathcal{N}}),$$

where each player's strategy set S_i enumerates all complete contingent plans, and u_i^Γ assigns to each profile s the corresponding expected payoff.

This mapping $G \mapsto \Gamma(G)$ transforms the dynamic and information-based structure of the extensive form into a static representation of strategic choices. When every information set is a singleton (perfect information), this transformation is one-to-one: each strategy corresponds to a unique path in the tree. When some information sets contain multiple nodes (imperfect or incomplete information), the normal form expands the players' strategy spaces to include conditional plans for all possible information realizations. Hence, G and $\Gamma(G)$ are *strategically equivalent*: they induce identical outcome distributions and expected payoffs for all profiles of pure or mixed strategies. In particular, the Nash equilibrium theory developed for normal-form games in Lectures 3 and 4 applies directly to $\Gamma(G)$.

5.3 ■ Mixed and Behavioral Strategies in Extensive-Form Games

The correspondence between the extensive-form game G and its normal-form representation $\Gamma(G)$ provides the foundation for defining *mixed strategies*. In its extensive form, a pure strategy specifies a complete contingent plan of action, that is, an assignment of one feasible action to every information set the player may encounter. By reducing the extensive form G to its strategic (normal) form $\Gamma(G)$, we obtain a representation in which each player's strategy set explicitly enumerates all such complete plans: $\Gamma(G) = (\mathcal{N}, (S_i)_{i \in \mathcal{N}}, (u_i^\Gamma)_{i \in \mathcal{N}})$, where $S_i = \prod_{\eta_i \in \mathcal{I}_i} A(\eta_i)$. Throughout these Lecture Notes, we reserve σ_i for a *mixed strategy*, which randomizes once, ex ante, over the complete plans in S_i , and β_i for a *behavioral strategy*, which randomizes locally at each information set in $\mathcal{I}_i$.

5.3.1 ■ Mixed Strategies

A *mixed strategy* for player i is a probability distribution over her pure-strategy set S_i . Since each pure strategy specifies a complete contingent plan in the extensive form, a mixed strategy represents an *ex ante* randomization in which player i selects one such plan at the beginning of the game and then follows it deterministically throughout play.

Definition 5.9 (Mixed Strategy). Let $\Gamma(G) = (\mathcal{N}, (S_i)_{i \in \mathcal{N}}, (u_i^\Gamma)_{i \in \mathcal{N}})$ denote the normal-form representation of a finite extensive-form game. A mixed strategy for player i is a probability distribution

$$\sigma_i : S_i \rightarrow [0, 1], \quad \sum_{s_i \in S_i} \sigma_i(s_i) = 1.$$

The set of all mixed strategies for player i is the simplex $\Delta(S_i)$.

A mixed-strategy profile $\sigma = (\sigma_i)_{i \in \mathcal{N}} \in \prod_{i \in \mathcal{N}} \Delta(S_i)$ induces a joint distribution over pure-strategy profiles $s = (s_i)_{i \in \mathcal{N}}$. The expected payoff to player i is then

$$U_i^\Gamma(\sigma) = \sum_{s \in \prod_{j \in \mathcal{N}} S_j} \left(\prod_{j \in \mathcal{N}} \sigma_j(s_j) \right) u_i^\Gamma(s).$$

Thus, a mixed strategy captures the idea of randomizing over full contingent plans in the strategic-form representation derived from the extensive form.

Definition 5.10 (Mixed-Strategy Nash Equilibrium). Let $\Gamma(G) = (\mathcal{N}, (S_i)_{i \in \mathcal{N}}, (U_i^\Gamma)_{i \in \mathcal{N}})$ be the normal-form representation of an extensive-form game. A mixed-strategy profile $\sigma = (\sigma_i)_{i \in \mathcal{N}}$, $\sigma_i \in \Delta(S_i)$, is a mixed-strategy Nash equilibrium if for every player $i \in \mathcal{N}$,

$$U_i^\Gamma(\sigma_i, \sigma_{-i}) \geq U_i^\Gamma(\sigma'_i, \sigma_{-i}) \quad \text{for all } \sigma'_i \in \Delta(S_i).$$

That is, given the mixed strategies chosen by the other players, σ_i is a best response for player i , and no unilateral deviation can yield a strictly higher expected payoff.

This is precisely the notion of a mixed-strategy Nash equilibrium introduced for normal-form games in Lectures 3 and 4, applied here to the normal form $\Gamma(G)$.

In two-player zero-sum games, the above condition simplifies and strengthens to the classical minimax (or saddle-point) formulation. We now formalize this in the following.

Definition 5.11 (Mixed-Strategy Saddle-Point Equilibrium). Consider a two-player zero-sum extensive-form game with normal-form representation $\Gamma(G) = (\{1, 2\}, (S_i)_{i=1,2}, (U_i^\Gamma)_{i=1,2})$, where the payoffs satisfy $U_2^\Gamma = -U_1^\Gamma$. Let $\sigma = (\sigma_1, \sigma_2)$, $\sigma_i \in \Delta(S_i)$, be a mixed-strategy profile.

We say that σ is a mixed-strategy saddle-point equilibrium if

$$U_1^\Gamma(\sigma_1, \sigma_2) \leq U_1^\Gamma(\sigma'_1, \sigma_2) \quad \text{for all } \sigma'_1 \in \Delta(S_1),$$

and

$$U_1^\Gamma(\sigma_1, \sigma_2) \geq U_1^\Gamma(\sigma_1, \sigma'_2) \quad \text{for all } \sigma'_2 \in \Delta(S_2).$$

Equivalently, σ satisfies the minimax identity

$$\min_{\sigma'_1 \in \Delta(S_1)} \max_{\sigma'_2 \in \Delta(S_2)} U_1^\Gamma(\sigma'_1, \sigma'_2) = U_1^\Gamma(\sigma_1, \sigma_2) = \max_{\sigma'_2 \in \Delta(S_2)} \min_{\sigma'_1 \in \Delta(S_1)} U_1^\Gamma(\sigma'_1, \sigma'_2).$$

At a saddle point, P1 (the minimizer) chooses a mixed strategy that minimizes her worst-case expected loss, while P2 (the maximizer) chooses a mixed strategy that maximizes Player 1's expected loss. No unilateral deviation by either player can yield a strictly better outcome according to their objective.

5.3.2 ■ Behavioral Strategies

While mixed strategies represent *ex ante randomization*, i.e., a player commits at the beginning of the game to a complete plan of action drawn from a probability distribution over pure strategies, it is often more natural in dynamic settings to allow for *local randomization* at each decision point. In such cases, a player may randomize independently at every information set, based only on the information available at that point in time. This leads to the concept of a *behavioral strategy*.

Definition 5.12 (Behavioral Strategy). Let $\mathcal{I}_i = \{\eta_i^1, \eta_i^2, \dots, \eta_i^{M_i}\}$ denote the collection of information sets for player i , and let $A(\eta_i^k)$ be the set of feasible actions available at information set η_i^k . A behavioral strategy for player i is a mapping

$$\beta_i : \mathcal{I}_i \rightarrow \bigcup_{\eta_i^k \in \mathcal{I}_i} \Delta(A(\eta_i^k)), \quad \beta_i(\eta_i^k) \in \Delta(A(\eta_i^k)) \quad \text{for all } \eta_i^k \in \mathcal{I}_i.$$

Equivalently, the behavioral strategy can be written in conditional form as

$$\beta_i = \begin{cases} \beta_{i, \eta_i^1} \in \Delta(A(\eta_i^1)) & \text{if at } \eta_i^1, \\ \beta_{i, \eta_i^2} \in \Delta(A(\eta_i^2)) & \text{if at } \eta_i^2, \\ \vdots & \vdots \\ \beta_{i, \eta_i^{M_i}} \in \Delta(A(\eta_i^{M_i})) & \text{if at } \eta_i^{M_i}. \end{cases}$$

Here, each β_{i, η_i^k} denotes a probability distribution over the action set $A(\eta_i^k)$, specifying how player i randomizes among feasible actions when the play of the game reaches information set η_i^k . Each component β_{i, η_i^k} thus defines a local decision rule at η_i^k , assigning probabilities to available actions conditional on the information available at that point and independently of randomizations made at other information sets.

Given an extensive-form game $G = (\mathcal{N}, \mathcal{V}, \mathcal{E}, x^0, (\mathcal{V}_i)_{i \in \mathcal{N} \cup \{0\}}, (\mathcal{I}_i)_{i \in \mathcal{N}}, (p_x)_{x \in \mathcal{V}_0}, u)$, recall that $O \subset \mathcal{V}$ denotes the set of terminal nodes. A behavioral strategy profile is written

$\beta = (\beta_i)_{i \in \mathcal{N}}$, where each β_i assigns, for every information set $\eta \in \mathcal{I}_i$, a probability distribution over the feasible actions available at η . These local randomizations determine the probability with which play moves from node to node, and hence the probability with which each path in the game tree is traversed.

To formalize this, we associate with β a probability $\pi_\beta(x)$ for each node $x \in \mathcal{V}$. The value $\pi_\beta(x)$ represents the probability that play reaches node x when players follow the behavioral strategies in β . We begin by setting $\pi_\beta(x^0) = 1$.

Now suppose $e = (x, x') \in \mathcal{E}$ is an edge connecting a node x to its successor x' . The probability of reaching x' is obtained by multiplying the probability of reaching its parent x by the probability that the decision maker at x selects edge e . Formally,

$$\pi_\beta(x') = \begin{cases} \pi_\beta(x) p_x(e), & \text{if } x \in \mathcal{V}_0 \text{ (Nature node),} \\ \pi_\beta(x) \beta_i(\eta)(e), & \text{if } x \in \mathcal{V}_i \text{ and } x \in \eta \in \mathcal{I}_i. \end{cases}$$

Because the extensive form is a rooted tree, every non-root node $x \in \mathcal{V}$ has a unique sequence of ancestors leading back to the root. Formally, for any node $x \neq x^0$, there exists a unique sequence of nodes $(x^0, x^1, \dots, x^m = x)$ such that each pair $(x^t, x^{t+1}) \in \mathcal{E}$ is an edge in the tree. This sequence is called the path from x^0 to x .

At each step of this path, the recursive rule specifies how to compute the probability of reaching the next node. That is, for each edge (x^t, x^{t+1}) along the path, we have

$$\pi_\beta(x^{t+1}) = \begin{cases} \pi_\beta(x^t) p_{x^t}(e), & \text{if } x^t \in \mathcal{V}_0, \\ \pi_\beta(x^t) \beta_i(\eta)(e), & \text{if } x^t \in \mathcal{V}_i \text{ and } x^t \in \eta \in \mathcal{I}_i, \end{cases}$$

where $e = (x^t, x^{t+1})$. Starting from the initial condition $\pi_\beta(x^0) = 1$, we apply this update rule successively for $t = 0, 1, \dots, m-1$. This yields

$$\pi_\beta(x) = \prod_{t=0}^{m-1} \Pr_\beta(x^t \rightarrow x^{t+1}),$$

where $\Pr_\beta(x^t \rightarrow x^{t+1})$ is either $p_{x^t}(e)$ or $\beta_i(\eta)(e)$, depending on who moves at x^t . Thus, the reach probability $\pi_\beta(x)$ is fully determined by multiplying the probabilities assigned to each edge along the unique path from the root to x . Because the tree is finite and because the path is unique, this procedure defines $\pi_\beta(x)$ unambiguously for every node.

In particular, each terminal node $o \in O$ receives a well-defined reach probability, and we write $\Pr_\beta(o) := \pi_\beta(o)$ for the probability that terminal node o is reached under the behavioral-strategy profile β .

The expected payoff to player i under β is then obtained by weighting each terminal payoff $u_i(o)$ by its probability of occurrence:

$$U_i(\beta) := \sum_{o \in O} \Pr_\beta(o) u_i(o). \quad (5.1)$$

Whenever useful, we write $U_i(\beta_i, \beta_{-i})$ instead of $U_i(\beta)$ to emphasize the role of player i 's own strategy versus the strategies of the other players.

Definition 5.13 (Behavioral-Strategy Nash Equilibrium). *Let G be an extensive-form game with information sets $(\mathcal{I}_i)_{i \in \mathcal{N}}$, and let $\beta = (\beta_i)_{i \in \mathcal{N}}$ be a profile of behavioral strategies. The profile β is a behavioral-strategy Nash equilibrium if, for every player $i \in \mathcal{N}$,*

$$U_i(\beta_i, \beta_{-i}) \geq U_i(\beta'_i, \beta_{-i}) \quad \text{for all behavioral strategies } \beta'_i.$$

Equivalently, given the behavioral strategies of the other players, β_i maximizes player i 's expected payoff U_i as defined in (5.1), and thus no unilateral deviation at any information set can strictly improve her outcome.

Definition 5.14 (Behavioral-Strategy Saddle-Point Equilibrium). Consider a two-player zero-sum extensive-form game G with $u_2 = -u_1$. Let $\beta = (\beta_1, \beta_2)$ be a profile of behavioral strategies. We say that β is a saddle-point equilibrium in behavioral strategies if

$$U_1(\beta_1, \beta_2) \leq U_1(\beta'_1, \beta_2) \quad \text{for all } \beta'_1$$

and

$$U_1(\beta_1, \beta_2) \geq U_1(\beta_1, \beta'_2) \quad \text{for all } \beta'_2.$$

Equivalently, (β_1, β_2) satisfies the minimax identity

$$\min_{\beta'_1} \max_{\beta'_2} U_1(\beta'_1, \beta'_2) = U_1(\beta_1, \beta_2) = \max_{\beta_2} \min_{\beta_1} U_1(\beta_1, \beta_2).$$

At a saddle point, P1 chooses a behavioral strategy that minimizes her worst-case loss, while P2 chooses a behavioral strategy that maximizes player 1's loss, and no unilateral deviation by either player can yield a strict improvement.

Example 5.15 Matching Pennies with Chance Moves (Continued).

Consider again the extensive-form game shown in Figure 5.3. Nature first selects a state $\omega \in \{\omega_1, \omega_2\}$, each with probability $\frac{1}{2}$. If $\omega = \omega_1$, P1 moves first at node x^{ω_1} , choosing between H and T . Then P2 moves at one of the nodes $\{x^{\omega_1, H}, x^{\omega_1, T}\}$ without knowing which node has been reached, since P2 does not observe Player 1's action. If $\omega = \omega_2$, the roles are reversed: P2 moves first at node x^{ω_2} , choosing between H and T , and P1 moves at one of the nodes $\{x^{\omega_2, H}, x^{\omega_2, T}\}$, again without observing Player 2's move.

Hence, each player has a single information set consisting of two decision nodes where the opponent's previous move is unobserved:

$$\eta_1 = \{x^{\omega_2, H}, x^{\omega_2, T}\}, \quad \eta_2 = \{x^{\omega_1, H}, x^{\omega_1, T}\}.$$

Each information set thus represents a situation of imperfect information in which the player must randomize over H or T without knowing the opponent's prior choice.

A behavioral strategy for P1 is defined as $\beta_1 = \{(p_1, 1 - p_1)\}$, $p_1 \in [0, 1]$, meaning that P1 plays H with probability p_1 and T with probability $1 - p_1$ whenever she moves at her information set η_1 (that is, after Nature chooses ω_2). Similarly, a behavioral strategy for P2 is $\beta_2 = \{(p_2, 1 - p_2)\}$, $p_2 \in [0, 1]$, meaning that P2 plays H with probability p_2 and T with probability $1 - p_2$ whenever she moves at η_2 (that is, after Nature chooses ω_1).

The behavioral strategy profile (β_1, β_2) thus specifies independent randomizations at each player's information set, conditional on Nature's realization. Unlike mixed strategies, which assign probabilities to complete plans of action, behavioral strategies prescribe local probabilistic decisions at each information set. If both players

select degenerate probabilities $p_i \in \{0, 1\}$, the profile reduces to a pure strategy pair consistent with the extensive-form representation. $\square$

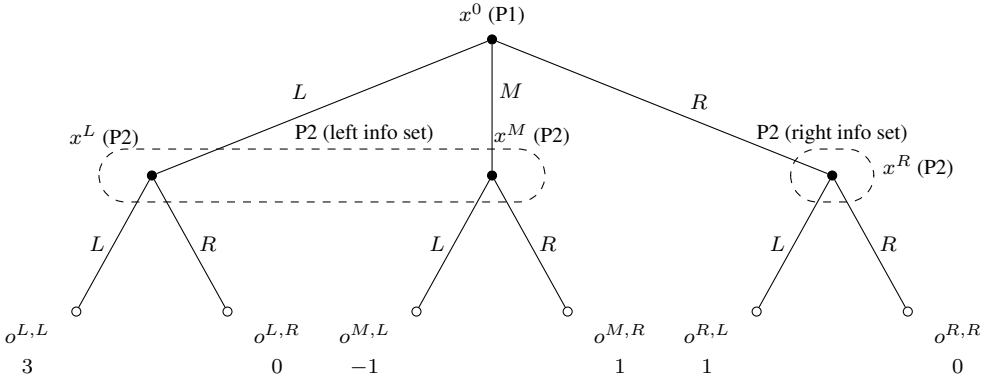


Figure 5.6: Zero-sum extensive-form game. Terminal payoffs are Player 1's utilities.

Example 5.16 Zero-sum extensive-form game with two information sets.

We analyze a two-player zero-sum extensive-form game illustrated in Figure 5.6, in which **P1 is the minimizer** and **P2 is the maximizer**. All terminal payoffs represent P1's utilities; hence, P2's utilities are their negations.

P1 moves first at the root node x^0 , choosing an action $a_1 \in \{L, M, R\}$. Depending on this choice, play transitions to one of P2's decision nodes:

- If $a_1 \in \{L, M\}$, play proceeds to P2's *left information set*, denoted by $\eta_\ell = \{x^L, x^M\}$.
- If $a_1 = R$, play proceeds to the *right information set*, denoted by $\eta_r = \{x^R\}$.

At each information set, P2 chooses between two available actions, L and R . Because P2 has two distinct information sets, denoted η_ℓ (left) and η_r (right), a complete pure strategy must specify an action at each.

The four pure strategies of P2 can be expressed in functional form as

$$s_2^1(\eta^2) = L, \quad (\text{play } L \text{ at both information sets}),$$

$$s_2^2(\eta^2) = R, \quad (\text{play } R \text{ at both information sets}),$$

$$s_2^3(\eta^2) = \begin{cases} L, & \text{if } \eta^2 = \eta_r, \\ R, & \text{otherwise,} \end{cases} \quad (\text{play } L \text{ only at the right information set}),$$

$$s_2^4(\eta^2) = \begin{cases} R, & \text{if } \eta^2 = \eta_r, \\ L, & \text{otherwise,} \end{cases} \quad (\text{play } R \text{ only at the right information set}).$$

For brevity, we introduce the shorthand notation $(L, L), (L, R), (R, L), (R, R)$ to represent these four pure strategies respectively.

Normal-form representation and mixed strategies. The induced normal-form payoff matrix (entries are payoffs to P1) is

	(L, L)	(L, R)	(R, L)	(R, R)
L	3	3	0	0
M	-1	-1	1	1
R	1	0	1	0

Since the game is zero-sum, P1 minimizes the expected payoff while P2 maximizes it. Let p_L, p_M, p_R denote P1's mixed-strategy probabilities for actions $\{L, M, R\}$, and let P2 mix over her four pure strategies $(L, L), (L, R), (R, L), (R, R)$ with probabilities $(\alpha, \beta, \gamma, \delta)$, where $\alpha, \beta, \gamma, \delta \geq 0$ and $\alpha + \beta + \gamma + \delta = 1$.

By inspection, columns (L, R) and (R, R) are dominated for P2, as they yield strictly smaller payoffs than (L, L) and (R, L) , respectively. Hence the game reduces to

	(L, L)	(R, L)
L	3	0
M	-1	1
R	1	1

Let q be the probability that P2 plays (L, L) and $1 - q$ the probability of (R, L) . The expected payoffs to P1 are

$$u_L = 3q, \quad u_M = 1 - 2q, \quad u_R = 1.$$

To determine the mixed-strategy saddle-point value, we determine the q that equalizes u_L and u_M and obtain $q^* = 0.2$, $v = 0.6$. At $q^* = 0.2$, $u_L = u_M = 0.6 < u_R = 1$, and thus P1 randomizes between L and M and never plays R .

To determine P1's equilibrium mixed strategy, we solve for probabilities that equalize P2's payoffs between her two active columns:

$$3p_L - p_M = p_M, \quad p_L + p_M + p_R = 1, \quad p_R = 0.$$

This gives $p_L^* = 0.4$, $p_M^* = 0.6$, $p_R^* = 0$. Hence the mixed-strategy equilibrium is

$$\sigma_1^* = (0.4, 0.6, 0), \quad \sigma_2^* = (0.2, 0, 0.8, 0), \quad v = 0.6.$$

At equilibrium, P1 randomizes between L and M to make P2 indifferent between choosing L at both information sets or switching to R only at the left one. The game value $v = 0.6$ represents P1's guaranteed expected loss and P2's corresponding gain.

Behavioral Strategy Analysis. P2's behavioral strategy can be represented as two conditional stochastic mappings:

At the left information set η_ℓ :

$$\beta_2(\eta^2) = \begin{cases} L, & \text{with probability } q, \quad \text{if } \eta^2 = \eta_\ell, \\ R, & \text{with probability } 1 - q, \quad \text{if } \eta^2 = \eta_\ell. \end{cases}$$

At the right information set η_r :

$$\beta_2(\eta^2) = \begin{cases} L, & \text{with probability } r, \quad \text{if } \eta^2 = \eta_r, \\ R, & \text{with probability } 1 - r, \quad \text{if } \eta^2 = \eta_r. \end{cases}$$

If P1 plays a pure action, the expected payoffs to P1 given (q, r) are obtained directly from the game tree:

$$\pi_L(q) = 3q, \quad \pi_M(q) = 1 - 2q, \quad \pi_R(r) = r + 1 - r = 1.$$

The minimizer P1 selects the action that minimizes the expected payoff. Note that $\pi_R(r) = 1$ is constant, while $\pi_L(q)$ and $\pi_M(q)$ vary with q . Equating $\pi_L(q) = \pi_M(q)$ gives $q^* = 0.2$, which yields $v = 0.6$. At this point, P1 is indifferent between L and M ; since $\pi_R(r) = 1 > v$, the action R is strictly dominated and not played.

Hence, the behavioral equilibrium parameters are $q^* = 0.2$, $r^* = 1$, and the corresponding value of the game is $v = 0.6$.

Determining P1's mixing probability. Let $x = \Pr(L)$ and $y = \Pr(M)$ denote P1's mixing probabilities over her feasible actions, with $\Pr(R) = 1 - x - y$. In equilibrium, we will find that R is never played and thus receives zero probability, but it remains part of the full strategy space. Accordingly, P1's behavioral strategy can be written as the stochastic mapping

$$\beta_1 = \begin{cases} L, & \text{with probability } x, \\ M, & \text{with probability } y, \\ R, & \text{with probability } 1 - x - y. \end{cases}$$

From the behavioral payoff analysis, the expected payoffs to P1 for each pure action are

$$\pi_L(q) = 3q, \quad \pi_M(q) = 1 - 2q, \quad \pi_R(r) = 1.$$

At the equilibrium parameter $q^* = 0.2$, $\pi_L(q^*) = \pi_M(q^*) = 0.6$, $\pi_R(r) = 1$. Since $\pi_R(r)$ yields a strictly larger payoff (and P1 is the minimizer), playing R would increase the loss. Hence, the action R is strictly dominated and receives probability zero in equilibrium: $\Pr(R) = 0$.

Equilibrium mixture over the active actions. Restricting to the active support $\{L, M\}$, let $x = \Pr(L)$ and $1 - x = \Pr(M)$. By the indifference principle, P2 must be indifferent between her active pure strategies (L, L) and (R, L) . Setting P1's expected payoffs equal,

$$3x + (-1)(1 - x) = 0x + 1(1 - x),$$

gives $x^* = 0.4$. Thus the equilibrium strategy of P1 is

$$\beta_1^* = \begin{cases} L, & \text{with probability 0.4,} \\ M, & \text{with probability 0.6,} \\ R, & \text{with probability 0.} \end{cases}$$

The inclusion of R with probability zero ensures the strategy space to remain complete and consistent with the extensive-form representation of the game.

Mixed-Behavioral Equivalence. Consider P2's mixed strategy over her four pure strategies $(L, L), (L, R), (R, L), (R, R)$, denoted by the probability vector

$$(\alpha, \beta, \gamma, \delta), \quad \alpha, \beta, \gamma, \delta \geq 0, \quad \alpha + \beta + \gamma + \delta = 1.$$

Each pure strategy prescribes an action at both information sets: the first component corresponds to the left set η_ℓ , and the second to the right set η_r . Therefore, the local probabilities that P2 plays L at each information set are obtained by summing over the strategies in which L is chosen at that set:

$$q = \alpha + \beta, \quad r = \alpha + \gamma.$$

Since the two information sets η_ℓ and η_r are mutually exclusive in play, the overall outcome distribution and expected payoff depend only on the pair (q, r) . Hence, any mixed strategy that induces the same pair (q, r) is outcome-equivalent to the behavioral strategy β_2 , defined by local probabilities q and r of playing L at the left and the right information sets, respectively.

Constructing an equivalent mixed strategy. Conversely, given any desired behavioral parameters $(q, r) \in [0, 1]^2$, a mixed strategy that reproduces these local probabilities can be constructed as

$$\alpha = \min\{q, r\}, \quad \beta = q - \alpha, \quad \gamma = r - \alpha, \quad \delta = 1 - q - r + \alpha.$$

This ensures that $\alpha + \beta = q$, $\alpha + \gamma = r$, and all components remain nonnegative.

At the equilibrium behavioral parameters $(q^*, r^*) = (0.2, 1)$, the corresponding mixed-strategy probabilities must satisfy $\alpha + \beta = 0.2$, $\alpha + \gamma = 1$, $\alpha + \beta + \gamma + \delta = 1$, $\alpha, \beta, \gamma, \delta \geq 0$. A simple feasible choice satisfying these constraints is $\alpha = 0.2$, $\beta = 0$, $\gamma = 0.8$, $\delta = 0$. Thus, the equilibrium mixed strategy for P2 is $\sigma_2^* = (0.2, 0, 0.8, 0)$, which induces the same behavioral probabilities $(q^*, r^*) = (0.2, 1)$ and achieves the game value $v = 0.6$.

□

Example 5.17 A Two-Person Dynamic Game.

This example previews the dynamic games of Lecture 16; following the control-theoretic convention adopted there; player indices appear as superscripts, as in the control variables v^1 and v^2 and the payoffs u^1 and u^2 . Let the random initial state x_0 be

$$x_0 = \begin{cases} -1, & \text{w.p. } \frac{1}{4}, \\ 0, & \text{w.p. } \frac{1}{2}, \\ 1, & \text{w.p. } \frac{1}{4}. \end{cases}$$

P1, with control variable $v^1 \in \{-1, +1\}$, acts first, moving the system to $x_1 = x_0 + v^1$. Then P2, with control variable $v^2 \in \{-1, +1\}$, acts and updates the state to $x_2 = x_1 + v^2$.

P1 does not observe the realization of x_0 , whereas P2 observes the signal

$$\text{sgn}(x_1) = \begin{cases} -1, & x_1 < 0, \\ 0, & x_1 = 0, \\ +1, & x_1 > 0. \end{cases}$$

Each player $i = 1, 2$ seeks to *maximize*, respectively,

$$u^1 = -g(x_2) - v^1, \quad u^2 = -g(x_2) + v^2,$$

where

$$g(x) = \begin{cases} -1, & x \in \{-3, -1, 2\}, \\ 0, & x \in \{0, 3\}, \\ 1, & x \in \{-2, 1\}. \end{cases}$$

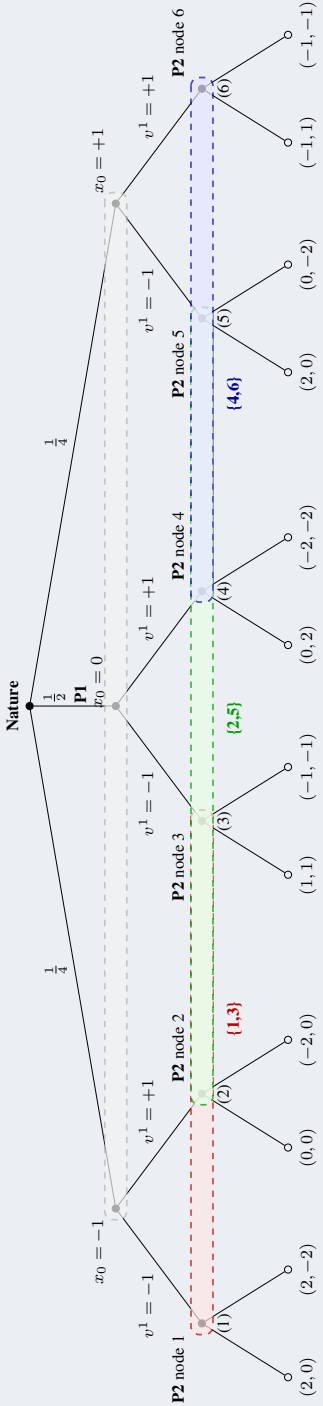


Figure 5.7: Dynamic game tree with shaded, color-coded information sets. Nature chooses $x_0 \in \{-1, 0, +1\}$ with probabilities $(\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$. P1 (gray shaded box) acts without observing x_0 . P2 observes $\text{sgn}(x_1)$, yielding three information sets: $\{1, 3\}$ (red, $\text{sgn}(x_1) = -1$), $\{2, 5\}$ (green, $\text{sgn}(x_1) = 0$), and $\{4, 6\}$ (blue, $\text{sgn}(x_1) = +1$). Leaf nodes show payoffs (u^1, u^2) (sign-reversed from cost).

The dynamics of the game and the associated information structure are illustrated in Figure 5.7. At the top of the tree, Nature randomly selects the initial state $x_0 \in \{-1, 0, +1\}$ with respective probabilities $(\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$. P1 then chooses an action $v^1 \in \{-1, +1\}$ without observing the realization of x_0 , as indicated by the single gray-shaded information set enclosing all three of Nature's successor nodes. This captures P1's uncertainty about the state of the world.

After P1's move, the system transitions to $x_1 = x_0 + v^1$, which determines the observation $\eta^2 = \text{sgn}(x_1)$ available to P2. Based on this signal, P2 selects an action $v^2 \in \{-1, +1\}$, but cannot distinguish between states that yield the same sign of x_1 . Accordingly, P2's decision nodes are grouped into three shaded and color-coded information sets corresponding to $\eta^2 = -1, 0, +1$, shown in red, green, and blue, respectively. Each terminal node represents the resulting pair of payoffs (u^1, u^2) , obtained after sign reversal from the original cost functions.

The figure highlights the sequential and informational asymmetry of the game: P1 acts under complete uncertainty about x_0 , while P2 responds with partial information, observing only the coarse sign of the intermediate state x_1 . P1 gains no information and thus has two pure strategies:

$$s_1^1 = -1, \quad s_2^1 = +1.$$

P2 has eight pure strategies $s_k^2(\eta^2)$, $k = 1, \dots, 8$, where $\eta^2 \in \{-1, 0, +1\}$ denotes the observation $\text{sgn}(x_1)$:

$$\begin{aligned} s_1^2(\eta^2) &= -1, & s_2^2(\eta^2) &= +1, \\ s_3^2(\eta^2) &= \begin{cases} -1, & \eta^2 = \eta_1^2, \\ +1, & \text{otherwise,} \end{cases} & s_4^2(\eta^2) &= \begin{cases} +1, & \eta^2 = \eta_1^2, \\ -1, & \text{otherwise,} \end{cases} \\ s_5^2(\eta^2) &= \begin{cases} -1, & \eta^2 = \eta_2^2, \\ +1, & \text{otherwise,} \end{cases} & s_6^2(\eta^2) &= \begin{cases} +1, & \eta^2 = \eta_2^2, \\ -1, & \text{otherwise,} \end{cases} \\ s_7^2(\eta^2) &= \begin{cases} -1, & \eta^2 = \eta_3^2, \\ +1, & \text{otherwise,} \end{cases} & s_8^2(\eta^2) &= \begin{cases} +1, & \eta^2 = \eta_3^2, \\ -1, & \text{otherwise.} \end{cases} \end{aligned}$$

The average payoffs (u^1, u^2) are:

P1	P2							
	s_1^2	s_2^2	s_3^2	s_4^2	s_5^2	s_6^2	s_7^2	s_8^2
s_1^1	(1, -1)	(1, 1)	($\frac{1}{2}$, -1)	($\frac{3}{2}$, 1)	($\frac{3}{2}$, 1)	($\frac{1}{2}$, -1)	(1, 1)	(1, -1)
s_2^1	(-1, -1)	($-\frac{3}{4}$, $\frac{5}{4}$)	($-\frac{3}{4}$, $\frac{5}{4}$)	(-1, -1)	($-\frac{1}{4}$, $\frac{3}{4}$)	($-\frac{3}{2}$, -1)	($-\frac{3}{2}$, -1)	($-\frac{1}{4}$, $\frac{3}{4}$)

By inspection, $u^1(s_1^1, s_k^2) > u^1(s_2^1, s_k^2)$, $\forall k$, and thus P1's best reply in any Nash equilibrium is $s^{1*} = s_1^1 = -1$. Fixing P1 at s_1^1 , the maximal u^2 in the first row equals 1, attained at $k \in \{2, 4, 5, 7\}$. Hence, the set of *pure-strategy Nash equilibria* is (s_1^1, s_k^2) , $k \in \{2, 4, 5, 7\}$, with equilibrium payoffs

$$u^1 = 1 \text{ for } k = 2, 7, \quad u^1 = \frac{3}{2} \text{ for } k = 4, 5, \quad u^2 = 1 \text{ in all cases.}$$

Behavioral Strategies. Let P2's behavioral strategy be the triple $(p_{-1}, p_0, p_{+1}) \in [0, 1]^3$, where $p_z = \Pr(v^2 = -1 \mid \eta^2 = z)$. Thus, $\Pr(v^2 = +1 \mid \eta^2 = z) = 1 - p_z$. Assuming that P1 plays $v^1 = -1$ surely, at $\eta^2 = -1$ P2 prefers $v^2 = +1$, and thus $p_{-1} = 0$. At $\eta^2 \in \{0, +1\}$, P2 is indifferent, and thus $p_0, p_{+1} \in [0, 1]$. Hence, the behavioral-strategy Nash equilibria satisfy

$$\begin{aligned} v^1 &= -1, & \Pr(v^2 = +1 \mid \eta^2 = -1) &= 1, \\ \Pr(v^2 = +1 \mid \eta^2 = 0), \Pr(v^2 = +1 \mid \eta^2 = +1) &\in [0, 1]. \end{aligned}$$

By Kuhn's theorem, which will be presented in the following Section 5.4, this behavioral set is outcome-equivalent to the mixed strategies supported on the four pure strategies $\{s_2^2, s_4^2, s_5^2, s_7^2\}$, each of which chooses $v^2 = +1$ whenever $\eta^2 = -1$. Therefore, in every Nash equilibrium:

$$v^1 = -1, \quad v^2 = +1 \text{ if } \eta^2 = -1, \text{ and possibly randomized otherwise.}$$

In the reduced normal form, these correspond to (row 1, col k), $k \in \{2, 4, 5, 7\}$, and every mixed equilibrium of P2 is supported on these columns. $\square$

5.4 ■ Kuhn's Theorem and Perfect Recall

Example 5.17 in the previous section illustrated that, for the player with multiple information sets, randomizing independently at each information set (behavioral strategy) produced the same outcome probabilities as randomizing once over all complete plans (mixed strategy). This observation suggests a deeper structural equivalence between the two representations of strategic randomization in extensive-form games. However, such equivalence does not arise automatically; it depends critically on how the game encodes a player's ability to remember her own past actions and the information available when those actions were chosen. The property that guarantees this consistency is known as *perfect recall*. Kuhn's Theorem formalizes this relationship by characterizing the precise conditions under which mixed and behavioral strategies lead to identical outcome distributions.

5.4.1 ■ Absent-Minded Driver Problem: Motivating Example

A well-known example of *imperfect recall* is the *absent-minded driver problem*, in which a driver encounters two identical intersections but cannot recall whether she has already passed the first one or not. Both intersections belong to the same information set, forcing her to use identical action probabilities at each.

Consider the game in Figure 5.8. The driver (a single player) moves at the root and again at two indistinguishable intersections. Let the terminal outcomes be o_0, o_1, o_2, o_3 as labeled in the figure.

Because the two intersections belong to a *single* information set, a pure strategy must prescribe (i) an action at the root, and (ii) a single action to be used at *both* intersection nodes. Thus the driver's pure strategies are the four plans $S = \{RR, RL, LR, LL\}$, where the first letter denotes the action at the root, and the second one denotes the action used at each intersection whenever reached.

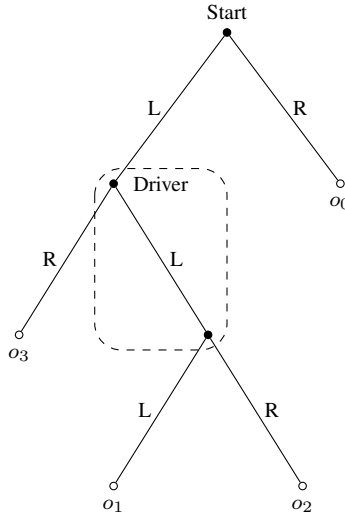


Figure 5.8: Absent-minded driver game: imperfect recall.

Mixed strategies. A mixed strategy is a probability distribution $\sigma \in \Delta(S)$. Let

$$\sigma(RR) = \alpha, \quad \sigma(RL) = \beta, \quad \sigma(LR) = \gamma, \quad \sigma(LL) = \delta, \quad \alpha + \beta + \gamma + \delta = 1.$$

The induced probabilities of terminal outcomes under σ are

$$P_\sigma(o_0) = \alpha + \beta, \quad P_\sigma(o_3) = \gamma, \quad P_\sigma(o_1) = \delta, \quad P_\sigma(o_2) = 0.$$

Explanation. If the root action is R (plans RR or RL), the play ends immediately at o_0 . If the root action is L and the intersection action is R (plan LR), the driver exits at the first intersection, yielding o_3 . If the intersection action is L (plan LL), the driver passes the first intersection and exits at the second, yielding o_1 . The outcome o_2 (exit right at the second intersection) is *unreachable* for any pure plan, and hence for any mixture.

Behavioral strategies. A behavioral strategy assigns independent randomizations at each information set: $r = \Pr(\text{play } R \text{ at root})$, $q = \Pr(\text{play } L \text{ at an intersection})$, where the same q applies to both intersections since they share a single information set. The induced probabilities of terminal outcomes are

$$\begin{aligned} P_{(r,q)}(o_0) &= r, \\ P_{(r,q)}(o_3) &= (1-r)(1-q), \\ P_{(r,q)}(o_1) &= (1-r)q^2, \\ P_{(r,q)}(o_2) &= (1-r)q(1-q). \end{aligned}$$

With probability r , the root action R terminates the game at o_0 . Otherwise, $(1-r)$, the driver takes L and reaches the first intersection. There she plays R with probability $(1-q)$ (ending at o_3) or plays L with probability q , in which case she reaches the second intersection and randomizes again, producing o_1 or o_2 .

Non-equivalence (imperfect recall). The two strategy representations generate different feasible sets of outcome distributions:

$$\{P_\sigma\}_{\sigma \in \Delta(S)} = \{(p_0, p_3, p_1, 0) : p_0, p_3, p_1 \geq 0, p_0 + p_3 + p_1 = 1\},$$

$$\{P_{(r,q)}\}_{(r,q) \in [0,1]^2} = \{(r, (1-r)(1-q), (1-r)q^2, (1-r)q(1-q)) : r, q \in [0, 1]\}.$$

Behavioral strategies can yield $P_{(r,q)}(o_2) > 0$ whenever $q \in (0, 1)$, which is impossible under any mixed strategy, since $P_\sigma(o_2) = 0$ for all σ . Conversely, mixed strategies can produce distributions with $P_\sigma(o_1) > 0$ and $P_\sigma(o_3) > 0$ simultaneously (by mixing between LL and LR), which no behavioral pair (r, q) can replicate without introducing $P_{(r,q)}(o_2) > 0$. Hence, in the absent-minded driver game, where recall is imperfect, mixed and behavioral strategies are *not* outcome equivalent.

Payoff analysis. Assign payoffs $u(o_k) = v_k$ for $k \in \{0, 1, 2, 3\}$. Under the behavioral strategy (r, q) , the expected payoff is

$$\mathbb{E}_{(r,q)}[u] = r v_0 + (1-r)[(1-q)v_3 + q(1-q)v_2 + q^2 v_1].$$

This expression shows how the absent-minded constraint, which requires the same q at both intersections, couples the driver's second-stage decisions and generates the non-linear terms q^2 and $q(1-q)$, which cannot arise from any mixture of pure plans.

Definition 5.18 (Perfect Recall). Let $G = (\mathcal{N}, \mathcal{V}, \mathcal{E}, x^0, (\mathcal{V}_i)_{i \in \mathcal{N} \cup \{0\}}, (p_x)_{x \in \mathcal{V}_0}, (\mathcal{I}_i)_{i \in \mathcal{N}}, O, u)$ be a finite extensive-form game. Player i is said to have perfect recall if, for any two decision nodes $x, y \in \mathcal{V}_i$ that belong to the same information set $\eta_i \in \mathcal{I}_i$, the following two conditions hold:

- (i) The sequence of information sets of player i encountered along the path $(x^0 \rightsquigarrow x)$ is identical to that along $(x^0 \rightsquigarrow y)$.
- (ii) The sequences of actions taken by player i at those prior information sets are also identical.

In other words, player i never forgets what they once knew (their own past information sets) and never forgets what they did (the actions chosen at those sets). Equivalently, at any node $x \in \mathcal{V}_i$, the information set $\eta_i(x)$ uniquely determines player i 's entire personal history of information and actions.

With this definition in place, we can formally state Kuhn's Theorem.

Theorem 5.19 (Kuhn, 1953 [69]). In any finite extensive-form game G in which every player has perfect recall, the following holds: for each mixed strategy σ_i of player i , there exists a behavioral strategy β_i that induces the same probability distribution over terminal histories, and conversely, for each behavioral strategy β_i , there exists a mixed strategy σ_i that is outcome-equivalent.

Kuhn's theorem implies that in any game with perfect recall, the mixed and behavioral representations of randomization are strategically interchangeable: both generate identical distributions over terminal outcomes for every profile of opponents' strategies. Hence, equilibrium analysis can be conducted without loss of generality in terms of behavioral strategies, which are often more convenient in dynamic settings, such as the dynamic extensive-form games of Lecture 16.

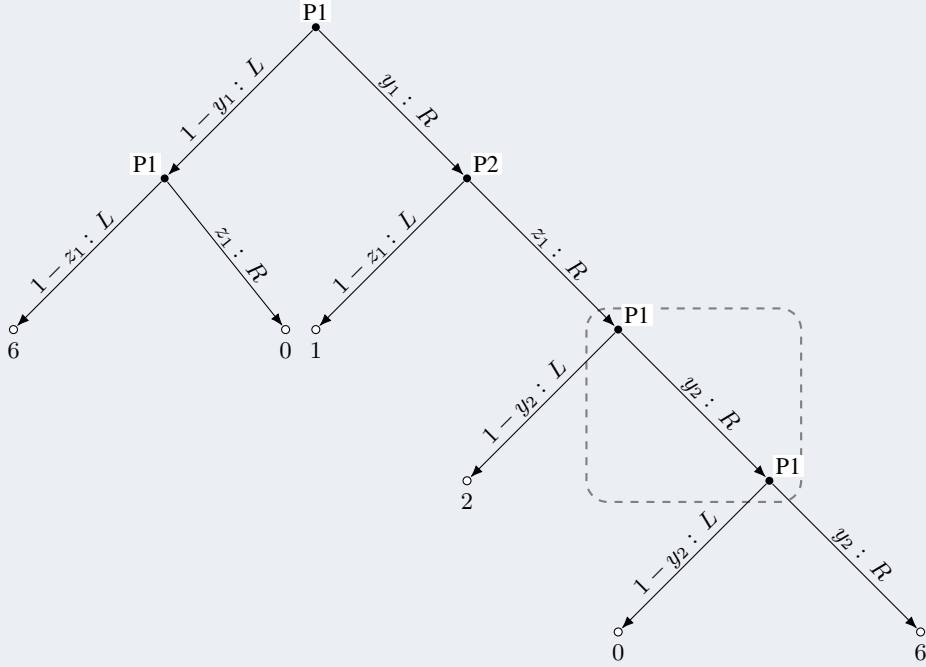
Example 5.20 Illustration of Mixed vs. Behavioral Strategies under Imperfect Recall.


Figure 5.9: Mapping a mixed strategy to an equivalent behavioral strategy under imperfect recall, with decision vertices (solid) and terminal vertices (hollow).

Consider a player who may have to make several sequential decisions. At the initial node, she chooses between two actions, L and R . If R is chosen, the play terminates immediately with a terminal payoff. If L is chosen, the game proceeds to a chance node where another agent (or Nature) selects a branch with probabilities $1 - z_1$ and z_1 . If the left branch ($1 - z_1$) is realized, the game ends; if the right branch (z_1) is realized, the same player acts again at a later decision node, choosing between L and R . The corresponding game tree is shown in Figure 5.9.

Let $y_1 \in [0, 1]$ denote the probability with which the player chooses L at her first decision node, and $y_2 \in [0, 1]$ the probability of choosing L at the later decision node(s), conditional on reaching them. The parameter $z_1 \in [0, 1]$ represents the probability that play transitions from the first L -branch to the right subtree. Tracing each path in Figure 5.9, the expected payoff is

$$\begin{aligned} J(y_1, y_2, z_1) &= 6(1 - y_1)(1 - z_1) + 0(1 - y_1)z_1 \\ &\quad + 1 \cdot y_1(1 - z_1) + 2 \cdot y_1 z_1(1 - y_2) \\ &\quad + 0 \cdot y_1 z_1 y_2(1 - y_2) + 6 \cdot y_1 z_1 y_2^2. \end{aligned}$$

Equivalently, dropping the zero terms and collecting factors, we may write $J(y_1, y_2, z_1) = 6(1 - y_1)(1 - z_1) + y_1(1 - z_1) + 2y_1 z_1(1 - y_2) + 6y_1 z_1 y_2^2$. The

player chooses (y_1, y_2) to minimize expected loss, while the opponent chooses z_1 to maximize it:

$$\max_{z_1} \min_{y_1, y_2} J(y_1, y_2, z_1).$$

Although the tree resembles a standard extensive form with two distinct decision stages for the same player, the dotted box in Figure 5.9 indicates that the two later decision nodes belong to a single information set. When the player moves inside this information set, she cannot tell whether she is encountering it for the first time or has already passed through it once, nor does she remember which action she previously took there. Consequently, the same local randomization y_2 must be used whenever this information set is reached. This violation of *perfect recall*, i.e., the player forgets her own past action and the history that led to the current node, is precisely what distinguishes this example from the standard case covered by Kuhn's theorem.

When perfect recall fails, **Kuhn's Theorem** no longer applies. Behavioral strategies, represented here by local probabilities (y_1, y_2) at the two information sets, need not be equivalent to mixed strategies over complete plans such as $\{LL, LR, RL, RR\}$, where the two letters describe the player's actions on the first and second visits to the later information set. Some mixed strategies induce correlations between the first and second moves at that information set that cannot be replicated by a single local probability y_2 , while some behavioral strategies generate outcome distributions that cannot be obtained from any mixture over pure plans. This example thus illustrates concretely how imperfect recall breaks the mixed-behavioral equivalence guaranteed by Kuhn's theorem in games with perfect recall.

□

5.5 ■ Further Notes and Reading

The notion of equivalent extensive forms, as introduced by F. B. Thompson (1952) [134], represents one of the earliest systematic efforts to formalize when distinct game trees encode the same strategic structure. Developed within the RAND research program, Thompson's analysis anticipates the more general treatment of information sets and strategic equivalence that later appeared in H. W. Kuhn's (1953) seminal work [69] on *Extensive Games and the Problem of Information*. Kuhn's contribution goes beyond issues of representation. By introducing the condition of perfect recall, he established the now-classic equivalence between mixed and behavioral strategies, a result that remains foundational for understanding randomization and consistency in dynamic play.

Subsequent refinements deepened the theoretical structure of extensive-form games. R. Selten [106, 107] extended Kuhn's framework through the definition of subgame-perfect and trembling-hand perfect equilibria, emphasizing the logical consistency of rational behavior at every decision node. These developments marked a transition from questions of representational adequacy to the internal coherence and stability of equilibrium concepts. Later work by J. C. Harsanyi [50] and R. Aumann [5] generalized the extensive-form framework to games of incomplete information, embedding it within an explicit epistemic theory of beliefs and common knowledge. This line of research culminated in the notion of sequential equilibrium due to Kreps and Wilson (1982) [68], which integrates belief consistency with equilibrium reasoning and closes the historical arc

initiated by Thompson's concern for representational equivalence.

Together, these readings trace the evolution of the extensive form from a tool for describing sequential play to a comprehensive language for reasoning about information, belief, and dynamic rationality. They remain essential for readers interested in the conceptual and methodological underpinnings of modern game-theoretic analysis.

Early treatments of extensive-form games were primarily conceptual, emphasizing representation, information structures, and equilibrium logic. Subsequent work introduced computational and algorithmic perspectives that transformed the field from an axiomatic theory into a framework for implementable reasoning. The tree structure of the extensive form naturally supports recursive solution methods, with backward induction producing efficient equilibria in perfect-information games by exploiting the acyclic structure of the game. More generally, structural results on recursive computation of equilibria in both single-controller and multi-controller dynamic games were developed in the context of dynamic and differential games; see Chapter 3 of Başar and Olsder [12] for systematic treatments of single-act and multi-act games based on dynamic programming and fixed-point characterizations.

In contrast, imperfect-information games pose significant computational challenges: equilibrium computation must account for information sets, belief hierarchies, and consistency constraints, leading to an exponential blow-up in the effective state space. Koller and Megiddo (1992) [65] established formal complexity bounds, demonstrating that even two-player zero-sum extensive-form games may require exponential resources when represented explicitly. This observation motivated the development of compact representations, most notably the sequence form introduced by von Stengel (1996) [140], which exploits the linear structure of behavioral strategies to yield polynomial-size formulations for equilibrium computation.

Modern algorithmic advances, such as counterfactual regret minimization (CFR) proposed by Zinkevich et al. (2007) [151] and policy-gradient reinforcement learning methods, extend these recursive and decomposition-based principles to high-dimensional settings, including large-scale poker and multi-agent decision systems. Together, these computational frameworks connect classical game-theoretic reasoning with optimization and learning, turning equilibrium analysis from a primarily conceptual construct into a scalable and practicable tool for contemporary strategic interaction.

5.6 ■ Exercises

Exercise 5.1 (Mixed versus Behavioral Strategies under Imperfect Recall). Consider the two-player zero-sum game of Figure 5.9. The minimizing player chooses R at the root with probability y_1 , and chooses R with probability y_2 at her later information set; because the two later decision nodes belong to a single information set, the same local probability y_2 applies on every visit. The maximizing opponent selects R with probability z_1 at each of his nodes. The resulting expected payoff (a loss to the minimizer) is

$$J(y_1, y_2, z_1) = 6(1 - y_1)(1 - z_1) + y_1(1 - z_1) + 2y_1z_1(1 - y_2) + 6y_1z_1y_2^2.$$

- (i) Show that the minimizer's later information set violates perfect recall, and enumerate her pure strategies as complete plans in $\{L, R\}^2$, where the first component is the action at the root and the second one is the action used on every visit to the later information set.
- (ii) Compute, for each pure plan, the expected payoff as a function of z_1 , and derive the mixed max-min value

$$v_{\text{mix}} = \max_{z_1 \in [0,1]} \min_{\sigma \in \Delta(\{L, R\}^2)} J(\sigma, z_1),$$

together with the maximizing z_1^* and an optimal mixture σ^* over pure plans.

(iii) Derive the behavioral max-min value

$$v_{\text{beh}} = \max_{z_1 \in [0,1]} \min_{(y_1, y_2) \in [0,1]^2} J(y_1, y_2, z_1)$$

by first minimizing J over y_2 for fixed (y_1, z_1) and then over y_1 ; report the optimizing local probability y_2^* and the maximizing z_1^* .

(iv) Implement the computations of parts (ii) and (iii) in Python or MATLAB. Discretize $z_1 \in [0, 1]$ with step 10^{-3} ; for each grid value of z_1 , evaluate the inner minimum over the four pure plans in the mixed case, and over a grid on $(y_1, y_2) \in [0, 1]^2$ with step 10^{-2} in the behavioral case. Plot both inner-minimum curves against z_1 on the same axes and report v_{mix} and v_{beh} to three decimal places. The expected output is a single plot with two curves and two distinct numerical values that agree with parts (ii) and (iii).

(v) Compare v_{mix} and v_{beh} , conclude that mixed and behavioral strategies are not outcome-equivalent in this game, and explain why this conclusion does not contradict Kuhn's theorem.

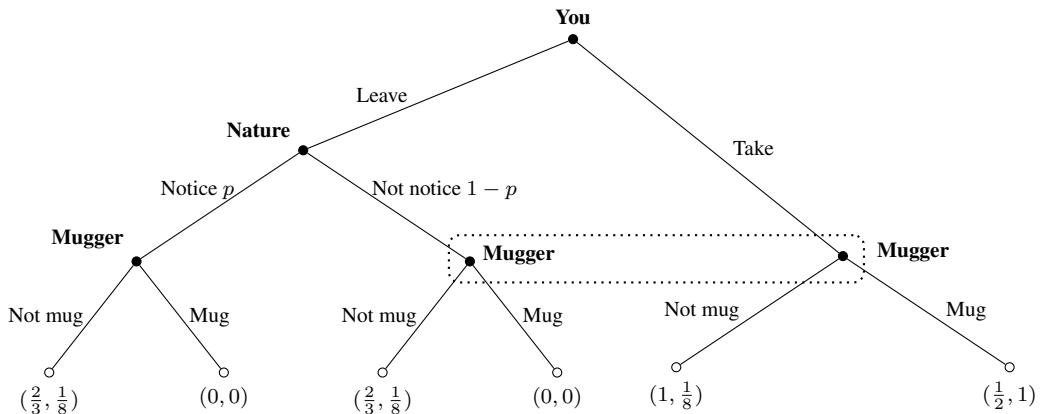


Figure 5.10: Extensive-form game with a clearly visible dotted information set enclosing the two Mugger nodes under *Not notice* and *Take*.

Exercise 5.2 (The Mugging Game with Observation Uncertainty). A pedestrian (“You”) is deciding whether to *Leave* a wallet on a park bench or *Take* it along. A nearby mugger observes your behavior and decides whether to *Mug* or *Not mug*. If you *Leave*, then *Nature* determines whether the mugger *notices* the wallet with probability p , or *does not notice* it with probability $1 - p$. If you *Take* the wallet, the mugger sees you but cannot tell whether you are alert or distracted. This uncertainty is represented by the mugger’s information set connecting the two decision nodes in the figure.

At each of his decision nodes, the mugger chooses between *Mug* and *Not mug*. The payoffs (u_Y, u_M) represent the utilities for You and the Mugger, respectively. If you leave the wallet and the mugger does not mug, you obtain a moderate payoff while the mugger gains a small benefit from inaction. If you take the wallet and the mugger does not mug, you obtain the highest payoff and the mugger receives a small positive return. Mugging results in confrontation or capture, giving both players low or zero payoffs.

The payoffs are summarized below:

Branch	Outcome	(u_Y, u_M)
Leave, Notice, Not mug	Mugger sees wallet but abstains	$(\frac{2}{3}, \frac{1}{8})$
Leave, Notice, Mug	Mugger attacks and is caught	$(0, 0)$
Leave, Not notice, Not mug	Mugger unaware, no attack	$(\frac{2}{3}, \frac{1}{8})$
Leave, Not notice, Mug	Blind mugging attempt	$(0, 0)$
Take, Not mug	You safe, no attack	$(1, \frac{1}{8})$
Take, Mug	Attack with higher risk and reward	$(\frac{1}{2}, 1)$

The extensive-form representation of the game is shown in Figure 5.10. The sequence of plays is as follows:

You $\rightarrow$ Nature (if Leave) $\rightarrow$ Mugger $\rightarrow$ Payoffs.

The mugger's information set indicates that he cannot distinguish between his two decision nodes when you take the wallet or when he fails to notice it.

- Construct the extensive-form game from the description above and verify that it matches Figure 5.10. Label each decision node, chance node, action, and payoff.
- Derive the pure, mixed, and behavioral strategies of each player, and characterize the Nash equilibrium strategies together with the corresponding expected payoffs.
- Analyze how imperfect information affects the equilibrium behavior of the players: determine how increasing p (the probability that the wallet is noticed) changes incentives and outcomes, and interpret the result in terms of deterrence and rational inattention.

Exercise 5.3 (Intrusion Detection Game with Incomplete Information and Chance). A defender (D) protects a critical server that may or may not be under attack. Nature (N) first selects the true state of the world, $\omega \in \{\omega_A, \omega_S\}$, $\Pr(\omega_A) = p$, $\Pr(\omega_S) = 1 - p$, where ω_A represents an *attack state* and ω_S represents a *safe state*.

An intrusion detection system (IDS) generates a noisy signal $s \in \{H, L\}$ observed by the defender but not by the attacker (A), with conditional probabilities

$$\Pr(H \mid \omega_A) = \alpha, \quad \Pr(H \mid \omega_S) = \beta, \quad 0 < \beta < \alpha < 1.$$

After observing s , the defender chooses an action $a_D \in \{I, G\}$, where I denotes *Inspect* and G denotes *Ignore*. If the state is ω_A , the attacker, who observes neither s nor a_D , decides whether to continue (Go) or withdraw (Stop).

The terminal payoffs (u_D, u_A) are specified as follows. Inspection incurs a cost $c > 0$ in either state. Under ω_A with the attacker playing Go: if the defender ignores, the system suffers a loss $L > 0$ and the attacker gains $G_A > 0$, yielding $(-L, G_A)$; if the defender inspects, the attack is detected with probability $\delta \in (0, 1)$, yielding $(-c - k, -L_A)$, where $k > 0$ is the recovery cost and $L_A > 0$ is the attacker's loss from a failed breach, and with probability $1 - \delta$ the attack goes undetected, yielding $(-c - L, G_A)$. If the attacker plays Stop, or if the state is ω_S , the defender receives $-c$ when she inspects and 0 otherwise, and the attacker receives 0.

The sequence of moves is as follows. Nature selects the state ω . The IDS emits a signal s conditional on ω . The defender observes s and chooses an action. If the state is ω_A , the attacker decides whether to continue or withdraw. The defender thus has two information sets, η_H and η_L , corresponding to the two IDS signals. The attacker has a single information set reached only under ω_A .

- (i) Construct the extensive-form game tree, indicating Nature's move, the IDS chance node, and both players' information sets. Label all terminal nodes with payoffs (u_D, u_A) .
- (ii) Derive the pure-strategy sets S_D and S_A of the defender and the attacker, and specify the corresponding mixed-strategy sets $\Delta(S_D)$ and $\Delta(S_A)$.
- (iii) Formulate the defender's and the attacker's behavioral strategies as families of local probability distributions, one for each information set.
- (iv) Show that every mixed strategy of the defender induces local action probabilities at η_H and η_L that are reproduced by an appropriately chosen behavioral strategy.
- (v) Prove that the defender's information structure satisfies perfect recall, and conclude from Kuhn's theorem that mixed and behavioral strategies are outcome-equivalent in this game.
- (vi) Implement, in Python or MATLAB, a program that enumerates the defender's four pure strategies $S_D = \{II, IG, GI, GG\}$ (the first letter is the action after signal H , the second after signal L) and the attacker's two pure strategies $S_A = \{Go, Stop\}$, and computes the induced normal form for the parameter values

$$(p, \alpha, \beta, \delta, c, L, G_A, k, L_A) = (0.3, 0.9, 0.2, 0.8, 1, 10, 5, 2, 4).$$

The expected output is the 4×2 expected-payoff bi-matrix, printed as a table with entries rounded to three decimal places, together with the set of pure-strategy Nash equilibria of the induced normal form, or a statement that none exists.

Exercise 5.4 (Inflation/Deflation of Information Sets and Game Equivalence). Two players move in an extensive form without chance. P1 moves at the root and, depending on her choice, either moves again on the left branch or allows P2 to move on the right branch. After P2's right-branch move, P1 moves again. In *Game 1* (Fig. 5.11), P1's two decision nodes that occur *after* P2 acts are in the same information set (P1 cannot distinguish which of those two nodes she is at). In *Game 2* (Fig. 5.12), we “inflate” the information by merging *all three* of P1's decision nodes into one information set, creating imperfect recall. The terminal labels a, b, c, d, e, f stand for outcomes with payoffs to be specified when needed.

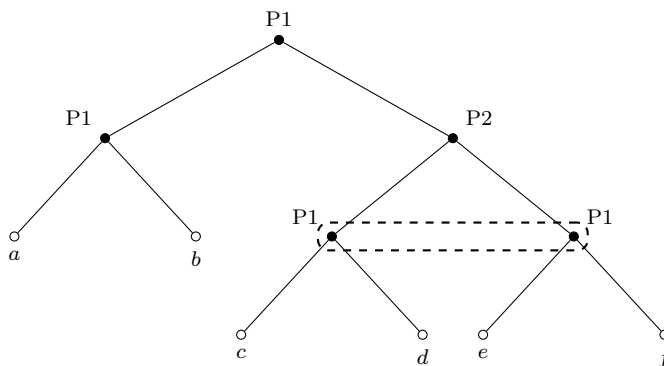


Figure 5.11: Game 1: P1's two right-branch nodes are in the same information set.

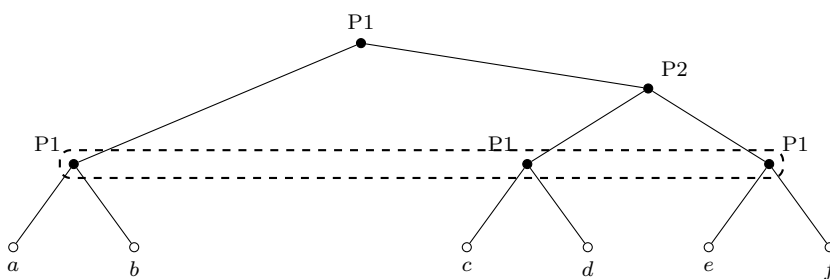


Figure 5.12: Game 2: All three of P1's decision nodes aligned horizontally within a single information set (imperfect recall).

Let Γ denote Game 1 and Γ' denote Game 2.

- (i) Identify the information sets of both players in Γ and Γ' , determine whether each game has *perfect recall*, and justify your answers.
- (ii) Formulate the reduced normal form (strategic form) of Γ and of Γ' , parameterizing the terminal payoffs to P1 as $u(a), u(b), \dots, u(f)$ (with corresponding payoffs for P2), and define the behavioral strategies of P1 at each information set.
- (iii) Compare Γ and Γ' : show that the merged information set in Γ' forces P1 to use a common local randomization at all three of her decision nodes, exhibit a payoff assignment $u(a), \dots, u(f)$ under which an outcome distribution attainable in Γ is not attainable in Γ' , and explain the connection to Kuhn's theorem.

Chapter 6

Stackelberg (Leader-Follower) Games

The Nash equilibrium solution concept studied in the preceding lectures provides a reasonable noncooperative equilibrium solution for nonzero-sum games when the roles of the players are symmetric, that is to say, when no single player dominates the decision process. There are, however, other types of noncooperative decision problems in which one of the players has the ability to enforce her strategy on the other player(s), and for such decision problems one has to introduce a *hierarchical* equilibrium solution concept. Following the original work of von Stackelberg [139], the player who holds the powerful position in such a decision problem is called the *leader*, and the other players who react (rationally) to the leader's decision (strategy) are called the *followers*.

This lecture develops the Stackelberg solution for two-person *finite* games. The development proceeds in two movements. We begin with the *extensive-form* view: a Stackelberg game is, before anything else, a two-stage game of perfect information, and its tree exhibits directly what commitment means and why the follower's contingent plans, rather than his actions, are the right strategic objects. Passing from the tree to the normal form then yields the compact description on which the rest of the lecture operates. The second movement is the *optimization* view: once the follower's rational reaction set is in hand, the leader's problem is a *bilevel* program, an optimization whose constraints are themselves the solution set of another optimization, and the equilibrium concepts, the computational methods, and the pathologies of the subject all follow from that structure.

Along the way we treat equilibria in pure strategies, the role of the follower's rational reaction set, the comparison between Stackelberg and Nash costs, equilibria in mixed and behavioral strategies (including the perhaps surprising fact that the leader can strictly benefit from randomized commitment) and the computation of optimal mixed commitments by linear programming, culminating in two applications of the theory: security games, the model underlying deployed randomized patrol and screening systems, and a case study on the optimal design and defense of networks, where commitment takes the form of built infrastructure and deterrence is achieved through robustness rather than randomization. There are, of course, cases of multiple levels of hierarchy in decision making, with many leaders and followers; for purposes of brevity and clarity in exposition, the core of the lecture treats hierarchical decision problems that incorporate only two players, one leader and one follower. The closing section extends the solution concept to a single leader facing *several* followers who, once the leader has committed to a strategy, play a Nash game among themselves: the leader then no longer merely anticipates a best response but *designs the equilibrium* of the followers' game, a viewpoint that connects Stackelberg theory with the theory of mechanism design.

Remark 6.1 (Cost Convention in This Lecture). Throughout this lecture both players *minimize* costs, in contrast with the payoff-maximization convention used elsewhere in Part I. Two reasons motivate the choice. Hierarchical problems arise most often in engineering and control settings, where the natural primitive is a cost to be minimized; and the continuous-kernel counterpart of this theory, developed in Lecture 10, is written in costs throughout, so that the two lectures read as one development. Every result below transcribes to the maximization convention by replacing each cost J^i with a payoff $u^i = -J^i$ and interchanging the minima and maxima that define the solution concepts.

6.1 ■ The Extensive-Form View

A hierarchical decision problem is, in the first instance, a *sequential* one: the leader acts, the follower sees what she did, and the follower then acts. That is precisely an extensive-form game of the kind developed in Lecture 5, and it is worth beginning there, because the tree makes visible two things that the matrix representation hides. It shows what *commitment* means, namely that the leader's move is made and observed before the follower moves; and it shows that the follower's strategic object is not an action but a *contingent plan*, one action for each thing the leader might have done.

Throughout, P1 is the leader and P2 is the follower, in keeping with the convention used in the remainder of this lecture and in Lecture 10.

Definition 6.2 (Two-Player Stackelberg Game). A two-player Stackelberg game is an extensive-form game

$$\mathcal{G}_S = (\mathcal{N}, \mathcal{V}, \mathcal{E}, x^0, (\mathcal{V}_i)_{i \in \mathcal{N}}, (\mathcal{I}_i)_{i \in \mathcal{N}}, \mathcal{O}, (J^i)_{i \in \mathcal{N}}),$$

where:

- $\mathcal{N} = \{1, 2\}$ is the set of players, P1 being the leader and P2 the follower.
- $(\mathcal{V}, \mathcal{E}, x^0)$ is a finite rooted directed tree of depth two, where x^0 is the unique root node.
- The decision nodes are partitioned as $\mathcal{V}_1 = \{x^0\}$ and $\mathcal{V}_2 = \{x^{a^1} : a^1 \in A^1(x^0)\}$, and there are no chance nodes, i.e., $\mathcal{V}_0 = \emptyset$.
- Each player $i \in \mathcal{N}$ has an action-set correspondence $A^i : \mathcal{V}_i \rightrightarrows \bar{A}^i$ that assigns to each node $x \in \mathcal{V}_i$ a nonempty feasible set $A^i(x) \subseteq \bar{A}^i$, where $\bar{A}^i$ is the underlying alphabet of all possible actions for player i . In particular:
 - At the root node $x^0 \in \mathcal{V}_1$, the leader chooses an action $a^1 \in A^1(x^0) \subseteq \bar{A}^1$.
 - For each $a^1 \in A^1(x^0)$, play moves to a unique follower node $x^{a^1} \in \mathcal{V}_2$, where the follower chooses an action $a^2 \in A^2(x^{a^1}) \subseteq \bar{A}^2$.
- Each directed edge $(x, y) \in \mathcal{E}$ represents a feasible transition resulting from an action taken at node x .
- The set of terminal (outcome) nodes is $\mathcal{O} = \{o(a^1, a^2) : a^1 \in A^1(x^0), a^2 \in A^2(x^{a^1})\}$.
- Each terminal node $o(a^1, a^2) \in \mathcal{O}$ yields a cost vector $J(a^1, a^2) = (J^1(a^1, a^2), J^2(a^1, a^2)) \in \mathbb{R}^2$, which each player seeks to minimize in her own component.
- The information partitions are $\mathcal{I}_1 = \{\eta^1\}$ and $\mathcal{I}_2 = \{\{x^{a^1}\}, a^1 \in A^1(x^0)\}$, implying that the follower perfectly observes the leader's action and hence the game is of perfect information. Thus, a Stackelberg game $\mathcal{G}_S$ is a two-stage extensive-form game of perfect information: the leader chooses $a^1 \in A^1(x^0)$ at the root, the follower observes this choice and selects $a^2 \in A^2(x^{a^1})$, and play terminates at $o(a^1, a^2)$ with costs $J^1(a^1, a^2)$ and $J^2(a^1, a^2)$.

Since the tree has depth two and no chance moves, it is a game of perfect information and therefore of perfect recall, so Kuhn's theorem of Lecture 5 applies: mixed and behavioral strategies are interchangeable here, a point made concrete in Section 6.2.1 and taken up again in Sec-

tion 6.6. The example below, which will serve as the running example of this lecture, exhibits the whole structure on the smallest interesting tree.

Example 6.3 Entry Deterrence: The Tree.

An incumbent firm, P1, can expand its production capacity (E) or not (N). A potential entrant, P2, observes that decision and then enters the market (I) or stays out (O). Expansion is costly to the incumbent, but it makes the post-entry market unprofitable for the entrant. The extensive form is the depth-two tree of Figure 6.1.

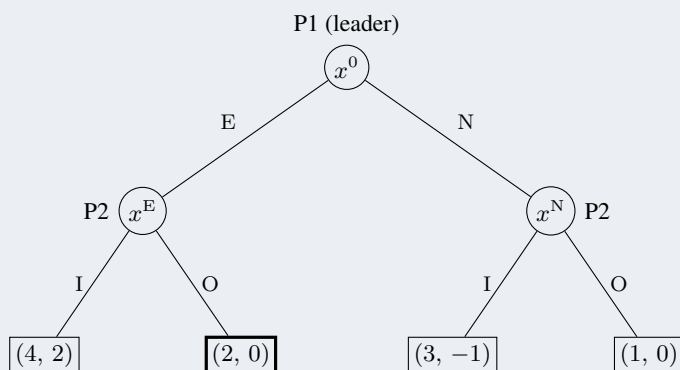


Figure 6.1: Extensive form of the entry-deterrence game. Each terminal node carries the cost pair (J^1, J^2) , and each player minimizes her own component. The Stackelberg outcome $\{E, O\}$, derived in Example 6.4 and Example 6.13, is boxed in bold.

Formally, $\mathcal{N} = \{1, 2\}$ with $A^1(x^0) = \{E, N\}$ and $A^2(x^{a^1}) = \{I, O\}$ for each a^1 , so that $\mathcal{O}$ consists of the four terminal nodes $o(a^1, a^2)$. The costs are collected in the two matrices

$$J^1 = \begin{array}{c|cc} & I & O \\ \hline E & 4 & 2 \\ N & 3 & 1 \end{array}, \quad J^2 = \begin{array}{c|cc} & I & O \\ \hline E & 2 & 0 \\ N & -1 & 0 \end{array},$$

where the row indexes $a^1 \in \{E, N\}$ and the column indexes $a^2 \in \{I, O\}$. Reading the tree from the bottom up: at x^E the entrant compares 0 with 2 and stays out; at x^N she compares -1 with 0 and enters. Anticipating this, the incumbent compares the cost 2 that follows E with the cost 3 that follows N, and expands. This is backward induction, and it is the whole of the Stackelberg solution in the case where the follower's response to each leader action is unique. Sections 6.3 onward develop what must replace it when that uniqueness fails. $\square$

6.2 ■ From the Tree to the Normal Form

The tree of Definition 6.2 can be reduced to a normal-form game in the manner of Lecture 5, and the reduction is worth performing once in full, because it is where the asymmetry between the two players first becomes visible in the notation. Write Γ^1 and Γ^2 for the pure-strategy spaces of P1 and P2; these are the spaces on which the whole of this lecture operates.

The leader moves once, at the root, so her pure strategies are simply her actions, $\Gamma^1 = A^1(x^0)$. The follower moves after observing that action, and each of her information sets $\{x^{a^1}\}$ is a singleton, so a pure strategy for her must prescribe an action at *every* node she might be asked to move at. Her pure strategies are therefore *contingent plans*

$$\gamma^2 : A^1(x^0) \rightarrow \bar{A}^2, \quad \gamma^2(a^1) \in A^2(x^{a^1}),$$

and there are $\prod_{a^1} |A^2(x^{a^1})|$ of them, in general exponentially many in the number of leader actions. The induced costs are

$$J^i(\gamma^1, \gamma^2) = J^i(a^1, \gamma^2(a^1)), \quad i \in \{1, 2\},$$

where a^1 is the action that the leader's pure strategy γ^1 selects. This asymmetry, one action for the leader against a full response schedule for the follower, is the formal residue of commitment, and it is what the solution concepts of the next section are built on.

Example 6.4 Entry Deterrence: The Normal Form.

Continuing Example 6.3, the entrant has four contingent plans, written as ordered pairs (response to E, response to N):

$$\Gamma^2 = \{\text{II}, \text{IO}, \text{OI}, \text{OO}\}.$$

Reading the terminal costs off Figure 6.1 gives the 2×4 normal form

P1	P2			
	II	IO	OI	OO
E	(4, 2)	(4, 2)	(2, 0)	(2, 0)
N	(3, -1)	(1, 0)	(3, -1)	(1, 0)

This single table repays study, because it separates three notions that the 2×2 picture runs together.

Simultaneous play. If the entrant could not observe the incumbent's move, she would choose an action rather than a plan, and the game would be the 2×2 one obtained from the matrices J^1, J^2 of Example 6.3. Its unique Nash equilibrium is $\{N, I\}$ with outcome $(3, -1)$: expansion is not a best response to entry ($4 > 3$), and entry pays against an unexpanded incumbent ($-1 < 0$). The incumbent accommodates.

Nash equilibria of the normal form. In the 2×4 game there are two pure Nash equilibria. One is $\{N, \text{II}\}$ (underlined), reproducing the simultaneous-play outcome $(3, -1)$. The other is $\{E, \text{OI}\}$ (boxed), with outcome $(2, 0)$: against the plan OI the incumbent prefers E ($2 < 3$), and against E the entrant is content to stay out ($0 < 2$). The enlargement of the follower's strategy set from actions to plans has created an equilibrium that did not exist before.

Which of the two survives. The equilibrium $\{N, \text{II}\}$ rests on the plan II, "enter whatever the incumbent does". That plan is not optimal at the node x^E , where entering costs 2 and staying out costs 0: it is a threat the entrant would not carry out. By contrast OI prescribes the optimal action at *both* of the entrant's nodes. Backward

induction on the tree selects $\{E, OI\}$, and this is the Stackelberg solution of Example 6.3. The general principle, that equilibria of the normal form may rely on plans which are irrational off the path actually taken, is the subject of equilibrium refinement in Lecture 7, and reappears as *subgame perfection* in Lecture 16; in games of perfect information such as this one, the Stackelberg solution and the subgame-perfect equilibrium coincide.

Note finally what the leader gains: her Stackelberg cost is 2, against a Nash cost of 3 in the simultaneous game. Capacity expansion *deters* entry, but only under commitment, since once the entrant has stayed out the incumbent would rather not have expanded ($1 < 2$). Section 6.3.1 shows that this improvement is not an accident of the numbers. $\square$

6.2.1 ■ Mixed and Behavioral Strategies on the Tree

Randomization can be introduced in the tree in two different places, and the distinction, familiar from Lecture 5, is worth setting down here because both objects reappear later: mixed strategies in Section 6.4 and behavioral strategies in Section 6.6.

Mixed strategies. A *mixed strategy* for player i is a probability distribution over her pure-strategy set Γ^i . Randomization thus takes place once, before play begins, over *complete plans of action*:

$$\mu^1 \in \Delta(\Gamma^1) = \Delta(A^1(x^0)), \quad \mu^2 \in \Delta(\Gamma^2).$$

Here μ^1 is a lottery over the leader's actions, while μ^2 is a lottery over the follower's contingent plans, that is, over mappings $\gamma^2 : A^1(x^0) \rightarrow \bar{A}^2$. The expected cost to player i is

$$\bar{J}^i(\mu^1, \mu^2) = \sum_{\gamma^1 \in \Gamma^1} \sum_{\gamma^2 \in \Gamma^2} \mu^1(\gamma^1) \mu^2(\gamma^2) J^i(\gamma^1, \gamma^2), \quad i \in \{1, 2\}.$$

Behavioral strategies. A *behavioral strategy* randomizes locally, at each information set, rather than over entire plans. Since $\mathcal{G}_S$ has singleton information sets, the leader's behavioral strategy is a single distribution $\hat{\gamma}^1 \in \Delta(A^1(x^0))$ applied at the root $\eta^1 = \{x^0\}$, and the follower's is a *family* of conditional distributions

$$\hat{\gamma}^2 = \{ \hat{\gamma}^2(\cdot | x^{a^1}) \in \Delta(A^2(x^{a^1})) \}_{a^1 \in A^1(x^0)},$$

one for each node she may be called upon to move at, drawn only after the leader's action has been observed.

The two descriptions are equivalent here. The game has perfect recall, so Kuhn's theorem applies: every mixed strategy is realization-equivalent to a behavioral one and conversely. The example below shows what that equivalence looks like in practice, and why the behavioral description is usually the more economical of the two.

Example 6.5 Entry Deterrence: Mixing over Plans versus Randomizing at Nodes.

Continuing Examples 6.3 and 6.4, suppose the entrant randomizes. A behavioral

strategy for her is a pair of local probabilities,

$$\hat{\gamma}^2(I \mid x^E) = p, \quad \hat{\gamma}^2(I \mid x^N) = q, \quad p, q \in [0, 1],$$

that is, “enter with probability p if the incumbent expands, and with probability q if he does not”. Two numbers describe her completely.

A mixed strategy, by contrast, is a distribution μ^2 over the four plans II, IO, OI, OO, and so lives in a three-dimensional simplex. The behavioral pair (p, q) corresponds to the *product* distribution

$$\mu^2(\text{II}) = pq, \quad \mu^2(\text{IO}) = p(1-q), \quad \mu^2(\text{OI}) = (1-p)q, \quad \mu^2(\text{OO}) = (1-p)(1-q),$$

which is a two-dimensional surface inside that simplex. The mixed strategies off the surface are those that *correlate* the entrant’s behavior at the two nodes; for instance $\mu^2(\text{II}) = \mu^2(\text{OO}) = \frac{1}{2}$, “either enter at both nodes or stay out at both”.

Kuhn’s theorem says that such correlation buys nothing. Only one of the two nodes is ever reached, since the leader’s action determines which, so the joint distribution of the entrant’s play at x^E and x^N never affects any outcome; only the two marginals do. The correlated strategy above is realization-equivalent to the behavioral strategy $p = q = \frac{1}{2}$, which is also the product distribution $\mu^2 = (\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$: against E all three induce entry with probability $\frac{1}{2}$, and against N likewise.

The economy is worth noting. The entrant’s behavioral strategies form a square $[0, 1]^2$, while her mixed strategies form a three-dimensional simplex that collapses onto it. In a tree with m leader actions and n follower actions at each node, the follower has n^m pure plans and hence a mixed-strategy simplex of dimension $n^m - 1$, against $m(n - 1)$ behavioral parameters. This gap is what makes the behavioral description the practical one, and it is the reason Section 6.6 states the behavioral-strategy Stackelberg solution directly in terms of local randomizations. $\square$

6.3 ■ Stackelberg Equilibria in Pure Strategies

The running example has already exhibited the Stackelberg solution in its simplest form, where the follower’s response to each leader action was unique and backward induction settled the matter. To develop the concept in general, and in particular to see what must replace backward induction when that uniqueness fails, we work with a larger bi-matrix game (A, B) , displayed under our standard convention as

$$A = \begin{array}{c|ccc} & \text{L} & \text{M} & \text{R} \\ \hline \text{L} & \boxed{0} & 2 & 3/2 \\ \text{M} & 1 & \underline{1} & 3 \\ \text{R} & -1 & 2 & 2 \end{array}, \quad B = \begin{array}{c|ccc} & \text{L} & \text{M} & \text{R} \\ \hline \text{L} & \boxed{-1} & 1 & -2/3 \\ \text{M} & 2 & \underline{0} & 1 \\ \text{R} & 0 & 1 & -1/2 \end{array} \quad (6.1)$$

where Player 1 (P1), the row player choosing among rows L, M, R, minimizes the entries of A , and Player 2 (P2), the column player, minimizes the entries of B . As stated in Remark 6.1, each player incurs a cost that she seeks to minimize, and all displayed matrices list costs throughout this lecture. This bi-matrix game clearly admits a unique Nash equilibrium in pure strategies, which is $\{M, M\}$ (underlined above), with the corresponding outcome being $(1, 0)$.

Let us now stipulate that the roles of the players are not symmetric and that P1 can enforce her strategy on P2. Then, before announcing her strategy, P1 has to take into account the possible responses of P2 (the follower), and in view of this, she has to decide on the strategy that is most favorable to her. For the decision problem whose possible cost pairs are given as the entries of A and B above, let us work out the reasoning that P1 (the leader) has to go through:

- If P1 chooses L, then P2 has a unique cost-minimizing response, which is L (row L of B is $(-1, 1, -2/3)$, minimized at the first entry), thereby yielding a cost of 0 to P1.
- If P1 chooses M, then P2's response is again unique (namely M), with the corresponding cost incurred to P1 being 1.
- If P1 picks R, then P2's unique response is R, and the cost to P1 is 2.

Since the lowest of these costs is the first one, it readily follows that L is the most reasonable choice for the leader (P1) in this hierarchical decision problem. We then say that L is the *Stackelberg strategy* of the leader (P1) in this game, and the pair $\{L, L\}$ (boxed in (6.1)) is the *Stackelberg solution* with P1 as the leader. Furthermore, the cost pair $(0, -1)$ is the *Stackelberg (equilibrium) outcome* of the game with P1 as the leader. It should be noted that this outcome is actually more favorable for *both* players than the unique Nash outcome $(1, 0)$; this latter feature, however, is not a general rule in such games. If, for example, P2 is the leader in the bi-matrix game (6.1), then (reasoning column by column on A to find P1's unique responses, and then choosing the best column for P2) the unique Stackelberg solution is $\{L, R\}$, with the corresponding outcome $(3/2, -2/3)$, which is clearly not favorable for P1 (now the follower) when compared with her unique Nash cost of 1. For P2 (the leader in this case), however, the Stackelberg outcome $-2/3$ is again better than his Nash outcome 0.

The Stackelberg solution concept introduced above within the context of the bi-matrix game (6.1) is applicable to all two-person finite games in normal form, but provided that they exhibit one feature which was inherent to the bi-matrix game (6.1) and was used implicitly in the derivation: *the follower's response to every strategy of the leader should be unique*. If this requirement is not satisfied, then there is ambiguity in the possible responses of the follower and thereby in the possible attainable cost levels of the leader. As an explicit example of such a decision situation, consider the bi-matrix game

$$A = \begin{array}{c|ccc} & L & M & R \\ \hline L & 0 & 1 & 3 \\ R & 2 & 2 & -1 \end{array}, \quad B = \begin{array}{c|ccc} & L & M & R \\ \hline L & 0 & 0 & 1 \\ R & -1 & 0 & -1 \end{array} \quad (6.2)$$

with P1 (the row player) acting as the leader. Here, if P1 chooses (and announces) L, P2 has *two* optimal responses, L and M (both yielding him a cost of 0), whereas if P1 picks R, P2 again has two optimal responses, L and R. Since this multiplicity of optimal responses for the follower results in a multiplicity of cost levels for the leader for each of her strategies, the Stackelberg solution concept introduced earlier cannot directly be applied here. However, this ambiguity in the attainable cost levels of the leader can be resolved if we stipulate that the leader's attitude is toward *securing* her possible losses against the choices of the follower within the class of his optimal responses, rather than toward taking risks. Then, under such a mode of play, P1's secured cost level corresponding to her strategy L would be $\max\{0, 1\} = 1$, and the one corresponding to R would be $\max\{2, -1\} = 2$. Hence, we declare $\gamma^{1*} = L$ as the unique Stackelberg strategy of P1 in the bi-matrix game (6.2) when she acts as the leader,² and the corresponding *Stackelberg cost* for P1 (the leader) is $J^{1*} = 1$. It should be noted that, in the actual play of the game, P1 could end up with a lower cost level, depending on whether the follower chooses his optimal response $\gamma^2 = L$ or $\gamma^2 = M$. Consequently, the outcome of the game could be either $(1, 0)$ or

²Of course, the "strategy" here could also be viewed as an "action" if what we have is a static game; but since we are dealing with normal forms (which could have an underlying extensive-form description), we will use the term "strategy" throughout, denoted by γ^i for Player i , with the cost to Player i denoted by J^i .

$(0, 0)$, and hence we cannot speak of a unique Stackelberg equilibrium *outcome* of the bi-matrix game (6.2) with P1 acting as the leader. Before concluding our discussion of this example, we finally note that the admissible Nash outcome of the bi-matrix game (6.2) is $(-1, -1)$, attained at $\{R, R\}$, which is more favorable for *both* players than the possible Stackelberg outcomes given above: here leadership hurts.

We now provide a precise definition of the Stackelberg solution concept introduced above within the context of the two bi-matrix games, so as to encompass all two-person finite games of the single-act and multi-act types, which do not incorporate any chance moves. For such a game, let Γ^1 and Γ^2 denote the pure-strategy spaces of P1 and P2, respectively, and let $J^i(\gamma^1, \gamma^2)$ denote the cost incurred to Player i corresponding to a strategy pair $\{\gamma^1 \in \Gamma^1, \gamma^2 \in \Gamma^2\}$.

Definition 6.6 (Rational Reaction Set of the Follower). *In a two-person finite game, the set $R^2(\gamma^1) \subset \Gamma^2$, defined for each $\gamma^1 \in \Gamma^1$ by*

$$R^2(\gamma^1) = \{\xi \in \Gamma^2 : J^2(\gamma^1, \xi) \leq J^2(\gamma^1, \gamma^2), \forall \gamma^2 \in \Gamma^2\}, \quad (6.3)$$

is the optimal response (rational reaction) set of P2 to the strategy $\gamma^1 \in \Gamma^1$ of P1.

Definition 6.7 (Stackelberg Equilibrium Strategy). *In a two-person finite game with P1 as the leader, a strategy $\gamma^{1*} \in \Gamma^1$ is called a Stackelberg equilibrium strategy for the leader if*

$$\max_{\gamma^2 \in R^2(\gamma^{1*})} J^1(\gamma^{1*}, \gamma^2) = \min_{\gamma^1 \in \Gamma^1} \max_{\gamma^2 \in R^2(\gamma^1)} J^1(\gamma^1, \gamma^2) =: J^{1*}. \quad (6.4)$$

The quantity J^{1} is the Stackelberg cost of the leader. If, instead, P2 is the leader, the same definition applies with the superscripts 1 and 2 interchanged.³*

Theorem 6.8 (Existence). *Every two-person finite game admits a Stackelberg strategy for the leader.*

Proof. Since Γ^1 and Γ^2 are finite sets, and $R^2(\gamma^1)$ is a nonempty subset of Γ^2 for each $\gamma^1 \in \Gamma^1$, the inner maximum and the outer minimum in (6.4) are taken over nonempty finite sets, and hence are attained; the result readily follows. $\square$

Remark 6.9 (Nonuniqueness Is Harmless). The Stackelberg strategy for the leader does not necessarily have to be unique. But nonuniqueness of the equilibrium strategy does not create any problem here (as it did in the case of Nash equilibria, which lack the ordered interchangeability enjoyed by saddle-point strategies in the matrix games of Lecture 2), since the Stackelberg *cost* for the leader is unique.

Remark 6.10 (Unique Responses). If $R^2(\gamma^1)$ is a singleton for each $\gamma^1 \in \Gamma^1$, then there exists a mapping $T^2 : \Gamma^1 \rightarrow \Gamma^2$ such that $\gamma^2 \in R^2(\gamma^1)$ implies $\gamma^2 = T^2\gamma^1$. This corresponds to the case in which the optimal response of the follower (T^2) is unique for every strategy of the leader, and it leads to the following simplified version of (6.4):

$$J^1(\gamma^{1*}, T^2\gamma^{1*}) = \min_{\gamma^1 \in \Gamma^1} J^1(\gamma^1, T^2\gamma^1) =: J^{1*}. \quad (6.5)$$

³The inner maximum in (6.4) corresponds to a *pessimistic* attitude on the part of the leader: when the follower is indifferent among several responses, the leader guards against the one least favorable to her. The other extreme is the *optimistic* attitude, in which the follower's indifference is broken in favor of the leader, obtained by replacing that inner maximum with a minimum. The two conventions are also known as the *weak* and *strong* Stackelberg solutions, respectively; we return to the distinction in Section 6.4 and again in Remark 6.36.

Here J^{1*} is no longer only a *secured* equilibrium cost level for the leader; it is the cost level that is actually *attained*.

From the follower's point of view, the equilibrium strategy in a Stackelberg game is any optimal response to the announced Stackelberg strategy of the leader. More precisely:

Definition 6.11 (Stackelberg Solution and Outcome). *Let $\gamma^{1*} \in \Gamma^1$ be a Stackelberg strategy for the leader (P1). Then any element $\gamma^{2*} \in R^2(\gamma^{1*})$ is an optimal strategy for the follower (P2) that is in equilibrium with γ^{1*} . The pair $\{\gamma^{1*}, \gamma^{2*}\}$ is a Stackelberg solution of the game with P1 as the leader, and the cost pair $(J^1(\gamma^{1*}, \gamma^{2*}), J^2(\gamma^{1*}, \gamma^{2*}))$ is the corresponding Stackelberg equilibrium outcome.*

Remark 6.12. In the preceding definition, the realized cost level $J^1(\gamma^{1*}, \gamma^{2*})$ could in fact be *lower* than the Stackelberg cost J^{1*} , a feature already observed within the context of the bi-matrix game (6.2), where the outcome (0, 0) is possible although $J^{1*} = 1$. However, if $R^2(\gamma^{1*})$ is a singleton, then these two cost levels have to coincide.

The following stylized example illustrates why the Stackelberg concept, and the commitment it presupposes, matter economically: a strategy that is worthless without commitment (because it is not part of any Nash equilibrium) can become the leader's optimal choice once it can be announced credibly.

Example 6.13 Entry Deterrence: The Reaction Set.

The running example of Section 6.1 is now readable in the language just introduced. With the incumbent as leader and the entrant as follower, the follower's rational reaction set is single-valued at each leader action,

$$R^2(E) = \{O\}, \quad R^2(N) = \{I\},$$

since staying out costs the entrant 0 against 2 after expansion, and entering costs him -1 against 0 after no expansion. The inner maximum in (6.4) is therefore taken over a singleton and does no work, and the leader simply compares $J^1(E, O) = 2$ with $J^1(N, I) = 3$. The Stackelberg strategy is $\gamma^{1*} = E$, the Stackelberg cost is $J^{1*} = 2$, and the Stackelberg solution is $\{E, O\}$ with outcome (2, 0).

The comparison with the simultaneous game is the substantive point. There the unique Nash equilibrium was $\{N, I\}$ with the incumbent's cost 3: without the power to move first she accommodates entry. Commitment is worth exactly one unit of cost to her here, and Section 6.3.1 shows that a leader whose follower has unique reactions can never do worse under commitment than at a Nash equilibrium. The example is a two-action distillation of the classical capacity-commitment models of entry deterrence. $\square$

6.3.1 ■ Comparison with Nash Costs

For a given two-person finite game, let J^{1*} again denote the Stackelberg cost of the leader (P1), and let J_N^1 denote any Nash equilibrium cost for the same player. We have already seen, within the context of the bi-matrix game (6.2), that J^{1*} is not necessarily lower than J_N^1 , in particular, when the optimal response of the follower is not unique. The following proposition provides one

sufficient condition under which the leader never does worse in a “Stackelberg game” than in a “Nash game.”

Proposition 6.14 (Leadership Cannot Hurt under Unique Responses). *For a given two-person finite game, let J^{1*} and J_N^1 be as defined above. If $R^2(\gamma^1)$ is a singleton for each $\gamma^1 \in \Gamma^1$, then*

$$J^{1*} \leq J_N^1.$$

Proof. Under the hypothesis of the proposition, assume to the contrary that there exists a Nash equilibrium solution $\{\gamma^{1\circ} \in \Gamma^1, \gamma^{2\circ} \in \Gamma^2\}$ whose corresponding cost to P1 is lower than J^{1*} , i.e.,

$$J^{1*} > J^1(\gamma^{1\circ}, \gamma^{2\circ}). \quad (6.6)$$

Since $R^2(\gamma^1)$ is a singleton for every γ^1 , let $T^2 : \Gamma^1 \rightarrow \Gamma^2$ be the unique response mapping introduced in Remark 6.10. Then clearly $\gamma^{2\circ} = T^2\gamma^{1\circ}$ (the follower’s Nash strategy is an optimal response to $\gamma^{1\circ}$), and if this is used in (6.6) together with (6.5), we obtain

$$\min_{\gamma^1 \in \Gamma^1} J^1(\gamma^1, T^2\gamma^1) = J^{1*} > J^1(\gamma^{1\circ}, T^2\gamma^{1\circ}),$$

which is a contradiction. $\square$

Remark 6.15 (The Singleton Condition Cannot Be Relaxed). One might be tempted to think that if a nonzero-sum game admits a unique Nash equilibrium solution and a unique Stackelberg strategy γ^{1*} for the leader, and further if $R^2(\gamma^{1*})$ is a singleton, then the inequality of Proposition 6.14 should still hold. This, however, is not true, as the following bi-matrix game demonstrates:

$$A = \begin{array}{c|cc} & L & R \\ \hline L & \underline{0} & 1 \\ R & -1 & \underline{2} \end{array}, \quad B = \begin{array}{c|cc} & L & R \\ \hline L & \underline{0} & 2 \\ R & \underline{1} & 1 \end{array}.$$

Here there exists a unique Nash equilibrium solution, $\{R, L\}$ (underlined), and a unique Stackelberg strategy $\gamma^{1*} = L$ for the leader (P1): announcing L elicits the unique response L at a cost of 0 to P1, whereas announcing R leaves the follower indifferent between L and R (row R of B is $(1, 1)$), so that P1’s secured cost is $\max\{-1, 2\} = 2$. Furthermore, the follower’s optimal response to $\gamma^{1*} = L$ is unique (namely $\gamma^2 = L$). However, $0 = J^{1*} > J_N^1 = -1$: the leader would rather play the Nash game. The culprit is the multivaluedness of R^2 at the *other* strategy R (the one supporting the favorable Nash outcome), which forces the pessimistic leader to discount it. This counterexample indicates that the sufficient condition of Proposition 6.14 cannot be relaxed any further in any satisfactory way.

6.4 ■ Stackelberg Equilibria in Mixed Strategies

The motivation behind introducing mixed strategies in the investigation of saddle-point and Nash equilibria was that such equilibria do not always exist in pure strategies, whereas within the enlarged class of mixed strategies existence can be ensured. In the case of the Stackelberg solution of two-person finite games, however, an equilibrium always exists in pure strategies (cf. Theorem 6.8), and thus, at the outset, there seems to be no need to introduce mixed strategies. Besides, since the leader dictates her strategy to the follower, it might at first seem unreasonable to imagine that the leader would ever employ a mixed strategy. Such an argument, however, is not always valid, and there are cases in which the leader can actually do better (in the average sense)

with a proper mixed strategy than the best she can do within the class of pure strategies. As an illustration of such a possibility, consider the bi-matrix game (A, B) displayed below:

$$A = \begin{array}{c|cc} & L & R \\ \hline L & 1 & 0 \\ R & 0 & 1 \end{array}, \quad B = \begin{array}{c|cc} & L & R \\ \hline L & 1/2 & 1 \\ R & 1 & 1/2 \end{array}. \quad (6.7)$$

If P1 acts as the leader, then the game admits two pure-strategy Stackelberg solutions, $\{L, L\}$ and $\{R, R\}$, the Stackelberg outcome in each case being $(1, 1/2)$. However, if the leader (P1) adopts the mixed strategy of picking L and R with equal probability $1/2$, then the average cost incurred to P1 equals $1/2$, quite independently of the follower's (pure or mixed) strategy, since each column of A then yields the expected cost $\frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 0 = \frac{1}{2}$. This value $\bar{J}^1 = 1/2$ is clearly lower than the leader's Stackelberg cost $J^{1*} = 1$ in pure strategies. It can further be shown to be the unique Stackelberg cost of the leader in mixed strategies: any deviation from $(1/2, 1/2)$ results in a higher value of $\bar{J}^1$, once the optimal responses of the follower are taken into account. Indeed, if the leader plays L with probability $y_1 > 1/2$, the follower's unique optimal response is L (his expected costs from columns L and R are $1 - y_1/2$ and $1/2 + y_1/2$, respectively), and the leader's average cost is $y_1 > 1/2$; symmetrically, $y_1 < 1/2$ elicits the response R at an average cost of $1 - y_1 > 1/2$.

The preceding result establishes the significance of mixed strategies in the investigation of Stackelberg equilibria of two-person nonzero-sum games, and demonstrates the possibility that a proper mixed-strategy Stackelberg solution could lead to a lower cost level for the leader than the Stackelberg cost level in pure strategies. To introduce the concept of a mixed-strategy Stackelberg equilibrium in mathematical terms, we take the two-person nonzero-sum finite game to be in normal form (without any loss of generality) and associate with it a bi-matrix game (A, B) . Abiding by the earlier notation and terminology, let Y and Z denote the mixed-strategy spaces (probability simplices) of P1 and P2, respectively, with their typical elements denoted by y and z .

Definition 6.16 (Mixed Rational Reaction Set). For a bi-matrix game (A, B) , the set

$$\bar{R}^2(y) = \{z^\circ \in Z : y'Bz^\circ \leq y'Bz, \forall z \in Z\} \quad (6.8)$$

is the optimal response (rational reaction) set of P2 in mixed strategies to the mixed strategy $y \in Y$ of P1.

Definition 6.17 (Mixed Stackelberg Equilibrium Strategy). In a bi-matrix game (A, B) with P1 acting as the leader, a mixed strategy $y^* \in Y$ is called a mixed Stackelberg equilibrium strategy for the leader if

$$\max_{z \in \bar{R}^2(y^*)} y^{*'}Az = \inf_{y \in Y} \max_{z \in \bar{R}^2(y)} y'Az =: \bar{J}^{1*}. \quad (6.9)$$

The quantity $\bar{J}^{1*}$ is the Stackelberg cost of the leader in mixed strategies.

It should be noted that the “maximum” in (6.9) always exists since, for each $y \in Y$, $y'Az$ is continuous in z and $\bar{R}^2(y)$ is a closed and bounded subset of Z (which is a finite-dimensional simplex). Hence $\bar{J}^{1*}$ is a well-defined quantity. The “infimum” in (6.9), however, cannot always be replaced by a “minimum,” unless the game admits a mixed Stackelberg equilibrium strategy for the leader. The following example demonstrates the possibility that a two-person finite game might not admit a mixed Stackelberg strategy even though $\bar{J}^{1*} < J^{1*}$.

Example 6.18 No Mixed Stackelberg Strategy, Yet Mixing Helps.

Consider the following modified version of the bi-matrix game (6.7):

$$A = \begin{array}{c|cc} & L & R \\ \hline L & 1 & 0 \\ R & 0 & 1 \end{array}, \quad B = \begin{array}{c|cc} & L & R \\ \hline L & 1/2 & 1 \\ R & 1 & 1/3 \end{array}.$$

With P1 as the leader, this bi-matrix game also admits two pure-strategy Stackelberg solutions, $\{L, L\}$ and $\{R, R\}$, with the pure-strategy Stackelberg cost for the leader being $J^{1*} = 1$. Now let the leader adopt the mixed strategy $y = (y_1, 1 - y_1)'$, under which the follower's average cost is

$$\bar{J}^2(y, z) = y' B z = \left(-\frac{7}{6} y_1 + \frac{2}{3}\right) z_1 + \frac{2}{3} y_1 + \frac{1}{3},$$

where $z = (z_1, 1 - z_1)'$ denotes any mixed strategy of P2. Minimizing over $z_1 \in [0, 1]$, the mixed-strategy optimal response set of P2 is readily determined as

$$\bar{R}^2(y) = \begin{cases} \{z = (1, 0)'\}, & y_1 > 4/7, \\ \{z = (0, 1)'\}, & y_1 < 4/7, \\ Z, & y_1 = 4/7. \end{cases}$$

Hence, for $y_1 > 4/7$ the follower chooses “column L” with probability 1, which leads to an average cost of $\bar{J}^1 = y_1$ for P1; for $y_1 < 4/7$, P2 chooses “column R” with probability 1, which leads to an average cost of $\bar{J}^1 = 1 - y_1$ for P1. Clearly the leader prefers to stay in the latter region; in fact, if she employs the mixed strategy $y = (4/7 - \epsilon, 3/7 + \epsilon)'$, where $\epsilon > 0$ is sufficiently small, her realized average cost is $\bar{J}^1 = 3/7 + \epsilon$, since P2 then responds with the unique pure strategy $\gamma^2 = R$. Since $\epsilon > 0$ can be taken arbitrarily small, we arrive at the conclusion that

$$\bar{J}^{1*} = \frac{3}{7} < 1 = J^{1*}.$$

In spite of this fact, the leader does *not* have a mixed Stackelberg strategy: for the only candidate $y^\circ = (4/7, 3/7)'$ we have $\bar{R}^2(y^\circ) = Z$, and therefore

$$\max_{z \in \bar{R}^2(y^\circ)} y^{\circ'} A z = \frac{4}{7},$$

which is higher than $\bar{J}^{1*} = 3/7$: exactly at the equalizing strategy, the pessimistic evaluation jumps up, and the infimum in (6.9) is not attained. Figure 6.2 displays the leader's worst-case average cost as a function of her mixing probability, making the discontinuity (and hence the nonexistence) visible at a glance. $\square$

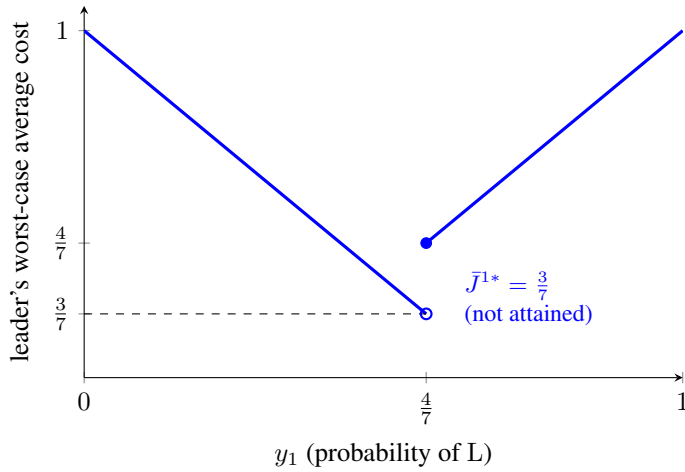


Figure 6.2: The leader's pessimistic (worst-case) average cost in Example 6.18, as a function of her mixing probability y_1 . On $[0, 4/7)$ the follower responds with R and the cost is $1 - y_1$; on $(4/7, 1]$ he responds with L and the cost is y_1 ; at $y_1 = 4/7$ he is indifferent, and the pessimistic evaluation jumps up to $4/7$ (filled dot). The infimum $\bar{J}^{1*} = 3/7$ (open dot) is approached from the left but attained by no strategy, while under the optimistic tie-breaking of the multiple-LP method it is attained exactly at $y_1 = 4/7$.

The preceding example substantiates the possibility that a mixed Stackelberg strategy might not exist for the leader, while she can still do better than her pure Stackelberg cost J^{1*} by employing some suboptimal mixed strategy (such as $y = (4/7 - \epsilon, 3/7 + \epsilon)'$ in the example, for sufficiently small $\epsilon > 0$). In fact, whenever $\bar{J}^{1*} < J^{1*}$, there always exists such an approximating mixed strategy for the leader. If $\bar{J}^{1*} = J^{1*}$, however, it is of course reasonable to employ the pure Stackelberg strategy, which always exists by Theorem 6.8. The following proposition verifies that $\bar{J}^{1*} < J^{1*}$ and $\bar{J}^{1*} = J^{1*}$ are the only two possible relations between $\bar{J}^{1*}$ and J^{1*} ; in other words, the inequality $\bar{J}^{1*} > J^{1*}$ never holds.

Proposition 6.19 (Mixing Never Hurts the Leader). *For every two-person finite game,*

$$\bar{J}^{1*} \leq J^{1*}. \quad (6.10)$$

Proof. Let Y_0 denote the subset of Y consisting of all one-point distributions, and define Z_0 analogously as the set of one-point distributions in Z . Note that Y_0 is equivalent to Γ^1 , and Z_0 to Γ^2 . Then, for each $y \in Y_0$,

$$\min_{z \in Z} y' Bz = \min_{z \in Z_0} y' Bz,$$

since any minimizing solution in Z can be replaced by an element of Z_0 . This further implies that, for each $y \in Y_0$, the elements of $\bar{R}^2(y)$ are probability distributions on $R^2(y)$, where the latter is the pure-strategy reaction set (6.3) (equivalently, (6.8) with Z replaced by Z_0). Now, since $Y_0 \subset Y$,

$$\bar{J}^{1*} = \inf_{y \in Y} \max_{z \in \bar{R}^2(y)} y' Az \leq \min_{y \in Y_0} \max_{z \in \bar{R}^2(y)} y' Az,$$

and further, because of the cited relation between $\bar{R}^2(\cdot)$ and $R^2(\cdot)$ (and since the maximum of the linear function $y' Az$ over probability distributions on $R^2(y)$ is attained at a one-point

distribution) the latter quantity equals

$$\min_{y \in Y_0} \max_{z \in R^2(y)} y'Az,$$

which, by definition, is J^{1*} , since $R^2(y)$ is equivalent to the pure-strategy optimal response set of the follower defined by (6.3). Hence $\bar{J}^{1*} \leq J^{1*}$. $\square$

The value of randomized commitment is not an artifact of abstract payoff tables; it is the operating principle of every audit, patrol, and screening system. The following inspection game is the canonical illustration.

Example 6.20 An Inspection Game.

An inspector (P1, the leader) either inspects (I) at a cost, or does not (N); an inspectee (P2, the follower) either complies (C) or cheats (H). Cheating is very costly to the inspector if undetected, and costly to the inspectee if caught:

$$A = \begin{array}{c|cc} & C & H \\ \hline I & 1 & 0 \\ N & 0 & 4 \end{array}, \quad B = \begin{array}{c|cc} & C & H \\ \hline I & 0 & 3 \\ N & 0 & -1 \end{array}.$$

Pure commitment. Announcing I elicits compliance (row I of B : $0 < 3$) at cost 1 to the inspector; announcing N elicits cheating at cost 4. Hence $J^{1*} = 1$; with deterministic announcements, the inspector must always inspect, and she incurs the inspection cost every time.

Randomized commitment. Let the inspector commit to inspecting with probability y_1 . The inspectee's expected cost from cheating is $3y_1 - (1 - y_1) = 4y_1 - 1$, versus 0 from complying, and hence he complies if and only if $y_1 > \frac{1}{4}$. On that region the inspector's average cost is just the inspection expense y_1 , which she drives down toward the deterrence threshold:

$$\bar{J}^{1*} = \frac{1}{4} < 1 = J^{1*}.$$

Exactly at $y_1 = \frac{1}{4}$ the inspectee is indifferent; under the pessimistic convention he cheats, the inspector's cost jumps to $4 \cdot \frac{3}{4} = 3$, and (as in Example 6.18) no mixed Stackelberg strategy exists; the inspector therefore uses $y_1 = \frac{1}{4} + \epsilon$, inspecting just often enough to keep deterrence strict. The example distills the economics of auditing: the *threat* of inspection, rather than inspection itself, provides the deterrence, and commitment to a randomized schedule allows the leader to secure deterrence at a quarter of the cost of blanket inspection. This is precisely the logic that the Stackelberg security games of Section 6.7 scale up to many targets and scarce defensive resources. $\square$

6.5 ■ The Bilevel View and the Computation of Optimal Commitments

The definitions of the preceding two sections all share one shape. Written out, the leader's problem is

$$\min_{\gamma^1 \in \Gamma^1} \max_{\gamma^2 \in R^2(\gamma^1)} J^1(\gamma^1, \gamma^2), \quad R^2(\gamma^1) = \arg \min_{\gamma^2 \in \Gamma^2} J^2(\gamma^1, \gamma^2), \quad (6.11)$$

an optimization problem *one of whose constraints is the solution set of another optimization problem*. Problems of this shape are called *bilevel*, and reading (6.11) in that light explains most of what has already been observed. The inner $\arg \min$ is a set rather than a point, which is why the outer problem needs the secured, worst-case maximum of Definition 6.7 to be well posed at all. The feasible set of the outer problem, the graph of R^2 , is in general not convex, being a union of pieces indexed by which follower response is optimal, which is why mixed-strategy Stackelberg solutions can fail to exist even though the data are finite (Example 6.18). And the leader's cost is evaluated at a point that another agent chooses, which is why no amount of smoothing of J^1 alone repairs matters.

This section turns that structure into an algorithm. The continuous-kernel counterpart, where the same bilevel programme is treated with the machinery of mathematical programs with equilibrium constraints, is developed in Lecture 10.

Computation of a mixed-strategy Stackelberg equilibrium (whenever it exists) is not as straightforward as in the case of pure-strategy equilibria, since the spaces Y and Z are not finite. The standard technique is first to determine the minimizing solution(s) of

$$\min_{z \in Z} y' B z$$

as functions of $y \in Y$. This leads to a decomposition of Y into subsets (regions), on each of which a reaction set for the follower is defined. (Note that in the analysis of Example 6.18, Y was decomposed into three such regions.) Then one minimizes $y' A z$ over $y \in Y$ subject to the constraints imposed by these reaction sets, and under the stipulation that the same quantity is maximized over the reaction sets whenever they are not singletons. This “brute-force” approach also provides approximating strategies for the leader whenever a mixed Stackelberg solution does not exist, together with the value of $\bar{J}^{1*}$.

The region-by-region structure just described can be organized into an efficient computational scheme, which has become the standard algorithm for computing optimal mixed commitments. The key observation is that the follower, facing a *known* mixed strategy y , always has a pure optimal response; hence Y decomposes into (at most n , the number of P2's pure strategies) polyhedral regions

$$Y_j := \{y \in Y : (B'y)_j \leq (B'y)_k, \forall k\}, \quad j = 1, \dots, n,$$

on which the pure response j is optimal for the follower. On each region one solves the *linear program*

$$(LP_j) : \quad \min_{y \in Y_j} (A'y)_j \quad \Longleftrightarrow \quad \begin{aligned} & \min_y \sum_i y_i A_{ij} \\ & \text{s.t. } \sum_i y_i B_{ik} \leq \sum_i y_i B_{ij}, \forall k, \\ & \quad \sum_i y_i = 1, \quad y \geq 0, \end{aligned} \quad (6.12)$$

i.e., the leader's best mixed strategy *subject to inducing the response j* ; the overall solution is obtained by taking the best among the n optimal values. This *multiple-LP* method is due to Conitzer

and Sandholm [27]. One caveat deserves emphasis, and it takes up the distinction drawn in the footnote to Definition 6.7: on the boundaries between regions the follower is indifferent, and the multiple-LP method implicitly assumes that ties are broken *in favor of the leader* (the “optimistic,” or *strong*, Stackelberg convention), whereas Definition 6.17 embodies the “pessimistic” (*weak*) convention through its inner maximum. Under the strong convention an optimal mixed commitment always exists (each LP_j attains its minimum on a compact polyhedron); under the weak convention it may not, as Example 6.18 showed.

Example 6.21 The Multiple-LP Method on Example 6.18.

For the game of Example 6.18, with $y = (y_1, 1 - y_1)'$:

- (LP_L) : inducing the response L requires $y_1 \geq 4/7$, and the leader's cost is $(A'y)_L = y_1$; the minimum over the region is attained at $y_1 = 4/7$, with value $4/7$.
- (LP_R) : inducing the response R requires $y_1 \leq 4/7$, and the leader's cost is $(A'y)_R = 1 - y_1$; the minimum is again attained at $y_1 = 4/7$, with value $3/7$.

The best of the two values is $3/7$, attained at $y^\circ = (4/7, 3/7)'$ with the follower breaking his indifference in the leader's favor (playing R). Thus the *strong* Stackelberg value $3/7$ is attained exactly where the *weak* value $\bar{J}^{1*} = 3/7$ was only approachable: the two conventions agree in value here, and differ only in whether an optimal strategy exists. This is no coincidence: the strong value always equals the limit of achievable weak values in finite games, and ϵ -optimal strategies under the weak convention can be extracted from the LP solution by perturbing y° into the interior of the winning region, exactly as done in Example 6.18. $\square$

6.6 ■ Stackelberg Equilibria in Behavioral Strategies

If the two-person finite game under consideration is a dynamic game in extensive form, a player can randomize in two operationally different ways: by a *mixed* strategy, a single grand lottery over complete pure strategies (contingent plans), drawn once at the outset and thereafter executed deterministically; or by a *behavioral* strategy, which assigns an independent local randomization to each of the player's information sets, resolved only if and when that information set is reached. Stackelberg equilibria within the class of behavioral strategies are introduced exactly as in Definitions 6.16 and 6.17, by replacing the mixed strategy sets with behavioral strategy sets. Hence, using the earlier terminology and notation, we have the following counterparts of those definitions.

Definition 6.22 (Behavioral Rational Reaction Set). *Given a two-person finite dynamic game with behavioral-strategy sets $(\hat{\Gamma}^1, \hat{\Gamma}^2)$ and average cost functions $(\hat{J}^1, \hat{J}^2)$, the set*

$$\hat{R}^2(\hat{\gamma}^1) = \left\{ \hat{\gamma}^{2^\circ} \in \hat{\Gamma}^2 : \hat{J}^2(\hat{\gamma}^1, \hat{\gamma}^{2^\circ}) \leq \hat{J}^2(\hat{\gamma}^1, \hat{\gamma}^2), \forall \hat{\gamma}^2 \in \hat{\Gamma}^2 \right\} \quad (6.13)$$

is the optimal response (rational reaction) set of P2 in behavioral strategies to the behavioral strategy $\hat{\gamma}^1 \in \hat{\Gamma}^1$ of P1.

Definition 6.23 (Behavioral Stackelberg Equilibrium Strategy). *In a two-person finite dynamic game with P1 acting as the leader, a behavioral strategy $\hat{\gamma}^{1*} \in \hat{\Gamma}^1$ is called a behavioral*

Stackelberg equilibrium strategy for the leader if

$$\sup_{\hat{\gamma}^2 \in \hat{R}^2(\hat{\gamma}^{1*})} \hat{J}^1(\hat{\gamma}^{1*}, \hat{\gamma}^2) = \inf_{\hat{\gamma}^1 \in \hat{\Gamma}^1} \sup_{\hat{\gamma}^2 \in \hat{R}^2(\hat{\gamma}^1)} \hat{J}^1(\hat{\gamma}^1, \hat{\gamma}^2) =: \hat{J}^{1*}. \quad (6.14)$$

The quantity $\hat{J}^{1*}$ is the Stackelberg cost of the leader in behavioral strategies.

With the definitions in hand, the first question to ask of them is whether they contain anything new: is the Stackelberg cost in behavioral strategies ever different from the one in mixed strategies? For games of *perfect recall* (no player ever forgets what she once knew or did), the answer is no.

Proposition 6.24 (Equivalence under Perfect Recall). *In a two-person finite dynamic game in which both players have perfect recall,*

$$\hat{J}^{1*} = \bar{J}^{1*}.$$

In particular, in the canonical two-stage game, in which the leader acts once and the follower responds, the two strategy classes for the leader coincide outright, and the equality is an identity.

Proof. By Kuhn's theorem [69], under perfect recall every mixed strategy of a player admits an *outcome-equivalent* behavioral strategy, and conversely: the two induce the same probability distribution over the terminal nodes of the tree against every (pure, mixed, or behavioral) strategy of the opponent. Since average costs are expectations over terminal nodes, an outcome-equivalent pair is indistinguishable in both players' cost functions. Fix a mixed strategy y of the leader and an outcome-equivalent behavioral strategy $\hat{\gamma}^1$. The pairs of cost levels the follower can attain against y within his mixed class coincide with those he can attain against $\hat{\gamma}^1$ within his behavioral class, and hence the reaction sets $\bar{R}^2(y)$ and $\hat{R}^2(\hat{\gamma}^1)$ correspond element by element up to outcome equivalence, and the leader's secured cost is the same under both:

$$\max_{z \in \bar{R}^2(y)} y'Az = \sup_{\hat{\gamma}^2 \in \hat{R}^2(\hat{\gamma}^1)} \hat{J}^1(\hat{\gamma}^1, \hat{\gamma}^2).$$

Since, again by Kuhn's theorem, every behavioral strategy of the leader arises, up to outcome equivalence, from some mixed strategy and conversely, the two classes generate the same set of secured cost levels, and taking infima yields $\hat{J}^{1*} = \bar{J}^{1*}$. As for the two-stage case: a player with a single information set has behavioral and mixed strategies that are literally the same objects, namely randomizations over her actions, which settles the claim for the leader; the follower's two classes do differ as objects (a mixed strategy for him is a lottery over reaction functions), but he moves knowing everything and therefore always has a *pure* optimal response, and hence the distinction is immaterial on his side as well. $\square$

Proposition 6.24 says that, at the level of attainable values, the behavioral formulation adds nothing to Section 6.4: under perfect recall it is a *reparameterization* of randomized commitment, not an enlargement of it. Three observations nevertheless make the behavioral viewpoint worth having, and they, rather than the definitions above, are the real content of this section.

First, *implementation*. A mixed strategy asks the leader to draw a complete contingent plan at time zero and to carry its realization in memory for the rest of the game; the outcome-equivalent behavioral strategy is executed by local coin flips, tossed at each information set only when it is reached. The deployed security systems of Section 6.7 operate in exactly this mode: no one samples a year of patrol schedules in advance; the randomization is re-executed day by day.

Proposition 6.24 is what licenses computing the optimal commitment in whichever parameterization is convenient (the normal form, via the linear programs of Section 6.5) and implementing it in the other, with no loss of commitment value.

Second, *the caveat*. Perfect recall is load-bearing. When the “leader” is in fact a team of agents pooled into a single player (a patrol force whose units cannot communicate during the day, a distributed controller), the pooled player has imperfect recall, Kuhn’s theorem fails, and behavioral strategies can be strictly weaker than mixed ones: the grand lottery correlates the units’ actions in ways that no profile of independent local lotteries can reproduce. The equality of Proposition 6.24 must then be given up, and which class is the right model becomes a substantive question about what can be centrally drawn and distributed in advance.

Third, and most instructive, *what is randomized must not be confused with what is observed*. Model the act of commitment itself as a move: write the two-stage interaction as an extensive-form game whose root move is the leader’s announcement, observed by the follower before he responds. A behavioral randomization *at the root of that game* is a lottery over announcements: the follower sees the realized announcement and reacts to it, and hence the leader’s secured cost is

$$\sum_{\gamma^1 \in \Gamma^1} y(\gamma^1) \max_{\gamma^2 \in R^2(\gamma^1)} J^1(\gamma^1, \gamma^2) \geq J^{1*},$$

an average of pure-commitment values that can never beat the best of them. The strict improvements of Section 6.4 arise only because the follower there observes the *distribution itself* and best-responds to it before it is resolved. In the inspection game of Example 6.20, secretly flipping a coin between the announcements “always inspect” and “never inspect” costs the inspector $y \cdot 1 + (1 - y) \cdot 4 \geq 1$, whereas committing to the publicly observable inspection *probability* $\frac{1}{4} + \epsilon$ costs about $\frac{1}{4}$. Randomizing over commitments is worthless; committing to a randomization is where the value lies.

The genuinely new phenomena in dynamic games thus concern not *how* the leader randomizes but *what* she commits to, and when; this is the subject of the closing remark.

Remark 6.25 (What Exactly Is Committed To?). In dynamic games, the hierarchical solution concept forks according to *what* the leader commits to. Under the *global* Stackelberg solution implicit in Definition 6.23, the leader announces, once and for all, an entire behavioral strategy (a contingent plan covering all her information sets), and the follower optimizes against this announcement. Alternatively, leadership may be exercised *stagewise*: at each stage the leader moves first in that stage only, without committing to future behavior; this yields the *feedback* Stackelberg solution, which is time-consistent but generally leaves the leader with a higher cost than global commitment. The interplay between these two notions, together with the striking ability of a leader who commits to *strategies as functions of the follower’s observed actions* to shape the follower’s incentives (so-called incentive, or inverse Stackelberg, designs), is a central theme of Lecture 22 on dynamic Stackelberg games and incentives. The credibility question raised by global commitment (whether the leader would actually carry out her announced plan at information sets that equilibrium play never reaches) is the same one that motivates Selten’s subgame-perfectness refinement of Nash equilibrium [107]; the feedback Stackelberg solution is, in this sense, the hierarchical sibling of subgame perfection.

6.6.1 ■ Relation to Mixed and Behavioral Equilibria

It is worth setting the Stackelberg solution beside the two equilibrium notions of Lecture 5 from which its ingredients are drawn. Like a behavioral equilibrium, it specifies play at the level of information sets and rests on best responses given what a player observes. It differs in imposing

one further layer of structure: the leader does not merely best respond to the follower, she *anticipates* his best response and optimizes against it. In this sense the Stackelberg solution refines the notion of equilibrium by enforcing optimality subgame by subgame, which is why, in the perfect-information tree of Section 6.1, it selected $\{E, OI\}$ and rejected the Nash equilibrium $\{N, II\}$ that rested on a threat the follower would not carry out. The general notion of refinement is taken up in Lecture 7, and subgame perfection in Lecture 16, where the Stackelberg solution of a perfect-information game is shown to coincide with its subgame-perfect equilibrium.

Two asymmetries with the Nash theory deserve emphasis. A pure-strategy Stackelberg solution always exists in a finite game, by Theorem 6.8, since backward induction terminates; no randomization is needed for existence, in sharp contrast with Nash equilibria. But the mixed-strategy Stackelberg solution need not exist at all (Example 6.18), and whether it does depends on whether the pessimistic or the optimistic convention is adopted for the follower's tie-breaking, as discussed in the footnote to Definition 6.7. Randomization, which the Nash theory needs for existence and which the Stackelberg theory does not, instead buys the leader something else: a strictly lower cost, as Example 6.18 also showed.

6.7 ■ An Application: Stackelberg Security Games

The mixed-commitment theory of Section 6.4 and the linear-programming machinery of Section 6.5 together constitute the mathematical core of *Stackelberg security games*, a class of models which, over the past two decades, has moved from the research literature into daily operational use at airports, on flights, and in harbors. The setting is that of a *defender* (P1, the leader) who must protect a set of n targets (terminals, flights, patrol areas) with $m < n$ security resources, against an *attacker* (P2, the follower) who selects a single target to attack. Two features of the security domain make the Stackelberg solution in mixed strategies, rather than any other solution concept of the preceding lectures, the appropriate model. First, leadership is not a modeling choice here but a fact of the environment: an attacker conducts surveillance over days or weeks before acting, and thereby observes the defender's *policy* (the long-run frequencies of her patrols and checkpoints) though not, on the day of the attack, its realization. The defender is thus committed to a mixed strategy that the follower observes before responding, which is precisely the informational structure of Definition 6.17. Second, commitment to a *pure* strategy is useless at best: any deterministic schedule leaves $n - m$ targets predictably unprotected, and the attacker simply strikes the best of them. Randomization is therefore not a refinement but a necessity; as in the inspection game of Example 6.20, the leader's power resides entirely in the calibrated *probabilities* of protection.

Definition 6.26 (Security Game). A security game is a two-person finite game specified by a set of targets $T = \{1, \dots, n\}$, an integer $m < n$ of identical, interchangeable defensive resources, and, for each target $t \in T$, four numbers satisfying

$$d_c(t) < d_u(t), \quad a_u(t) < a_c(t).$$

A pure strategy of the defender (P1) is a subset $S \subseteq T$ of targets to cover, with $|S| \leq m$; a pure strategy of the attacker (P2) is a single target $t \in T$. The incurred costs depend only on the attacked target and on whether it is covered:

$$J^1(S, t) = \begin{cases} d_c(t), & t \in S, \\ d_u(t), & t \notin S, \end{cases} \quad J^2(S, t) = \begin{cases} a_c(t), & t \in S, \\ a_u(t), & t \notin S. \end{cases}$$

The two strict inequalities say that coverage of the attacked target benefits the defender and

harms the attacker; beyond this, the cost structure is unrestricted.⁴ In particular, the game is *not* assumed to be zero-sum (and in applications it never is, since the two players value targets differently), so that, by the discussion in Section 6.10 below, commitment has genuine value here.

Written out as a bi-matrix game, a security game has one row for each of the $\sum_{k \leq m} \binom{n}{k}$ coverage subsets: the normal form is exponentially large, and the multiple-LP method of Section 6.5, applied literally, would require solving n linear programs, each with exponentially many variables. The special structure of Definition 6.26, however, collapses this. Since the costs depend on S only through the coverage status of the attacked target, a mixed strategy $y = (y_S)_S$ of the defender enters both players' expected costs only through the induced *coverage vector* $x = (x_1, \dots, x_n)$,

$$x_t = \sum_{S: t \in S} y_S, \quad t \in T, \quad (6.15)$$

whose t th entry is the probability that target t is covered: if the attacker strikes target t , the expected costs are

$$\bar{J}^1 = x_t d_c(t) + (1 - x_t) d_u(t), \quad \bar{J}^2 = x_t a_c(t) + (1 - x_t) a_u(t). \quad (6.16)$$

The coverage vectors realizable through (6.15) form the polytope

$$X = \left\{ x \in [0, 1]^n : \sum_{t \in T} x_t \leq m \right\}, \quad (6.17)$$

and (crucially) *every* element of X is realizable, so that the defender may optimize directly over X in place of the exponential simplex of mixtures.

Remark 6.27 (Implementing Marginal Coverage). That every $x \in X$ arises from some mixed strategy over coverage subsets can be seen by a classical splitting argument: lay intervals of lengths $x_1, \dots, x_n$ end to end on the segment $[0, m]$, cut the segment into m unit-length pieces, draw a single uniform random variable θ on $[0, 1]$, and cover those targets whose intervals contain a point of $\{\theta, \theta+1, \dots, \theta+m-1\}$. Since $x_t \leq 1$, no target is selected twice; at most m targets are selected; and target t is covered with probability exactly x_t . The argument uses the assumption that the m resources are interchangeable and unconstrained. When, instead, each resource must execute a *schedule* covering several targets at once (an air marshal flies a round-trip itinerary; a patrol boat follows a route), not every marginal vector is implementable, and both the reduction and its computational benefits can fail; see the notes at the end of this section.

The multiple-LP method now specializes as follows. The follower's pure strategies are the n targets, and hence there are n response regions and n linear programs; and, by (6.16), each program may be posed directly over the coverage polytope X : for each $t \in T$, solve

$$\begin{aligned} (\text{SLP}_t) : \quad & \min_{x \in X} x_t d_c(t) + (1 - x_t) d_u(t) \\ & \text{s.t. } x_t a_c(t) + (1 - x_t) a_u(t) \leq x_{t'} a_c(t') + (1 - x_{t'}) a_u(t'), \quad \forall t' \in T, \end{aligned} \quad (6.18)$$

i.e., the defender's best coverage *subject to steering the attack to target t* ; the optimal commitment is the best of the n optimal values, under the strong (optimistic) convention of Section 6.5. The exponential objects have disappeared: what remains are n linear programs in n variables each, solvable at the scale of thousands of targets [63]. At the optimum, the constraints of (6.18)

⁴The security-games literature states the model in terms of utilities to be maximized; we retain here the cost (minimization) convention of this lecture.

typically bind at several targets: on the *attack set* (the set of targets at which the attacker's expected cost is minimal) the optimal coverage *equalizes* the attacker's expected costs, in exactly the way the inspector of Example 6.20 tuned her inspection probability to the deterrence threshold.

Example 6.28 A Randomized Patrol.

A defender with a single patrol unit ($m = 1$) protects $n = 3$ targets; an attack on a covered target is fully thwarted ($d_c(t) = 0$ for every t), and the remaining data are

	$d_u(t)$	$a_u(t)$	$a_c(t)$
$t = 1$	10	-5	5
$t = 2$	6	-3	3
$t = 3$	4	-1	1

so that both players rank target 1 highest and target 3 lowest.

Pure commitment fails. If the defender announces coverage of target 1, the attacker strikes target 2 (his best uncovered option: $-3 < -1$), at a cost of 6 to the defender; announcing coverage of target 2 or of target 3 sends the attacker to target 1, at a cost of 10. Hence $J^{1*} = 6$: a deterministic schedule protects one target and concedes the best of the rest.

Randomized commitment. Under a coverage vector $x = (x_1, x_2, x_3) \in X$, the attacker's expected costs at the three targets are, by (6.16),

$$10x_1 - 5, \quad 6x_2 - 3, \quad 2x_3 - 1.$$

Consider (SLP₃): the objective $4(1 - x_3)$ decreases in x_3 , while raising x_3 makes target 3 less attractive and forces x_1 and x_2 upward through the steering constraints; at the optimum all three constraints bind ($x_1 + x_2 + x_3 = 1$ and the attacker's three expected costs are equal), which yields the unique optimizer

$$x^* = \left(\frac{10}{23}, \frac{9}{23}, \frac{4}{23} \right),$$

under which the attacker's expected cost equals $-\frac{15}{23}$ at *every* target (the attack set is all of T) and the defender's cost, when target 3 is attacked, is $4(1 - \frac{4}{23}) = \frac{76}{23} \approx 3.3$. The other two programs turn out to have the same unique optimizer x^* ; the three programs differ only in which member of the attack set the indifferent attacker is presumed to select, with values $\frac{130}{23}$ (target 1) and $\frac{84}{23}$ (target 2). The strong Stackelberg cost is therefore $\frac{76}{23}$, attained at x^* with the tie broken toward target 3.

Three features of the solution deserve notice. (i) *Coverage follows value, but the attack is steered to the cheapest loss*: $x_1^* > x_2^* > x_3^*$, and yet the intended attack is on target 3, the defender protects the valuable targets heavily precisely so that the attack lands where it hurts least. (ii) *Strong versus weak, once more*: at x^* itself the pessimistic evaluation is $\frac{130}{23}$, but by shifting a probability mass ϵ of coverage from target 3 to targets 1 and 2 the defender makes target 3 the attacker's *strictly* preferred choice at a cost of $\frac{76}{23} + 4\epsilon$; hence the weak Stackelberg cost is again the same value $\frac{76}{23}$, approached but not attained, exactly the situation of Examples 6.18 and 6.21. (iii) *The value of randomization*: the mixed commitment nearly halves the pure-commitment cost, $\frac{76}{23} \approx 3.3$ against $J^{1*} = 6$. $\square$

Remark 6.29 (Does the Attacker Really Observe?). The Stackelberg model presumes that the attacker observes the defender’s randomized policy before responding, an assumption that surveillance makes plausible but that can never be verified in the field. A structural result of Korzhyk et al. [66] removes much of the worry: in security games of the form of Definition 6.26 (interchangeable resources, no schedule constraints), every strong Stackelberg strategy of the defender is also her strategy in some Nash equilibrium of the *simultaneous-move* game. The optimal randomized schedule is thus defensible whether or not the surveillance actually took place (commitment cannot hurt), although the *realized* cost may differ, since an unobserving, indifferent attacker need not break ties in the defender’s favor. This robustness is a property of the security-game structure, not of Stackelberg games at large: in general bi-matrix games the leader’s optimal commitment and her Nash strategies can be entirely unrelated, as the examples of Section 6.3 show.

The passage from Example 6.28 to a deployed system is a matter of scale and enrichment rather than of concept. Resources with schedule constraints (itineraries, routes) void the marginal-coverage reduction of Remark 6.27, and computing the optimal commitment can then become NP-hard; uncertainty about the attacker’s preferences is handled by Bayesian formulations over attacker *types*, solved by mixed-integer programming [93]; and robust variants hedge between the strong and weak conventions and against imperfect attacker rationality. Systems built on these foundations include ARMOR, which has randomized checkpoint placement and canine patrols at Los Angeles International Airport since 2007; IRIS, which schedules the federal air marshals; and PROTECT, which randomizes the coast guard’s harbor patrols [124]. Each computes, at heart, the object defined in this lecture: a mixed Stackelberg equilibrium strategy for the leader.

6.8 ■ A Case Study: Stackelberg Network Design and Defense

In a security game, the leader’s commitment is informational: a randomized policy whose long-run frequencies the attacker learns through surveillance, and whose entire power resides in its unpredictability. In many infrastructure problems the commitment is of the opposite, physical kind. The leader *builds* a network (a communication backbone among production sites, a supply network, a power grid), and the built object is there for the adversary to inspect at leisure before striking; nothing about it can be kept random. Deterrence must then be achieved not through unpredictability but through *robustness*: the network is designed so that, once it stands, no attack within the adversary’s means is worth mounting. This section presents a case study of this second kind of Stackelberg reasoning, the network design and defense game of Bravard, Charroin, and Touati [20], whose main result identifies, in closed form, three “polar” architectures that can arise at equilibrium, together with the simple cost-ratio comparisons that select among them.

The model. A *Designer* (P1, the leader) operates $n \geq 5$ facilities, the nodes $N = \{1, \dots, n\}$. The nodes are *complementary*: think of production units each manufacturing one part of a single product, so that the operation generates its value (normalized to 1) only if every node can communicate with every other one. At the first stage, the Designer chooses which pairs of nodes to join by links, and builds each link in one of two grades: *non-protected*, at cost c_L per link, or *protected*, at cost c_P per link, where $0 < c_L < c_P$. A protected link cannot be severed by the adversary (a buried or hardened line); a non-protected link can. At the second stage, an *Adversary* (P2, the follower) observes the built network $g = (E_P, E_{NP})$, where E_P and E_{NP} denote the sets of protected and non-protected links, and removes a set $E^A \subseteq E_{NP}$ of non-protected links, at cost c_A per attacked link. The *residual network* g^R is g with the links of E^A deleted.

Retaining the cost-minimization convention of this lecture, the players' costs are

$$\begin{aligned} J^1(g, E^A) &= c_P |E_P| + c_L |E_{NP}| + \mathbb{1}\{g^R \text{ is disconnected}\}, \\ J^2(g, E^A) &= c_A |E^A| - \mathbb{1}\{g^R \text{ is disconnected}\} : \end{aligned} \quad (6.19)$$

the Designer pays for what she builds and loses the (unit) value of the operation if the residual network is disconnected; the Adversary pays for what he attacks and gains that unit if he succeeds in disconnecting. Three standing assumptions keep the problem nontrivial: $c_P < 1/(n-1)$ and $c_L < 2/(n(n-1))$, so that connectivity is worth buying at either grade; and $c_A \in [1/(n-3), 1)$, so that the integer

$$k := \lfloor 1/c_A \rfloor, \quad 1 \leq k \leq n-3, \quad (6.20)$$

is the largest number of links the Adversary could ever profitably attack (an attack on j links can improve on inaction only if it disconnects *and* $c_A j < 1$, i.e., $j \leq k$).

The follower's reaction, and a reduction. Say that g *resists* k attacks if no removal of k or fewer non-protected links disconnects it. The follower's rational reaction set (6.3) is then immediate: if g resists k attacks, every nonempty attack is money thrown away, and $R^2(g) = \{\emptyset\}$ is a *singleton*; if g does not resist k attacks, the Adversary's best responses are the cheapest disconnecting attacks, and the Designer's realized cost exceeds 1, which is worse than building nothing at all. Under the standing assumptions, resistant networks cheaper than 1 exist (Theorem 6.30 exhibits them), and hence the leader's problem (6.4) reduces to a combinatorial design problem:

build, at minimal cost, a network that survives every attack on k links.

Two contrasts with the security games of Section 6.7 are worth recording before solving it. First, on the designs that matter the follower's optimal response is unique, and hence by Proposition 6.14 leadership cannot hurt, and no secured-cost subtleties arise. Second, randomization has no role: the Adversary responds to the *realized* network, not to the distribution from which it was drawn, and hence a mixed commitment offers the leader nothing; the pure-strategy theory of Section 6.3 suffices. Where the patrol randomizes because she cannot be everywhere, the Designer builds redundancy because she cannot hide.

Connectivity and the cost of resistance. A network on n nodes is κ -link-connected if it remains connected after the removal of any $\kappa - 1$ links. A network consisting solely of non-protected links thus resists k attacks if and only if it is $(k+1)$ -link-connected. A classical result of Harary [49] identifies the cheapest such networks: since every node must have degree at least κ (otherwise the adversary severs all links at some node), a κ -link-connected network has at least $\lceil \kappa n/2 \rceil$ links, and this bound is attained by the *Harary networks* $H_{\kappa,n}$, obtained by placing the n nodes on a circle, joining each to its $\lfloor \kappa/2 \rfloor$ nearest neighbors on each side, and adding, when κ is odd, roughly $n/2$ "diameters" across the circle. Protection enters the count as follows. If the Designer builds $p \geq 1$ protected links, at an optimum they form a forest (a protected cycle wastes a link), merging the n nodes into $n-p$ unbreakable components; contracting each component to a single super-node, the network resists k attacks precisely when the contracted network of non-protected links is $(k+1)$ -link-connected. Applying Harary's count to the contracted network, Bravard, Charroin, and Touati show that the minimal number of non-protected links needed in combination with p protected ones is

$$n_1(p, k) = \left\lceil \frac{(n-p)(k+1)}{2} \right\rceil \quad (6.21)$$

for p outside an intermediate window (inside the window, the requirement that links join *distinct* pairs of nodes forces a slightly larger count, a correction that only reinforces the conclusion

below), and $n_1(n-1, k) = 0$. The Designer's problem is now one-dimensional: choose p to minimize

$$C^*(p, k) = p c_P + n_1(p, k) c_L. \quad (6.22)$$

Each protected link buys its way out of roughly $(k+1)/2$ non-protected links, and hence the frontier (6.22) is essentially affine in p , and an affine function on an interval is minimized at an endpoint: the optimal design is *polar*, either no protection at all or (almost) full protection, with the ceiling in (6.21) contributing a parity refinement that occasionally rewards exactly one protected link. This is the content of the main result.

Theorem 6.30 (Bravard–Charroin–Touati [20]). *Under the standing assumptions, in every Stackelberg equilibrium the Adversary does not attack, and the Designer builds one of the following architectures.*

(1) $n(k+1)$ even.

- (a) If $\frac{c_P}{c_L} < \frac{n}{n-1} \cdot \frac{k+1}{2}$, a spanning tree of $n-1$ protected links and no non-protected links.
- (b) If $\frac{c_P}{c_L} > \frac{n}{n-1} \cdot \frac{k+1}{2}$, a minimal $(k+1)$ -link-connected network of $n(k+1)/2$ non-protected links (e.g., the Harary network $H_{k+1,n}$) and no protected links.

(2) $n(k+1)$ odd (i.e., n odd and k even).

- (a) If $\frac{c_P}{c_L} < \frac{n-1}{n-2} \cdot \frac{k+1}{2}$, a spanning tree of $n-1$ protected links.
- (b) If $\frac{n-1}{n-2} \cdot \frac{k+1}{2} < \frac{c_P}{c_L} < \frac{k+2}{2}$, exactly one protected link together with $(n-1)(k+1)/2$ non-protected links (this intermediate interval is nonempty if and only if $k < n-3$).
- (c) If $\frac{c_P}{c_L} > \frac{k+2}{2}$, a minimal $(k+1)$ -link-connected network of $\lceil n(k+1)/2 \rceil$ non-protected links.

At the threshold values both adjacent architectures are optimal. The leader's Stackelberg cost is $J^{1*} = \min_p C^*(p, k)$, attained at $p = n-1$, $p = 0$, or (in case 2(b)) $p = 1$; the follower's equilibrium cost is 0.

The thresholds are exactly the slopes of the cost frontier between the polar designs: $\frac{n}{n-1} \cdot \frac{k+1}{2}$ is the average number of non-protected links that each of the $n-1$ protected links of the tree replaces, and the theorem simply compares this saving with the price ratio c_P/c_L . The curious intermediate regime 2(b) is a pure parity effect. When $n(k+1)$ is odd, the Harary count $\lceil n(k+1)/2 \rceil$ rounds *up*: the degree requirement cannot be met exactly, and one node carries a wasted $(k+2)$ nd link. The *first* protected link absorbs this defect, replacing $(k+2)/2$ non-protected links, strictly more than the average; every further protected link saves less. When $n(k+1)$ is even there is no defect to absorb, the first protected link saves no more than the average, and the intermediate regime disappears: the design is all-or-nothing.

Example 6.31 Seven Facilities Against Two Link Cuts.

Let $n = 7$ and $c_A = 0.4$, so that $k = \lfloor 1/0.4 \rfloor = 2$: the Adversary will never attack more than two links. Since $n(k+1) = 21$ is odd, all three regimes of Theorem 6.30 are alive, with thresholds $\frac{n-1}{n-2} \cdot \frac{k+1}{2} = \frac{6}{5} \cdot \frac{3}{2} = \frac{9}{5}$ and $\frac{k+2}{2} = 2$. The three candidate architectures are displayed in Figure 6.3:

- *Full protection* ($p = 6$): a spanning tree of protected links, here a star centered at node 1; nothing can be severed. Cost: $6c_P$.
- *One protected link* ($p = 1$): protect the link $\{1, 2\}$, and treat the welded pair as a single super-node; surrounding the resulting 6 super-nodes with a copy of $H_{3,6}$ (a hexagon plus its three main diagonals) requires $(n-1)(k+1)/2 = 9$ non-protected links. Removing any two of them leaves the contracted network connected ($H_{3,6}$ is 3-link-connected), and the protected link cannot be removed at all. Cost: $c_P + 9c_L$.
- *No protection* ($p = 0$): the Harary network $H_{3,7}$, a 7-cycle plus four near-diameters, with $\lceil 21/2 \rceil = 11$ non-protected links; the rounding forces one node (node 4 in the figure) to carry a fourth link. Cost: $11c_L$.

Comparing costs pairwise recovers the thresholds of the theorem: the tree beats the hybrid if and only if $6c_P < c_P + 9c_L$, i.e., $c_P/c_L < 9/5$; the hybrid beats the Harary network if and only if $c_P < 2c_L$. Concretely, take $c_L = 0.04$ (each grade of link is cheap relative to the unit value of connectivity):

c_P	c_P/c_L	$6c_P$	$c_P + 9c_L$	$11c_L$	optimal design
0.06	1.5	0.360	0.420	0.440	protected tree
0.075	1.875	0.450	0.435	0.440	one protected link
0.10	2.5	0.600	0.460	0.440	Harary network $H_{3,7}$

In each row the Stackelberg cost J^{1*} (boldface) is well below 1: designing for resistance beats conceding. To see that the follower's threat truly binds, suppose the Designer, in the third regime, tried to economize by dropping the chord $\{4, 7\}$ from $H_{3,7}$: the design would cost $10c_L = 0.40 < 0.44$, but node 7 would then be held by only two links, $\{6, 7\}$ and $\{7, 1\}$, and the Adversary would cut both at cost $2c_A = 0.8 < 1$, isolating node 7. The Designer's realized cost would be $0.40 + 1 = 1.40$: the cheapest network is not the cheapest *resistant* network, and the $(k+1)$ -link-connectivity requirement is nothing but the follower's rational reaction made structural. $\square$

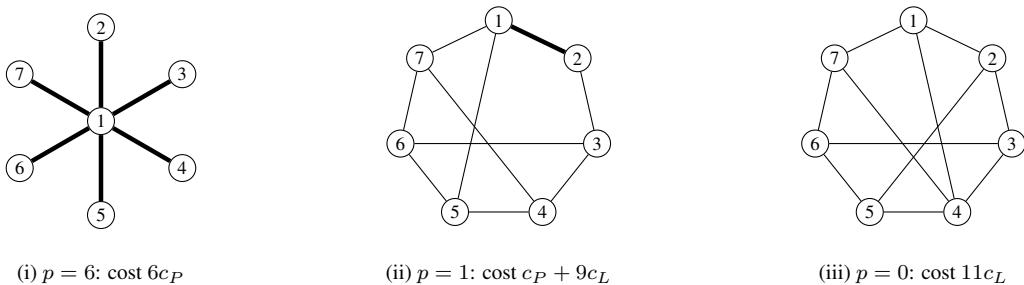


Figure 6.3: The three polar architectures of Example 6.31 ($n = 7$, $k = 2$). Thick links are protected; thin links are non-protected. (i) A spanning tree of 6 protected links. (ii) One protected link $\{1, 2\}$ plus 9 non-protected links forming $H_{3,6}$ on the six components obtained by welding nodes 1 and 2 together. (iii) The Harary network $H_{3,7}$ with 11 non-protected links; every node has degree 3 except node 4, which absorbs the parity defect with degree 4. Each network survives every attack on $k = 2$ non-protected links, and in equilibrium the Adversary therefore does not attack.

Remark 6.32 (Deterrence by Design). As in the entry-deterrence game of Example 6.13, all the action is off the equilibrium path: the attack never occurs, yet the anticipation of it (the follower's rational reaction set evaluated at every candidate design) dictates the entire architecture, down to the number and placement of individual links. The comparison with Section 6.7 is instructive. The patrol, endowed with $m < n$ resources, *cannot* fully deter, and hence she randomizes and the attack occurs in equilibrium; the Designer, free to build as much redundancy as she is willing to pay for, deters completely, and the Stackelberg calculation moves from the probability simplex to a combinatorial choice among network topologies. The model also rewards a second look at its boundaries: if the Designer's stock of protected links is exogenously capped, genuinely mixed architectures, with many protected *and* many non-protected links, arise in equilibrium; and if protection is imperfect (a protected link survives an attack only with a given probability), the polar architectures of Theorem 6.30 persist as long as the failure probability is small relative to the attack cost [20]. Pointers to the surrounding literature, including the node-attack counterpart of the model and its extensions to multi-layer and dynamic infrastructure networks, are collected in Section 6.10.

6.9 ■ One Leader, Many Followers: Designing the Followers' Equilibrium

In every hierarchical problem treated so far, the leader faced a single follower. In most applications she faces a population: a regulator announcing a rule to competing firms, a platform posting fees to interacting users, a toll authority setting prices for drivers who congest one another, a grid operator publishing tariffs to power producers coupled through the same network. The followers in these problems do not merely react to the leader; they interact strategically *with one another*, and what the leader's announcement elicits is not a best response but an *equilibrium*. The natural generalization of the Stackelberg solution replaces the follower's rational reaction set of Definition 6.6 by the set of Nash equilibria of the game the leader's announcement induces among the followers. The leader, in choosing her announcement, is thus choosing *which Nash equilibrium the followers will be playing*: she designs the followers' equilibrium as her response.

It is convenient in this section to index the leader by 0. Consider, then, an $(N + 1)$ -person finite game with a leader P_0 , whose pure-strategy set is Γ^0 , and followers $P_1, \dots, P_N$, with

pure-strategy sets $\Gamma^1, \dots, \Gamma^N$; the cost incurred to Player i ($i = 0, 1, \dots, N$) under the strategy profile $(\gamma^0, \gamma^1, \dots, \gamma^N)$ is $J^i(\gamma^0, \gamma^1, \dots, \gamma^N)$. The order of play is hierarchical: the leader announces $\gamma^0 \in \Gamma^0$ first; the followers observe the announcement and then choose their strategies *simultaneously*. Each announcement γ^0 therefore induces an N -person finite game $G(\gamma^0)$ among the followers, with cost functions $J^i(\gamma^0, \cdot, \dots, \cdot)$, and we take the followers' play in $G(\gamma^0)$ to be a Nash equilibrium, in mixed strategies whenever pure ones fail to exist. Letting Z^i denote the mixed-strategy simplex of follower i and $\bar{J}^i$ the multilinear (expected-cost) extension of J^i , we have the following counterparts of Definitions 6.16 and 6.7: the followers' reaction set is now made up of mixed strategies, as in Definition 6.16, while the leader still announces a pure strategy, as in Definition 6.7. A more general formulation, not pursued here, would allow the leader to randomize as well.

Definition 6.33 (Nash Reaction Set of the Followers). For each $\gamma^0 \in \Gamma^0$, the set

$$\bar{\mathcal{N}}(\gamma^0) = \left\{ (z^1, \dots, z^N) \in Z^1 \times \dots \times Z^N : \bar{J}^i(\gamma^0, z^i, z^{-i}) \leq \bar{J}^i(\gamma^0, \zeta^i, z^{-i}), \right. \\ \left. \forall \zeta^i \in Z^i, \forall i = 1, \dots, N \right\}, \quad (6.23)$$

where z^{-i} collects the strategies of the followers other than i , is the Nash reaction set of the followers to the announcement γ^0 : the set of (mixed-strategy) Nash equilibria of the induced game $G(\gamma^0)$.

Definition 6.34 (Stackelberg–Nash Equilibrium Strategy). A strategy $\gamma^{0*} \in \Gamma^0$ is a Stackelberg–Nash equilibrium strategy for the leader if

$$\max_{z \in \bar{\mathcal{N}}(\gamma^{0*})} \bar{J}^0(\gamma^{0*}, z) = \min_{\gamma^0 \in \Gamma^0} \max_{z \in \bar{\mathcal{N}}(\gamma^0)} \bar{J}^0(\gamma^0, z) =: J^{0*}, \quad (6.24)$$

where $z = (z^1, \dots, z^N)$. The quantity J^{0*} is the Stackelberg–Nash cost of the leader. Any $z^* \in \bar{\mathcal{N}}(\gamma^{0*})$ is a profile of follower strategies in equilibrium with γ^{0*} , and (γ^{0*}, z^*) is a Stackelberg–Nash solution of the hierarchical game.

For $N = 1$ the induced “game” is a one-player decision problem, the set (6.23) reduces to the follower's (mixed-strategy) best-response set, and Definition 6.34 recovers the secured Stackelberg solution of Section 6.3. The existence theorem also carries over, and for the same compactness reasons.

Theorem 6.35 (Existence). Every $(N + 1)$ -person finite game admits a Stackelberg–Nash equilibrium strategy for the leader.

Proof. For each $\gamma^0 \in \Gamma^0$, the induced finite game $G(\gamma^0)$ admits a Nash equilibrium in mixed strategies by Nash's theorem, and hence $\bar{\mathcal{N}}(\gamma^0)$ is nonempty. It is also compact: it suffices in (6.23) to check the deviations ζ^i ranging over the (finitely many) pure strategies of follower i , so $\bar{\mathcal{N}}(\gamma^0)$ is carved out of the compact product $Z^1 \times \dots \times Z^N$ by finitely many weak inequalities between continuous functions, and is therefore a closed subset of a compact set. Since $\bar{J}^0(\gamma^0, \cdot)$ is continuous, the inner maximum in (6.24) is attained; since Γ^0 is finite, the outer minimum also exists. $\square$

Remark 6.36 (Multiplicity Is Now Essential). With a single follower, a multivalued reaction set was a knife-edge phenomenon, produced by exact ties in the follower's costs (Section 6.3).

With several followers it is the *generic* situation: induced games routinely possess several Nash equilibria (every coordination structure among the followers produces them), and no perturbation of the costs makes the multiplicity disappear. The inner maximum in (6.24) is therefore no longer a tie-breaking convention but a substantive modeling choice: the pessimistic leader guards against the followers' *equilibrium selection*, demanding a guarantee that holds whichever equilibrium they coordinate on. The optimistic variant (replacing the maximum by a minimum) credits the leader with the power to steer the followers to her favorite equilibrium, for instance through a public recommendation issued alongside the announcement. The example below shows the two conventions parting ways, and shows what the leader must pay, in her own cost, to secure the optimistic outcome against a pessimistic reading. As in Section 6.4, the leader may also benefit from announcing a mixed strategy, with the same caveat that an optimal one need not exist; we confine the discussion to pure announcements.

Example 6.37 Subsidizing a Joint Project.

A regulator (P0) wants two firms (P1 and P2) to contribute to a joint project, a shared logistics facility, say. Each firm either contributes (C) or defects (D). The regulator moves first and announces a per-contributor subsidy σ , which we restrict to three levels: $\sigma \in \{0, 1.2, 1.6\}$. Given the announcement, the firms play the symmetric bi-matrix game with costs

$$A^1(\sigma) = \begin{array}{c|cc} & C & D \\ \hline C & 1 - \sigma & 3 - \sigma \\ D & 0 & 1.5 \end{array}, \quad A^2(\sigma) = A^1(\sigma)',$$

(P1 the row player; each firm minimizes): contributing costs a firm 1 if the other contributes too and 3 if it is left contributing alone, defection is free if the other carries the project and costs 1.5 if the project dies, and the subsidy σ rewards contribution regardless of what the other firm does. The regulator values the project and pays the subsidies: with m contributors her cost is

$$J^0 = 4 - 2m + \sigma m.$$

The three announcements induce three quite different games among the firms.

- $\sigma = 0$: defection is strictly dominant ($0 < 1$ and $1.5 < 3$), a prisoners' dilemma; the unique Nash equilibrium is (D, D), and $\mathcal{N}(0)$ is a singleton. The project dies and the regulator's cost is 4.
- $\sigma = 1.2$: against C, contributing is now strictly better ($1 - 1.2 = -0.2 < 0$); against D, defecting still is ($1.5 < 3 - 1.2 = 1.8$). The induced game is a *coordination game* with three Nash equilibria: the pure profiles (C, C) and (D, D), and a mixed profile in which each firm contributes with probability q solving $1.8 - 2q = 1.5 - 1.5q$, i.e., $q = 0.6$. The regulator's cost at the three equilibria is, respectively,

$$2\sigma = 2.4, \quad 4, \quad 4 + 2q(\sigma - 2) = 3.04,$$

so her secured cost is $\max\{2.4, 4, 3.04\} = 4$: the modest subsidy makes cooperation *an* equilibrium, but it cannot guarantee it, and the pessimistic evaluation is no better than doing nothing.

- $\sigma = 1.6$: contributing is strictly dominant ($-0.6 < 0$ and $1.4 < 1.5$); the unique Nash equilibrium is (C, C), and the regulator's cost is $2\sigma = 3.2$.

Hence $J^{0*} = \min\{4, 4, 3.2\} = 3.2$, and the Stackelberg–Nash strategy is the generous subsidy $\gamma^{0*} = 1.6$: the regulator overpays, relative to the cheapest subsidy supporting cooperation, precisely in order to make cooperation the *only* equilibrium. An optimistic regulator, confident that the firms would coordinate on the equilibrium she prefers, would announce $\sigma = 1.2$ and expect a cost of 2.4; the difference $3.2 - 2.4 = 0.8$ is the premium the pessimistic leader pays for not relying on favorable equilibrium selection. $\square$

Remark 6.38 (From Playing Games to Designing Them). The leader of this section does something qualitatively different from the leader of Section 6.3: she does not choose a move *in* the followers’ game; she chooses *which game the followers play*, and then lets Nash equilibrium sort out the rest. Read γ^0 as “the rules”, an auction format, a tax schedule, a market protocol, a subsidy scheme, and the correspondence $\gamma^0 \mapsto \bar{N}(\gamma^0)$ becomes the central object of *mechanism design*: the map from institutions to the equilibrium outcomes they induce. The two conventions of Remark 6.36 reappear there as the two standard notions of implementation: the optimistic convention corresponds to *weak* implementation (some equilibrium of the mechanism achieves the designer’s goal), the pessimistic convention to *full* implementation (all of them do), the theory of which is due to Maskin [77]. Example 6.37 illustrates the third classical route, *dominant-strategy* implementation, which removes the selection problem altogether, typically at a price. What distinguishes mechanism design proper from the complete-information hierarchy of this section is private information: when the followers’ costs are known only to themselves, the leader must design the game so that the agents reveal what she needs to know, and the revelation principle of Myerson [83] reduces this search to direct mechanisms in which truth-telling is an equilibrium, turning the design problem into an optimization under incentive-compatibility constraints; a systematic treatment is given by Borgers [18]. Finally, the leader’s design power grows dramatically when she can commit to a *strategy*, a rule conditioning her own action on the followers’ observed behavior, rather than to a fixed announcement; such incentive (inverse Stackelberg) designs are the subject of Lecture 22.

6.10 ■ Further Notes and Reading

This lecture has developed the Stackelberg solution concept for two-person finite games, beginning from the extensive form (the two-stage tree of Section 6.1, its reduction to a normal form in which the follower’s strategies are contingent plans, and the reading of the solution as backward induction) and passing to the bilevel optimization view of Section 6.5, in which the leader minimizes over the graph of the follower’s reaction correspondence. Between the two lie: existence in pure strategies (in sharp contrast with Nash equilibria, no randomization is needed for existence); the secured, worst-case treatment of multivalued follower responses; the comparison with Nash costs (Proposition 6.14 and the counterexample of Remark 6.15); the strict benefit that a leader may derive from randomized commitment (with Proposition 6.19 showing that mixing can never hurt); the application of the mixed-commitment theory to security games (Section 6.7); a case study on the optimal design and defense of networks under link attacks (Section 6.8), where commitment is physical and deterrence is achieved through robustness rather than randomization; the passage to behavioral strategies in extensive-form games, which under perfect recall is, by Kuhn’s theorem, a reparameterization of mixed commitment rather than an enlargement of it (Proposition 6.24), the substance lying instead in what the leader commits to and when; and the extension to one leader and many followers (Section 6.9), where the followers play a Nash game among themselves, the leader designs her equilibrium, and the solution concept makes contact

with mechanism design. The treatment follows Chapter 3 of Başar and Olsder [12], where the reader will also find multi-level hierarchies with many leaders and followers, and many further examples. The systematic study of the Stackelberg concept in nonzero-sum games was initiated by Simaan and Cruz [117, 116]; for the surrounding landscape of noncooperative solution concepts in finite games, see also Fudenberg and Tirole [40].

Zero-sum games: leadership is worthless. When the game is zero-sum ($B = -A$), the follower's reaction set against a pure or mixed announcement is exactly his set of best replies in the matrix game, and the leader's (mixed) Stackelberg cost (6.9) reduces to $\min_y \max_z y'Az$, the upper value in mixed strategies which, by the minimax theorem of Lecture 2 (due to von Neumann; see [138]), equals the value of the game. Hence in zero-sum games the Stackelberg solution collapses to the saddle-point solution, announced commitment confers no advantage (and no disadvantage), and the entire hierarchy question is moot. The Stackelberg concept acquires independent interest precisely in genuine nonzero-sum games, where the examples of this lecture have shown all possibilities: leadership may benefit both players (game (6.1)), hurt the leader relative to Nash play (Remark 6.15), or call for randomized commitment (Section 6.4).

Computation and complexity. The multiple-LP method of Section 6.5 computes an optimal mixed commitment (under the strong/optimistic convention) in time polynomial in the size of the bi-matrix game [27]; remarkably, computing an optimal strategy *to commit to* is computationally easier than computing a Nash equilibrium of the same game, which is PPAD-complete. For games with structured strategy spaces (schedules, patrols, resource allocations), compact formulations and decomposition methods extend the approach to exponentially large strategy sets, as illustrated by the coverage-vector reduction of Section 6.7.

Stackelberg security games. The security-game model of Section 6.7, the algorithms that scale it to schedule constraints and Bayesian attacker types, and the deployed systems built on it are the subject of Tambe [124]; Sinha et al. [118] survey the decade that followed. The Stackelberg-versus-Nash comparison in security games is settled by Korzhyk et al. [66] (cf. Remark 6.29), and the general theory of leadership games in mixed strategies (including the comparison of the leader's commitment value with her equilibrium payoffs in the simultaneous-move game) is developed by von Stengel and Zamir [141].

Network design and defense. The network design game of Section 6.8 is due to Bravard, Charroin, and Touati [20], who also treat the two variations mentioned in Remark 6.32: a capped stock of protected links, under which equilibrium networks genuinely mix protected and non-protected links, and imperfect protections that survive an attack only with a given probability. The extremal graph theory underlying the minimal κ -link-connected networks is Harary's [49]. The node-attack counterpart of the model, in which the Designer immunizes nodes and the Adversary removes unprotected ones, was formulated and solved by Dziubiński and Goyal [31]; there the polar architectures are the protected star and the unprotected minimally $(k + 1)$ -node-connected network. The Stackelberg design-and-defense paradigm has been carried into modern infrastructure settings by Chen, Touati, and Zhu: [25] designs optimal secure two-layer networks (an IoT device layer riding on a cloud or communication layer) facing link attacks, and [24] develops a dynamic game of secure and resilient infrastructure network design in which construction, attack, and recovery unfold over time; both works show how the polar-architecture logic of Theorem 6.30 extends to multi-layer and dynamic environments.

Many followers and mechanism design. The one-leader, many-follower solution concept of Section 6.9 goes back to the hierarchical equilibria of Simaan and Cruz [117] and is developed, together with its infinite-game and dynamic counterparts, in Başar and Olsder [12]. The mechanism-design reading of the framework has a large literature of its own, recognized by the 2007 Nobel Memorial Prize to Hurwicz, Maskin, and Myerson: Maskin [77] characterizes the

social objectives that are *fully* implementable in Nash equilibrium (the pessimistic convention of Definition 6.34 taken to its logical conclusion); Myerson [83] introduces the revelation principle and solves the optimal auction problem, the paradigmatic instance of a leader designing a game for privately informed followers; and Börgers [18] is a modern textbook account. The security games of Section 6.7 also admit many-follower variants, with multiple attackers or a population of adversary types; see Sinha et al. [118].

Toward infinite games. All the concepts of this lecture (the rational reaction set, the secured Stackelberg cost, the leader-versus-Nash comparison, and the (possible) benefit of randomized commitment) extend to games with continuous action spaces, where new phenomena arise: existence of a Stackelberg strategy is no longer automatic (the infimum defining the Stackelberg cost may fail to be attained), and ϵ -optimal strategies take center stage. That theory is developed in Lecture 10, on continuous-kernel games and Stackelberg equilibria.

6.11 ■ Exercises

Exercise 6.1 (Stackelberg Analysis of a Bi-matrix Game). Consider the bi-matrix game

$$A = \begin{array}{c|ccc} & L & M & R \\ \hline L & 2 & 0 & 3 \\ M & 1 & 2 & 0 \\ R & 3 & 1 & 1 \end{array}, \quad B = \begin{array}{c|ccc} & L & M & R \\ \hline L & 1 & 0 & 2 \\ M & 2 & 1 & 3 \\ R & 0 & 2 & 1 \end{array},$$

where both players are minimizers, P1 chooses rows, and P2 chooses columns.

- (i) Find all pure-strategy Nash equilibria of the game.
- (ii) Compute the follower's rational reaction set $R^2(\gamma^1)$ for each row, and determine the Stackelberg strategy, cost, and outcome with P1 as the leader.
- (iii) Repeat part (ii) with P2 as the leader (compute $R^1(\gamma^2)$ column by column).
- (iv) Compare the cost pairs obtained (Nash, P1-led Stackelberg, P2-led Stackelberg) and discuss who benefits from leadership in this game.
- (v) Write a program, in Python or MATLAB, that takes a pair of cost matrices (A, B) as input and (1) enumerates the leader's pure strategies, computing for each the follower's reaction set and the leader's secured cost, so as to output the Stackelberg strategy and cost for either choice of leader, and (2) enumerates all cells to output the pure-strategy Nash equilibria. Run the program on the game above and on the bi-matrix game (6.1) of the text, and verify your answers to parts (i) to (iii) as well as the values reported in Section 6.3. (Expected output: for the game above, the unique Nash equilibrium $\{L, M\}$ with outcome $(0, 0)$, P1-led Stackelberg strategy L with cost 0, and P2-led Stackelberg strategy M with cost 0, so that here leadership reproduces the Nash outcome; for the game (6.1), the unique Nash equilibrium $\{M, M\}$ with outcome $(1, 0)$, P1-led Stackelberg cost 0 with outcome $(0, -1)$, and P2-led Stackelberg cost $-2/3$ with outcome $(3/2, -2/3)$.)

Exercise 6.2 (Zero-Sum Games: The Value of Leadership). Let $(A, -A)$ be an arbitrary two-person zero-sum finite game, with P1 (row player) minimizing $y'Az$ and P2 maximizing it.

- (i) Show that the leader's *pure-strategy* Stackelberg cost with P1 as the leader equals the upper value $\bar{V}(A) = \min_i \max_j a_{ij}$ of the matrix game.

- (ii) Show that the leader's *mixed-strategy* Stackelberg cost (6.9) equals the value $V_m(A)$ of the game in mixed strategies, and conclude, using the Minimax Theorem of Lecture 2, that in zero-sum games neither player gains or loses by moving first with an observable mixed commitment.
- (iii) Reconcile part (ii) with the fact that, in nonzero-sum games, the leader's mixed Stackelberg cost can be strictly below her cost at every Nash equilibrium. What feature of the zero-sum structure eliminates the value of commitment?

Exercise 6.3 (Multivalued Responses and Secured Costs). Consider the bi-matrix game (6.2) of the text, with P1 as the leader.

- (i) Verify all the claims made in the text: the reaction sets $R^2(L) = \{L, M\}$ and $R^2(R) = \{L, R\}$, the Stackelberg strategy $\gamma^{1*} = L$ with secured cost $J^{1*} = 1$, and the Nash equilibria of the game together with their admissibility.
- (ii) Now adopt the *optimistic* convention, replacing the maximum in (6.4) by a minimum over $\gamma^2 \in R^2(\gamma^1)$. Compute the optimistic Stackelberg strategy and cost of the leader, and compare with the pessimistic values of part (i).
- (iii) Suppose the leader can accompany her announcement with a small side payment $\delta > 0$ to the follower, contingent on the follower selecting a particular response in his reaction set. Show that for any $\delta > 0$ the leader can then guarantee herself a cost within δ of the optimistic value of part (ii), and discuss the implication for which of the two conventions is the more plausible model in applications.

Exercise 6.4 (When Does Mixing Help the Leader?). Consider the parameterized family of bi-matrix games

$$A = \begin{array}{c|cc} & L & R \\ \hline L & 1 & 0 \\ R & 0 & 1 \end{array}, \quad B(\beta) = \begin{array}{c|cc} & L & R \\ \hline L & 1/2 & 1 \\ R & 1 & \beta \end{array},$$

with P1 as the leader and $\beta \in (0, 1)$ a parameter. (The case $\beta = 1/2$ is the game (6.7); the case $\beta = 1/3$ is Example 6.18.)

- (i) For every $\beta \in (0, 1)$, compute the pure-strategy Stackelberg cost J^{1*} and the decomposition of the leader's mixed-strategy space Y into the follower's response regions.
- (ii) Compute the mixed Stackelberg cost $\bar{J}^{1*}$ as a function of β , and determine for which β a mixed Stackelberg equilibrium strategy exists (in the pessimistic sense of Definition 6.17) and for which it fails to exist.
- (iii) Verify Proposition 6.19 on this family, and identify the values of β at which $\bar{J}^{1*} = J^{1*}$.

Exercise 6.5 (The Multiple-LP Method). Consider the bi-matrix game

$$A = \begin{array}{c|ccc} & L & M & R \\ \hline L & 0 & 2 & 1 \\ R & 2 & 0 & 1 \end{array}, \quad B = \begin{array}{c|ccc} & L & M & R \\ \hline L & 0 & 2 & 1/2 \\ R & 2 & 0 & 1/2 \end{array},$$

with P1 (rows) as the leader, minimizing.

- (i) Parameterize the leader's mixed strategies by $y = (y_1, 1 - y_1)'$, and determine the follower's response regions $Y_L, Y_M, Y_R \subset [0, 1]$ in which each of his three pure strategies is optimal.

- (ii) Solve the three linear programs (6.12) and determine the leader's optimal mixed commitment and its value under the strong (optimistic) convention.
- (iii) Determine the pessimistic mixed Stackelberg cost $\bar{J}^{1*}$ of Definition 6.17 for this game, and compare it with the value found in part (ii) and with the pure Stackelberg cost J^{1*} .
- (iv) Implement the multiple-LP method (6.12) in Python (for instance with `scipy.optimize.linprog`) or in MATLAB (with `linprog`), and run it on this game to verify your answers to parts (i) and (ii). (Expected output: the three programs LP_L , LP_M , LP_R have optimal values 0, 0, and 1, attained on the response regions $Y_L = [3/4, 1]$, $Y_M = [0, 1/4]$, and $Y_R = [1/4, 3/4]$ at $y_1 = 1$, $y_1 = 0$, and any point of Y_R , respectively, so that the strong Stackelberg value is 0.)

Exercise 6.6 (A Stackelberg Security Game). A defender with a single patrol unit ($m = 1$) protects three targets, in the model of Definition 6.26, with $d_c(t) = 0$ for every t and

	$d_u(t)$	$a_u(t)$	$a_c(t)$
$t = 1$	6	-3	3
$t = 2$	3	-3	3
$t = 3$	2	1	2

(target 3 is inherently fortified: attacking it is costly to the attacker even when it is unpatrolled).

- (i) Compute the defender's pure-strategy Stackelberg cost J^{1*} .
- (ii) Write down and solve the three linear programs (6.18). Show that (SLP_3) is *infeasible*, and interpret this fact; then conclude that the optimal coverage vector is $x^* = (1/2, 1/2, 0)$, with strong Stackelberg cost $3/2$ and attack set $\{1, 2\}$. Explain why it is optimal to leave target 3 entirely unpatrolled, even though the defender has coverage probability to spare.
- (iii) Show that under the pessimistic (weak) convention of Definition 6.17 the value $3/2$ is not attained, and exhibit a family of ϵ -optimal coverage vectors.
- (iv) Show that x^* is also the defender's strategy in a Nash equilibrium of the simultaneous-move game, by exhibiting an attacker mixed strategy that completes the equilibrium (this illustrates Remark 6.29). Compute the defender's expected cost at this Nash equilibrium, and explain why it differs from the strong Stackelberg cost of part (ii) even though the defender's strategy is the same.

Exercise 6.7 (Leader-Follower Roles and Social Cost). Construct (or find) 2×2 bi-matrix games, with both players minimizing, exhibiting each of the following phenomena, or argue that no such game exists:

- (i) Both players strictly prefer the P1-led Stackelberg outcome to every Nash outcome.
- (ii) Both players strictly prefer every Nash outcome to every possible P1-led Stackelberg outcome (under the pessimistic convention).
- (iii) Each player strictly prefers to be the *follower*: the P1-led Stackelberg outcome gives P2 a strictly lower cost than the P2-led one gives him, and vice versa.
- (iv) The sum of the players' costs at the P1-led Stackelberg solution is strictly lower than at every Nash equilibrium of the game.

For each constructed game, exhibit the reaction sets and verify the claimed solutions carefully.

Exercise 6.8 (Commitment in an Extensive-Form Game). Consider a two-stage finite dynamic game: at stage 1, P1 chooses $a_1 \in \{T, B\}$, observed by P2; at stage 2, P2 chooses $a_2 \in \{l, r\}$; then P1 chooses $a_3 \in \{t, b\}$ after observing (a_1, a_2) . The cost pairs (J^1, J^2) are given by

$$\begin{array}{llll} (T, l, t) : (0, 2) & (T, l, b) : (3, 0) & (T, r, t) : (2, 1) & (T, r, b) : (1, 3) \\ (B, l, t) : (2, 0) & (B, l, b) : (1, 1) & (B, r, t) : (4, 2) & (B, r, b) : (0, 1). \end{array}$$

- (i) Write down the pure-strategy spaces Γ^1 (note that P1's strategy specifies a_1 and a rule for a_3 as a function of (a_1, a_2)) and Γ^2 , and compute the Stackelberg solution in the normal form, with P1 as the leader committing to a complete strategy.
- (ii) Compute the feedback (stagewise) solution obtained by backward induction, in which no player commits to future behavior, and compare P1's cost with her global Stackelberg cost from part (i).
- (iii) Explain, in view of Remark 6.25, why the global commitment value can be strictly better for the leader than the feedback value, and identify which of P1's commitments in part (i) is not credible without commitment power.

Chapter 7

Equilibrium Refinement

This lecture examines the limitations of the Nash equilibrium concept, developed in Lecture 3 for finite static (normal-form) games, and motivates the need for refinement. While Nash equilibrium provides a foundational notion of strategic stability, many games admit several equilibria, some of which rely on weakly dominated strategies or require players to assume that certain actions, although available, will never be played. Because the Nash concept treats all equilibria as equally plausible, it lacks predictive sharpness and can admit outcomes that are fragile to small mistakes or modeling noise.

Refinement concepts address this issue by strengthening the internal consistency of equilibrium behavior. We begin with the idea of *perturbations*, or *trembles*, which represent the possibility that players may choose every action with a small but positive probability. In a normal-form game, this means that every action, whether equilibrium-prescribed or not, is played with positive probability in the perturbed environment. Selten's trembling-hand perfect equilibrium [107] selects Nash equilibria that remain optimal in the limit as tremble magnitudes vanish. We then introduce Myerson's ε -perfect equilibrium [81], a parametric version that simplifies the analysis of perfection, as well as his *proper equilibrium*, which further restricts allowable mistakes by requiring that worse deviations occur with proportionally smaller probability. These notions create a hierarchy of increasingly demanding equilibrium concepts, each designed to rule out non-robust or behaviorally implausible Nash equilibria.

Only after developing these refinements for static games do we extend the ideas to dynamic (extensive-form) settings, where the timing of moves and the structure of information become essential. In that context, refinements incorporate not only robustness to trembles but also requirements on how players form beliefs at decision points reached with small probability. We discuss quasi-perfect equilibrium (van Damme [136]), sequential equilibrium (Kreps and Wilson [68]), and strategic stability (Kohlberg and Mertens [64]), each imposing consistency of strategies and beliefs in the face of small perturbations to either behavior or the game form itself. Collectively, these refinements clarify which equilibria are credible and robust in environments where small mistakes, uncertainty, and dynamic reasoning play a central role.

7.1 ■ Motivation for Refinements of Nash Equilibrium

Finite normal-form games, introduced in Lecture 3, often possess multiple Nash equilibria (NE) in pure or mixed strategies, and these equilibria can result in very different payoff outcomes for the players. In general, these equilibria are not interchangeable: each may correspond to a distinct distribution of benefits or costs, and they are not necessarily ordered in a way that makes

one clearly superior to the others for all players. The phenomenon of multiple equilibria is not confined to simple static games; it frequently appears in dynamic and sequential settings as well, where it emerges from the informational richness of the underlying decision problem and the particular structure of the players' payoff or cost matrices. Because of this multiplicity, the NE concept by itself often lacks predictive precision. Some equilibria are sustained by strategies that are weakly dominated, which raises doubts about their plausibility as actual strategic choices. Others are fragile in the sense that even minute perturbations of the payoffs can destroy the equilibrium property, causing at least one player to prefer deviating. Moreover, the definition of NE contains no mechanism for discriminating between equilibria that are stable under such perturbations and those that are not. All Nash equilibria are treated as equally viable under the basic definition, regardless of their robustness or vulnerability.

Throughout this lecture, the player set is $\mathcal{N} := \{1, \dots, N\}$; each player $i \in \mathcal{N}$ has a finite action set A_i with generic action a_i , and a mixed strategy of player i is an element $\sigma_i \in \Delta(A_i)$ of the probability simplex over A_i . Unless stated otherwise, players maximize their utility functions u_i ; a few matrix-game examples are instead phrased in terms of cost matrices that the players minimize, and we say so explicitly in each such case.

This limitation motivates the development of refinement concepts. These refinements introduce additional criteria, often grounded in robustness to perturbations or in behavioral assumptions about the players, that allow one to discard less plausible equilibria and retain those with stronger theoretical or practical justification. The result is a smaller and more selective set of equilibria, chosen in a rational and principled manner to better capture the outcomes one might expect in real strategic interactions.

Example 7.1 A 2×2 Game with Different Trembles.

Consider the 2×2 minimization game in which both players have the same cost matrices:

$$A = \begin{array}{c|cc} & L & R \\ \hline U & 0 & 1 \\ D & 1 & 1 \end{array}, \quad B = \begin{array}{c|cc} & L & R \\ \hline U & 0 & 1 \\ D & 1 & 1 \end{array}.$$

The row player (P1) chooses U or D , and the column player (P2) chooses L or R . There are two pure-strategy Nash equilibria: (U, L) with costs $(0, 0)$ and (D, R) with costs $(1, 1)$. Both are admissible in the Nash sense, but (U, L) is robust to perturbations while (D, R) is not.

Suppose now that P1, when intending to play a pure strategy, selects the other strategy with probability $\varepsilon_1 > 0$, while P2, when intending to play a pure strategy, selects the other strategy with probability $\varepsilon_2 > 0$. These trembles occur independently for the two players.

The row tremble matrix for P1 is

$$T^{(1)} = \begin{pmatrix} 1 - \varepsilon_1 & \varepsilon_1 \\ \varepsilon_1 & 1 - \varepsilon_1 \end{pmatrix},$$

and the column tremble matrix for P2 is

$$T^{(2)} = \begin{pmatrix} 1 - \varepsilon_2 & \varepsilon_2 \\ \varepsilon_2 & 1 - \varepsilon_2 \end{pmatrix}.$$

The perturbed payoff matrix for P1 is

$$A_{\varepsilon_1, \varepsilon_2} = T^{(1)} A (T^{(2)})^\top.$$

A direct computation yields

$$A_{\varepsilon_1, \varepsilon_2} = \begin{array}{c|cc} & L & R \\ \hline U & (1 - \varepsilon_1)\varepsilon_2 + \varepsilon_1(1 - \varepsilon_2) + \varepsilon_1\varepsilon_2 & 1 - (1 - \varepsilon_1)\varepsilon_2 \\ D & 1 - \varepsilon_1(1 - \varepsilon_2) & 1 - \varepsilon_1\varepsilon_2 \end{array},$$

where each entry is the expected cost of the corresponding intended strategy pair; only the outcome (U, L) has zero cost, and hence each entry equals one minus the probability that the realized action pair is (U, L) .

Similarly, the perturbed payoff matrix for P2 is

$$B_{\varepsilon_1, \varepsilon_2} = T^{(1)} B (T^{(2)})^\top,$$

which in this symmetric-cost example coincides with $A_{\varepsilon_1, \varepsilon_2}$.

For any small $\varepsilon_1, \varepsilon_2 > 0$ with $\varepsilon_1, \varepsilon_2 < \frac{1}{2}$, the perturbed game has a unique Nash equilibrium at (U, L) . As $(\varepsilon_1, \varepsilon_2) \rightarrow (0, 0)$, the costs at (U, L) converge to $(0, 0)$, whereas (D, R) fails to survive. This robustness property is precisely what trembling-hand perfection formalizes. $\square$

7.2 ■ Trembling-Hand Perfect Equilibrium

In real strategic situations, players may occasionally make small mistakes in executing their intended strategies. These mistakes, referred to as *trembles*, imply that each pure strategy is chosen with at least some small positive probability, even if the player does not intend to use it. The concept of a *trembling-hand perfect equilibrium*, introduced by Selten [107], formalizes robustness to such execution errors. A strategy profile is perfect if it remains optimal for every player under every sufficiently small tremble of the other players. Thus, a perfect equilibrium does not depend on finely tuned beliefs about low-probability actions or on fragile assumptions about what players will never do.

Definition 7.2 (Trembling-Hand Perfect Equilibrium). Let $\Gamma = (\mathcal{N}, \{\Delta(A_i)\}_{i \in \mathcal{N}}, \{u_i\}_{i \in \mathcal{N}})$ be a finite normal-form game. A mixed-strategy profile $\sigma^* \in \times_{i \in \mathcal{N}} \Delta(A_i)$ is a trembling-hand perfect equilibrium if there exist a sequence of tremble vectors $\{\varepsilon_k\}_{k=1}^\infty$ and a sequence of totally mixed strategy profiles $\{\sigma^{\varepsilon_k}\}_{k=1}^\infty$ such that:

- (i) For each k , σ^{ε_k} is a Nash equilibrium of a perturbed game Γ_{ε_k} in which every pure action is required to be played with at least a small probability:

$$\sigma_i^{\varepsilon_k}(a_i) \geq \varepsilon_k(a_i) > 0, \quad \sum_{a_i \in A_i} \varepsilon_k(a_i) < 1, \quad \forall i \in \mathcal{N}.$$

Thus, each σ^{ε_k} is a Nash equilibrium of a fully mixed perturbation of the original game.

- (ii) The sequence of perturbations satisfies $\varepsilon_k \rightarrow 0$ and the corresponding equilibria converge:

$$\sigma^{\varepsilon_k} \rightarrow \sigma^* \quad \text{as } k \rightarrow \infty.$$

The key requirement is that as the tremble probabilities vanish, the limiting profile σ^* remains a Nash equilibrium of the unperturbed game. This ensures that σ^* is robust to arbitrarily small execution errors by the other players.

This refinement eliminates equilibria in which a player's strategy prescribes suboptimal actions at information sets that are reached with probability zero in equilibrium, since trembles make every such information set reachable with positive probability. Weakly dominated strategies cannot appear in a trembling-hand perfect equilibrium: under perturbations, even small mistakes by others may create profitable deviations if a player relies on a weakly dominated strategy. Thus, trembling-hand perfection rules out equilibria that depend on incredible threats or implausible assumptions that certain contingencies will never arise. A perfect equilibrium is therefore robust, credible, and structurally well-behaved in the presence of vanishingly small mistakes.

Example 7.3 Trembling-Hand Perfection in a 2×2 Minimization Game.

Again consider the game of Example 7.1, the 2×2 game in which *both players minimize cost*:

$$A = B = \begin{array}{c|cc} & L & R \\ \hline U & 0 & 1 \\ D & 1 & 1 \end{array}, \quad A_1 = \{U, D\}, \quad A_2 = \{L, R\}.$$

The Nash equilibria are (U, L) and (D, R) .

To test for trembling-hand perfection, impose strictly positive lower bounds on each pure action: $\varepsilon_1(U), \varepsilon_1(D) > 0$ with $\varepsilon_1(U) + \varepsilon_1(D) < 1$, and $\varepsilon_2(L), \varepsilon_2(R) > 0$ with $\varepsilon_2(L) + \varepsilon_2(R) < 1$. Define the perturbed mixed-strategy sets

$$\Delta_1(\varepsilon_1) = \{\sigma_1 \in \Delta(A_1) : \sigma_1(U) \geq \varepsilon_1(U), \sigma_1(D) \geq \varepsilon_1(D)\},$$

$$\Delta_2(\varepsilon_2) = \{\sigma_2 \in \Delta(A_2) : \sigma_2(L) \geq \varepsilon_2(L), \sigma_2(R) \geq \varepsilon_2(R)\}.$$

In the perturbed game Γ_ε , all strategies are totally mixed.

P1. For any $\sigma_2 \in \Delta_2(\varepsilon_2)$, the expected cost of U is $C_1(U, \sigma_2) = \sigma_2(R)$, and the cost of D is $C_1(D, \sigma_2) = 1$. Since $\sigma_2(R) \leq 1 - \varepsilon_2(L) < 1$, we have $C_1(U, \sigma_2) < C_1(D, \sigma_2)$ for all feasible σ_2 . Thus U strictly dominates D in every perturbed game, and hence any perturbed equilibrium must place the maximal feasible probability on U , namely $\sigma_1(U) = 1 - \varepsilon_1(D)$, $\sigma_1(D) = \varepsilon_1(D)$.

P2. For any $\sigma_1 \in \Delta_1(\varepsilon_1)$, the expected cost of L is $C_2(L, \sigma_1) = \sigma_1(D)$, and the cost of R is $C_2(R, \sigma_1) = 1$. Since $\sigma_1(D) \leq 1 - \varepsilon_1(U) < 1$, we have $C_2(L, \sigma_1) < C_2(R, \sigma_1)$ for all feasible σ_1 . Thus L strictly dominates R , and hence any perturbed equilibrium must place the maximal feasible probability on L , namely $\sigma_2(L) = 1 - \varepsilon_2(R)$, $\sigma_2(R) = \varepsilon_2(R)$.

Equilibrium of the perturbed game. For every perturbation with strictly positive lower bounds, the perturbed game Γ_ε therefore has a *unique* Nash equilibrium $\sigma^\varepsilon = (\sigma_1^\varepsilon, \sigma_2^\varepsilon)$, which places the maximal feasible probability on U and L . As $\varepsilon \rightarrow 0$, these perturbed equilibria converge to (U, L) . Hence (U, L) is a *trembling-hand perfect equilibrium* of the original game.

Elimination of (D, R) . The equilibrium (D, R) cannot be the limit of equilibria of perturbed games, because D and R are strictly dominated in every Γ_ε and therefore always receive the minimal feasible probability, which vanishes as $\varepsilon \rightarrow 0$. No totally mixed perturbed equilibrium ever approaches (D, R) , and hence this profile fails Selten's criterion.

Thus trembling-hand perfection selects (U, L) and eliminates (D, R) . $\square$

Example 7.4 Trembling-Hand Perfection in a Non-Team 3×2 Game.

The game of Examples 7.1 and 7.3 has identical cost matrices, $A = B$, and is therefore a team problem, whose unique globally optimal solution is the perfect equilibrium (U, L) . One may reasonably ask whether the argument leans on that team structure, and whether a 2×2 game without it would make the point more convincingly. It would not, because no such game exists: Exercise 7.4 asks the reader to show that a 2×2 game with two pure-strategy Nash equilibria, one of which fails perfection, is necessarily strategically equivalent to a team problem. The smallest games in which the team structure can be avoided are therefore 3×2 , and it is enough to give P1 one further action. Consider the 3×2 game in which *both players minimize cost*,

$$A = \begin{array}{c|cc} & L & R \\ \hline U & 0 & 1 \\ D & 1 & 1 \\ T & 0 & 2 \end{array}, \quad B = \begin{array}{c|cc} & L & R \\ \hline U & 0 & 1 \\ D & 1 & 1 \\ T & 1 & 0 \end{array}.$$

The upper 2×2 block is the game of Example 7.3; the row T is new.

The game is not a team problem. With the mixed second differences taken relative to the cell (U, L) , as in Exercise 7.4, the only nonzero cells are

$$\tilde{A}(D, R) = -1, \quad \tilde{A}(T, R) = 1, \quad \tilde{B}(D, R) = -1, \quad \tilde{B}(T, R) = -2.$$

Were the game strategically equivalent to a team problem, $\tilde{B}$ would be a positive multiple of $\tilde{A}$, and to a zero-sum game, a negative multiple. The two cells give the ratios 1 and -2 , so no multiple works and the game is neither. It is a genuine nonzero-sum game.

The equilibria are unchanged. Column by column, L gives P1 the costs $(0, 1, 0)$ and R gives $(1, 1, 2)$; row by row, P2's costs are $(0, 1)$ at U, $(1, 1)$ at D and $(1, 0)$ at T. Checking the six cells leaves exactly two pure-strategy Nash equilibria, (U, L) with

costs (0, 0) and (D, R) with costs (1, 1). The profile (T, L) fails because P2 prefers R at T, and (T, R) fails because P1 prefers U against R.

The conclusion is unchanged. Row D is weakly dominated by U, which is at least as cheap in both columns and strictly cheaper in L, so (D, R) uses a weakly dominated action and is not trembling-hand perfect. Neither U nor L is weakly dominated, L because P2 strictly prefers it at U while strictly preferring R at T, so (U, L) is perfect. Perfection again selects (U, L) and eliminates (D, R), exactly as in Example 7.3, but now in a game with no team structure to appeal to.

It is worth noting what the extra row does and does not do. Row T is itself weakly dominated by U, so P1 would never use it, and it belongs to no equilibrium. Its presence nevertheless destroys the proportionality of the two second differences, and with it the team structure. The team interpretation of Example 7.3 is therefore genuinely incidental to the perfection argument: it can be removed by an action that no one plays. $\square$

Theorem 7.5 (Selten, 1975, [107]). *Every finite normal-form game admits at least one trembling-hand perfect equilibrium.*

Proof. Let $G = (\mathcal{N}, (A_i)_{i \in \mathcal{N}}, (u_i)_{i \in \mathcal{N}})$ be a finite normal-form game, where each A_i is finite and $u_i : A = \prod_{j \in \mathcal{N}} A_j \rightarrow \mathbb{R}$ is player i 's payoff. Let $\Delta(A_i)$ denote the mixed-strategy simplex of player i , and let $\Delta := \prod_i \Delta(A_i)$.

Fix $\varepsilon > 0$ small enough that $\varepsilon \leq 1/|A_i|$ for every i . Define the ε -restricted (or *perturbed*) mixed-strategy sets

$$\Delta_i^\varepsilon := \{\sigma_i \in \Delta(A_i) : \sigma_i(a_i) \geq \varepsilon \text{ for all } a_i \in A_i\}, \quad \Delta^\varepsilon := \prod_i \Delta_i^\varepsilon.$$

Each Δ_i^ε is nonempty, compact, and convex (the uniform distribution is feasible). Let $G(\varepsilon)$ be the perturbed game obtained by restricting each player to Δ_i^ε ; multilinear payoffs ensure continuity and quasiconcavity in own strategy.

For each player i , the best-reply correspondence $\text{BR}_i^\varepsilon : \Delta_{-i}^\varepsilon \rightrightarrows \Delta_i^\varepsilon$ is nonempty, convex-valued, and upper hemicontinuous (it is the maximizer of a continuous linear function on a compact convex polytope). Hence, the joint correspondence $\prod_i \text{BR}_i^\varepsilon$ satisfies Kakutani's fixed-point theorem, invoked here exactly as in the existence proof for Nash equilibria in Lecture 3, and admits a fixed point $\sigma^\varepsilon \in \Delta^\varepsilon$. Thus each perturbed game $G(\varepsilon)$ has a Nash equilibrium σ^ε , which is *completely mixed* because of the lower-bound constraints.

Let (ε_k) be any sequence with $\varepsilon_k \downarrow 0$. For each k , let σ^{ε_k} be a Nash equilibrium of $G(\varepsilon_k)$ as above. Since the unrestricted strategy space Δ is compact, the sequence σ^{ε_k} has a convergent subsequence; relabeled so that $\sigma^{\varepsilon_k} \rightarrow \sigma^* \in \Delta$.

Suppose, to obtain a contradiction, that σ^* is not a Nash equilibrium of G . Then there exists i and $\tau_i \in \Delta(A_i)$ such that

$$u_i(\tau_i, \sigma_{-i}^*) > u_i(\sigma_i^*, \sigma_{-i}^*) + 2\delta \quad \text{for some } \delta > 0.$$

By continuity of u_i and $\sigma^{\varepsilon_k} \rightarrow \sigma^*$, for all sufficiently large k ,

$$u_i(\tau_i, \sigma_{-i}^{\varepsilon_k}) > u_i(\sigma_i^*, \sigma_{-i}^{\varepsilon_k}) + \delta.$$

Now let $\tau_i^{\varepsilon_k}$ be the projection of τ_i into $\Delta_i^{\varepsilon_k}$ obtained by mixing τ_i with the uniform distribution by the minimal amount required to satisfy the lower bounds ε_k ; then $\tau_i^{\varepsilon_k} \rightarrow \tau_i$. For large

k ,

$$u_i(\tau_i^{\varepsilon_k}, \sigma_{-i}^{\varepsilon_k}) > u_i(\sigma_i^*, \sigma_{-i}^{\varepsilon_k}) + \frac{\delta}{2}.$$

Since $\sigma_i^{\varepsilon_k} \rightarrow \sigma_i^*$, again by continuity we obtain, for all sufficiently large k ,

$$u_i(\tau_i^{\varepsilon_k}, \sigma_{-i}^{\varepsilon_k}) > u_i(\sigma_i^{\varepsilon_k}, \sigma_{-i}^{\varepsilon_k}),$$

contradicting the fact that σ^{ε_k} is a Nash equilibrium of the perturbed game $G(\varepsilon_k)$, where $\tau_i^{\varepsilon_k}$ is a feasible deviation. Therefore σ^* must be a Nash equilibrium of G .

By Selten's definition, a trembling-hand perfect equilibrium is precisely a strategy profile that arises as the limit of a sequence of completely mixed Nash equilibria of increasingly small perturbations. Here, σ^{ε_k} is completely mixed for each k , $\varepsilon_k \downarrow 0$, and $\sigma^{\varepsilon_k} \rightarrow \sigma^*$. Hence σ^* is a trembling-hand perfect equilibrium of G .

This proves that every finite normal-form game admits at least one trembling-hand perfect equilibrium. $\square$

Selten's existence result extends beyond normal-form games to all finite *extensive-form games with perfect recall*. The key device in this generalization is the *agent normal form* (ANF) introduced in [107]. Let Γ be a finite extensive-form game with perfect recall, as formalized in Lecture 5, and let $\mathcal{I}_i$ denote the collection of information sets of player i (in the notation of Lecture 5, the elements of $\mathcal{I}_i$ are the information sets η_i^k ; here we write I_i for a generic element of $\mathcal{I}_i$). For each player i and each information set $I_i \in \mathcal{I}_i$, the ANF introduces a distinct *agent* (i, I_i) whose action set is precisely the set of feasible actions $A_i(I_i)$ at I_i .

All agents belonging to player i share player i 's terminal payoff function u_i from Γ ; thus the utility of an agent (i, I_i) is defined to be u_i evaluated at the terminal node reached in the underlying extensive-form game. The resulting ANF is therefore a finite normal-form game whose pure strategies correspond to complete action choices at each information set.

By the existence theorem for trembling-hand perfect equilibria in finite normal-form games, the ANF admits a trembling-hand perfect equilibrium $\hat{\sigma}$ in mixed strategies. Under perfect recall, *Kuhn's theorem* guarantees a bijection between mixed strategies in the ANF and behavioral strategies in Γ , via the notion of *realization equivalence*. In particular, each mixed strategy in the ANF induces a unique behavior strategy in Γ that assigns to every information set I_i the same distribution over paths as the ANF mixed strategy does over the corresponding agent's actions.

Consequently, the trembling-hand perfect equilibrium $\hat{\sigma}$ of the ANF induces a behavior strategy profile σ in the extensive-form game Γ . Furthermore, Selten's perturbation in the ANF, requiring that every agent assign a strictly positive minimum probability to each action, and then letting these lower bounds tend to zero, corresponds exactly to introducing small *trembles at every information set* in the extensive form. Thus, the limit of a sequence of completely mixed equilibria of perturbed ANF games yields a behavior strategy profile in Γ that satisfies Selten's definition of a *trembling-hand perfect equilibrium* for extensive-form games.

Hence, every finite extensive-form game with perfect recall admits a trembling-hand perfect equilibrium.

Example 7.6 Selten's Trembling Perfection via the Agent Normal Form.

Consider a two-player extensive-form game with perfect recall, where both players are maximizers. P1 moves first and chooses either A or B . If A is chosen, the game ends immediately with payoffs $(2, 1)$. If instead B is chosen, the play proceeds to P2, who chooses between L and R , leading respectively to the terminal payoffs

$u(B, L) = (0, 0)$ and $u(B, R) = (3, 2)$.

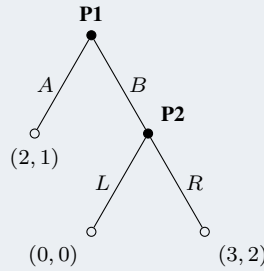


Figure 7.1: A simple two-stage extensive-form game.

Agent Normal Form (ANF). The extensive form contains two information sets:

$$I_1 = \{\text{P1's root}\}, \quad I_2 = \{\text{P2's node after } B\}.$$

The *Agent Normal Form* introduces one agent for each information set:

$$(1, I_1): \{A, B\}, \quad (2, I_2): \{L, R\}.$$

Each agent chooses its action simultaneously in the ANF, and all agents belonging to player i share the same payoff function u_i from the extensive form. The resulting normal-form payoff table is:

P1	P2	
	L	R
A	(2, 1)	(2, 1)
B	(0, 0)	(3, 2)

The unique Nash equilibrium of this ANF is (B, R) , the backward-induction outcome of the extensive form.

Trembles in the ANF. Let agent $(1, I_1)$ play A with probability p and B with probability $1 - p$, and agent $(2, I_2)$ play L with probability q and R with $1 - q$. In an ε -perturbed ANF every pure action must have probability at least $\varepsilon > 0$, and thus $p, q \in [\varepsilon, 1 - \varepsilon]$.

P1's expected payoffs are

$$U_1(A) = 2, \quad U_1(B) = 3(1 - q).$$

Hence P1 prefers B iff $q < \frac{1}{3}$, prefers A iff $q > \frac{1}{3}$, and is indifferent at $q = \frac{1}{3}$.

P2's expected payoffs are

$$U_2(L) = p, \quad U_2(R) = 2 - p.$$

Thus $U_2(R) > U_2(L)$ for every $p < 1$; R strictly dominates L for all feasible $p \leq 1 - \varepsilon$.

Equilibrium of the perturbed ANF. In any perturbed game, P2 must assign maximal mass to R subject to the lower bound:

$$q^\varepsilon = \varepsilon, \quad 1 - q^\varepsilon = 1 - \varepsilon.$$

Given $q^\varepsilon = \varepsilon < 1/3$, P1 strictly prefers B , and thus the perturbed equilibrium must similarly place maximal mass on B :

$$p^\varepsilon = \varepsilon, \quad 1 - p^\varepsilon = 1 - \varepsilon.$$

Thus the perturbed ANF has the completely mixed Nash equilibrium

$$\hat{\sigma}^\varepsilon = ((\varepsilon, 1 - \varepsilon), (\varepsilon, 1 - \varepsilon)).$$

Limit as $\varepsilon \rightarrow 0$. As $\varepsilon \downarrow 0$, $\hat{\sigma}^\varepsilon \rightarrow (B, R)$. By Selten's definition, this limit of completely mixed equilibria is a *trembling-hand perfect equilibrium* of the ANF.

Since the game has perfect recall, *Kuhn's theorem* provides a one-to-one, realization-equivalent correspondence between mixed strategies in the ANF and behavior strategies in the extensive-form game Γ . Let σ^ε be the behavior strategy corresponding to $\hat{\sigma}^\varepsilon$. Then $\sigma = \lim_{\varepsilon \rightarrow 0} \sigma^\varepsilon$ is realization-equivalent to (B, R) and satisfies Selten's trembling condition at each information set. Therefore σ is a *trembling-hand perfect equilibrium* of Γ .

□

7.3 ■ ε -Perfect Equilibrium

Myerson introduced in [81] the notion of an ε -*perfect equilibrium* as a parametric refinement of Nash equilibrium that renders the robustness requirement of trembling-hand perfection more transparent. The idea is to fix a small $\varepsilon > 0$ and perturb the strategy space so that (i) every pure strategy must be played with probability of at least ε , and (ii) any action that is strictly worse than some alternative must be played with probability at most an ε -fraction of that better action. These two restrictions ensure that every strategy is used with positive probability, while inferior strategies are placed at systematically lower probability levels.

Definition 7.7 (ε -Perfect Equilibrium [81]). Let $\Gamma = (\mathcal{N}, \{\Delta(A_i)\}_{i \in \mathcal{N}}, \{u_i\}_{i \in \mathcal{N}})$ be a finite normal-form game. A mixed strategy profile $\sigma \in \times_{i \in \mathcal{N}} \Delta(A_i)$ is an ε -perfect equilibrium if:

- (i) σ is a Nash equilibrium of Γ ;
- (ii) For every player i and every $a_i \in A_i$, $\sigma_i(a_i) \geq \varepsilon$;
- (iii) For any $a_i, a'_i \in A_i$ such that $u_i(a'_i, \sigma_{-i}) > u_i(a_i, \sigma_{-i})$, $\sigma_i(a_i) \leq \varepsilon \sigma_i(a'_i)$.

Condition (iii) requires that strictly inferior actions be played with probability at most an ε -fraction of strictly better actions. The parameter ε therefore regulates the permissible extent of suboptimal play.

Myerson showed that ε -perfect equilibria characterize trembling-hand perfection in the limit:

Proposition 7.8 (Myerson, 1978). *A mixed-strategy Nash equilibrium σ^* is trembling-hand perfect if, and only if, there exists a sequence $\{\varepsilon_k\}$ with $\varepsilon_k \rightarrow 0$ and a corresponding sequence of ε_k -perfect equilibria $\{\sigma^{\varepsilon_k}\}$ such that $\sigma^{\varepsilon_k} \rightarrow \sigma^*$.*

The ε -perfect equilibrium concept thus offers a practical test for trembling-hand perfection. Given an equilibrium, one perturbs each player's strategy so that all pure actions receive probability of at least ε , verifies the proportionality bound in (iii), and examines whether the perturbed profiles converge back to the original equilibrium as $\varepsilon \rightarrow 0$. If they do, the equilibrium is trembling-hand perfect; if not, it is eliminated by the refinement.

Example 7.9 ε -Perfect Equilibrium in a 2×2 Game.

Consider again the 2×2 game of Examples 7.1 and 7.3 in which both players minimize the cost matrix

$$A = B = \begin{array}{c|cc} & L & R \\ \hline U & 0 & 1 \\ D & 1 & 1 \end{array}.$$

The pure-strategy Nash equilibria are (U, L) and (D, R) . We test each for ε -perfection.

The equilibrium (U, L) . Fix $\varepsilon \in (0, 1)$. Because an ε -perfect equilibrium requires every pure strategy to be played with strictly positive probability, we perturb the equilibrium by assigning

$$\begin{aligned} \sigma_1^\varepsilon(U) &= \frac{1}{1+\varepsilon}, & \sigma_1^\varepsilon(D) &= \frac{\varepsilon}{1+\varepsilon}, \\ \sigma_2^\varepsilon(L) &= \frac{1}{1+\varepsilon}, & \sigma_2^\varepsilon(R) &= \frac{\varepsilon}{1+\varepsilon}. \end{aligned}$$

Given σ_2^ε , P1's expected costs are

$$C_1(U) = \frac{1}{1+\varepsilon} \cdot 0 + \frac{\varepsilon}{1+\varepsilon} \cdot 1 = \frac{\varepsilon}{1+\varepsilon}, \quad C_1(D) = \frac{1}{1+\varepsilon} \cdot 1 + \frac{\varepsilon}{1+\varepsilon} \cdot 1 = 1.$$

Thus U strictly dominates D at the perturbed profile. Condition (iii) in Definition 7.7 requires

$$\sigma_1^\varepsilon(D) \leq \varepsilon \sigma_1^\varepsilon(U) \iff \frac{\varepsilon}{1+\varepsilon} \leq \frac{\varepsilon}{1+\varepsilon},$$

which holds with equality for all $\varepsilon \in (0, 1)$. A symmetric argument applies to P2: L is strictly better than R , and the corresponding inequality holds. Hence $(\sigma_1^\varepsilon, \sigma_2^\varepsilon)$ satisfies the ratio condition for every $\varepsilon \in (0, 1)$, and letting $\varepsilon \rightarrow 0$ yields (U, L) . Therefore (U, L) is trembling-hand perfect.

The equilibrium (D, R) . Now perturb (D, R) analogously:

$$\sigma_1^\varepsilon(D) = \frac{1}{1+\varepsilon}, \quad \sigma_1^\varepsilon(U) = \frac{\varepsilon}{1+\varepsilon}, \quad \sigma_2^\varepsilon(R) = \frac{1}{1+\varepsilon}, \quad \sigma_2^\varepsilon(L) = \frac{\varepsilon}{1+\varepsilon}.$$

Given σ_2^ε , P1's expected costs are

$$C_1(U) = \frac{1}{1+\varepsilon} \cdot 1 + \frac{\varepsilon}{1+\varepsilon} \cdot 0 = \frac{1}{1+\varepsilon}, \quad C_1(D) = 1.$$

Thus U is now strictly better than D in the perturbed game. But condition (iii) would require

$$\sigma_1^\varepsilon(D) \leq \varepsilon \sigma_1^\varepsilon(U) \iff 1 \leq \varepsilon^2,$$

which is impossible for $\varepsilon \in (0, 1)$. The same issue arises symmetrically for P2. Since any totally mixed profile near (D, R) must place probability close to 1 on D , while the ratio condition forces the probability of D below ε , no sequence of ε -perfect equilibria can converge to (D, R) , and the strategy pair fails trembling-hand perfection. □

Remark 7.10. The equivalence holds because ε -perfect equilibria model players as making *small, structured mistakes*, whereas trembling-hand perfection requires that an equilibrium remains stable under all such vanishing perturbations. In an ε -perfect equilibrium, every action satisfies $\sigma_i(a_i) \geq \varepsilon$, but, crucially, any strictly worse action a_i must also satisfy condition (iii) in Definition 7.7. This ratio constraint imposes a strict *hierarchy* among actions with different payoff rankings, effectively forcing inferior actions to appear only as *higher-order trembles*. To see this, suppose that player i has actions $a_i^{(1)}, a_i^{(2)}, a_i^{(3)}$ with

$$u_i(a_i^{(1)}, \sigma_{-i}) > u_i(a_i^{(2)}, \sigma_{-i}) > u_i(a_i^{(3)}, \sigma_{-i}).$$

Condition (iii) then implies

$$\sigma_i(a_i^{(2)}) \leq \varepsilon \sigma_i(a_i^{(1)}), \quad \sigma_i(a_i^{(3)}) \leq \varepsilon \sigma_i(a_i^{(2)}) \leq \varepsilon^2 \sigma_i(a_i^{(1)}).$$

Thus an optimal action $a_i^{(1)}$ may be played with probability of order 1, a strictly worse action $a_i^{(2)}$ with probability of order ε , and a still worse action $a_i^{(3)}$ with probability of order ε^2 . In general, an action that is k steps down in the payoff ranking must receive probability on the order of ε^k . This creates a disciplined perturbation structure in which the likelihood of an action decays multiplicatively with its degree of suboptimality: the worse an action is, the faster its probability must vanish as $\varepsilon \rightarrow 0$. Such differential decay rates encode the idea that players primarily choose best responses, while increasingly poor actions arise only through increasingly improbable trembles.

As $\varepsilon \rightarrow 0$, the noise disappears but the relative magnitudes of $\sigma_i(a_i)$ preserve the best-response ordering. If an equilibrium is trembling-hand perfect, then for each ε one can construct an ε -perfect equilibrium reflecting how players best-respond under small trembles, and these profiles converge to the equilibrium. If an equilibrium is not trembling-hand perfect, no sequence σ^{ε_k} with $\varepsilon_k \rightarrow 0$ can satisfy the ratio constraint, since some inferior action would necessarily retain too much probability in the limit. Thus, the ε -perfect conditions, and especially the ratio bound, provide an algebraic representation of “vanishing but disciplined mistakes,” and the limits of such sequences identify exactly the trembling-hand perfect equilibria.

7.4 ■ Proper Equilibrium

Selten's trembling-hand perfect equilibrium assumes that players make small mistakes (*trembles*) but does not distinguish between large and small mistakes: every pure strategy must be played with some small positive probability, even if one strategy is only slightly worse while another one is severely dominated. As a result, trembling-hand perfection may permit equilibria in which badly performing or even dominated strategies retain disproportionately large probability.

Myerson's *proper equilibrium* [81] sharpens this by imposing *structured mistakes*: the worse a strategy is relative to a better one, the faster its probability must vanish. If a_i yields a strictly lower payoff than a'_i , then the probability of playing a_i must be at most an ε -fraction of the probability of playing a'_i . Thus, more costly mistakes are dramatically less likely than small ones.

Definition 7.11 (ε -Proper Equilibrium). Let $\Gamma = (\mathcal{N}, \{A_i\}_{i \in \mathcal{N}}, \{u_i\}_{i \in \mathcal{N}})$ be a finite normal-form game. A mixed strategy profile $\sigma \in \times_{i \in \mathcal{N}} \Delta(A_i)$ is an ε -proper equilibrium if for every player i and every pair of actions $a_i, a'_i \in A_i$,

$$u_i(a'_i, \sigma_{-i}) > u_i(a_i, \sigma_{-i}) \implies \sigma_i(a_i) \leq \varepsilon \sigma_i(a'_i).$$

It is important to observe that the key difference between perfect and proper equilibrium lies not in the hierarchical decay of inferior strategies, which both refinements share, but in the *lower bound* imposed on strategy probabilities. An ε -perfect equilibrium requires

$$\sigma_i(a_i) \geq \varepsilon, \quad \text{for all } a_i,$$

and thus every pure strategy must be played with probability of at least ε , no matter how inferior it is. This uniform lower bound creates a “floor” that prevents extremely poor strategies from disappearing too quickly; their decay is capped at order ε , even when the payoff structure would dictate that they should be played far less frequently.

An ε -proper equilibrium imposes *no such floor*. A strategy that is substantially worse than the available alternatives may be driven down to order ε^2 , ε^3 , or even smaller, depending on how many strategies dominate it. Proper equilibrium therefore preserves the hierarchical decay of inferior actions *without* enforcing a uniform minimum probability. This allows the refinement to eliminate equilibria in which severely dominated or near-dominated strategies retain probabilities that arise only because of the ε -perfect lower bound rather than from any plausible structure of mistakes.

This distinction also clarifies the refinement relationship. Every ε -proper equilibrium automatically satisfies the requirements of ε -perfection, since proportional decay implies the necessary absolute decay. However, the uniform lower bound demanded by ε -perfection does not follow from proportionality. Consequently, every proper equilibrium is trembling-hand perfect in the limit as $\varepsilon \rightarrow 0$, but the reverse need not hold: trembling-hand perfection may allow inferior strategies to retain artificially large probabilities due to the imposed lower bound, whereas proper equilibrium eliminates such behavior. Proper equilibrium thus yields a strictly stronger and more behaviorally disciplined refinement.

Example 7.12 Proper Equilibrium in a 3×2 Game.

Consider the 3×2 game in which P1 has strategies U, M, D and P2 has strategies

L , R , and both players maximize their payoffs. P1's payoff matrix is

$$A = \begin{array}{c|cc} & L & R \\ \hline U & 2 & 0 \\ M & 1 & 1 \\ D & 0 & 0 \end{array}, \quad \text{and P2's payoff matrix is} \quad B = \begin{array}{c|cc} & L & R \\ \hline U & 1 & 0 \\ M & 0 & 1 \\ D & 0 & 0 \end{array}.$$

Here, D is strictly dominated by M for P1, and L is P2's unique best response when P1 plays U . Thus (U, L) is the natural candidate to test for the refinement concepts.

Trembling-hand perfect sequences. Trembling-hand perfection requires considering sequences of completely mixed strategy profiles whose perturbations vanish. Fix a sequence $\{\varepsilon_k\} \downarrow 0$ and let P2 mix according to

$$\sigma_2^{\varepsilon_k}(L) = 1 - \varepsilon_k, \quad \sigma_2^{\varepsilon_k}(R) = \varepsilon_k.$$

Given this, P1's unique payoff-maximizing reply is U . To satisfy the requirement that *every* pure strategy receive at least probability ε_k , one valid sequence for P1 is

$$\sigma_1^{\varepsilon_k}(U) = 1 - 2\varepsilon_k, \quad \sigma_1^{\varepsilon_k}(M) = \varepsilon_k, \quad \sigma_1^{\varepsilon_k}(D) = \varepsilon_k.$$

Each $\sigma_1^{\varepsilon_k}$ is fully mixed, each profile $(\sigma_1^{\varepsilon_k}, \sigma_2^{\varepsilon_k})$ is a Nash equilibrium of the ε_k -perturbed game, and the sequence converges to (U, L) . Hence (U, L) is trembling-hand perfect.

Why trembling-hand perfection is not enough. However, the same sequence shows the weakness of perfect equilibrium. The dominated action D receives exactly the same probability as M :

$$\sigma_1^{\varepsilon_k}(D) = \sigma_1^{\varepsilon_k}(M) = \varepsilon_k.$$

Thus, the “large” mistake D and the “smaller” mistake M occur with equal probability, even though M strictly dominates D . Trembling-hand perfection allows this because it requires all mistakes to be small but does not require worse mistakes to be *less* frequent than better ones.

Proper equilibrium sequences. Proper equilibrium imposes exactly this missing proportional discipline. Whenever M strictly dominates D , the ε_k -proper condition requires

$$\sigma_1^{\varepsilon_k}(D) \leq \varepsilon_k \sigma_1^{\varepsilon_k}(M).$$

But the trembling-hand perfect sequence violates this for all k :

$$\sigma_1^{\varepsilon_k}(D) = \varepsilon_k = \sigma_1^{\varepsilon_k}(M) \implies 1 \leq \varepsilon_k,$$

which is impossible for small ε_k . Thus the sequence is ε_k -perfect but not ε_k -proper.

A proper-equilibrium sequence. To satisfy properness, the strictly dominated action D must vanish *faster* than the milder mistake M , and M in turn must vanish relative to the best reply U . A sequence satisfying these requirements is

$$\sigma_1^{\varepsilon_k}(U) = \frac{1}{1 + \varepsilon_k + \varepsilon_k^2}, \quad \sigma_1^{\varepsilon_k}(M) = \frac{\varepsilon_k}{1 + \varepsilon_k + \varepsilon_k^2}, \quad \sigma_1^{\varepsilon_k}(D) = \frac{\varepsilon_k^2}{1 + \varepsilon_k + \varepsilon_k^2},$$

with $\sigma_2^{\varepsilon_k}(L) = \frac{1}{1 + \varepsilon_k}$ and $\sigma_2^{\varepsilon_k}(R) = \frac{\varepsilon_k}{1 + \varepsilon_k}$. Against $\sigma_2^{\varepsilon_k}$, the actions of P1 are ordered as U , then M , then D , and the sequence satisfies $\sigma_1^{\varepsilon_k}(M) = \varepsilon_k \sigma_1^{\varepsilon_k}(U)$ and $\sigma_1^{\varepsilon_k}(D) = \varepsilon_k \sigma_1^{\varepsilon_k}(M)$; similarly, against $\sigma_1^{\varepsilon_k}$, action L is strictly better than R for P2, and $\sigma_2^{\varepsilon_k}(R) = \varepsilon_k \sigma_2^{\varepsilon_k}(L)$. Hence each profile is ε_k -proper, and the sequence converges to (U, L) . Thus (U, L) is not only trembling-hand perfect but also proper.

This example illustrates the distinction: trembling-hand perfection controls only the *absolute* size of mistakes, while proper equilibrium controls their *relative* frequency, ensuring that worse mistakes are exponentially rarer than milder ones. The first sequence shows why perfection alone is insufficient; the second one shows how proper equilibrium restores the correct dominance structure among off-path strategies. $\square$

The ratio condition in Definition 7.11 can be represented by a logit or exponential response with a vanishing noise parameter. Fix a sequence $\{\lambda_k\}_{k \in \mathbb{N}}$ with $\lambda_k \rightarrow \infty$, set $\varepsilon_k := e^{-\lambda_k}$, and for each k define player i 's mixed strategy $\sigma_i^{\varepsilon_k}$ by

$$\sigma_i^{\varepsilon_k}(a) \propto \exp(\lambda_k u_i(a, \sigma_{-i}^{\varepsilon_k})), \quad a \in A_i,$$

i.e.,

$$\sigma_i^{\varepsilon_k}(a) = \frac{\exp(\lambda_k u_i(a, \sigma_{-i}^{\varepsilon_k}))}{\sum_{b \in A_i} \exp(\lambda_k u_i(b, \sigma_{-i}^{\varepsilon_k}))}.$$

For any two actions $a, a' \in A_i$, this implies the exact ratio identity

$$\frac{\sigma_i^{\varepsilon_k}(a)}{\sigma_i^{\varepsilon_k}(a')} = \exp(\lambda_k (u_i(a, \sigma_{-i}^{\varepsilon_k}) - u_i(a', \sigma_{-i}^{\varepsilon_k}))).$$

If a' is strictly better than a at $\sigma_{-i}^{\varepsilon_k}$, i.e. $u_i(a', \sigma_{-i}^{\varepsilon_k}) > u_i(a, \sigma_{-i}^{\varepsilon_k})$, then the payoff difference $\Delta_k(a', a) := u_i(a', \sigma_{-i}^{\varepsilon_k}) - u_i(a, \sigma_{-i}^{\varepsilon_k}) > 0$ and

$$\frac{\sigma_i^{\varepsilon_k}(a)}{\sigma_i^{\varepsilon_k}(a')} = \exp(-\lambda_k \Delta_k(a', a)).$$

If we normalize payoffs so that every strictly positive payoff difference is at least 1, i.e.

$$u_i(a', \sigma_{-i}) > u_i(a, \sigma_{-i}) \quad \Rightarrow \quad u_i(a', \sigma_{-i}) - u_i(a, \sigma_{-i}) \geq 1,$$

then

$$\frac{\sigma_i^{\varepsilon_k}(a)}{\sigma_i^{\varepsilon_k}(a')} \leq \exp(-\lambda_k).$$

Hence, recalling that

$$\varepsilon_k = e^{-\lambda_k},$$

we obtain

$$u_i(a', \sigma_{-i}^{\varepsilon_k}) > u_i(a, \sigma_{-i}^{\varepsilon_k}) \quad \Rightarrow \quad \sigma_i^{\varepsilon_k}(a) \leq \varepsilon_k \sigma_i^{\varepsilon_k}(a'),$$

which is exactly the ε_k -proper ratio condition.

This observation serves two complementary purposes. Formally, it shows that the ratio-based characterization of proper equilibrium can be implemented by a sequence of logit-type (exponential) response profiles with a vanishing noise parameter $\lambda_k \rightarrow \infty$. Intuitively, it provides a concrete construction: as λ_k grows, players place exponentially more weight on better replies, and the probabilities of worse actions shrink at an exponential rate, automatically generating the ε_k -proper inequalities with $\varepsilon_k = e^{-\lambda_k} \rightarrow 0$. Proper equilibrium itself is still defined purely by the ratio condition and the limit $k \rightarrow \infty$; the exponential form is a convenient tool for producing sequences that satisfy these requirements.

Lemma 7.13 (Exponential sequences and ε -proper equilibrium). *Let $\Gamma = (\mathcal{N}, \{A_i\}_{i \in \mathcal{N}}, \{u_i\}_{i \in \mathcal{N}})$ be a finite normal-form game. Normalize payoffs so that every strictly positive payoff difference is at least 1, i.e.,*

$$u_i(a'_i, a_{-i}) > u_i(a_i, a_{-i}) \implies u_i(a'_i, a_{-i}) - u_i(a_i, a_{-i}) \geq 1, \quad \forall i \in \mathcal{N}. \quad (7.1)$$

Let $\{\lambda_k\}_{k \in \mathbb{N}}$ be a sequence with $\lambda_k \rightarrow \infty$, set $\varepsilon_k := e^{-\lambda_k}$, and suppose that for each k we have a profile $\sigma^{\varepsilon_k} \in \times_{i \in \mathcal{N}} \Delta(A_i)$ such that for every player i and every action $a_i \in A_i$,

$$\sigma_i^{\varepsilon_k}(a_i) = \frac{\exp(\lambda_k u_i(a_i, \sigma_{-i}^{\varepsilon_k}))}{\sum_{b_i \in A_i} \exp(\lambda_k u_i(b_i, \sigma_{-i}^{\varepsilon_k}))}. \quad (7.2)$$

Then for every player i and every pair of actions $a_i, a'_i \in A_i$,

$$u_i(a'_i, \sigma_{-i}^{\varepsilon_k}) > u_i(a_i, \sigma_{-i}^{\varepsilon_k}) \implies \sigma_i^{\varepsilon_k}(a_i) \leq \varepsilon_k \sigma_i^{\varepsilon_k}(a'_i). \quad (7.3)$$

In particular, each σ^{ε_k} is an ε_k -proper profile in the sense of the ratio condition. Moreover, if a mixed strategy profile σ^ is a Nash equilibrium and there exists a subsequence $\sigma^{\varepsilon_{k_\ell}} \rightarrow \sigma^*$, then σ^* is a proper equilibrium.*

Proof. Fix a player $i \in \mathcal{N}$ and $k \in \mathbb{N}$. By (7.2), for any two actions $a_i, a'_i \in A_i$ we have

$$\begin{aligned} \frac{\sigma_i^{\varepsilon_k}(a_i)}{\sigma_i^{\varepsilon_k}(a'_i)} &= \frac{\exp(\lambda_k u_i(a_i, \sigma_{-i}^{\varepsilon_k}))}{\sum_{b_i \in A_i} \exp(\lambda_k u_i(b_i, \sigma_{-i}^{\varepsilon_k}))} \cdot \frac{\sum_{b_i \in A_i} \exp(\lambda_k u_i(b_i, \sigma_{-i}^{\varepsilon_k}))}{\exp(\lambda_k u_i(a'_i, \sigma_{-i}^{\varepsilon_k}))} \\ &= \exp\left(\lambda_k (u_i(a_i, \sigma_{-i}^{\varepsilon_k}) - u_i(a'_i, \sigma_{-i}^{\varepsilon_k}))\right). \end{aligned}$$

Now suppose that $u_i(a'_i, \sigma_{-i}^{\varepsilon_k}) > u_i(a_i, \sigma_{-i}^{\varepsilon_k})$. Let

$$\Delta_k(a'_i, a_i) := u_i(a'_i, \sigma_{-i}^{\varepsilon_k}) - u_i(a_i, \sigma_{-i}^{\varepsilon_k}).$$

By the normalization assumption (7.1), we have $\Delta_k(a'_i, a_i) \geq 1$. Then,

$$\begin{aligned} \frac{\sigma_i^{\varepsilon_k}(a_i)}{\sigma_i^{\varepsilon_k}(a'_i)} &= \exp(-\lambda_k \Delta_k(a'_i, a_i)) \\ &\leq \exp(-\lambda_k), \end{aligned}$$

where the inequality uses $\Delta_k(a'_i, a_i) \geq 1$. Recalling the definition $\varepsilon_k := e^{-\lambda_k}$, we obtain

$$\frac{\sigma_i^{\varepsilon_k}(a_i)}{\sigma_i^{\varepsilon_k}(a'_i)} \leq \varepsilon_k,$$

or equivalently

$$\sigma_i^{\varepsilon_k}(a_i) \leq \varepsilon_k \sigma_i^{\varepsilon_k}(a'_i).$$

This is exactly the ε_k -proper ratio condition (7.3). Since this holds for every player i and every pair of actions a_i, a'_i , each profile σ^{ε_k} is ε_k -proper in the sense of the definition of ε -proper equilibrium (without yet imposing any Nash requirement).

Next, note that $\lambda_k \rightarrow \infty$ implies

$$\varepsilon_k = e^{-\lambda_k} \rightarrow 0 \quad \text{as } k \rightarrow \infty.$$

Suppose now that there exists a subsequence $\{\sigma^{\varepsilon_{k_\ell}}\}_{\ell \in \mathbb{N}}$ that converges to some profile σ^* , and that σ^* is a Nash equilibrium of Γ . By what we have already shown, each $\sigma^{\varepsilon_{k_\ell}}$ is ε_{k_ℓ} -proper, with $\varepsilon_{k_\ell} \rightarrow 0$. By the standard limit characterization of proper equilibrium (stated, for instance, as: “a mixed-strategy Nash equilibrium is proper if, and only if, it is the limit of a sequence of ε_k -proper profiles with $\varepsilon_k \rightarrow 0$ ”), this is exactly the condition required for σ^* to be a proper equilibrium.

For completeness, let us briefly justify this last step. By definition, a mixed-strategy profile σ^* is a proper equilibrium if σ^* is a Nash equilibrium; and there exists a sequence $\{\varepsilon_k\}$ with $\varepsilon_k \rightarrow 0$ and a sequence of ε_k -proper profiles σ^{ε_k} such that $\sigma^{\varepsilon_k} \rightarrow \sigma^*$.

In our setting, σ^* is assumed to be in Nash equilibrium by hypothesis, and we have explicitly constructed such a sequence $\{\sigma^{\varepsilon_k}\}$ with $\varepsilon_k = e^{-\lambda_k} \rightarrow 0$ and $\sigma^{\varepsilon_k} \rightarrow \sigma^*$. Therefore, σ^* satisfies the definition of a proper equilibrium. This completes the proof. $\square$

7.5 ■ Other Refinements

While trembling-hand perfection and proper equilibrium are the most common refinements in normal-form games, several additional concepts have been proposed to address subtleties that arise in *extensive-form* games (introduced in Lecture 5), where the temporal structure of play, information sets, and belief formation play a central role. In extensive-form games, a player’s strategy specifies an action at each information set, and equilibrium analysis must ensure that these actions are not only optimal ex ante but also optimal at every point along the game tree, given the player’s beliefs about how the game has unfolded. Refinements in this setting strengthen equilibrium requirements by imposing *sequential rationality* and *consistency of beliefs* throughout the game.

7.5.1 ■ Quasi-Perfect Equilibrium

Quasi-perfect equilibrium, introduced by van Damme [136], is a refinement of trembling-hand perfection tailored specifically to extensive-form games. The motivation arises from the observation that, in sequential settings, the normal-form representation blurs the informational and temporal structure of the game, and thus treats a player’s own future deviations in a way that is inconsistent with the idea of rational planning. Quasi-perfection restores this structure by imposing a perturbation-based criterion that respects both the decision tree and perfect recall.

The key behavioral asymmetry underlying quasi-perfect equilibrium concerns how players model deviations. Each player assumes that *other players* may make small mistakes (*trembles*), but never themselves. This differs from trembling-hand perfection, in which all players (including oneself) may tremble with small probability. By ruling out self-trembles, quasi-perfect equilibrium captures an introspective form of rationality: a player evaluates every continuation decision under the conviction that she will follow their own strategy, but that opponents may err. This resolves certain pathologies of trembling-hand perfection in extensive forms, such as

equilibria supported by beliefs about one's own hypothetical mistakes, while preserving the robustness to others' trembles.

Definition 7.14 (Quasi-Perfect Equilibrium [136]). Let Γ be a finite extensive-form game with perfect recall, and let σ^* be a behavioral strategy profile. We say that σ^* is a quasi-perfect equilibrium if there exists a sequence of fully mixed behavioral strategy profiles $\{\sigma^k\}_{k=1}^\infty$ such that $\sigma^k \rightarrow \sigma^*$ and the following condition holds: for every player i and every information set $I_i \in \mathcal{I}_i$,

$$\sigma_i^*(\cdot \mid I_i) \in \arg \max_{\tau_i \in \Delta(A(I_i))} u_i(\tau_i, \sigma_{-i}^k \mid I_i) \quad \text{for all sufficiently large } k.$$

Beliefs μ^k at each information set are induced from σ^k via Bayes' rule (since σ^k is fully mixed) and converge to a belief system μ^* associated with σ^* .

Thus, even at information sets that are not reached under σ^* , each player's beliefs about the history of play arise from consistent perturbations in which every node is reached with positive probability.

This definition ensures that at every information set I_i , the continuation strategy $\sigma_i^*(\cdot \mid I_i)$ is a best response to beliefs generated by small trembles of *other* players only. Consequently, quasi-perfect equilibrium enforces a form of *sequential rationality* that does not rely on conditioning on one's own hypothetical mistakes. It rules out equilibria that require self-contradictory or non-credible off-path beliefs and strengthens trembling-hand perfection in dynamic settings.

Quasi-perfect equilibrium lies conceptually between trembling-hand perfection and sequential equilibrium (introduced next). Like trembling-hand perfection, it is grounded in perturbations of strategies; like sequential equilibrium, it ensures optimality at every information set and beliefs derived from consistent perturbations. The distinctive feature is that beliefs are constructed under the restriction that players contemplate only *others'* trembles, and not their own. This subtle but important modification eliminates equilibria supported by internally inconsistent reasoning about off-path play. Quasi-perfect equilibrium thus enhances the analysis of dynamic games by ensuring robustness, introspective consistency, and credibility of strategies throughout the game tree.

7.5.2 ■ Sequential Equilibrium

Sequential equilibrium, introduced by Kreps and Wilson [68], is a refinement of Nash equilibrium specifically tailored to extensive-form games with imperfect information. It augments the equilibrium concept by requiring that players form consistent beliefs about how the game has unfolded at every information set and act optimally given those beliefs. Unlike Nash equilibrium, sequential equilibrium evaluates optimality *locally* at each information set, including those reached with zero probability in equilibrium play. A detailed analysis of sequential equilibrium, including examples, belief-system construction, and its relation to other dynamic refinements, will be presented in **Lecture 16 on Extensive-Form Dynamic Games**.

Definition 7.15 (Sequential Equilibrium [68]). Let Γ be a finite extensive-form game with perfect recall. A sequential equilibrium is a pair (σ, μ) consisting of:

- (a) a behavioral strategy profile $\sigma = (\sigma_i)_{i \in \mathcal{N}}$, where each σ_i assigns to every information set $I_i \in \mathcal{I}_i$ of player i a probability distribution over the feasible actions $A(I_i)$; and
- (b) a belief system μ , where for each information set I_i of player i , the belief $\mu(I_i)$ is a probability distribution over the nodes contained in I_i .

The pair (σ, μ) must satisfy:

(i) Sequential rationality. For every player i and every information set I_i ,

$$\sigma_i(\cdot \mid I_i) \in \arg \max_{\tau_i \in \Delta(A(I_i))} \mathbb{E}_{\mu(I_i), \sigma_{-i}} [u_i \text{ given } (\tau_i, \sigma_{-i})],$$

i.e., player i 's continuation strategy at I_i maximizes her expected payoff given her belief $\mu(I_i)$ and the continuation strategies of the other players.

(ii) Consistency. There exists a sequence of totally mixed behavioral strategy profiles $\{\sigma^k\}_{k=1}^\infty$ converging to σ such that the beliefs μ arise as the limit of beliefs μ^k derived from σ^k using Bayes' rule wherever possible. Formally, for every information set I_i :

- if I_i is reached with positive probability under σ , then $\mu(I_i)$ is obtained directly from σ by Bayes' rule;
- if I_i is reached with zero probability under σ , then $\mu(I_i) = \lim_{k \rightarrow \infty} \mu^k(I_i)$, where each $\mu^k(I_i)$ is computed from σ^k by Bayes' rule.

Sequential equilibrium refines Nash equilibrium by ensuring that strategies are optimal and beliefs are coherent at every information set, including those not reached in equilibrium play. This eliminates equilibria supported by arbitrary or non-credible beliefs and ensures that strategies remain dynamically consistent throughout the game tree. From a conceptual perspective, sequential equilibrium bridges behavioral and informational refinements: it requires players to act optimally given their information (behavioral coherence) and requires beliefs to be grounded in limits of perturbed play (informational coherence). Every sequential equilibrium is a perfect equilibrium in Selten's sense, but not conversely; sequential equilibrium ties both rationality and beliefs explicitly to the extensive-form structure of the game. The interplay between sequential rationality and belief consistency introduced here reappears in Lecture 17, where the closely related notion of perfect Bayesian equilibrium is developed for signaling games with incomplete information.

7.5.3 ■ Strategic Stability

The theory of *strategic stability*, introduced by Kohlberg and Mertens [64], provides one of the most structurally robust refinements of Nash equilibrium. A central aspect of their analysis is that robustness in this sense cannot in general be borne by a single equilibrium point: the refinement selects *sets* of equilibria, the *strategically stable sets*. Whereas refinements such as trembling-hand perfection, quasi-perfect equilibrium, and sequential equilibrium emphasize local behavioral consistency or robustness to execution errors, strategic stability emphasizes *global stability* of equilibrium under small perturbations of the game and *invariance* across strategically equivalent representations. The central idea is that equilibrium predictions should depend only on the strategic essence of the game, not on arbitrary numerical or representational details.

Many classical refinements can be sensitive to the way a game is represented or parameterized. For example, converting a normal-form game into an equivalent extensive form, or perturbing payoffs by arbitrarily small amounts, may change the equilibria selected by these refinements. Kohlberg and Mertens addressed these issues by proposing a refinement based on topological and structural stability: equilibria should persist under all strategic perturbations and should vary continuously with the game.

Definition 7.16 (Strategically Stable Set [64]). Let Γ be a finite normal-form game with Nash equilibrium set $NE(\Gamma)$. A strategically stable set $E(\Gamma)$ is a minimal closed, connected subset of $NE(\Gamma)$ satisfying:

- (i) **Invariance.** If two games Γ and Γ' are strategically equivalent, i.e., they induce identical preference orderings over mixed strategies for every player, then $E(\Gamma) = E(\Gamma')$.
- (ii) **Continuity.** The correspondence $E(\Gamma)$ is upper hemicontinuous in payoffs: small perturbations to Γ lead to only small changes in $E(\Gamma)$.
- (iii) **Essentiality.** Every equilibrium $\sigma^* \in E(\Gamma)$ is essential, meaning that it persists under small perturbations. Formally, there exists a neighborhood $\mathcal{O}$ of Γ such that every $\Gamma' \in \mathcal{O}$ admits an equilibrium $\sigma' \in NE(\Gamma')$ arbitrarily close to σ^* .

The set $E(\Gamma)$ is then said to be strategically stable, and any equilibrium $\sigma^* \in E(\Gamma)$ is called a stable equilibrium.

The invariance requirement ensures that the refinement does not depend on the representation of the game; continuity ensures robustness under small numerical perturbations; and essentiality guarantees persistence under payoff shocks. Together, these properties identify the equilibria that are structurally robust features of the game's strategic geometry. Minimality is essential here: without it the whole set $NE(\Gamma)$ would qualify trivially, since some equilibrium of every nearby game lies close to $NE(\Gamma)$; requiring a *minimal* such set is what turns stability into a genuine refinement.

Strategic stability thus synthesizes behavioral and structural robustness. Behaviorally, every stable equilibrium is a Nash equilibrium and, in fact, a proper equilibrium. Structurally, it requires that equilibrium predictions vary continuously with the game and do not hinge on fragile or knife-edge conditions. This yields the refinement chain

Stable Equilibrium $\subset$ Proper Equilibrium $\subset$ Perfect Equilibrium $\subset$ Nash Equilibrium,

with all inclusions being strict in general.

Example 7.17 Coordination Game and Strategic Stability.

Consider the classical 2×2 coordination game, in which both players *maximize* their payoffs:

	P2	
	C	D
P1		
C	(2, 2)	(0, 0)
D	(0, 0)	(1, 1)

The game has two pure-strategy Nash equilibria, (C, C) and (D, D) , each reflecting successful coordination on one of the two available actions. In addition, there is a mixed-strategy equilibrium in which both players choose C with probability $p^* = 1/3$ and D with probability $2/3$. This mixed-strategy equilibrium arises from the indifference condition for P1: if P2 chooses C with probability p , then the expected payoff from choosing C is $2p$, while that from choosing D is $1 - p$, and equating these yields $p = 1/3$. By symmetry, P2 must also mix with this probability.

To examine strategic stability, we perturb the game by altering the payoff at (D, D)

through a parameter ε :

P1	P2	
	C	D
C	$(2, 2)$	$(0, 0)$
D	$(0, 0)$	$(1 + \varepsilon, 1 + \varepsilon)$

$\varepsilon > -1.$

This creates a family of nearby games, permitting us to test whether each equilibrium persists under changes in payoffs. The parameter ε need not be small: the analysis below holds over the entire range $\varepsilon > -1$.

Begin with the equilibrium (C, C) . If P2 chooses C , then P1 compares $u_1(C, C) = 2$ with $u_1(D, C) = 0$, and since $2 > 0$ independently of ε , P1 always prefers C when P2 plays C . Consequently, (C, C) persists as an equilibrium for every perturbation of the payoff at (D, D) . The equilibrium (D, D) is almost as robust: if P2 chooses D , P1 compares $u_1(D, D) = 1 + \varepsilon$ with $u_1(C, D) = 0$, so D remains a best response precisely when $\varepsilon \geq -1$. Both pure-strategy equilibria therefore survive on the whole range $\varepsilon > -1$, and not merely for small ε .

The mixed-strategy equilibrium behaves differently. In the perturbed game, P1 is indifferent when $2p = (1 + \varepsilon)(1 - p)$, leading to the equilibrium mixing probability

$$p(\varepsilon) = \frac{1 + \varepsilon}{3 + \varepsilon}.$$

This value changes continuously with ε and reduces to $1/3$ at $\varepsilon = 0$. Although a symmetric perturbed game still admits a mixed-strategy equilibrium as long as $p(\varepsilon) \in [0, 1]$, the equilibrium depends critically on an exact equality between expected payoffs. Any slight asymmetry between players, or any perturbation affecting different payoffs unequally, breaks the indifference condition and causes the mixed-strategy equilibrium to disappear entirely. Even in symmetric perturbations, the mixed-strategy equilibrium is fragile, since maintaining the required equality demands knife-edge conditions that are unlikely to hold once small modeling noise, trembles, or measurement errors are introduced.

Comparing the behavior of the equilibria under perturbation reveals the essential idea behind strategic stability. Both pure-strategy equilibria survive every sufficiently small perturbation: neither relies on fragile equalities, and each continues to satisfy the best-response requirements in the perturbed games. By contrast, the mixed-strategy equilibrium fails to survive generic perturbations, since its existence depends on precise indifference conditions that vanish under even minimal perturbations of the payoff structure. This example therefore illustrates a general phenomenon in finite games: pure equilibria that correspond to robust coordination points tend to be strategically stable, while mixed-strategy equilibria, although mathematically valid as Nash equilibria, are structurally fragile and typically fail the strategic stability test. $\square$

The perturbation logic underlying strategic stability is not confined to static games. In dynamic games the same idea reappears, with the perturbation applied to the *system dynamics*

rather than to the payoffs: injecting an arbitrarily small stochastic disturbance into the dynamics destroys those equilibria that rely on the exact, deterministically predictable evolution of the state, retaining only those that remain optimal under every realization. As we will see in Lecture 20, this device resolves the *informational nonuniqueness* of Nash equilibria in dynamic games with full or partial memory, singling out the feedback (Markov) equilibria from an otherwise uncountable family.

7.6 ■ Further Notes and Reading

Beyond static equilibrium refinements, several complementary mechanisms explain how equilibria are selected or emerge in practice. These approaches link equilibrium analysis with dynamic, behavioral, and sociological perspectives on coordination and learning. They act as *selection principles* that refine the multiplicity of Nash equilibria by identifying those outcomes that are dynamically stable, behaviorally attainable, or socially focal.

Learning-based selection In many environments, players do not begin with full rational foresight but gradually adapt their strategies through feedback and experimentation. Learning models therefore introduce a *process-based refinement* of equilibrium: rather than asking which profiles satisfy mutual best responses at once, they ask which equilibria are *reachable* under plausible adaptive dynamics. In reinforcement learning, strategies that yield higher payoffs are chosen more frequently over time [103]. In best-response dynamics and fictitious play [21, 101], players repeatedly best respond to the empirical distribution of opponents' past actions, and convergence, when it occurs, selects a subset of Nash equilibria consistent with bounded rational adaptation. Evolutionary and learning-based dynamics [143, 39] thus filter out fragile or coordination-unstable equilibria, yielding those that are locally attracting under the dynamics. The asymptotically stable equilibria of such adaptive systems correspond to behaviorally plausible outcomes and can therefore be viewed as dynamically refined equilibria.

Focal-point selection While learning dynamics appeal to temporal adaptation, focal-point selection appeals to cognitive and social salience. As Schelling [105] noted, when multiple equilibria exist, players often coordinate on outcomes that are distinguished by symmetry, convention, or cultural precedent. Such *focal points* serve as equilibrium refinements grounded in shared expectations and common knowledge rather than payoff perturbations. They explain how communication, labeling, or framing can collapse indeterminacy in coordination games. Thus, focal-point reasoning refines the equilibrium set by introducing a sociological filter: only those equilibria that are cognitively salient and mutually recognizable are likely to be observed.

Evolutionary stability In population games, evolutionary pressures provide a natural mechanism for equilibrium selection. The evolutionarily stable strategy (ESS) of Maynard Smith and Price [120] identifies equilibria that can withstand small perturbations in population composition: no mutant strategy can invade a population playing an ESS. This introduces a *dynamic robustness* criterion analogous to trembling-hand perfection but operating at the population level. Under replicator dynamics, strategy frequencies evolve proportionally to their relative payoffs, and only ESS or asymptotically stable rest points persist in the long run. Hence, evolutionary selection provides a dynamic foundation for refinement: it discards Nash equilibria that are neutrally or locally unstable, preserving those that are evolutionarily or adaptively resilient.

These perspectives connect equilibrium refinements with *dynamic stability* and *behavioral adaptation*. While classical refinements such as trembling-hand perfection or strategic stability

impose *static* robustness, which requires equilibria to persist under payoff perturbations or representation changes, learning, evolutionary, and focal-point mechanisms introduce *process-based* and *social* robustness. They explain not only which equilibria can exist but also which ones are likely to emerge when rationality is bounded, information is incomplete, or behavior evolves over time. In this sense, they complement the structural and topological stability emphasized by Kohlberg and Mertens [64] with empirical and adaptive notions of equilibrium emergence.

The topic of refinement will be revisited later in the context of *infinite dynamic games*, where equilibrium selection plays a central role in resolving *informational non-uniqueness* of Nash equilibria. In such settings, informational asymmetries, feedback effects, and forward-looking strategic interactions can generate continuum sets of equilibria, even under standard convexity and smoothness assumptions. This phenomenon has been extensively documented in noncooperative dynamic and differential games, where equilibrium outcomes may fail to be unique without additional stability, robustness, or consistency requirements [9, 8]. Accordingly, stronger equilibrium selection principles, such as time consistency, robustness to perturbations, and admissibility under dynamic deviations, are often invoked to distinguish equilibria that are behaviorally or dynamically meaningful [10].

7.7 ■ Exercises

Exercise 7.1 (Refinement of Equilibria in a Bi-matrix Game). Consider the bi-matrix game

$$A = \begin{bmatrix} 1 & 3 \\ 1 & 2 \\ 2 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 2 & 1 \end{bmatrix},$$

where A gives the payoffs to P1 and B gives the payoffs to P2, and both players maximize their payoffs. The rows, indexed 1, 2, 3, correspond to the pure strategies of P1, and the columns, indexed 1, 2, correspond to the pure strategies of P2.

- (i) Compute all pure-strategy Nash equilibria of the game.
- (ii) Determine which of the equilibria found in part (i) are trembling-hand perfect. For each perfect equilibrium, exhibit a sequence of totally mixed strategy profiles that satisfies Selten's definition; for each equilibrium that fails the test, prove that no such sequence exists.
- (iii) Determine which of the equilibria found in part (i) are proper. Verify or refute the ε -proper ratio condition of Definition 7.11 along candidate sequences, and explain how strictly inferior replies constrain the admissible tremble probabilities in each case.
- (iv) Implement the following tremble test in Python or MATLAB. For each pure-strategy equilibrium found in part (i) and each $\varepsilon \in \{10^{-1}, 10^{-2}, 10^{-3}\}$, assign to each player the totally mixed strategy that places probability ε on every non-equilibrium pure strategy and the remaining probability mass on the equilibrium strategy. Compute the expected payoff of each pure strategy of each player against the opponent's perturbed strategy, and report a table with one row per pair (equilibrium, ε) indicating whether each player's equilibrium strategy remains a best response. The expected output is a table with six rows that confirms your classification in part (ii).

Exercise 7.2 (Trembling-Hand Perfection in an Extensive-Form Game). Consider the following two-player extensive-form game. P1 moves first and chooses between actions L and R . If P1 chooses L , the game ends with payoffs $(2, 1)$. If P1 chooses R , then P2 observes the move and chooses between l and r , with terminal payoffs

$$u(R, l) = (0, 0), \quad u(R, r) = (1, 2),$$

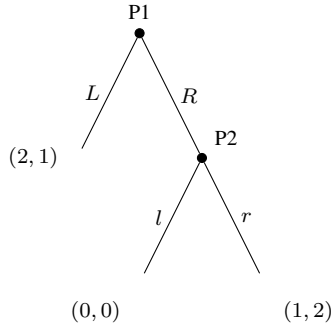


Figure 7.2: Extensive-form representation of the game with labeled decision nodes and terminal histories.

where the first entry is the payoff to P1 and the second entry is the payoff to P2. Both players maximize their payoffs. The extensive form is illustrated in Figure 7.2.

- (i) Derive the strategic-form representation of the game and compute all pure-strategy Nash equilibria.
- (ii) Show that (L, l) , in which P1 plays L and P2 plans to play l at the node reached after R , is a Nash equilibrium that is not subgame perfect.
- (iii) Compute, for $\varepsilon \in (0, 1)$, the expected payoff of each pure strategy of P1 against the perturbed strategy of P2 that plays l with probability ε and r with probability $1 - \varepsilon$, and the expected payoff of each pure strategy of P2 against the perturbed strategy of P1 that plays L with probability $1 - \varepsilon$ and R with probability ε .
- (iv) Prove that (L, r) is a trembling-hand perfect equilibrium by verifying that it remains a mutual best response under the perturbations of part (iii) for all sufficiently small $\varepsilon > 0$. Prove also that (L, l) is not trembling-hand perfect by showing that l fails to be a best response in every perturbed game in which P1 plays R with positive probability.
- (v) Implement the tremble test of part (iii) in Python or MATLAB. For each $\varepsilon \in \{0.2, 0.1, 0.05, 0.01\}$, tabulate the four expected payoffs computed in part (iii). The expected output is a table with columns ε , $u_1(L)$, $u_1(R)$, $u_2(l)$, and $u_2(r)$, confirming that $u_1(L) > u_1(R)$ and $u_2(r) > u_2(l)$ for every listed value of ε .

Exercise 7.3 (Strategic Stability in Finite Zero-Sum Games). Consider a finite two-player zero-sum normal-form game defined by the triplet $G = \langle \mathcal{N}, (S_i)_{i \in \mathcal{N}}, (u_i)_{i \in \mathcal{N}} \rangle$, where $\mathcal{N} = \{1, 2\}$ is the set of players; $S_1 = \{1, \dots, m\}$ and $S_2 = \{1, \dots, n\}$ are the finite action sets; $u_1 : S_1 \times S_2 \rightarrow \mathbb{R}$ is the payoff function of P1, and $u_2 = -u_1$. Let $A \in \mathbb{R}^{m \times n}$ denote the payoff matrix of P1, and let $\Delta(S_i)$ be the mixed-strategy simplex of Player i .

- (i) **Saddle-point characterization.** Show that there exists a value v^* and mixed strategies $(x^*, y^*) \in \Delta(S_1) \times \Delta(S_2)$ such that

$$v^* = \max_{x \in \Delta(S_1)} \min_{y \in \Delta(S_2)} u_1(x, y) = \min_{y \in \Delta(S_2)} \max_{x \in \Delta(S_1)} u_1(x, y).$$

Prove that the set of all saddle-point triplets

$$\mathcal{E}(G) = \left\{ (x^*, y^*, v^*) \in \Delta(S_1) \times \Delta(S_2) \times \mathbb{R} : u_1(x, y^*) \leq v^* \leq u_1(x^*, y) \right. \\ \left. \forall x \in \Delta(S_1), y \in \Delta(S_2) \right\}$$

is nonempty, convex, and connected.

- (ii) **Invariance under strategic equivalence.** Let G' be the zero-sum game with payoff functions

$$u'_1(s_1, s_2) = \alpha u_1(s_1, s_2) + \gamma_1, \quad u'_2(s_1, s_2) = -u'_1(s_1, s_2),$$

for some $\alpha > 0$ and constant $\gamma_1 \in \mathbb{R}$. Show that $\mathcal{E}(G')$ coincides with $\mathcal{E}(G)$ up to the induced affine transformation of the value:

$$(x^*, y^*, v^*) \in \mathcal{E}(G) \iff (x^*, y^*, \alpha v^* + \gamma_1) \in \mathcal{E}(G').$$

Conclude that the set of saddle-point strategies is invariant under positive affine transformations of the payoff function.

- (iii) **Continuity and stability.** Let $(u_1^k)_{k \geq 1}$ be a sequence of payoff functions converging uniformly to u_1 , and let G^k denote the zero-sum game with payoff function u_1^k . Prove that if $(x^k, y^k, v^k) \in \mathcal{E}(G^k)$ for each k , then any limit point (x, y, v) of the sequence satisfies $(x, y, v) \in \mathcal{E}(G)$. (Hint: use Berge's maximum theorem for continuity of argmax correspondences.)
- (iv) **Strategic stability characterization.** Kohlberg and Mertens [64] define a *strategically stable set* as a nonempty, closed, connected subset of Nash equilibria that is:
- (a) invariant under strategic equivalence, and
 - (b) upper hemicontinuous in payoffs.

Prove, using parts (i) to (iii), that the set $\mathcal{E}(G)$ of saddle-point triplets satisfies all of these properties and hence constitutes a Kohlberg-Mertens strategically stable set. Conclude that in any finite zero-sum normal-form game $G = \langle \mathcal{N}, (S_i)_{i \in \mathcal{N}}, (u_i)_{i \in \mathcal{N}} \rangle$, the saddle-point equilibrium (and hence the Nash equilibrium) is *strategically stable*.

Exercise 7.4 (No Non-Team 2×2 Illustration of Imperfection). The game used in Examples 7.1 and 7.3 has identical cost matrices and is therefore a team problem. This exercise shows that this is not an accident of the numbers: at 2×2 it cannot be avoided.

Let (A, B) be a 2×2 game in which both players minimize, with P1 choosing rows $\{U, D\}$ and P2 columns $\{L, R\}$. Define the *mixed second differences*

$$d_1 = A(U, L) - A(U, R) - A(D, L) + A(D, R), \quad d_2 = B(U, L) - B(U, R) - B(D, L) + B(D, R).$$

- (i) Show that d_1 and d_2 are unchanged when a function of the opponent's action is added to a player's costs, and that they are multiplied by $\alpha_i > 0$ when player i 's costs are scaled by α_i . Conclude that the pair (d_1, d_2) carries the entire strategic content of a 2×2 game, in the sense that two such games are strategically equivalent precisely when their pairs are positive multiples of one another.
- (ii) Deduce that the game is strategically equivalent to a team problem exactly when d_1 and d_2 are nonzero of the same sign, and to a zero-sum game exactly when they are nonzero of opposite signs.

- (iii) Suppose now that the game has two pure-strategy Nash equilibria, (U, L) and (D, R) , and that (D, R) is not trembling-hand perfect. Using the fact that in a two-player game an equilibrium is perfect if and only if neither player uses a weakly dominated strategy, show that D is weakly dominated for $P1$ (or, symmetrically, R for $P2$), and hence that $A(U, R) = A(D, R)$ and $A(U, L) < A(D, L)$.
- (iv) Conclude that $d_1 < 0$, and that the two equilibrium conditions for $P2$ force $d_2 \leq 0$. Deduce that the game is strategically equivalent to a team problem unless $d_2 = 0$, and show that $d_2 = 0$ occurs only in the degenerate case in which $P2$ is indifferent between her two columns at both rows.
- (v) Conclude that any 2×2 game exhibiting the failure of trembling-hand perfection is strategically equivalent to a team problem, so that a non-team illustration requires a larger game. Verify that the 3×2 game of Example 7.4 is such a game, by computing its second differences relative to the cell (U, L) and checking that they are not proportional.

Chapter 8

Correlated Equilibrium

Lecture 3 introduced the concept of *Nash equilibrium*, which characterizes stable outcomes in strategic interactions where no player can gain by unilaterally deviating from her strategy. Nash equilibrium provides a foundational notion of rationality and mutual consistency, yet it assumes that players' mixed strategies are statistically independent. This independence requirement can be overly restrictive in environments where players have access to common signals, observations, or pre-play communication that may induce correlation in their actions.

This lecture introduces the concept of **correlated equilibrium (CE)**, a generalization of Nash equilibrium that relaxes the independence assumption and allows players to coordinate their strategies through shared information. The central idea is that a mediator, or equivalently a public correlation device, draws a recommendation from a known joint distribution and privately advises each player on which action to take. Each player can then decide whether to follow or ignore the recommendation. A distribution is a correlated equilibrium if following the recommendation is rational for every player, given that others do the same.

This extension broadens the predictive and prescriptive power of equilibrium analysis. Correlated equilibria can achieve higher efficiency and fairness than Nash equilibria, capturing outcomes that require only *correlation*, not explicit cooperation. Moreover, the set of correlated equilibria forms a convex polytope that strictly contains the set of Nash equilibria, making it amenable to linear and geometric analysis.

The lecture proceeds by first discussing a motivational example of coordination in the *Game of Chicken*, followed by presenting the formal definition of correlated equilibrium and showing how the incentive constraints can be written as linear inequalities. We then come back to the *Game of Chicken*, and illustrate the concept through it, exploring both geometric and probabilistic interpretations. Finally, we extend the idea to *coarse correlated equilibrium (CCE)*, which further relaxes informational assumptions, and connect these equilibrium notions to dynamic models of learning and adaptation.

8.1 ■ Motivation: Coordination in the Game of Chicken

The *Game of Chicken* illustrates a fundamental challenge in non-cooperative decision-making: the tension between individual rationality and collective coordination in high-stakes interactions. Consider two drivers approaching an intersection from opposite directions. Each must independently decide whether to go straight through the intersection (G) or to yield (Y). If both proceed, a costly collision occurs. If one yields while the other proceeds, the latter receives a higher payoff. If both yield, they suffer a minor inconvenience.

The strategic situation is captured by the following payoff matrix:

P1	P2	
	G	Y
G	(−10, −10)	(5, 0)
Y	(0, 5)	(−1, −1)

There are two pure-strategy Nash equilibria: (G, Y) and (Y, G). Each represents a stable outcome where one driver proceeds and the other one yields. However, both equilibria are asymmetric and raise fairness concerns. Which driver should yield? In real-world interactions, such as traffic control, network routing, or multi-agent decision systems, such asymmetry may result in inefficiency, risk, or even deadlock.

A symmetric *mixed-strategy Nash equilibrium* offers an alternative. Let P1 choose G with probability y and P2 choose G with probability z . A best-response analysis yields:

$$\text{BR}_1(z) = \begin{cases} [0, 1] & \text{if } z = \frac{3}{8}, \\ 0 & \text{if } z > \frac{3}{8}, \\ 1 & \text{if } z < \frac{3}{8}, \end{cases} \quad \text{BR}_2(y) = \begin{cases} [0, 1] & \text{if } y = \frac{3}{8}, \\ 0 & \text{if } y > \frac{3}{8}, \\ 1 & \text{if } y < \frac{3}{8}. \end{cases} \quad (8.1)$$

At equilibrium, both players choose G with probability $3/8$ and Y with probability $5/8$. The resulting joint distribution over outcomes is:

P1	P2	
	G	Y
G	9/64	15/64
Y	15/64	25/64

This equilibrium, however, still assigns positive probability (9/64) to the worst-case outcome (G, G), a collision. From the standpoint of societal welfare or system design, this outcome is unacceptable. It raises a natural question:

Can rational agents coordinate to avoid catastrophic outcomes while preserving incentives and fairness?

Consider instead the following correlated distribution, which assigns equal probability to the two safe, efficient outcomes:

P1	P2	
	G	Y
G	0	1/2
Y	1/2	0

Here, each driver receives a signal that perfectly coordinates their actions: if one is told to go, the other is told to yield. The distribution is symmetric and guarantees safety by eliminating collisions. Such an arrangement cannot emerge from independent randomization; it requires correlation through a shared random device or mediator (e.g., a traffic light or an AI-based coordination signal).

This leads to the concept of a **correlated equilibrium** (Aumann, 1974) [6]: a generalization of Nash equilibrium in which players' actions may be conditionally dependent through a common signal. Each player's strategy is to follow the signal's recommendation whenever doing so maximizes her expected payoff, given their belief that others will also follow theirs. Aumann

emphasized that the significance of correlated equilibrium goes beyond technical generalization. It responds to a fundamental question he repeatedly raised in his writings on strategic conflict: *How can rational agents avoid mutually destructive outcomes when their incentives are only partially aligned?* [4, 6]. In his view, many conflicts persist not because agents are irrational, but because they lack access to shared signals or institutions capable of aligning expectations. Correlation, therefore, formalizes the informational structure required for peace: when players condition their behavior on a common signal and have no incentive to deviate, catastrophic outcomes can be systematically eliminated. This insight connects the traffic-light metaphor and the Game of Chicken to broader themes in conflict avoidance, crisis stability, and cooperative behavior in adversarial environments. Under correlated equilibrium, a simple publicly observed signal, whether a traffic light, a treaty-based monitoring device, or an AI coordination protocol, enables rational players to achieve outcomes that are safer, more efficient, and more equitable than those obtainable through independent randomization.

8.2 ■ Correlated Equilibrium

8.2.1 ■ Definition and Interpretation

In a Nash equilibrium, each player independently randomizes over her own strategy set, generating a product distribution over joint actions. This independence requirement constrains the set of feasible equilibrium outcomes: players cannot condition their behavior on shared signals, nor can they exploit correlation to avoid mutually harmful outcomes. In contrast, a *correlated equilibrium* (CE) permits players' actions to be statistically dependent, coordinated through a common random signal or mediator. The mediator draws an action profile according to a joint distribution π and privately recommends to each player her prescribed action a_i . Each player then evaluates whether it is optimal to follow this recommendation given that the others follow theirs. If obedience is optimal for every player and every possible recommendation, the distribution π is a correlated equilibrium. Allowing such correlated signals strictly enlarges the equilibrium set beyond Nash: it enables richer coordination structures, supports equilibria that are unattainable through independent randomization, and can substantially improve social welfare. Throughout this lecture, we adopt the utility-maximization convention of the finite-game setting: each player i evaluates outcomes through a payoff (utility) function u_i and seeks to maximize their expected utility.

Definition 8.1 (Correlated Equilibrium: Ex ante Formulation). Let $\mathcal{G} = \langle \mathcal{N}, (A_i)_{i \in \mathcal{N}}, (u_i)_{i \in \mathcal{N}} \rangle$ be a finite N -player normal-form game, where $\mathcal{N} = \{1, \dots, N\}$ is the set of players, A_i is the finite action set of player i , and $u_i : \prod_{j \in \mathcal{N}} A_j \rightarrow \mathbb{R}$ is player i 's payoff function. A probability distribution

$$\pi : \prod_{j \in \mathcal{N}} A_j \rightarrow [0, 1]$$

is a correlated equilibrium if for every player $i \in \mathcal{N}$ and every pair of actions $a_i, a'_i \in A_i$,

$$\sum_{a_{-i} \in A_{-i}} \pi(a_i, a_{-i}) u_i(a_i, a_{-i}) \geq \sum_{a_{-i} \in A_{-i}} \pi(a_i, a_{-i}) u_i(a'_i, a_{-i}). \quad (8.2)$$

This *ex ante* inequality (8.2) guarantees that whenever the mediator recommends a_i to player i , the expected payoff from obeying this recommendation is at least as large as the payoff from deviating to any alternative a'_i , assuming that all other players follow their own recommendations. The comparison is made *ex ante*; that is, before player i conditions on any particular signal and

before the mediator's recommendation is observed. Under this unconditional perspective, expectations are taken directly with respect to the joint distribution π , and the resulting best-response requirement is captured precisely by the linear constraint (8.2).

Equivalently, the same incentive requirement can be expressed from an *interim* point of view, meaning after player i learns the specific recommendation a_i but before taking an action. Conditional on receiving a_i , player i updates her belief about the opponents' actions using Bayes' rule:

$$\pi(a_{-i} \mid a_i) = \frac{\pi(a_i, a_{-i})}{\sum_{a'_{-i} \in A_{-i}} \pi(a_i, a'_{-i})}, \quad \text{whenever } \sum_{a_{-i}} \pi(a_i, a_{-i}) > 0.$$

The mediator's signal serves as information about what the others are likely to do. The appropriate incentive constraint at this *interim* stage requires that, under the posterior belief $\pi(\cdot \mid a_i)$, player i weakly prefers to follow the recommendation rather than deviate:

$$\sum_{a_{-i} \in A_{-i}} \pi(a_{-i} \mid a_i) u_i(a_i, a_{-i}) \geq \sum_{a'_{-i} \in A_{-i}} \pi(a_{-i} \mid a_i) u_i(a'_i, a_{-i}), \quad \forall a_i, a'_i \in A_i. \quad (8.3)$$

Multiplying both sides of (8.3) by the normalizing factor $\sum_{a'_{-i}} \pi(a_i, a'_{-i})$, which is strictly positive whenever a_i is recommended with positive probability, yields exactly the ex ante linear condition (8.2). Thus, the ex ante and interim formulations are mathematically equivalent, even though they represent distinct informational perspectives.

The interim formulation (8.3) captures the behavioral essence of correlated equilibrium: at the moment the recommendation is observed, player i treats the signal as information, updates her belief via Bayes' rule, and finds that obeying is a best response to this posterior. This observation leads naturally to the following interim definition of correlated equilibrium.

Definition 8.2 (Correlated Equilibrium: Interim Formulation). Let $\mathcal{G} = \langle \mathcal{N}, (A_i)_{i \in \mathcal{N}}, (u_i)_{i \in \mathcal{N}} \rangle$ be a finite normal-form game, and let $\pi \in \Delta\left(\prod_{j \in \mathcal{N}} A_j\right)$ be a probability distribution over action profiles. For each player i and each action $a_i \in A_i$, define the marginal probability $\pi(a_i) = \sum_{a_{-i} \in A_{-i}} \pi(a_i, a_{-i})$.

We say that π is a correlated equilibrium (in interim form) if for every player $i \in \mathcal{N}$ and every pair of actions $a_i, a'_i \in A_i$, the following condition holds:

(i) If $\pi(a_i) > 0$, then the posterior belief $\pi(a_{-i} \mid a_i) = \frac{\pi(a_i, a_{-i})}{\pi(a_i)}$ satisfies the interim incentive constraint

$$\sum_{a_{-i} \in A_{-i}} \pi(a_{-i} \mid a_i) u_i(a_i, a_{-i}) \geq \sum_{a_{-i} \in A_{-i}} \pi(a_{-i} \mid a_i) u_i(a'_i, a_{-i}). \quad (8.4)$$

(ii) If $\pi(a_i) = 0$, then no condition is imposed for the action a_i , because the mediator never recommends it and the corresponding posterior is undefined. In this case, (8.4) is vacuously satisfied.

A correlated equilibrium corresponds exactly to the set of distributions implementable by a trusted mediator who:

- (i) draws an action profile $a = (a_1, \dots, a_N)$ from π ,
- (ii) privately recommends a_i to each player i , and

- (iii) induces obedience because each player finds her recommendation sequentially rational given her posterior $\pi(\cdot \mid a_i)$.

Thus, correlated equilibrium strictly generalizes Nash equilibrium by replacing independent randomization with correlated signals while preserving rational best-response behavior.

Example 8.3 Game of Chicken.

Consider again the two-player *Game of Chicken*. Each player $i \in \{1, 2\}$ chooses an action from $A_i = \{G, Y\}$ with payoff matrices

P1	P2	
	G	Y
G	$(-10, -10)$	$(5, 0)$
Y	$(0, 5)$	$(-1, -1)$

Let $\pi_{ij} = \pi(a_1 = i, a_2 = j)$ denote the joint probabilities, where $i, j \in \{G, Y\}$. These probabilities satisfy $\pi_{ij} \geq 0$ and $\pi_{GG} + \pi_{GY} + \pi_{YG} + \pi_{YY} = 1$.

In a correlated equilibrium, each player must find it optimal to follow her recommended action. For P1, if the mediator recommends G, P1 must prefer G to deviating to Y:

$$u_1(G, G)\pi_{GG} + u_1(G, Y)\pi_{GY} \geq u_1(Y, G)\pi_{GG} + u_1(Y, Y)\pi_{GY}.$$

If P1 is recommended Y, she must prefer Y to G:

$$u_1(Y, G)\pi_{YG} + u_1(Y, Y)\pi_{YY} \geq u_1(G, G)\pi_{YG} + u_1(G, Y)\pi_{YY}.$$

Substituting the numerical payoffs yields

$$\begin{aligned} -10\pi_{GG} + 6\pi_{GY} &\geq 0, \\ 10\pi_{YG} - 6\pi_{YY} &\geq 0. \end{aligned}$$

Symmetrically, P2's incentive constraints are

$$\begin{aligned} -10\pi_{GG} + 6\pi_{YG} &\geq 0, \\ 10\pi_{GY} - 6\pi_{YY} &\geq 0. \end{aligned}$$

Collecting all inequalities, the correlated equilibrium polytope $\mathcal{C}$ is characterized by

$$A\pi \geq 0, \quad \mathbf{1}^\top \pi = 1, \quad \pi \geq 0,$$

where $\pi = (\pi_{GG}, \pi_{GY}, \pi_{YG}, \pi_{YY})^\top$, $\mathbf{1} = (1, 1, 1, 1)^\top$, and

$$A = \begin{bmatrix} -10 & 6 & 0 & 0 \\ 0 & 0 & 10 & -6 \\ -10 & 0 & 6 & 0 \\ 0 & 10 & 0 & -6 \end{bmatrix}.$$

Each row corresponds to an incentive constraint for one player under a specific recommendation.

The feasible region $\mathcal{C} \subset \Delta(\{GG, GY, YG, YY\})$ defined by these linear incentive constraints forms a convex polytope of all joint action distributions satisfying mutual incentive compatibility. The blue region in Figure 8.1 depicts this polytope within the probability simplex. Its extreme points correspond to correlated equilibria, including the efficient symmetric point

$$(\pi_{GY}, \pi_{YG}) = (0.5, 0.5), \quad \text{with payoffs } (2.5, 2.5),$$

which strictly dominates the mixed-strategy Nash equilibrium (red point), whose expected payoffs are $(-0.625, -0.625)$. Geometrically, the CE polytope $\mathcal{C}$ lies strictly within the ambient simplex $\Delta(\{GG, GY, YG, YY\})$, cut out by the supporting hyperplanes defined by the incentive inequalities; each hyperplane removes distributions at which one player would wish to deviate. The intersection of these half-spaces yields the light-blue region visible in Figure 8.1, representing all feasible correlated equilibria.

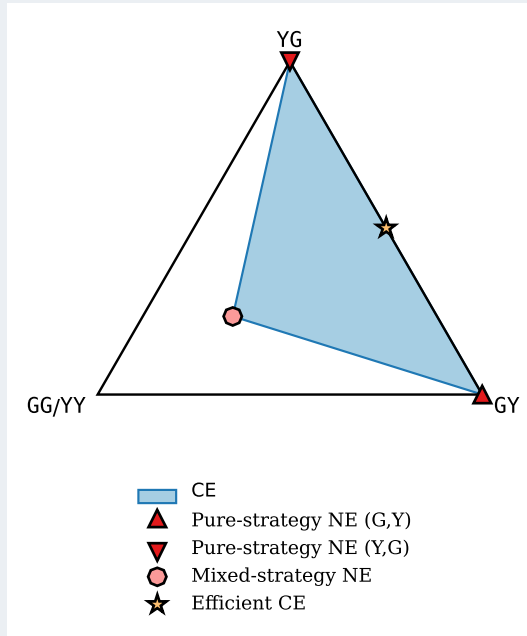


Figure 8.1: Geometry of the *correlated equilibrium* (CE) set in the Game of Chicken, shown via a projection of the four-dimensional probability simplex. The blue region represents the CE polytope defined by the linear incentive constraints for both players. The two pure-strategy Nash equilibria (G, Y) and (Y, G) appear as vertices on the boundary, while the unique mixed-strategy Nash equilibrium lies in the interior. Allowing correlation enlarges the equilibrium set beyond Nash equilibria, yielding efficient outcomes such as the symmetric $(0.5, 0.5)$ distribution that strictly improves the expected payoffs of both players.

□

Remark 8.4 (Social Norms, Conventions, and Informal Correlation). The mediator interpretation is mathematically convenient but not essential. A correlated equilibrium can arise even in the absence of an explicit mediator, through informal structures that implicitly correlate expectations among agents. Formally, consider a publicly observable variable or cue $S \in \mathcal{S}$, which may represent a social norm, a focal point, a cultural signal, an environmental condition, or any commonly understood contextual feature. Suppose that each player i uses a strategy $\sigma_i : \mathcal{S} \rightarrow \Delta(A_i)$, mapping the observed cue S into a mixed action over A_i . If S is drawn from a distribution $\mu \in \Delta(\mathcal{S})$, then the induced joint distribution over action profiles is

$$\pi(a) = \sum_{s \in \mathcal{S}} \mu(s) \prod_{i \in \mathcal{N}} \sigma_i(a_i | s),$$

which is a correlated distribution even though the players never communicate and no mediator is present.

Informal mechanisms operate precisely through such mappings. A social convention, such as driving on the right side of the road, can be viewed as a commonly understood deterministic strategy $\sigma_i(s)$ conditioned on a cue s that identifies the societal norm. Similarly, a focal point (in Schelling's sense) corresponds to a salient cue S that all players interpret in the same way, inducing a correlated profile π without explicit coordination. Cultural rules, shared heuristics, and contextual precedents function analogously: they provide a publicly recognized variable S such that players condition their actions on the same underlying information structure.

Aumann emphasized that much of real-world coordination emerges from these implicit mechanisms rather than from centralized enforcement or explicit signals [6]. Thus, the correlated equilibrium concept captures both formal institutional correlation (e.g., mediators, traffic lights, algorithmic signals) and decentralized, norm-driven coordination shaped by shared cues in the environment.

8.2.2 ■ Coarse Correlated Equilibrium: Example from the Game of Chicken

A related relaxation of the correlated equilibrium concept is the *coarse correlated equilibrium* (CCE). Unlike the correlated equilibrium, which assumes that each player observes her private recommendation before deciding whether to deviate, the CCE removes this informational advantage. Players must decide whether to commit to following the mediator's recommendation *before* any signal is revealed.

Definition 8.5 (Coarse Correlated Equilibrium). Let $\mathcal{G} = \langle \mathcal{N}, (A_i)_{i \in \mathcal{N}}, (u_i)_{i \in \mathcal{N}} \rangle$ be a finite N -player normal-form game. A probability distribution π over the joint action space $\prod_i A_i$ is a coarse correlated equilibrium if, for every player $i \in \mathcal{N}$ and every alternative pure strategy $a'_i \in A_i$,

$$\sum_{a \in \prod_i A_i} \pi(a) u_i(a) \geq \sum_{a_{-i} \in A_{-i}} \pi(a_{-i}) u_i(a'_i, a_{-i}),$$

where $\pi(a_{-i}) = \sum_{a_i \in A_i} \pi(a_i, a_{-i})$ denotes the marginal distribution over the opponents' joint actions.

Intuitively, the inequality asserts that no player can improve her expected payoff by committing *ex ante* to a fixed deviation prior to observing any recommendation. In contrast, a correlated equilibrium enforces incentive compatibility *after* the signal is observed (i.e., conditionally). Consequently, every correlated equilibrium is a coarse correlated equilibrium, but not conversely. The CCE set forms a larger convex polytope in the space of joint action distributions.

Example 8.6 Game of Chicken (Continued).

Consider again the classical *Game of Chicken*, where each player chooses either to go straight (G) or to yield (Y). In a *coarse correlated equilibrium*, players decide whether to participate in the mediator's randomization *before* seeing their signal. Formally, π is a CCE if for each player i and each pure deviation $a'_i \in A_i$,

$$\sum_{a \in A_1 \times A_2} \pi(a) u_i(a) \geq \sum_{a_{-i} \in A_{-i}} \pi(a_{-i}) u_i(a'_i, a_{-i}).$$

These inequalities state that no player can gain by committing ex ante to a fixed alternative strategy a'_i . For brevity, write $p_{ij} := \pi(i, j)$ for $i, j \in \{G, Y\}$.

P1 compares the expected payoff $\mathbb{E}_\pi[u_1]$ obtained by following the recommendations to the payoff of committing ex ante to either G or Y:

$$\begin{aligned} \text{Deviation to G: } \mathbb{E}_\pi[u_1] &\geq \sum_{a_2} \pi(a_2) u_1(G, a_2) \\ &= -10(p_{GG} + p_{YG}) + 5(p_{GY} + p_{YY}), \end{aligned}$$

$$\text{Deviation to Y: } \mathbb{E}_\pi[u_1] \geq \sum_{a_2} \pi(a_2) u_1(Y, a_2) = -(p_{GY} + p_{YY}).$$

Substituting $\mathbb{E}_\pi[u_1] = -10p_{GG} + 5p_{GY} - p_{YY}$ and simplifying yields the two linear inequalities:

$$\begin{aligned} 5p_{YG} - 3p_{YY} &\geq 0, \\ 3p_{GY} - 5p_{GG} &\geq 0. \end{aligned}$$

P2's incentive constraints. By symmetry, P2's CCE inequalities are:

$$\begin{aligned} 5p_{GY} - 3p_{YY} &\geq 0, \\ 3p_{YG} - 5p_{GG} &\geq 0. \end{aligned}$$

Combining both players' constraints, normalization, and nonnegativity, the set of coarse correlated equilibria is described by the following linear system:

$$\begin{aligned} 3p_{GY} - 5p_{GG} &\geq 0, & 3p_{YG} - 5p_{GG} &\geq 0, \\ 5p_{GY} - 3p_{YY} &\geq 0, & 5p_{YG} - 3p_{YY} &\geq 0, \\ p_{GG} + p_{GY} + p_{YG} + p_{YY} &= 1, & p_{GG}, p_{GY}, p_{YG}, p_{YY} &\geq 0. \end{aligned}$$

Let $p = (p_{GG}, p_{GY}, p_{YG}, p_{YY})^\top$. Then the inequalities can be written compactly as

$$Ap \geq 0, \quad \mathbf{1}^\top p = 1, \quad p \geq 0,$$

where

$$A = \begin{bmatrix} -5 & 3 & 0 & 0 \\ -5 & 0 & 3 & 0 \\ 0 & 5 & 0 & -3 \\ 0 & 0 & 5 & -3 \end{bmatrix}, \quad \mathbf{1} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}.$$

The convex polytope defined by these linear constraints represents all joint distributions π that constitute *coarse correlated equilibria* of the Game of Chicken. Every correlated equilibrium belongs to this region; in general, the CCE polytope may also contain additional distributions that satisfy only the ex ante incentive constraints. $\square$

Proposition 8.7 (Relationship between CE and CCE). *For any finite game $\mathcal{G}$, every correlated equilibrium is also a coarse correlated equilibrium: $\text{CE} \subseteq \text{CCE}$. However, the inclusion is in general strict.*

Proof. Let π be a correlated equilibrium of $\mathcal{G} = \langle \mathcal{N}, (A_i)_{i \in \mathcal{N}}, (u_i)_{i \in \mathcal{N}} \rangle$. By definition, for each player i and each deviation $a'_i \in A_i$, the conditional expected payoff from following the mediator's recommendation a_i is at least as large as that from deviating to a'_i , given the recommendation signal a_i :

$$\sum_{a_{-i}} \pi(a_{-i} \mid a_i) u_i(a_i, a_{-i}) \geq \sum_{a_{-i}} \pi(a_{-i} \mid a_i) u_i(a'_i, a_{-i}), \quad \forall a_i \in A_i. \quad (1)$$

Multiplying both sides of (1) by $\pi(a_i)$ and summing over a_i yields

$$\sum_{a_i, a_{-i}} \pi(a_i, a_{-i}) u_i(a_i, a_{-i}) \geq \sum_{a_i, a_{-i}} \pi(a_i, a_{-i}) u_i(a'_i, a_{-i}), \quad \forall a'_i \in A_i. \quad (2)$$

The right-hand side of (2) can be rewritten as $\sum_{a_{-i}} \pi(a_{-i}) u_i(a'_i, a_{-i})$, where $\pi(a_{-i}) = \sum_{a_i} \pi(a_i, a_{-i})$ is the marginal distribution of the opponents' actions. Hence inequality (2) is precisely the definition of a *coarse correlated equilibrium*:

$$\sum_a \pi(a) u_i(a) \geq \sum_{a_{-i}} \pi(a_{-i}) u_i(a'_i, a_{-i}), \quad \forall i, \forall a'_i.$$

Therefore every correlated equilibrium satisfies the CCE inequalities, and thus $\text{CE} \subseteq \text{CCE}$. $\square$

Example 8.8 CE and CCE in the Game of Chicken.

In the Game of Chicken, consider the family of joint distributions

$$\pi_\varepsilon : \quad \pi(\text{G}, \text{Y}) = \pi(\text{Y}, \text{G}) = \frac{1}{2} - \varepsilon, \quad \pi(\text{G}, \text{G}) = \pi(\text{Y}, \text{Y}) = \varepsilon, \quad 0 \leq \varepsilon \ll 1.$$

The four CCE inequalities

$$3\pi_{\text{GY}} \geq 5\pi_{\text{GG}}, \quad 3\pi_{\text{YG}} \geq 5\pi_{\text{GG}}, \quad 5\pi_{\text{GY}} \geq 3\pi_{\text{YY}}, \quad 5\pi_{\text{YG}} \geq 3\pi_{\text{YY}}$$

hold for π_ε precisely when $\varepsilon \leq 3/16$. In general, a coarse correlated equilibrium (CCE) imposes only *ex ante* incentive constraints, whereas a correlated equilibrium (CE) additionally requires *conditional* incentive constraints at each information set, so the CE polytope may be strictly smaller. For the Game of Chicken, however, the payoff structure has a special feature: every CE incentive constraint is a positive scalar multiple of the corresponding CCE constraint. For instance, P1's conditional

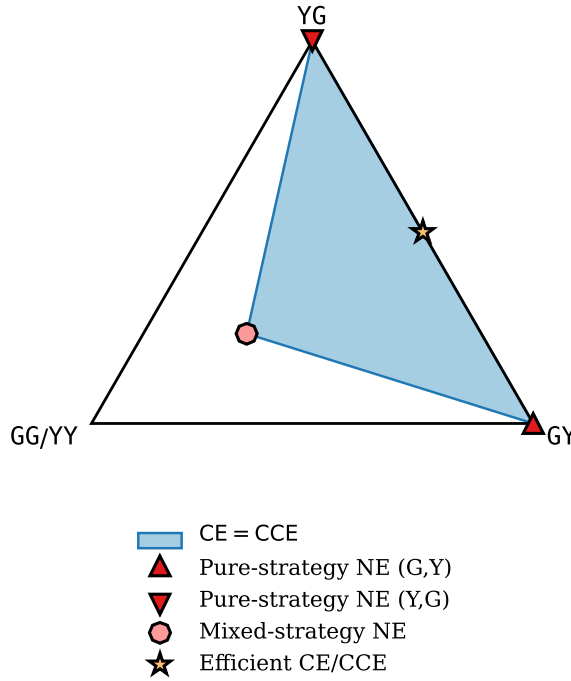


Figure 8.2: Geometry of the correlated equilibrium (CE) and coarse correlated equilibrium (CCE) sets in the Game of Chicken, shown in the same projection of the probability simplex as Figure 8.1. In this game, every CE incentive constraint is a positive scalar multiple of the corresponding CCE constraint, and hence the two polytopes coincide: $CE = CCE$. The pure- and mixed-strategy Nash equilibria and the efficient symmetric correlated equilibrium are marked as in Figure 8.1.

constraint $-10\pi_{GG} + 6\pi_{GY} \geq 0$ is exactly twice the CCE constraint $3\pi_{GY} - 5\pi_{GG} \geq 0$, and the remaining constraints match in the same way. Since multiplying a linear inequality by a positive constant does not change its feasible region, the two polytopes coincide in this game: $CE = CCE$, as illustrated in Figure 8.2. In particular, every π_ε with $\varepsilon \leq 3/16$ is both a CCE and a CE. The weaker CCE constraints already guarantee all conditional no-deviation requirements, and hence the additional CE restrictions impose no further refinement; the strictness of the inclusion $CE \subseteq CCE$ asserted in general must therefore be exhibited in other games. $\square$

8.3 ■ Extensive-Form Interpretation of Correlated Equilibrium

The correlated equilibrium of a normal-form game can be represented as the sequential outcome of an extensive-form game in which a chance move by *Nature* (or a *mediator*) draws a hidden state of the world ω and privately recommends actions to the players. Each player then chooses

whether to follow the recommendation or deviate. This dynamic representation reveals the informational structure underlying correlation and connects the static CE conditions to the principles of sequential rationality and obedience.

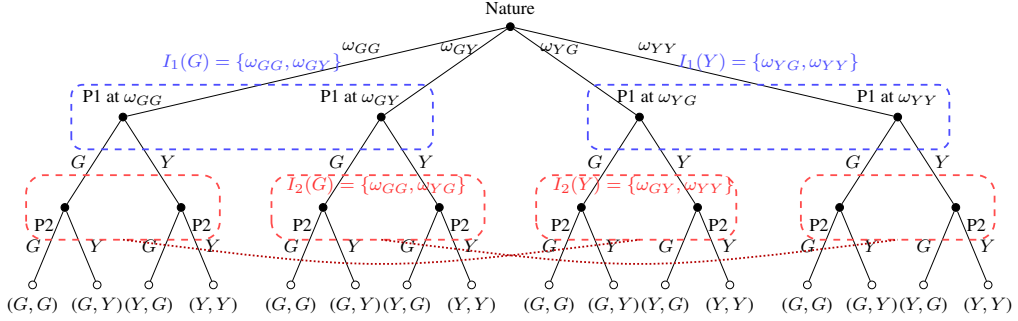


Figure 8.3: Extensive-form representation of the Game of Chicken with a correlation device. Nature draws ω_{ij} with probability π_{ij} , producing private signals $(t_1, t_2) = (i, j)$. Blue dashed boxes represent P1's information sets $I_1(G) = \{\omega_{GG}, \omega_{GY}\}$ and $I_1(Y) = \{\omega_{YG}, \omega_{YY}\}$. Red dashed boxes represent P2's information sets $I_2(G) = \{\omega_{GG}, \omega_{YG}\}$ and $I_2(Y) = \{\omega_{GY}, \omega_{YY}\}$, with dotted connectors joining disjoint branches that belong to the same set. Sixteen terminal hollow nodes represent all possible outcomes (a_1, a_2) .

Figure 8.3 illustrates the extensive-form representation of the *Game of Chicken with a Correlation Device*. Recall the payoff matrix of the Game of Chicken. Let Nature (or the mediator) draw a random state $\omega \in \Omega = \{\omega_{GG}, \omega_{GY}, \omega_{YG}, \omega_{YY}\}$, with probabilities $\pi(\omega_{ij}) = \pi_{ij}$, where $i, j \in \{G, Y\}$. The state ω_{ij} represents a pair of recommendations: P1 is advised to play i and P2 to play j . Each player observes only her own recommendation signal but not the opponent's.

The extensive form unfolds as follows:

- (1) **Nature's move.** Nature (the mediator) randomly selects a state $\omega \in \Omega$ according to π . Each branch emanating from the root node in Figure 8.3 corresponds to one such realized state $\omega_{GG}, \omega_{GY}, \omega_{YG}$, or ω_{YY} .
- (2) **P1's decision.** P1 observes her private recommendation $t_1(\omega) = i \in \{G, Y\}$, shown as the first component of the branch label, and chooses an action $a_1 \in \{G, Y\}$. The blue dashed rectangles in the figure represent P1's two information sets:

$$I_1(G) = \{\omega_{GG}, \omega_{GY}\}, \quad I_1(Y) = \{\omega_{YG}, \omega_{YY}\}.$$

Within each rectangle, P1 cannot distinguish which of the two possible states has occurred, since both share the same first component of the recommendation.

- (3) **P2's decision.** P2 observes her private recommendation $t_2(\omega) = j \in \{G, Y\}$, and then selects an action $a_2 \in \{G, Y\}$. The red dashed rectangles in Figure 8.3 depict P2's information sets:

$$I_2(G) = \{\omega_{GG}, \omega_{YG}\}, \quad I_2(Y) = \{\omega_{GY}, \omega_{YY}\}.$$

Since these sets are realized across different branches of Nature's initial move, they appear disjoint in the tree; the dotted red arcs visually connect the two separate components of each information set.

- (4) **Terminal outcomes.** Once both players act, the game terminates in one of 16 possible outcome nodes (a_1, a_2) , each shown as a hollow circle at the bottom of the tree. The payoffs at each terminal node correspond to $u_1(a_1, a_2)$ and $u_2(a_1, a_2)$ from the original Chicken Game matrix.

The correlated extensive form in Figure 8.3 is represented by

$$\Gamma^{\text{corr}} = (\mathcal{N}, V, E, x_0, (V_i)_{i \in \mathcal{N} \cup \{0\}}, (\mathcal{I}_i)_{i \in \mathcal{N}}, (\rho_x)_{x \in V_0}, O, (u_i)_{i \in \mathcal{N}}),$$

where $\mathcal{N} = \{1, 2\}$ is the set of players and 0 denotes Nature (the mediator). Nature draws a state $\omega \in \Omega = \{\omega_{GG}, \omega_{GY}, \omega_{YG}, \omega_{YY}\}$ with probability $\pi_{ij} = \pi(\omega_{ij})$, corresponding to the recommendation pair $(t_1, t_2) = (i, j)$ for $i, j \in \{G, Y\}$. The node labels $x_{1,\omega}$ and x_{2,ω,a_1} identify, respectively, P1 and P2 decision nodes that follow a given history. The node sets are $V_0 = \{x_0\}$ for Nature, $V_1 = \{x_{1,\omega} : \omega \in \Omega\}$ for P1, $V_2 = \{x_{2,\omega,a_1} : \omega \in \Omega, a_1 \in \{G, Y\}\}$ for P2, and $O = \{z_{\omega,a_1,a_2} : \omega \in \Omega, a_1, a_2 \in \{G, Y\}\}$ for terminal nodes, so that $|O| = 16$. The edge set $E = \{(x_0, x_{1,\omega})\} \cup \{(x_{1,\omega}, x_{2,\omega,a_1})\} \cup \{(x_{2,\omega,a_1}, z_{\omega,a_1,a_2})\}$ encodes the sequential structure of moves from Nature to P1, then to P2, and finally to the terminal outcomes. At each decision node $x \in V_1 \cup V_2$, the feasible actions are $\mathcal{A}(x) = \{G, Y\}$ with labels $\lambda(x, x') \in \{G, Y\}$, and Nature's kernel satisfies $\rho_{x_0}(x_{1,\omega_{ij}}) = \pi_{ij}$.

P1's information partition is $\mathcal{I}_1 = \{I_1(G), I_1(Y)\}$, where $I_1(G) = \{x_{1,\omega_{GG}}, x_{1,\omega_{GY}}\}$ and $I_1(Y) = \{x_{1,\omega_{YG}}, x_{1,\omega_{YY}}\}$. P2's partition is $\mathcal{I}_2 = \{I_2(G), I_2(Y)\}$ with $I_2(G) = \{x_{2,\omega_{GG},a_1}, x_{2,\omega_{YG},a_1} : a_1 \in \{G, Y\}\}$ and $I_2(Y) = \{x_{2,\omega_{GY},a_1}, x_{2,\omega_{YY},a_1} : a_1 \in \{G, Y\}\}$. Each $I_2(\cdot)$ links disjoint branches corresponding to identical private recommendations $t_2 \in \{G, Y\}$. At every terminal node $z_{\omega,a_1,a_2} \in O$, payoffs are given by $u_i(z_{\omega,a_1,a_2}) = u_i^{\text{Chicken}}(a_1, a_2)$ for $i \in \{1, 2\}$, so that utilities depend only on the realized actions (a_1, a_2) and not on the underlying recommendation ω .

The key feature of this construction is that each player acts under *imperfect information*. P1 knows only the first component of ω , and P2 only the second. Hence, as shown by the dashed boxes, each player's information set merges two distinct nodes corresponding to different Nature realizations that are observationally indistinguishable from their perspective. The dotted connectors between P2's red boxes in the figure emphasize that, although her information sets appear on separate branches, they are strategically equivalent: P2 faces the same decision problem in both parts of each connected set.

A correlation device introduces a pre-play signal structure by letting Nature draw a state $\omega_{ij} \in \Omega = \{\omega_{GG}, \omega_{GY}, \omega_{YG}, \omega_{YY}\}$ with probability π_{ij} . The state ω_{ij} itself is the recommendation: it prescribes action i to P1 and action j to P2. Each player observes only the corresponding coordinate of ω_{ij} , and thus the induced information partitions are

$$I_1(i) = \{\omega_{iG}, \omega_{iY}\}, \quad I_2(j) = \{\omega_{Gj}, \omega_{Yj}\}.$$

A correlated equilibrium can be interpreted as the design of a signal distribution π such that, when Nature draws a recommendation ω_{ij} from this distribution, the induced behavioral strategies form an incentive-compatible profile. In particular, each player's behavioral strategy obeys the action prescribed by the recommendation encoded in ω_{ij} :

$$\sigma_1(a_1 \mid I_1(i)) = \begin{cases} 1, & a_1 = i, \\ 0, & a_1 \neq i, \end{cases} \quad \sigma_2(a_2 \mid I_2(j)) = \begin{cases} 1, & a_2 = j, \\ 0, & a_2 \neq j. \end{cases}$$

More formally, to capture incentive compatibility and sequential rationality in the correlated extensive form, P1 forms a posterior belief about P2's recommended action conditional on being at the information set $I_1(i)$. By Bayes' rule, this posterior is

$$\Pr(\omega_{ij} \mid I_1(i)) = \frac{\pi_{ij}}{\pi_{iG} + \pi_{iY}},$$

and an analogous expression holds for P2. Given this posterior, obedience to the recommendation is sequentially rational. For any deviation $a'_1 \in A_1$, the expected payoff from following the recommendation i must be at least as large as the payoff from deviating:

$$\mathbb{E}[u_1(i, a_2) \mid I_1(i)] \geq \mathbb{E}[u_1(a'_1, a_2) \mid I_1(i)],$$

with a symmetric condition for P2. These are precisely the interim correlated-equilibrium incentive constraints. Consequently, when the correlation device samples ω according to π and the players follow their behavioral strategies σ_i at each information set, no profitable deviation exists at any point in the game, and the induced distribution over terminal action profiles coincides with the correlated-equilibrium distribution π .

8.4 ■ Bayesian Interpretation of Correlated Equilibrium

Aumann's seminal interpretation of correlated equilibrium [6] situates it within the framework of a *Bayesian game*, in which the *correlation device* is modeled as a random *state of the world*. From this viewpoint, a correlated equilibrium in a complete-information game is precisely a Bayesian Nash equilibrium of a related incomplete-information game, where the players' private information is generated by the correlation device. This interpretation provides a bridge between the extensive-form representation of the correlation mechanism (as in Figure 8.3) and its probabilistic foundations.

Let Ω be a finite set of states of the world, each representing a possible joint recommendation drawn by Nature. A common prior $p \in \Delta(\Omega)$ specifies the correlation device's distribution. In each play of the game, Nature selects a state $\omega \in \Omega$ according to p . Each player $i \in \mathcal{N}$ receives a private *signal* (or *type*) $t_i \in T_i$ generated by a signal function

$$t_i = \tau_i(\omega), \quad \tau_i : \Omega \rightarrow T_i.$$

The mappings τ_i are common knowledge, but only the realized type t_i is privately observed by player i . Thus, the players share a common prior over the joint signal structure but act under incomplete information about ω .

Example 8.9 The Game of Chicken (Continued).

In the correlated version of the Game of Chicken (Figure 8.3), we can identify:

$$\Omega = \{\omega_{GG}, \omega_{GY}, \omega_{YG}, \omega_{YY}\}, \quad p(\omega_{ij}) = \pi_{ij}, \quad i, j \in \{G, Y\}.$$

Here ω_{ij} encodes a correlated recommendation in which P1 is advised to play i and P2 is advised to play j . The players' signal functions are simply the coordinate projections:

$$\tau_1(\omega_{ij}) = i, \quad \tau_2(\omega_{ij}) = j.$$

In Figure 8.3, the dashed boxes represent the induced information partitions:

$$\begin{aligned} I_1(G) &= \{\omega_{GG}, \omega_{GY}\}, & I_1(Y) &= \{\omega_{YG}, \omega_{YY}\}; \\ I_2(G) &= \{\omega_{GG}, \omega_{YG}\}, & I_2(Y) &= \{\omega_{GY}, \omega_{YY}\}. \end{aligned}$$

Each information set corresponds to the set of states that produce the same private signal under τ_i . The dotted red lines connect disjoint branches that belong to the same information set for P2, highlighting that these nodes are informationally indistinguishable from that player's perspective. $\square$

Given a type $t_i \in T_i$, player i chooses an action $a_i \in A_i$ according to a (possibly mixed) *behavioral strategy*

$$\sigma_i : T_i \rightarrow \Delta(A_i), \quad \sigma_i(a_i | t_i) = \Pr(a_i \text{ played} | t_i).$$

The profile $(\sigma_i)_{i \in \mathcal{N}}$, together with the prior p , induces a joint distribution $\gamma \in \Delta(A_1 \times \cdots \times A_N)$ over action profiles:

$$\gamma(a_1, a_2) = \sum_{\omega \in \Omega} p(\omega) \sigma_1(a_1 | \tau_1(\omega)) \sigma_2(a_2 | \tau_2(\omega)).$$

This coincides with the correlation-device representation

$$\gamma(a_1, a_2) = \sum_{m_1 \in M_1} \sum_{m_2 \in M_2} \mu(m_1, m_2) \sigma_1(a_1 | m_1) \sigma_2(a_2 | m_2),$$

under the identification $m_i = \tau_i(\omega)$ and μ as the distribution of (m_1, m_2) induced by p . Hence, the correlation structure can be viewed equivalently as: (i) a random signal pair (m_1, m_2) drawn from a correlation device, or (ii) the realization of a common prior p over Ω with type mappings (τ_i) .

From the Bayesian perspective, the correlated-equilibrium constraints correspond exactly to the incentive-compatibility conditions of a Bayesian Nash equilibrium. For every player i , type $t_i \in T_i$, and deviation $a'_i \in A_i$,

$$\sum_{a_{-i} \in A_{-i}} \Pr(a_{-i} | t_i) u_i(a_i, a_{-i}) \geq \sum_{a_{-i} \in A_{-i}} \Pr(a_{-i} | t_i) u_i(a'_i, a_{-i}),$$

where the posterior belief about others' actions given one's type is

$$\Pr(a_{-i} | t_i) = \frac{\sum_{\omega: \tau_i(\omega)=t_i} p(\omega) \sigma_{-i}(a_{-i} | \tau_{-i}(\omega))}{\sum_{\omega': \tau_i(\omega')=t_i} p(\omega')}.$$

At equilibrium, following the recommended action maximizes expected utility conditional on one's private signal, showing the no-deviation inequalities represented by the dashed information sets in Figure 8.3.

Under this interpretation, the correlation device acts as a publicly known stochastic environment that generates privately observed but correlated types. Players update beliefs using the common prior p and play optimal actions based on those beliefs. Thus, a correlated equilibrium in the original complete-information game corresponds exactly to a Bayesian Nash equilibrium in the incomplete-information game $(\Omega, p, (\tau_i)_{i \in \mathcal{N}})$.

8.5 ■ Properties of Correlated Equilibria

Proposition 8.10 (Convexity and Compactness of CE). *The set of correlated equilibria CE of any finite normal-form game is a convex and compact subset of the probability simplex over joint actions.*

Proof. The set CE is defined as the set of all probability distributions π over the joint action space $\prod_{i \in \mathcal{N}} A_i$ that satisfy a finite collection of linear inequalities:

$$\sum_{a_{-i}} \pi(a_i, a_{-i}) u_i(a_i, a_{-i}) \geq \sum_{a_{-i}} \pi(a_i, a_{-i}) u_i(a'_i, a_{-i}) \quad \forall i \in \mathcal{N}, \forall a_i, a'_i \in A_i,$$

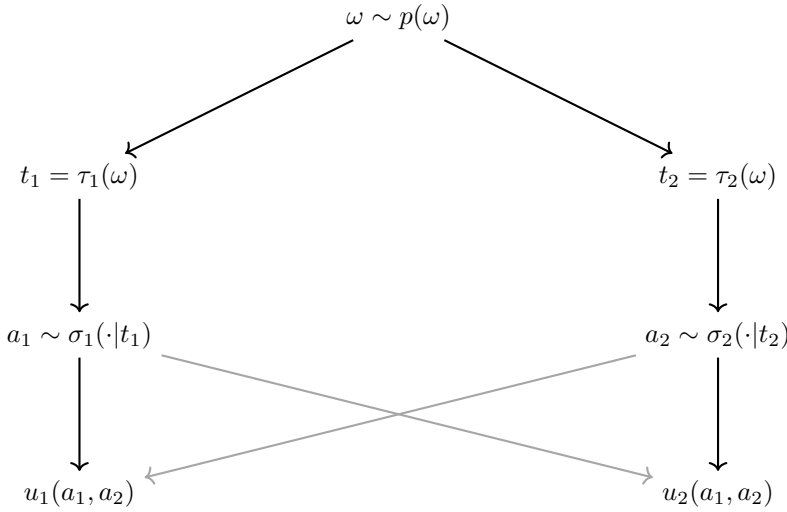


Figure 8.4: Bayesian interpretation of a correlated equilibrium. Nature draws a state ω from the common prior p , which determines the private signals t_1 and t_2 via τ_1 and τ_2 . Each player then selects $a_i \sim \sigma_i(\cdot|t_i)$, producing the joint outcome (a_1, a_2) and utilities $u_i(a_1, a_2)$.

along with the standard simplex constraints:

$$\sum_{a \in \prod_i A_i} \pi(a) = 1, \quad \pi(a) \geq 0 \quad \forall a.$$

These constraints are linear, and hence define a convex polyhedron inside the simplex. The simplex itself is compact and convex, and the intersection of convex sets is convex. Moreover, the intersection of a compact set with a closed set is compact. Thus, CE is convex and compact. $\square$

Proposition 8.11 (Inclusion of Nash Equilibria). *Every mixed-strategy Nash equilibrium of a finite game is also a correlated equilibrium. That is, $NE \subseteq CE$.*

Proof. Let $\sigma = (\sigma_1, \dots, \sigma_N)$ be a mixed-strategy Nash equilibrium. Define the product distribution $\pi = \sigma_1 \otimes \dots \otimes \sigma_N$ over the joint action space. Since σ is a Nash equilibrium, each player's strategy σ_i is a best response to σ_{-i} . That is, for any deviation $a'_i \in A_i$:

$$\mathbb{E}_{a_{-i} \sim \sigma_{-i}}[u_i(a_i, a_{-i})] \geq \mathbb{E}_{a_{-i} \sim \sigma_{-i}}[u_i(a'_i, a_{-i})] \quad \text{for } a_i \in \text{supp}(\sigma_i).$$

Since the product distribution π satisfies the CE constraints in expectation, and the support of π corresponds to strategy profiles played with positive probability, π is a correlated equilibrium. $\square$

Proposition 8.12 (Obedience Principle). *In a correlated equilibrium, none of the players has any incentive to unilaterally deviate from the action recommended by the correlation device.*

Proof. Let π be a correlated equilibrium. Suppose that a player i receives a recommended action a_i . The obedience condition requires that i prefers to follow a_i over any other $a'_i \in A_i$,

conditional on the recommendation a_i . This is formalized by the CE condition:

$$\sum_{a_{-i}} \pi(a_i, a_{-i}) u_i(a_i, a_{-i}) \geq \sum_{a_{-i}} \pi(a_i, a_{-i}) u_i(a'_i, a_{-i}),$$

which ensures that the expected utility of playing a_i is at least as good as switching to a'_i . Hence, the recommendation is incentive-compatible. $\square$

Proposition 8.13 (Computability via Linear Programming). *The set of correlated equilibria can be efficiently computed via linear programming.*

Proof. The CE constraints consist of linear inequalities over a probability distribution π on a finite set. Let A denote the matrix formed by the incentive constraints and probability constraints, and b the corresponding vector. Then $\pi \in \mathbb{R}^d$ (where $d = |\prod_i A_i|$) satisfies:

$$A\pi \leq b, \quad \pi \geq 0, \quad \sum_a \pi(a) = 1.$$

This is a standard linear program. The feasible region is a polytope, and any vertex of this polytope corresponds to an extreme point of the CE set. Hence, one can solve the optimization problems over CE using polynomial-time algorithms for LP. $\square$

Proposition 8.14 (Interpretation as Coordination Devices). *A correlated equilibrium can be implemented using a coordination mechanism such as a public signal or mediator that privately recommends actions to each player.*

Proof. By definition, a correlated equilibrium π prescribes a joint distribution over actions. One can interpret the sampling of a profile $a = (a_1, \dots, a_N)$ from π as a recommendation mechanism: a mediator draws $a \sim \pi$ and privately recommends a_i to player i . The CE condition ensures that none of the players has any incentive to disobey the recommendation. Therefore, π is implementable by a public correlation device and yields outcomes that are stable against unilateral deviations, conditional on the recommendation. $\square$

Example 8.15 Feasible Correlated Equilibria in the Game of Chicken.

Consider the Game of Chicken. The following distributions $\pi = (\pi_{GG}, \pi_{GY}, \pi_{YG}, \pi_{YY})$ represent valid correlated equilibria:

No.	Description	π_{GG}	π_{GY}	π_{YG}	π_{YY}
1	Balanced mix (safe and fair)	0	0.55	0.40	0.05
2	Collision-free symmetric	0	0.50	0.50	0
3	Skewed in favor of P1	0	0.70	0.30	0
4	Symmetric mixed NE	9/64	15/64	15/64	25/64
5	Pure (G, Y)	0	1	0	0
6	Pure (Y, G)	0	0	1	0

Here, Solution 4 corresponds to the symmetric mixed-strategy Nash equilibrium. Solutions 5 and 6 are pure-strategy equilibria. Solution 2 avoids collisions entirely while maintaining fairness, something that is not achievable with uncorrelated strategies. $\square$

Proposition 8.16 (Existence of Correlated Equilibria). *Every finite normal-form game admits at least one correlated equilibrium.*

Proof. By Nash's theorem, every finite normal-form game admits a mixed-strategy Nash equilibrium. By Proposition 8.11, every mixed-strategy Nash equilibrium is also a correlated equilibrium. Hence at least one correlated equilibrium exists. $\square$

8.6 ■ No-Regret Learning

The preceding sections have shown that the correlated equilibrium provides a powerful static characterization of coordination in games such as the Game of Chicken. It represents a joint distribution over actions from which no player wishes to deviate, given the informational structure induced by a mediator. However, an important question remains:

Can such equilibria arise endogenously without an external mediator or centralized coordination device?

In many real-world multi-agent systems, such as autonomous vehicles, distributed networks, and adaptive economic agents, players learn and adapt over time, responding only to their own experiences. They may not know the payoffs of others, the structure of the game, or even the equilibrium concept itself. Yet, through repeated interaction, the population often converges to patterns of behavior that resemble correlated equilibria.

This phenomenon is captured by the theory of *no-regret learning* [52], which connects repeated play to equilibrium concepts in static games; these learning dynamics and their convergence guarantees are developed in detail in Lecture 30. The key insight is that when every player adjusts her strategy to minimize a suitable form of long-term regret, the empirical distribution of play converges to an equilibrium set. The specific equilibrium notion depends on the *type of regret* being minimized (for which precise definitions will be given shortly, in Subsection 8.6.1):

- Minimizing **external regret** leads to convergence to the set of *coarse correlated equilibria* (CCE).
- Minimizing **internal regret**, a stronger requirement, leads to convergence to the set of *correlated equilibria* (CE).

Thus, coordination can emerge as an unintended consequence of individually myopic adaptation.

Formally, consider a finite normal-form game $\mathcal{G} = \langle \mathcal{N}, (A_i)_{i \in \mathcal{N}}, (u_i)_{i \in \mathcal{N}} \rangle$ that is played repeatedly over discrete time steps $t = 1, 2, \dots$. Here $\mathcal{N} = \{1, \dots, N\}$ denotes the set of players; each player $i \in \mathcal{N}$ has a finite set of actions, A_i ; and $u_i : A \rightarrow \mathbb{R}$ is the payoff function of player i , where $A := \prod_{i \in \mathcal{N}} A_i$ represents the joint action space.

At each round t , each player i selects an action $a_i^t \in A_i$, leading to the joint action $a^t = (a_1^t, \dots, a_N^t)$. The sequence of joint actions

$$h_t = (a^1, a^2, \dots, a^t)$$

constitutes the history of play up to time t .

In a no-regret learning process, players adjust their strategies based solely on their realized payoffs, without observing others' utilities or strategies, so as to avoid the ex post thought:

“If only I had played differently.”

Depending on whether this counterfactual comparison is made against one fixed action (external regret) or against a conditional switch of actions (internal regret), the learning process converges respectively to a CCE or a CE.

Hence, the static notion of correlation introduced via an explicit mediator in earlier sections can emerge here endogenously as a limiting outcome of decentralized dynamic learning.

8.6.1 ■ Regret and Counterfactual Payoffs

To quantify regret, we introduce the notion of *counterfactual payoffs*: the hypothetical rewards a player would have received had she acted differently in the past.

Definition 8.17 (Counterfactual Payoff). *Let $j, k \in A_i$ be two actions available to player i . The counterfactual payoff to switching from j to k at time τ is*

$$W_i^\tau(j, k) := \begin{cases} u_i(k, a_{-i}^\tau), & \text{if } a_i^\tau = j, \\ u_i(a_i^\tau, a_{-i}^\tau), & \text{otherwise,} \end{cases}$$

where a_{-i}^τ denotes the action profile of all other players at stage τ .

If player i played j at round τ , then $W_i^\tau(j, k)$ represents the payoff they *would have received* had they played k instead. If they did not play j , the counterfactual payoff coincides with their actual payoff.

Example 8.18 Rock-Paper-Scissors.

Consider repeated play of Rock-Paper-Scissors. If a player frequently loses by choosing Rock against an opponent who plays Paper, the regret for not having switched from Rock to Scissors becomes large. The counterfactual payoff captures the hypothetical improvement from making that switch in those specific rounds. $\square$

Definition 8.19 (Average Payoff Difference). The average gain from switching from j to k over t rounds is

$$D_i^t(j, k) := \frac{1}{t} \sum_{\tau=1}^t W_i^\tau(j, k) - \frac{1}{t} \sum_{\tau=1}^t u_i(a^\tau).$$

Definition 8.20 (External Regret). The external regret of player i for not committing in advance to a single action k is

$$R_i^{t, \text{ext}}(k) := \max \left\{ \frac{1}{t} \sum_{\tau=1}^t [u_i(k, a_{-i}^\tau) - u_i(a_i^\tau, a_{-i}^\tau)], 0 \right\}.$$

Player i has no external regret if $\lim_{t \rightarrow \infty} R_i^{t, \text{ext}}(k) = 0$ for all $k \in A_i$.

Definition 8.21 (Internal Regret). For each pair of actions $j, k \in A_i$, the internal regret of player i for not switching from j to k is defined as

$$R_i^{t, \text{int}}(j, k) := \max \left\{ \frac{1}{t} \sum_{\tau: a_i^\tau = j} [u_i(k, a_{-i}^\tau) - u_i(j, a_{-i}^\tau)], 0 \right\}.$$

Player i has no internal regret if $\lim_{t \rightarrow \infty} R_i^{t, \text{int}}(j, k) = 0$ for all $j, k \in A_i$.

External regret compares a player's realized average payoff to the best *fixed* action she could have chosen from the start (*ex ante*). Internal regret instead evaluates performance *conditionally*: for each action that was played, it measures the potential gain from consistently substituting another action. Minimizing external regret leads to CCE, where deviations are evaluated before observing recommendations, while minimizing internal regret leads to the more refined set of *correlated equilibria* (CE), where deviations are conditional on observed signals.

8.7 ■ From No-Regret Learning to Equilibrium

No-regret learning provides a decentralized mechanism through which equilibrium behavior can emerge endogenously in repeated games. Each player adjusts future actions based on accumulated regret from past plays, using only her own payoffs and without any explicit coordination or knowledge of others' utilities.

8.7.1 ■ Regret-Based Adaptive Updates

A simple way to implement no-regret learning is to let each player randomize among actions in proportion to their past regrets.

Definition 8.22 (Regret-Based Update Rule). Suppose player i played action j at time t . The probability of playing another action $k \in A_i \setminus \{j\}$ at time $t+1$ is

$$P_i^{t+1}(k) = \frac{R_i^t(j, k)}{\mu_i^t}, \quad P_i^{t+1}(j) = 1 - \sum_{k \neq j} P_i^{t+1}(k),$$

where the normalization constant

$$\mu_i^t := \sum_{k \neq j} R_i^t(j, k)$$

ensures that the probabilities sum to one. If $\mu_i^t = 0$, then $P_i^{t+1}(j) = 1$.

The greater a player's regret for not having played k , the higher is the probability of choosing k in the future. Actions with no positive regret receive zero probability. When the regrets $R_i^t(j, k)$ correspond to *internal* (pairwise-switching) regrets, this process is called *regret matching* [52] and is known to converge to the set of correlated equilibria. When $R_i^t(j, k)$ measure *external* regret with respect to fixed actions, convergence instead occurs to the broader set of coarse correlated equilibria.

8.7.2 ■ Empirical Distribution of Play

The evolution of the system can be summarized by the empirical joint distribution of play.

Definition 8.23 (Empirical Joint Distribution). *The empirical frequency of joint actions up to time t is*

$$z_t(a) := \frac{1}{t} |\{ \tau \leq t : a^\tau = a \}|, \quad a \in A.$$

This distribution represents how often each action profile has been observed in play. It provides a natural candidate for the limiting outcome of the learning process.

8.7.3 ■ Equilibrium Convergence under No-Regret Learning

The connection between adaptive learning dynamics and equilibrium concepts has been a central theme in game theory and multi-agent learning. Early results by Dean Foster and Rakesh Vohra [37] and by Sergiu Hart and Andreu Mas-Colell [52] established that when players repeatedly update their strategies to minimize long-run regret, the empirical distribution of play converges to well-defined equilibrium sets. Specifically, minimizing *external regret* leads to convergence toward the set of CCE, whereas eliminating the stronger notion of *internal regret* ensures convergence to the narrower set of CE. This convergence holds under minimal information assumptions: players need only observe their realized payoffs, and not others' strategies or utilities.

Theorem 8.24 (External vs. Internal Regret and Equilibrium Convergence).

Let $\mathcal{G} = \langle \mathcal{N}, (A_i)_{i \in \mathcal{N}}, (u_i)_{i \in \mathcal{N}} \rangle$ be a finite normal-form game played repeatedly over stages $t = 1, 2, \dots$, and let z_T denote the empirical distribution of joint actions up to time T .

- (i) **External Regret $\Rightarrow$ Coarse Correlated Equilibrium.** *If each player's external regret satisfies*

$$\lim_{T \rightarrow \infty} R_i^{T, \text{ext}}(k) = 0 \quad \text{for all } i \in \mathcal{N}, k \in A_i,$$

then every accumulation point z_∞ of $(z_T)_{T \geq 1}$ is a coarse correlated equilibrium (CCE) of $\mathcal{G}$.

- (ii) **Internal Regret $\Rightarrow$ Correlated Equilibrium.** *If each player's internal regret satisfies*

$$\lim_{T \rightarrow \infty} R_i^{T, \text{int}}(j, k) = 0 \quad \text{for all } i \in \mathcal{N}, j, k \in A_i,$$

then every accumulation point z_∞ of $(z_T)_{T \geq 1}$ is a correlated equilibrium (CE) of $\mathcal{G}$, i.e.,

$$\sum_{a \in A: a_i = j} z_\infty(a) [u_i(k, a_{-i}) - u_i(j, a_{-i})] \leq 0 \quad \text{for all } i, j, k.$$

Eliminating external regret guarantees stability against unconditional deviations, so that no player wishes that she had committed to a single fixed action from the start. Eliminating internal regret, a stronger condition, guarantees stability against conditional deviations, ensuring that no player systematically prefers switching from one action to another in the contexts where it was played. Consequently, minimizing external regret leads to convergence toward the set of coarse correlated equilibria, while eliminating internal regret ensures convergence to the smaller set of correlated equilibria:

Type of Regret	Limiting Equilibrium Concept
External regret	$\text{CCE}(\mathcal{G})$
Internal regret	$\text{CE}(\mathcal{G})$

No-regret learning requires only minimal feedback. Each player needs to observe only her own past actions and realized payoffs, without knowing others' utilities or strategies. This property makes regret-based dynamics particularly suitable for distributed and adaptive multi-agent systems, where coordination must emerge without centralized control or complete information.

8.8 ■ Regret Matching in the Game of Chicken

We now illustrate the regret-matching procedure through the classic *Game of Chicken* we started this Lecture with. Each player repeatedly chooses between two actions: going straight (G) or yielding (Y). At every round t , let $a_i^t \in \{G, Y\}$ denote the action of player i and $a_{-i}^t \in \{G, Y\}$ the action of the opponent. The realized joint action is then

$$a^t = (a_i^t, a_{-i}^t),$$

and player i observes both a^t and their payoff $u_i(a^t) = u_i(a_i^t, a_{-i}^t)$.

8.8.1 ■ Instantaneous Regret Updates

For each possible alternative action $a \in \{G, Y\}$, player i compares the payoff they would have obtained by playing a instead of the realized action a_i^t , given the observed opponent action a_{-i}^t . The *instantaneous regret increment* for action a at round t is therefore

$$\Delta R_i^t(a) = u_i(a, a_{-i}^t) - u_i(a_i^t, a_{-i}^t), \quad a \in \{G, Y\}.$$

In the Game of Chicken, the payoff table for player i is

$$u_i(G, G) = -10, \quad u_i(Y, G) = 0, \quad u_i(G, Y) = 5, \quad u_i(Y, Y) = -1.$$

Case 1: Opponent plays G. If $a_{-i}^t = G$, then

$$u_i(G, G) = -10, \quad u_i(Y, G) = 0.$$

Hence:

$$\begin{aligned} (a_i^t, a_{-i}^t) = (G, G) : \quad & \Delta R_i^t(G) = 0, \quad \Delta R_i^t(Y) = 10, \\ (a_i^t, a_{-i}^t) = (Y, G) : \quad & \Delta R_i^t(G) = -10, \quad \Delta R_i^t(Y) = 0. \end{aligned}$$

Case 2: Opponent plays Y. If $a_{-i}^t = Y$, then

$$u_i(G, Y) = 5, \quad u_i(Y, Y) = -1.$$

Thus:

$$\begin{aligned} (a_i^t, a_{-i}^t) = (G, Y) : \quad & \Delta R_i^t(G) = 0, \quad \Delta R_i^t(Y) = -6, \\ (a_i^t, a_{-i}^t) = (Y, Y) : \quad & \Delta R_i^t(G) = 6, \quad \Delta R_i^t(Y) = 0. \end{aligned}$$

8.8.2 ■ Regret-Matching Dynamics

Let $R_i^t(a) = \sum_{\tau=1}^t \Delta R_i^\tau(a)$, $a \in \{G, Y\}$, denote the cumulative regret for deviating to action a up to round t , and define $[x]_+ = \max\{x, 0\}$. The regret-matching rule uses only the current regret vector $(R_i^t(G), R_i^t(Y))$ to choose the next mixed action. The probability that player i assigns to action a at round $t+1$ is

$$p_i^{t+1}(a) = \begin{cases} \frac{[R_i^t(a)]_+}{[R_i^t(G)]_+ + [R_i^t(Y)]_+}, & \text{if } [R_i^t(G)]_+ + [R_i^t(Y)]_+ > 0, \\ \frac{1}{2}, & \text{otherwise,} \end{cases} \quad a \in \{G, Y\}.$$

This update rule increases the probability of actions with positive cumulative regret and decreases the weight on actions with nonpositive regret. Because regret is computed for each possible one-shot substitution $a_i^t \rightarrow a$, these dynamics minimize *internal regret*. By the main result of [52], the empirical distribution of play generated by internal regret-matching converges to the set of correlated equilibria of the stage game.

ALGORITHM 8.1. Internal-Regret Matching for the Game of Chicken (Player i).

- 1: **Input:** Payoff matrix of the Game of Chicken; optional exploration rate $\varepsilon \in [0, 0.05]$.
- 2: **Initialize:** $R_i^0(G) \leftarrow 0$, $R_i^0(Y) \leftarrow 0$, $p_i^1(G) \leftarrow \frac{1}{2}$, $p_i^1(Y) \leftarrow \frac{1}{2}$.
- 3: **for** $t = 1, 2, \dots$ **do**
- 4: Sample $a_i^t \sim p_i^t$; observe opponent's action a_{-i}^t and payoff $u_i(a_i^t, a_{-i}^t)$.
- 5: Compute instantaneous increments $\Delta R_i^t(G), \Delta R_i^t(Y)$.
- 6: Update cumulative regrets: $R_i^t(a) \leftarrow R_i^{t-1}(a) + \Delta R_i^t(a)$ for $a \in \{G, Y\}$.
- 7: Form weights $w(a) \leftarrow [R_i^t(a)]_+$.
- 8: **if** $w(G) + w(Y) > 0$ **then**
- 9: $p_i^{t+1}(a) \leftarrow (1 - \varepsilon) \frac{w(a)}{w(G) + w(Y)} + \varepsilon \cdot \frac{1}{2}$ for $a \in \{G, Y\}$.
- 10: **else**
- 11: $p_i^{t+1}(G) \leftarrow \frac{1}{2}$, $p_i^{t+1}(Y) \leftarrow \frac{1}{2}$.
- 12: **end if**
- 13: **end for**

8.8.3 ■ Convergence and Emergent Coordination

Let $(a^t)_{t \geq 1}$ denote the realized joint play under regret matching, and define the empirical (Cesàro) distribution

$$z_T(a) := \frac{1}{T} \sum_{t=1}^T \mathbf{1}\{a^t = a\}, \quad a \in \{(G, G), (G, Y), (Y, G), (Y, Y)\}.$$

Theorem 8.25 (Regret Matching Implies Convergence to CE). *If both players follow internal-regret matching, then with probability 1 the empirical distribution converges to the set of correlated equilibria:*

$$\text{dist}(z_T, \text{CE}) \xrightarrow[T \rightarrow \infty]{a.s.} 0.$$

Equivalently, every accumulation point z_∞ of $(z_T)_{T \geq 1}$ lies in CE.

Corollary 8.26 (Typical Limiting Behavior). *Under symmetric regret matching (with or without small exploration), every accumulation point z_∞ of the empirical distribution lies on the “safe”*

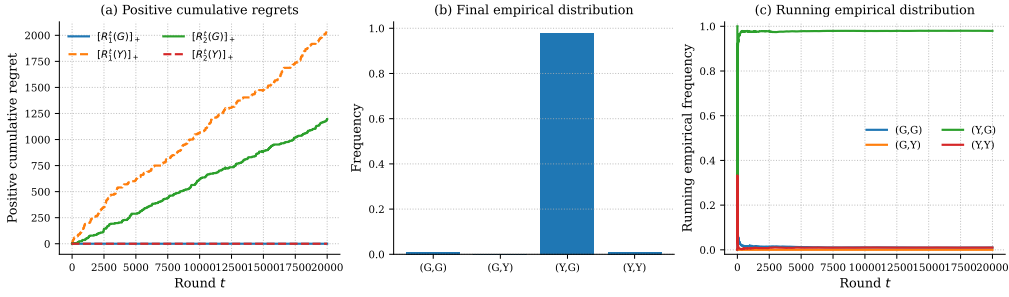


Figure 8.5: Internal-regret matching dynamics in the Game of Chicken for $\varepsilon = 0.02$ over $T = 20,000$ rounds. (a) Positive cumulative regrets $[R_i^t(a)]_+$. (b) Final empirical distribution over joint actions. (c) Running empirical frequencies $z_t(a)$.

correlated-equilibrium face spanned by the off-diagonal outcomes (G, Y) and (Y, G) :

$$z_\infty \in \text{co}\{\delta_{GY}, \delta_{YG}\}, \quad z_\infty(GG) = 0, \quad z_\infty(GY) + z_\infty(YG) \approx 1.$$

Over time, the joint play concentrates on the two efficient asymmetric outcomes where one player yields and the other one goes straight. The dangerous crash outcome (G, G) is eliminated, as it generates large positive regret for not yielding. The mutual-yield outcome (Y, Y) occurs only occasionally because its low payoff incentivizes players to alternate going straight. Consequently, the empirical play oscillates between (G, Y) and (Y, G) , producing a self-organized coordination pattern.

If each player employs a no-regret algorithm satisfying $R_i(T) = \mathcal{O}(\sqrt{T})$, the average incentive to deviate is at most $\mathcal{O}(1/\sqrt{T})$. In this game, the empirical crash frequency is bounded above by a constant multiple of $(R_1(T) + R_2(T))/10$, where 10 is the crash-penalty margin. Hence, under standard regret rates, the probability of a crash decays as $\mathcal{O}(1/\sqrt{T})$, becoming negligible in long-run play.

8.8.4 ■ Numerical Illustration of Regret-Matching Dynamics

We conclude with a numerical simulation of internal-regret matching in the Game of Chicken, using $T = 20,000$ rounds and a fixed exploration rate $\varepsilon = 0.02$. At each round, both players sample actions from their mixed strategies p_i^t , and regrets are updated using realized and counterfactual payoffs. The resulting trajectories are shown in Figure 8.5.

The final empirical distribution over the four joint outcomes (G, G) , (G, Y) , (Y, G) , and (Y, Y) is $(0.0051, 0, 0.9894, 0.0055)$, as shown in Figure 8.5(b). Almost all mass concentrates on (Y, G) , while (G, G) and (Y, Y) occur only with negligible frequency. This is fully consistent with Corollary 8.26: internal-regret dynamics drive the empirical distribution toward the correlated-equilibrium face $\text{co}\{\delta_{GY}, \delta_{YG}\}$, and early randomness selects one of its extreme points.

Figure 8.5(c) reports the running empirical frequencies $z_t(a)$. After a short exploratory phase, the trajectory locks into (Y, G) , while the frequencies of (G, G) , (G, Y) , and (Y, Y) decay steadily toward zero. Each of these outcomes generates persistent positive regret for at least one player, and regret-matching consequently suppresses them asymptotically.

Figure 8.5(a) shows the cumulative positive regrets. P1's regret for not yielding, $[R_1^t(Y)]_+$, grows steadily, indicating that whenever P1 plays G, switching to Y would have improved payoffs. Similarly, P2's regret for not going straight, $[R_2^t(G)]_+$, increases persistently, showing that G is

the profitable deviation for P2. All other regret components remain near zero, indicating the absence of profitable internal deviations in the limit.

8.9 ■ Further Notes and Reading

Aumann's notion of *correlated equilibrium* provides a rigorous foundation for how communication, institutions, and shared signals reduce the likelihood of destructive conflict. Although formulated as an epistemic refinement of Nash equilibrium, Aumann's original examples, such as traffic-light coordination and variants of the Game of Chicken, illustrate how correlation devices allow rational players to avoid inefficient or harmful Nash outcomes [4, 6]. This interpretation aligns with Schelling's theory of brinkmanship and crisis signaling in international conflict [105] and subsequent work showing how monitoring, mediation, and credible communication help prevent accidental war [85, 96, 33]. Current multi-agent and AI research extends these insights: learning processes frequently converge to correlated equilibria [37, 23], and CE provides a coordination mechanism for mitigating conflict and ensuring stable behavior in distributed autonomous systems [36, 73]. Thus, correlated equilibrium captures a unifying principle: structured communication, whether human or machine, supports peace, coordination, and stability in strategic environments.

The connection between correlated equilibrium and learning was formalized by Hart and Mas-Colell [52], who proved that any no-regret learning process drives the empirical distribution of play to the correlated-equilibrium set. This result unified a wide class of adaptive procedures, including regret matching, multiplicative weights, and mirror descent, under a single convergence framework. Subsequent work by Fudenberg and Levine [39] and by Cesa-Bianchi and Lugosi [23] established broader relationships between repeated play, online learning, and equilibrium selection, showing that correlated equilibrium can emerge even when players lack detailed knowledge of their opponents' payoffs or strategies.

Correlated learning is particularly suited to *coordination environments*, where multiple agents must align decisions for mutual benefit while avoiding costly conflicts or inefficiencies. In wireless communication networks, for example, transmitters can learn to select channels dynamically to minimize interference and converge to efficient correlated equilibria that approximate distributed scheduling [76]. Similar mechanisms underlie adaptive coordination in traffic and mobility management, where autonomous agents learn safe and efficient routing patterns via feedback and limited signaling. In environmental and resource-sharing games [47, 54], agents may gradually learn to balance exploitation and sustainability, while in cyber-defense and distributed control systems [149, 92, 70], correlated learning enables defenders or controllers to coordinate security or actuation policies under uncertainty and limited observability.

8.10 ■ Exercises

Exercise 8.1 (Correlated Equilibria and Learning in a Channel Interference Game). Two wireless transmitters share a common communication environment with two available channels:

$$A_i = \{\text{Low (L)}, \text{High (H)}\}, \quad i \in \{1, 2\}.$$

Each transmitter selects one channel to maximize its throughput. If both choose the same channel, interference degrades performance for both; if they choose different channels, each achieves higher throughput. The payoff matrix (row player first) is:

P1	P2	
	L	H
L	(1, 1)	(4, 2)
H	(2, 4)	(1, 1)

When both users transmit on the same band (L, L) or (H, H), mutual interference reduces their achievable rates to a low level (1, 1). When they occupy different bands, (L, H) or (H, L), the network operates efficiently, but the channel allocations are asymmetric: the user on the lower-interference band obtains a slightly higher rate. This setting reflects a coordination problem in distributed spectrum access, where efficient coexistence depends on avoiding collisions without centralized scheduling.

- (i) Determine all pure-strategy and mixed-strategy Nash equilibria. Identify the equilibria that correspond to interference-free operation and determine whether they are Pareto efficient. Compute the collision probability under the mixed-strategy Nash equilibrium and interpret its inefficiency.
- (ii) Derive the correlated-equilibrium (CE) incentive inequalities. Writing π_{LL} , π_{LH} , π_{HL} , and π_{HH} for the probabilities of the four joint actions, show that the obedience constraints are

$$\begin{aligned}
 \text{P1: } & 1 \pi_{LL} + 4 \pi_{LH} \geq 2 \pi_{LL} + 1 \pi_{LH}, \\
 & 2 \pi_{HL} + 1 \pi_{HH} \geq 1 \pi_{HL} + 4 \pi_{HH}, \\
 \text{P2: } & 1 \pi_{LL} + 4 \pi_{HL} \geq 2 \pi_{LL} + 1 \pi_{HL}, \\
 & 2 \pi_{LH} + 1 \pi_{HH} \geq 1 \pi_{LH} + 4 \pi_{HH}.
 \end{aligned}$$

Simplify these inequalities and combine them with the probability constraints $\pi_{ij} \geq 0$ and $\sum_{ij} \pi_{ij} = 1$ to describe the CE polytope.

- (iii) Formulate the problem of maximizing the total network throughput $u_1 + u_2$ over the CE polytope of part (ii) as a linear program, and implement and solve it numerically in Python (for example, with `scipy.optimize.linprog`) or MATLAB (`linprog`). Verify that the optimal value equals 6 and that it is attained by the symmetric distribution $\pi_{LH} = \pi_{HL} = \frac{1}{2}$, $\pi_{LL} = \pi_{HH} = 0$, which places all probability mass on the interference-free outcomes. Interpret this equilibrium as a distributed time-sharing or coordination policy.
- (iv) Implement the *regret-matching* dynamics in which both users adaptively update their channel-selection probabilities according to

$$p_i^{t+1}(a_i) = \frac{[R_i^t(a_i)]_+}{[R_i^t(\mathbf{L})]_+ + [R_i^t(\mathbf{H})]_+}, \quad R_i^t(a_i) = \sum_{\tau=1}^t (u_i(a_i, a_{-i}^\tau) - u_i(a_i^\tau, a_{-i}^\tau)).$$

Simulate the dynamics for $T = 20,000$ rounds and plot the empirical distribution

$$z_T(a) = \frac{1}{T} \sum_{t=1}^T \mathbf{1}\{a^t = a\}$$

over time. Explain why z_T converges to the interference-free region of the CE set, and relate this behavior to adaptive frequency selection in cognitive radio networks.

- (v) Discuss how correlated equilibria formalize implicit spectrum coordination through repeated interaction. Explain how lightweight communication or random beaconing can serve as the correlation device, and how no-regret learning provides a decentralized mechanism for achieving near-optimal coexistence in shared-spectrum environments.

Exercise 8.2 (Mixed-Strategy Nash Equilibrium as a Correlated Equilibrium). Consider a two-player normal-form game $\mathcal{G} = \langle \{1, 2\}, (A_1, A_2), (u_1, u_2) \rangle$, where $A_i = \{a_i^1, a_i^2\}$ is the finite action set of player i and $u_i : A_1 \times A_2 \rightarrow \mathbb{R}$ denotes player i 's payoff function.

- (i) Recall that a *mixed-strategy Nash equilibrium (MSNE)* consists of independent probability distributions $\sigma_1 \in \Delta(A_1)$ and $\sigma_2 \in \Delta(A_2)$ such that, for each player $i \in \{1, 2\}$,

$$\mathbb{E}_{a_{-i} \sim \sigma_{-i}}[u_i(a_i^*, a_{-i})] \geq \mathbb{E}_{a_{-i} \sim \sigma_{-i}}[u_i(a_i, a_{-i})] \quad \forall a_i \in A_i,$$

where a_i^* denotes any action in the support of σ_i . Explain in words why this condition implies that no player can gain by unilaterally deviating from her mixed strategy.

- (ii) Recall that a *correlated equilibrium (CE)* is a joint distribution $\pi \in \Delta(A_1 \times A_2)$ that satisfies, for each $i \in \{1, 2\}$ and for all $a_i, a_i' \in A_i$,

$$\sum_{a_{-i} \in A_{-i}} \pi(a_i, a_{-i}) u_i(a_i, a_{-i}) \geq \sum_{a_{-i} \in A_{-i}} \pi(a_i, a_{-i}) u_i(a_i', a_{-i}).$$

Interpret this inequality in terms of a player's incentive to follow a recommendation issued by a mediator.

- (iii) Prove that the joint distribution

$$\pi(a_1, a_2) := \sigma_1^*(a_1) \sigma_2^*(a_2),$$

induced by a mixed-strategy Nash equilibrium (σ_1^*, σ_2^*) , is a correlated equilibrium by verifying that it satisfies the CE inequalities above.

- (iv) Determine whether the converse holds: is every correlated equilibrium necessarily representable as the product of independent mixed strategies? Provide a brief justification or a counterexample.

Exercise 8.3 (Correlated Equilibrium via Signaling in a Bi-Matrix Game). Consider the following bi-matrix game between two *minimizing* players:

$$P_1 = \begin{bmatrix} 0 & 5 \\ 1 & 4 \end{bmatrix}, \quad P_2 = \begin{bmatrix} 4 & 5 \\ 1 & 0 \end{bmatrix}.$$

Player 1 chooses between U (up) and D (down); Player 2 chooses between L (left) and R (right). The game admits two pure-strategy Nash equilibria:

$$(U, L) \quad \text{with cost } (0, 4), \quad (D, R) \quad \text{with cost } (4, 0).$$

It also admits a mixed-strategy equilibrium where both players randomize uniformly between their two actions, $(0.5, 0.5)$, with expected costs $(2.5, 2.5)$.

We now explore how a *correlation device* can generate a correlated equilibrium that Pareto-dominates the Nash outcomes.

- (i) **Formulate the extensive form.** Represent the game with a random signaling device that selects one of three equally likely states $\Omega = \{A, B, C\}$. Player 1 observes the signal perfectly if $\omega = A$, but cannot distinguish between B and C ; conversely, Player 2 observes the signal perfectly if $\omega = C$, but cannot distinguish between A and B . Both players choose their actions simultaneously based on their private observations.
- (a) Draw the *extensive-form representation* of this game, clearly indicating Nature's randomization and the players' information sets.
 - (b) Derive the equivalent *normal form* of this Bayesian game.
 - (c) Verify that the two pure-strategy Nash equilibria of the original game, (U, L) and (D, R) , remain Nash equilibria in this expanded normal form.
- (ii) **Construct a correlated equilibrium.** Consider the following strategy pair associated with the signaling device:
- Player 1 plays U if the signal is A , and D otherwise.
 - Player 2 plays R if the signal is C , and L otherwise.
- (a) Show that this strategy profile constitutes a Nash equilibrium of the game with the signaling device.
 - (b) Compute the expected cost for both players under this strategy profile, and verify that it equals $(\frac{5}{3}, \frac{5}{3})$, which is strictly lower (better) for each player than both the equal-weight average $(2, 2)$ of the two pure-strategy Nash equilibrium costs and the mixed-strategy Nash equilibrium cost $(2.5, 2.5)$.
- (iii) **Verify the induced correlated equilibrium.** Show that the signaling device induces a correlated equilibrium in which the joint action distribution is:

$$\Pr[(U, L)] = \frac{1}{3}, \quad \Pr[(D, L)] = \frac{1}{3}, \quad \Pr[(D, R)] = \frac{1}{3},$$

corresponding to stage costs of $(0, 4)$, $(1, 1)$, and $(4, 0)$, respectively. Verify that this distribution satisfies the correlated-equilibrium incentive constraints for both players, with the inequalities reversed to reflect cost minimization.

Exercise 8.4 (Bayesian and Extensive-Form Interpretations of Coarse Correlated Equilibrium).

The *coarse correlated equilibrium* (CCE) generalizes the correlated equilibrium (CE) by allowing players to correlate their actions through a public random device but without conditioning on private signals. In a CCE, players decide whether to commit to the mediator's recommendation *before* seeing it, whereas in a CE, they observe their private signal before choosing whether to obey. We illustrate both the **Bayesian** and **extensive-form** views of a CCE using the *Game of Chicken*. A mediator recommends the joint actions (G, Y) and (Y, G) each with probability $\frac{1}{2}$:

$$\pi(G, Y) = \pi(Y, G) = \frac{1}{2}, \quad \pi(G, G) = \pi(Y, Y) = 0.$$

Each player must decide *ex ante* whether to commit to follow the mediator's recommendation before learning what it will be.

- (i) Show that the distribution π satisfies the *CCE inequalities* for both players:

$$\sum_{a \in A} \pi(a) u_i(a) \geq \sum_{a \in A} \pi(a) u_i(a'_i, a_{-i}), \quad \forall i, a'_i \in A_i.$$

Explain why, in general, a distribution satisfying the CCE inequalities need not satisfy the *CE inequalities*, which allow conditioning on the received signal, and determine whether this particular distribution π is also a correlated equilibrium.

- (ii) Model the CCE as a sequential game. Draw or describe the game tree, labeling Nature's move, the players' decisions, the information sets, and the terminal payoffs. Compare this tree with the CE tree (Figure 8.3), in which private signals are revealed before play.
- (iii) Represent the same process as a Bayesian game. Conclude that π forms a Bayesian Nash equilibrium and thus a CCE of the complete-information game.

Bibliography

- [1] G. A. Akerlof. The market for “lemons”: Quality uncertainty and the market mechanism. In *Uncertainty in economics*, pages 235–251. Elsevier, 1978.
- [2] T. Alpcan and T. Başar. *Network Security: A Decision and Game-Theoretic Approach*. Cambridge University Press, Cambridge, UK, 2010.
- [3] A.-M. Ampère. *Essai sur la Philosophie des Sciences Mathématiques*. Imprimerie Royale, Lyon, 1782.
- [4] R. J. Aumann. Subjectivity and correlation in randomized strategies. *Journal of Mathematical Economics*, 1(1):67–96, 1974.
- [5] R. J. Aumann. Agreeing to disagree. *The Annals of Statistics*, 4(6):1236–1239, 1976.
- [6] R. J. Aumann. Correlated equilibrium as an expression of bayesian rationality. *Econometrica*, 55(1):1–18, 1987.
- [7] T. Başar and G. Zaccour, editors. *Handbook of Dynamic Game Theory*. Springer, Cham, Switzerland, 2018. Volumes I and II.
- [8] T. Başar. On the uniqueness of the Nash solution in linear–quadratic differential games. *International Journal of Game Theory*, 5(2–3):65–90, 1976.
- [9] T. Başar. Informationally nonunique equilibrium solutions in differential games. *SIAM Journal on Control*, 15(4):636–660, July 1977.
- [10] T. Başar. Time consistency and robustness of equilibria in noncooperative dynamic games. In F. Van der Ploeg and A. de Zeeuw, editors, *Dynamic Policy Games in Economics*, pages 9–54. North-Holland, Amsterdam, 1989.
- [11] T. Başar and P. Bernhard. *H-infinity Optimal Control and Related Minimax Design Problems: A Dynamic Game Approach*. Birkhäuser, Boston, MA, 2nd edition, Aug. 1995.
- [12] T. Başar and G. J. Olsder. *Dynamic Noncooperative Game Theory*. SIAM, Philadelphia, PA, 2nd edition, 1999.
- [13] U. Berger. Fictitious play in $2 \times n$ games. *Journal of Economic Theory*, 120(2):139–154, 2005.
- [14] K. Binmore. *Fun and Games: A Text on Game Theory*. D. C. Heath and Company, 1992.
- [15] K. Binmore. *Game Theory and the Social Contract: Playing by Nash*. MIT Press, Cambridge, MA, 1998.
- [16] E. Borel. *La Théorie du Jeu et les Équations Intégrales à Noyau Symétrique*. Gauthier-Villars, Paris, 1921.

- [17] E. Borel and J. Ville. *Applications de la Théorie des Probabilités aux Jeux de Hasard*. Gauthier-Villars, Paris, 1938.
- [18] T. Börgers. *An Introduction to the Theory of Mechanism Design*. Oxford University Press, Oxford, UK, 2015.
- [19] A. Brandenburger. The power of paradox: Some recent developments in interactive epistemology. *International Journal of Game Theory*, 35(4):465–492, 2007.
- [20] C. Bravard, L. Charroin, and C. Touati. Optimal design and defense of networks under link attacks. *Journal of Mathematical Economics*, 68:62–79, 2017.
- [21] G. W. Brown. Iterative solutions of games by fictitious play. In T. C. Koopmans, editor, *Activity Analysis of Production and Allocation*, pages 374–376. Wiley, New York, 1951.
- [22] C. Camerer. *Behavioral Game Theory: Experiments in Strategic Interaction*. Princeton University Press, Princeton, NJ, 2003.
- [23] N. Cesa-Bianchi and G. Lugosi. *Prediction, Learning, and Games*. Cambridge University Press, Cambridge, UK, 2006.
- [24] J. Chen, C. Touati, and Q. Zhu. A dynamic game approach to strategic design of secure and resilient infrastructure network. *IEEE Transactions on Information Forensics and Security*, 15:462–474, 2020.
- [25] J. Chen, C. Touati, and Q. Zhu. Optimal secure two-layer IoT network design. *IEEE Transactions on Control of Network Systems*, 7(1):398–409, 2020.
- [26] J. Chen and Q. Zhu. *A game-and decision-theoretic approach to resilient interdependent network analysis and design*. Springer, 2019.
- [27] V. Conitzer and T. Sandholm. Computing the optimal strategy to commit to. In *Proceedings of the 7th ACM Conference on Electronic Commerce (EC)*, pages 82–90, 2006.
- [28] C. Daskalakis, P. W. Goldberg, and C. H. Papadimitriou. The complexity of computing a nash equilibrium. *SIAM Journal on Computing*, 39(1):195–259, 2009.
- [29] G.-L. L. de Buffon. *Essai d'Arithmétique Morale*. Imprimerie Royale, Paris, 1777.
- [30] G. Debreu. Representation of a preference ordering by a numerical function. In *Mathematical Economics: Twenty Papers of Gerard Debreu*, pages 105–110. Cambridge University Press, 2013.
- [31] M. Dziubiński and S. Goyal. Network design and defence. *Games and Economic Behavior*, 79:30–43, 2013.
- [32] J. C. Engwerda. *LQ Dynamic Optimization and Differential Games*. John Wiley & Sons, Chichester, UK, 2005.
- [33] J. D. Fearon. Rationalist explanations for war. *International Organization*, 49(3):379–414, 1995.
- [34] E. Fehr and K. M. Schmidt. Fairness, competition, and cooperation. *Quarterly Journal of Economics*, 115(3):817–868, 2000.
- [35] P. C. Fishburn. *Utility Theory for Decision Making*. Wiley, 1970.
- [36] F. Forges. A note on correlated equilibrium and communication equilibrium. *International Journal of Game Theory*, 21(3):297–305, 1993.
- [37] D. P. Foster and R. V. Vohra. Calibrated learning and correlated equilibrium. *Games and Economic Behavior*, 21(1-2):40–55, 1997.

- [38] D. Fudenberg and D. K. Levine. Consistency and cautious fictitious play. *Journal of Economic Dynamics and Control*, 19(5–7):1065–1089, 1995.
- [39] D. Fudenberg and D. K. Levine. *The Theory of Learning in Games*. MIT Press, Cambridge, MA, 1998.
- [40] D. Fudenberg and J. Tirole. *Game Theory*. MIT Press, Cambridge, MA, 1991.
- [41] R. A. Gatenby, A. S. Silva, R. J. Gillies, and B. R. Frieden. Adaptive therapy. *Cancer Research*, 69(11):4894–4903, 2009.
- [42] I. L. Glicksberg. A further generalization of the Kakutani fixed point theorem, with application to Nash equilibrium points. *Proceedings of the American Mathematical Society*, 3(1):170–174, 1952.
- [43] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio. Generative adversarial nets. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 27, pages 2672–2680, 2014.
- [44] O. A. Gross and R. E. Wagner. A continuous colonel blotto game. *RAND Corporation Research Memorandum*, (RM-408), 1950.
- [45] Z. Han, D. Niyato, W. Saad, and T. Başar. *Game Theory for Next Generation Wireless and Communication Networks: Modeling, Analysis, and Design*. Cambridge University Press, 2019.
- [46] Z. Han, D. Niyato, W. Saad, T. Başar, and A. Hjørungnes. *Game Theory in Wireless and Communication Networks: Theory, Models, and Applications*. Cambridge University Press, 2011.
- [47] Z. Han, C. Pandana, and K. R. Liu. Distributive opportunistic spectrum access for cognitive radio using correlated equilibrium and no-regret learning. In *2007 IEEE Wireless Communications and Networking Conference*, pages 11–15. IEEE, 2007.
- [48] L. P. Hansen and T. J. Sargent. *Robustness*. Princeton University Press, Princeton, NJ, 2008.
- [49] F. Harary. The maximum connectivity of a graph. *Proceedings of the National Academy of Sciences*, 48(7):1142–1146, 1962.
- [50] J. C. Harsanyi. Games with incomplete information played by “Bayesian” players, parts I–III. *Management Science*, 14(3):159–182, 320–334, 486–502, 1967–1968. Part I: The Basic Model; Part II: Bayesian Equilibrium Points; Part III: The Basic Probability Distribution of the Game.
- [51] J. C. Harsanyi. Games with randomly disturbed payoffs: A new rationale for mixed-strategy equilibrium points. *International Journal of Game Theory*, 2(1):1–23, 1973.
- [52] S. Hart and A. Mas-Colell. A simple adaptive procedure leading to correlated equilibrium. *Econometrica*, 68(5):1127–1150, 2000.
- [53] A. Hatcher. *Algebraic Topology*. Cambridge University Press, 2002.
- [54] S. Haykin. Cognitive radio: brain-empowered wireless communications. *IEEE Journal on Selected Areas in Communications*, 23(2):201–220, 2005.
- [55] J. Hofbauer and K. Sigmund. *Evolutionary Games and Population Dynamics*. Cambridge University Press, Cambridge, 1998.
- [56] Y. Huang, J. Chen, L. Huang, and Q. Zhu. Dynamic games for secure and resilient control system design. *National Science Review*, 7(7):1125–1141, 2020.
- [57] R. Isaacs. *Differential Games: A Mathematical Theory with Applications to Warfare and Pursuit, Control and Optimization*. Wiley, New York, 1965.

- [58] R. Isaacs. *Differential Games*. Krieger, New York, NY, 2nd edition, 1975. First edition: Wiley, New York, 1965.
- [59] J. J. F. Nash. Equilibrium points in n-person games. *Proc. National Academy of Sciences*, 36(1):48–49, Jan. 1950.
- [60] J. J. F. Nash. Non-cooperative games. *Annals of Mathematics*, 54(2):286–295, Sept. 1951.
- [61] S. Kakutani. A generalization of Brouwer’s fixed point theorem. *Duke Mathematical Journal*, 8(3):457–459, 1941.
- [62] C. A. Kamhoua, C. D. Kiekintveld, F. Fang, and Q. Zhu. *Game theory and machine learning for cyber security*. John Wiley & Sons, 2021.
- [63] C. Kiekintveld, M. Jain, J. Tsai, J. Pita, F. Ordóñez, and M. Tambe. Computing optimal randomized resource allocations for massive security games. In *Proceedings of the 8th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, pages 689–696, 2009.
- [64] E. Kohlberg and J.-F. Mertens. On the strategic stability of equilibria. *Econometrica*, 54:1003–1037, 1986.
- [65] D. Koller and N. Megiddo. The complexity of two-person zero-sum games in extensive form. In *Proceedings of the 24th Annual ACM Symposium on Theory of Computing (STOC)*, pages 263–272. ACM, 1992.
- [66] D. Korzhyk, Z. Yin, C. Kiekintveld, V. Conitzer, and M. Tambe. Stackelberg vs. Nash in security games: An extended investigation of interchangeability, equivalence, and uniqueness. *Journal of Artificial Intelligence Research*, 41:297–327, 2011.
- [67] E. Koutsoupias and C. Papadimitriou. Worst-case equilibria. *Proceedings of the 16th Annual Symposium on Theoretical Aspects of Computer Science (STACS)*, pages 404–413, 1999.
- [68] D. M. Kreps and R. Wilson. Sequential equilibria. *Econometrica*, 50(4):863–894, 1982.
- [69] H. W. Kuhn. Extensive games and the problem of information. In H. W. Kuhn and A. W. Tucker, editors, *Contributions to the Theory of Games, Volume II*, volume 28 of *Annals of Mathematics Studies*, pages 193–216. Princeton University Press, Princeton, NJ, 1953.
- [70] T. Li, G. Peng, Q. Zhu, and T. Başar. The confluence of networks, games, and learning: A game-theoretic framework for multiagent decision making over networks. *IEEE Control Systems Magazine*, 42(4):35–67, 2022.
- [71] M. L. Littman. Markov games as a framework for multi-agent reinforcement learning. In *Proceedings of the 11th International Conference on Machine Learning*, pages 157–163, 1994.
- [72] S. Maharjan, Q. Zhu, Y. Zhang, S. Gjessing, and T. Başar. Dependable demand response management in the smart grid: A Stackelberg game approach. *IEEE Transactions on Smart Grid*, 4(1):120–132, 2013.
- [73] S. Mannor and N. Shimkin. Reinforcement learning and correlated equilibria. *Neural Computation*, 15(3):747–763, 2003.
- [74] M. H. Manshaei, Q. Zhu, T. Alpcan, T. Başar, and J.-P. Hubaux. Game theory meets network security and privacy. *ACM Computing Surveys*, 45(3):1–39, 2013.
- [75] P. Marbach. Priority service and max-min fairness. In *Proceedings. Twenty-First Annual Joint Conference of the IEEE Computer and Communications Societies*, volume 1, pages 266–275. IEEE, 2002.

- [76] J. R. Marden, G. Arslan, and J. S. Shamma. Cooperative control and potential games. *IEEE Transactions on Systems, Man, and Cybernetics, Part B (Cybernetics)*, 39(6):1393–1407, 2009.
- [77] E. Maskin. Nash equilibrium and welfare optimality. *Review of Economic Studies*, 66(1):23–38, 1999.
- [78] D. Monderer and L. Shapley. Potential games. *Games and Economic Behavior*, 14:124–143, 1996.
- [79] D. Monderer and L. S. Shapley. Fictitious play property for games with identical interests. *Journal of Economic Theory*, 68(1):258–265, 1996.
- [80] J. R. Munkres. *Topology*. Prentice Hall, 2nd edition, 2000.
- [81] R. B. Myerson. Refinements of the Nash equilibrium concept. *International Journal of Game Theory*, 7(2):73–80, 1978.
- [82] R. B. Myerson. Incentive compatibility and the bargaining problem. *Econometrica*, 47(1):61–73, 1979.
- [83] R. B. Myerson. Optimal auction design. *Mathematics of Operations Research*, 6(1):58–73, 1981.
- [84] R. B. Myerson. *Game Theory: Analysis of Conflict*. Harvard University Press, Cambridge, MA, 1991.
- [85] B. Nalebuff. Brinkmanship and nuclear deterrence. *Conflict Management and Peace Science*, 9(2):19–30, 1989.
- [86] N. Nisan, T. Roughgarden, E. Tardos, and V. V. Vazirani, editors. *Algorithmic Game Theory*. Cambridge University Press, Cambridge, UK, 2007.
- [87] M. A. Nowak. *Evolutionary Dynamics: Exploring the Equations of Life*. Harvard University Press, Cambridge, MA, 2006.
- [88] M. A. Nowak. Five rules for the evolution of cooperation. *Science*, 314(5805):1560–1563, 2006.
- [89] R. S. A. of Sciences. The crafoord prize in biosciences 1999: John maynard smith, ernst mayr, george c. williams. <https://www.crafoordprize.se/prize/biology-1999>, 1999. Accessed: 2025-09-02.
- [90] M. Osborne and A. Rubinstein. *A Course in Game Theory*. MIT Press, 1994.
- [91] G. Owen. *Game Theory*. Emerald Group Publishing Limited, Bingley, UK, 4 edition, 2013.
- [92] Y. Pan and Q. Zhu. Extending no-regret hopping in FMCW radar interference avoidance. *ACM SIGMETRICS Performance Evaluation Review*, 53(2):131–133, 2025.
- [93] P. Paruchuri, J. P. Pearce, J. Marecki, M. Tambe, F. Ordóñez, and S. Kraus. Playing games for security: An efficient exact algorithm for solving Bayesian Stackelberg games. In *Proceedings of the 7th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, pages 895–902, 2008.
- [94] J. Pawlick and Q. Zhu. *Game theory for cyber deception*. Birkhäuser, 2021.
- [95] L. S. Pontryagin, V. G. Boltyanskii, R. V. Gamkrelidze, and E. F. Mishchenko. *The Mathematical Theory of Optimal Processes*. Interscience Publishers, New York, 1962.
- [96] R. Powell. *Nuclear Deterrence Theory: The Search for Credibility*. Cambridge University Press, 1990.
- [97] B. Radunovic and J.-Y. Le Boudec. A unified framework for max-min and min-max fairness with applications. *IEEE/ACM Transactions on networking*, 15(5):1073–1083, 2007.

- [98] T. E. S. Raghavan. Zero-sum two-person games. In R. J. Aumann and S. Hart, editors, *Handbook of Game Theory with Economic Applications*, volume 2, pages 735–768. Elsevier, 1994.
- [99] M. T. Ribeiro, S. Singh, and C. Guestrin. "why should i trust you?": Explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD '16)*, pages 1135–1144, New York, NY, USA, 2016. ACM.
- [100] B. Roberson. The colonel blotto game. *Economic Theory*, 29(1):1–24, 2006.
- [101] J. Robinson. An iterative method of solving a game. *Annals of Mathematics*, 54(2):296–301, 1951.
- [102] R. W. Rosenthal. A class of games possessing pure-strategy Nash equilibria. *International Journal of Game Theory*, 2(1):65–67, 1973.
- [103] A. E. Roth and I. Erev. Learning in extensive-form games: Experimental data and simple dynamic models in the intermediate term. *Games and Economic Behavior*, 8(1):164–212, 1995.
- [104] W. Rudin. *Principles of Mathematical Analysis*. McGraw–Hill, New York, NY, 3rd edition, 1976.
- [105] T. C. Schelling. *The Strategy of Conflict*. Harvard University Press, Cambridge, MA, 1960.
- [106] R. Selten. Spieltheoretische behandlung eines oligopolmodells mit nachfrageträgheit. *Zeitschrift für die gesamte Staatswissenschaft / Journal of Institutional and Theoretical Economics*, 121(2):301–324, 1965.
- [107] R. Selten. Reexamination of the perfectness concept for equilibrium points in extensive games. *International Journal of Game Theory*, 4(1):25–55, 1975.
- [108] J. S. Shamma and G. Arslan. Unified convergence proofs of continuous-time fictitious play. *IEEE Transactions on Automatic Control*, 49(7):1137–1142, July 2004.
- [109] J. S. Shamma and G. Arslan. Dynamic fictitious play, dynamic gradient play, and distributed convergence to Nash equilibria. *IEEE Transactions on Automatic Control*, 50(3):312–327, Mar. 2005.
- [110] L. S. Shapley. Some topics in two-person games. In M. Dresher and A. W. Tucker, editors, *Advances in Game Theory*, pages 1–29. Princeton University Press, Princeton, NJ, 1964.
- [111] Y. Shoham and K. Leyton-Brown. *Multiagent Systems: Algorithmic, Game-Theoretic, and Logical Foundations*. Cambridge University Press, Cambridge, UK, 2009.
- [112] A. Shrikumar, P. Greenside, and A. Kundaje. Learning important features through propagating activation differences. In D. Precup and Y. W. Teh, editors, *Proceedings of the 34th International Conference on Machine Learning (ICML '17)*, volume 70 of *Proceedings of Machine Learning Research*, pages 3145–3153. PMLR, 2017.
- [113] D. Silver, A. Huang, C. J. Maddison, A. Guez, L. Sifre, G. van den Driessche, J. Schrittwieser, I. Antonoglou, V. Panneershelvam, M. Lanctot, S. Dieleman, D. Grewe, J. Nham, N. Kalchbrenner, I. Sutskever, T. Lillicrap, M. Leach, K. Kavukcuoglu, T. Graepel, and D. Hassabis. Mastering the game of go with deep neural networks and tree search. *Nature*, 529(7587):484–489, 2016.
- [114] D. Silver, T. Hubert, J. Schrittwieser, I. Antonoglou, M. Lai, A. Guez, M. Lanctot, L. Sifre, D. Kumaran, T. Graepel, T. Lillicrap, K. Simonyan, and D. Hassabis. A general reinforcement learning algorithm that masters chess, shogi, and go through self-play. *Science*, 362(6419):1140–1144, 2018.
- [115] D. Silver, J. Schrittwieser, K. Simonyan, I. Antonoglou, A. Huang, A. Guez, T. Hubert, L. Baker, M. Lai, A. Bolton, Y. Chen, T. Lillicrap, F. Hui, L. Sifre, G. van den Driessche, T. Graepel, and D. Hassabis. Mastering the game of go without human knowledge. *Nature*, 550(7676):354–359, 2017.

- [116] M. Simaan and J. B. Cruz, Jr. Additional aspects of the Stackelberg strategy in nonzero-sum games. *Journal of Optimization Theory and Applications*, 11(6):613–626, 1973.
- [117] M. Simaan and J. B. Cruz, Jr. On the Stackelberg strategy in nonzero-sum games. *Journal of Optimization Theory and Applications*, 11(5):533–555, 1973.
- [118] A. Sinha, F. Fang, B. An, C. Kiekintveld, and M. Tambe. Stackelberg security games: Looking beyond a decade of success. In *Proceedings of the 27th International Joint Conference on Artificial Intelligence (IJCAI)*, pages 5494–5501, 2018.
- [119] J. M. Smith. *Evolution and the Theory of Games*. Cambridge University Press, Cambridge, UK, 1982.
- [120] J. M. Smith and G. R. Price. The logic of animal conflict. *Nature*, 246:15–18, 1973.
- [121] J. M. Smith and E. Szathmáry. *The Origins of Life: From the Birth of Life to the Origin of Language*. Oxford University Press, Oxford, 2003.
- [122] K. Staňková, J. S. Brown, W. S. Dalton, and R. A. Gatenby. Optimizing cancer treatment using game theory: A review. *JAMA Oncology*, 5(1):96–103, 2019.
- [123] R. K. Sundaram. *A First Course in Optimization Theory*. Cambridge University Press, Cambridge, UK, 1996.
- [124] M. Tambe. *Security and Game Theory: Algorithms, Deployed Systems, Lessons Learned*. Cambridge University Press, Cambridge, UK, 2011.
- [125] P. D. Taylor and L. B. Jonker. Evolutionary stable strategies and game dynamics. *Mathematical Biosciences*, 40(1–2):145–156, 1978.
- [126] The Nobel Prize. The Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel 1994. <https://www.nobelprize.org/prizes/economic-sciences/1994/summary/>, 1994. John C. Harsanyi, John F. Nash Jr., Reinhard Selten.
- [127] The Nobel Prize. The Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel 2004. <https://www.nobelprize.org/prizes/economic-sciences/2004/summary/>, 2004. Finn E. Kydland, Edward C. Prescott.
- [128] The Nobel Prize. The Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel 2005. <https://www.nobelprize.org/prizes/economic-sciences/2005/summary/>, 2005. Robert J. Aumann, Thomas C. Schelling.
- [129] The Nobel Prize. The Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel 2007. <https://www.nobelprize.org/prizes/economic-sciences/2007/summary/>, 2007. Leonid Hurwicz, Eric S. Maskin, Roger B. Myerson.
- [130] The Nobel Prize. The Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel 2012. <https://www.nobelprize.org/prizes/economic-sciences/2012/summary/>, 2012. Alvin E. Roth, Lloyd S. Shapley.
- [131] The Nobel Prize. The Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel 2016. <https://www.nobelprize.org/prizes/economic-sciences/2016/summary/>, 2016. Oliver Hart, Bengt Holmström.
- [132] The Nobel Prize. The Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel 2017. <https://www.nobelprize.org/prizes/economic-sciences/2017/summary/>, 2017. Richard Thaler.

- [133] The Nobel Prize. The Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel 2020. <https://www.nobelprize.org/prizes/economic-sciences/2020/summary/>, 2020. Paul R. Milgrom, Robert B. Wilson.
- [134] F. B. Thompson. Equivalence of games in extensive form. Technical Report RM-759, RAND Corporation, Santa Monica, CA, 1952. RAND Memorandum.
- [135] D. M. Topkis. Equilibrium points in nonzero-sum n -person submodular games. *SIAM Journal on Control and Optimization*, 17(6):773–787, 1979.
- [136] E. van Damme. A relationship between perfect equilibria in extensive games and proper equilibria in normal form. In R. Selten, editor, *Reexamination of the perfectness concept for equilibrium points in extensive games*, volume 13, pages 1–13. 1984.
- [137] J. von Neumann. Zur theorie der gesellschaftsspiele. *Mathematische Annalen*, 100:295–320, 1928.
- [138] J. von Neumann and O. Morgenstern. *Theory of Games and Economic Behavior*. Princeton University Press, Princeton, NJ, 2nd edition, 1947. First edition: 1944.
- [139] H. von Stackelberg. *Marktform und Gleichgewicht*. Springer, Vienna, 1934. English translation: *Market Structure and Equilibrium*, Springer, Berlin, 2011.
- [140] B. von Stengel. Efficient computation of behavior strategies. *Games and Economic Behavior*, 14(2):220–246, 1996.
- [141] B. von Stengel and S. Zamir. Leadership games with convex strategy sets. *Games and Economic Behavior*, 69(2):446–457, 2010.
- [142] N. N. Vorob’ev. Game theory: Lectures for economists and systems scientists. *Springer Series in Statistics*, 1977. Translated from Russian; original title: *Teoriya Igr*.
- [143] J. W. Weibull. *Evolutionary Game Theory*. MIT Press, Cambridge, MA, 1997.
- [144] B. Wölfl, H. te Rietmole, M. Salvioi, A. Kaznatcheev, F. Thuijsman, J. S. Brown, B. Burgering, and K. Staňková. The contribution of evolutionary game theory to understanding and treating cancer. *Dynamic Games and Applications*, 12(2):313–342, 2022.
- [145] S. Yüksel and T. Başar. *Stochastic teams, games, and control under information constraints*. Springer, 2024.
- [146] J. Zhang, J. J. Cunningham, J. S. Brown, and R. A. Gatenby. Integrating evolutionary dynamics into treatment of metastatic castrate-resistant prostate cancer. *Nature Communications*, 8:1816, 2017.
- [147] Q. Zhu and T. Basar. Game-theoretic methods for robustness, security, and resilience of cyberphysical control systems: games-in-games principle for optimal cross-layer resilient control systems. *IEEE Control Systems Magazine*, 35(1):46–65, 2015.
- [148] Q. Zhu and T. Başar. Revisiting game-theoretic control in socio-technical networks: Emerging design frameworks and contemporary applications. *IEEE Control Systems Letters*, 2025.
- [149] Q. Zhu, Z. Han, and T. Başar. No-regret learning in collaborative spectrum sensing with malicious nodes. In *2010 IEEE International Conference on Communications*, pages 1–6. IEEE, 2010.
- [150] Q. Zhu and Z. Xu. Cross-layer design for secure and resilient cyber-physical systems. *Cham, Switzerland: Springer*, 2020.
- [151] M. Zinkevich, M. Johanson, M. Bowling, and C. Piccione. Regret minimization in games with incomplete information. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 20, pages 1729–1736, 2007.